

I SEMESTER

S. No	Course code	Course Title	Credit load
1	AGR 101	Fundamentals of Agronomy and Agricultural Heritage	1+1
2	BIC 101	Fundamentals of Plant Biochemistry	2+1
3	SAC 101	Fundamentals of Soil Science	2+1
4	FOR 111	Introduction to Forestry	1+1
5	ENG 101	Comprehension & Communication Skills in English	1+1
6	HOR 111	Fundamentals of Horticulture	1+1
7	MAT 111	Elementary Mathematics	1+1
8	PBG 101	Introduction to Agricultural Botany	1+1
9	AEX101	Rural Sociology & Educational Psychology	2+0
10	TAM101/ ENG103	,yf;fpa';fspy; ntshz;ika[k; mwptpay; jkpH; gadhf;fKk; / Development Education Development Education	0+1
11	NSS/NCC 101	NSS/NCC	0+1*
12	PED 101	Physical Education	0+1*
13	PED102	Yoga for human excellence	0+1*
			12+9=21
		*Non-gradual courses compulsory courses	

AGR 101 Fundamentals of Agronomy and Agricultural Heritage (1+1)

Unit - I: Importance of agriculture

Agriculture - Definition - Importance and scope - Branches of agriculture - Evolution of human and agriculture - History of agricultural development in the World and India.

Unit - II: Agricultural heritage

Agriculture heritage - Agriculture in ancient India - Chronological agricultural technology development in India - Kautilya's Arthashastra - Sangam literature - Kambar Eazhupathu - Development of scientific Agriculture - National and International Agricultural Research Institutes in India - Indian agriculture.

Unit - III: Agroclimatic zones, crops and soils

Agronomy - Definition - Importance and scope - Agro-climatic zones of Tamil Nadu - Agro ecological zones of India - Crops and their classification - Economic and agronomic - Major crops of India and Tamil Nadu - Major soils of Tamil Nadu - Factors affecting crop production - climatic - edaphic - biotic - physiographic and socio economic factors.

Unit - IV: Tillage and after cultivation

Tillage - Definition - Types - Objectives - Modern concepts of tillage - Main field preparations - Seeds - seed rate - sowing methods - Crop establishment methods - Planting geometry and its effect on growth and yield - After cultivation - Thinning - Gap filling - Weeds - Definition - Weed control methods.

Unit - V: Cropping and farming systems

Manures and fertilizers (organic, in-organic, green manure) - time and method of application - Irrigation - Principles and concepts - Cropping patterns and cropping systems - Sustainable agriculture - integrated farming systems - Organic agriculture - Principles and concepts - Dry farming - Principles and concepts.

Practical:

Visit to college farm - Study of farm features and measurements - identification of crops and seeds - working out seed rate - Study of seed treatment practices - Study of tillage implements; practicing ploughing, puddling operations, practicing seeding different methods of sowing and planting - Study and practicing inter-cultivation implements; Practicing fertilizer applications - Participation in ongoing field operations.

Theory Lecture Schedule:

1. Agriculture - Definition - Importance and scope - Branches of agriculture - Evolution of man and agriculture.
2. Indian agriculture - Indian economy - National income - per capita income - Agricultural income in GDP - Women in agriculture and empowerment.
3. History of agricultural development in the world and India. Agriculture heritage - Agriculture in ancient India.
4. Agriculture heritage - Agriculture in ancient India.
5. Chronological agricultural technology development in India. Kautilya's Arthashastra - Sangam literature - rainfall prediction - ITK - Tamil Almanac.

6. Development of scientific agriculture - National and International Agricultural Research Institutes.
7. Agronomy - definition - meaning and scope. Agro-climatic zones of India and Tamil Nadu - Agro ecological zones of India and Tamil Nadu.
8. Crops and major soils - classification - Economic and agricultural importance in Tamil Nadu and India.
9. **Mid Semester Examination**
10. Factors affecting crop production - climatic - edaphic - biotic- physiographic and socio economic factors.
11. Tillage - Definition - objectives - types of tillage - modern concepts of tillage - main field preparation.
12. Seeds - Seed rate - sowing methods - Germination - Crop stand establishment - Planting geometry.
13. Weeds - Definition - harmful and beneficial effects of weeds - crop weed competition and management of weeds - IWM.
14. Role of manures and fertilizers in crop production - Inter cultivation - Thinning - gap filling and other intercultural operations.
15. Irrigation - time and methods - Modern techniques of irrigation - Drainage and its importance.
16. Cropping patterns and cropping system - intensive cropping - sustainable agriculture – IFS.
17. Organic / eco - friendly agriculture - Dry farming- principles and concepts.

Practical schedule:

1. Visit to college farm to observe wetland farming system and identification of crops.
2. Visit to college farm to observe garden land and dry land farming systems and identification of crops.
3. Identification of seeds, manures, fertilizers, green manures and green leaf manures.
4. Identification of tools and implements.
5. Acquiring skill in handling primary and secondary tillage implements.
6. Practicing different methods of land configuration for raising nursery for wet land crops.
7. Practicing different methods of land configuration for raising nursery for garden land crops.
8. Practicing different methods of seed treatments, methods of sowing and seeding implements.
9. Working out seed rates and practicing thinning, gap filling operations for optimum crop stand and intercultural operations.
10. Working out manure and fertilizer requirement of crops.
11. Practicing methods of application: manures and fertilizers and incorporation of green manure and green leaf manure.
12. Identification of weeds, weeding practices and handling of weeding tools and implements.
13. Observing various irrigation methods.
14. Practicing harvesting operations in major field crops.
15. Participation in on-going field operations during on campus / off campus visit.
16. Visit to nearby Agricultural Research station.
17. **Final Practical Examination.**

References:

- Yellamananda Reddy, T. and G.H. Sankara Reddi. 1997. Principles of Agronomy. Kalyani Publishers, New Delhi.
- Sankaran, S. and V.T. Subbiah Mudaliar. 1997. Principles of Agronomy. The Bangalore Printing and Publishing Co. Ltd., Bangalore.
- Reddy, S.R. Principles of Agronomy. 2016. Kalyani Publishers, New Delhi.
- Somasundaram, E. 2017. Agronomy: Principles and Practices. New India Publishing agency, New Delhi.
- ICAR. 2015. Handbook of Agriculture. Indian Council of Agricultural Research, New Delhi.

E-References:

1. <http://icar.res.in>
2. www.webcast.gov.in
3. www.icar.org.in/nasm.html

BIC 101 Fundamentals of Biochemistry (2+1)

UNIT I

Carbohydrates

Carbohydrates - occurrence and classification. Structure of monosaccharides, **oligosaccharides** and polysaccharides. Physical and chemical properties of carbohydrates – optical isomerism, optical activity, mutarotation, reducing property, reaction with acids and alkalies. **Glycoconjugates - Glycoproteins and Lectin - structure and significance.**

UNIT II

Lipids

Lipids - occurrence and classification. Storage lipids - fatty acids, triacyl glycerol, essential fatty acids, waxes. **Structural lipids - role of lipids in biological membrane - glycolipids** and phospholipids - types and importance; Sterols - basic structure and their importance. Physical and chemical constants of oils. Rancidity of oils.

UNIT III

Proteins and Enzymes

Amino acids - classification and structure. Essential amino acids. Properties of amino acids - amphoteric nature and isomerism. Classification of proteins based on functions and solubility. Structure of proteins: primary structure, secondary structure, tertiary structure and quaternary structure - **protein folding and denaturation.** Properties and reactions of proteins. Enzymes - Properties, classification and nomenclature. Mechanism of enzyme action. Factors affecting enzyme activity. Enzyme inhibition - Competitive, Non-competitive and Uncompetitive inhibition; **Allosteric enzymes.** Coenzymes, cofactors and isoenzyme.

UNIT IV

Metabolism

Carbohydrate metabolism - breakdown of starch by amylases, glycolysis, TCA cycle and pentose phosphate pathway. Respiration - electron transport chain and oxidative phosphorylation. Bioenergetics of glucose. Lipid metabolism - lipases and phospholipases. Beta-oxidation of fatty acids and bioenergetics. Biosynthesis of fatty acids and triacyl glycerol. General catabolic pathway for amino acids - transamination, deamination and decarboxylation. Ammonia assimilating enzymes. Metabolic inter-relationship.

UNIT V

Secondary metabolites

Secondary metabolites - occurrence, classification and functions of phenolics, terpenes and alkaloids.

Lecture schedule:

1. Introduction to Biochemistry, Carbohydrates – occurrence and classification R2: 1-4, 66-72.
2. Structure of monosaccharides. R2: 75-82.
3. Structure of oligosaccharides and polysaccharides. R2: 82-90.
4. Physical properties of carbohydrates - Mutarotation, optical activity, isomerism. R2: 73-78.
5. Chemical reactions of carbohydrates. R2: 90-95.
6. Glycoproteins and lectin - structure and significance. R1: 316-321.
7. Lipids - occurrence and classification. R2: 99-100.
8. Storage lipids - Fatty acids and triacyl glycerol. Essential fatty acids, waxes. R2: 101-106.
9. Structural lipids - Glycolipids and phospholipids - types and importance. R2: 107-111.

10. Sterols - basic structure and their importance. R2: 111-114.
11. Physical and chemical constants of oils. Rancidity of oils. R2: 114-119.
12. Amino acids - Classification and structure. R2: 17-21.
13. Properties of amino acids - amphoteric nature, isomerism, essential amino acids. R2: 21-26.
14. Classification of proteins based on function and solubility. R2: 26-31.
15. Structure of protein - Primary, secondary, tertiary and quaternary structure. R2: 31-41.
16. Protein folding, physical and chemical properties of proteins. R2: 41-43, R1: 52-55.
17. **Mid Semester Examination**
18. Enzymes - Properties, classification and nomenclature. R2: 123-127.
19. Mechanism of enzyme action. R2: 129-131.
20. Factors affecting enzyme activity. R2: 131-136.
21. Enzyme inhibition - competitive, non-competitive, uncompetitive and allosteric enzymes. R2: 136-137, R1: 224-225.
22. Coenzymes, cofactors and isoenzyme. R2: 127-129, 138.
23. Carbohydrate metabolism - breakdown of starch by amylases, Glycolysis - Reactions and bioenergetics. R2:159-164.
24. TCA cycle - Reactions and bioenergetics. R2: 164-168.
25. Pentose phosphate pathway - Reactions . R2: 174-177.
26. Respiration - electron transport chain and oxidative phosphorylation. R2: 170-173.
27. Lipid metabolism - lipases and phospholipases. R2: 205-208.
28. Beta-oxidation of fatty acids and bioenergetics. R2: 208-212.
29. Biosynthesis of fattyacids and triacylglycerol. R2: 213- 220.
30. Transamination, deamination and decarboxylation of amino acids. R2: 224-231.
31. Ammonia assimilating enzymes - GS, GOGAT and GDH. R2: 231-233.
32. Metabolic inter-relationship. R2: 287-289.
33. Secondary metabolites - occurrence, classification and functions of phenolics. R2: 274-276.
34. Occurrence, classification and functions of terpenes and alkaloids. R2: 277-280.

Practical Schedule:

1. Qualitative analysis of carbohydrates
2. Estimation of starch
3. Estimation of amylose
4. Determination of reducing sugars
5. Qualitative analysis of amino acids
6. Sorenson's formal titration of amino acids
7. Estimation of amino acids by Ninhydrin method
8. Estimation of protein by Biuret method
9. Determination of free fatty acid of an oil
10. Determination of iodine number of an oil
11. Estimation of ascorbic acid by dye method
12. Assay of amylase
13. Estimation of total phenols
14. **Extraction and estimation of lycopene and carotenoids**
15. Separation of amino acids by paper chromatography
16. Separation of **phenols** by thin layer chromatography
17. **Final Practical Examination**

References

1. Berg JM, Tymoczko JL and Stryer L, (2007), Biochemistry, 7th Ed. Wiley Eastern Ltd. ISBN:0-7167-8724-5.
2. Thayumanavan, B, Krishnaveni, S and Parvathi, K, (2004), Biochemistry for Agricultural Sciences, Galgotia Publications Pvt Ltd., New Delhi. ISBN :81-7515-459-4.

E-References:

1. Cox, MM and Nelson, DL. (2011), Principles of Biochemistry, Fourth (Indian edition) Macmillian, Worth Publishers. <http://bcs.whfreeman.com/lehninger6e> - Web links/ Tutorials/ Lecture companion Art
2. Harper's illustrated Biochemistry -<https://freemedebbooks.files.wordpress.com/2014/01/harpers-illustrated-biochemistry-28th-edition.pdf>
3. J M Berg, J L Tymoczko and L Stryer , Biochemistry, Sixth Edition - <http://www.irb.hr/users/precali/> Znanost.o.Moru/ Biokemija/Literatura/Lubert%20Stryer%20-%20Biochemistry.pdf
4. Sadasivam, S and Manickam, A. (2009), Biochemical Methods, 3rd Edn, New Age International.
5. Wilson, K. and Walker, J.M. (2000), Principles and techniques of Practical Biochemistry, 5th Edn. – Cambridge University Press.
6. www.ncbi.nlm.nih.gov

SAC 101 Fundamentals of Soil Science (2+1)

Unit I

Soil as a natural body, Pedological and edaphological concepts of soil. Components of soil. Soil genesis: Composition of Earth's crust- soil forming rocks and minerals – Primary and secondary minerals. Weathering of rocks and minerals. Factors of soil formation. Soil forming processes. Soil Profile.

Unit II

Soil physical properties: Soil texture, structure, density and porosity, soil colour, consistence and plasticity. Soil water retention, movement and availability. Soil air, composition, gaseous exchange-problem and its effect on crop growth. Source, amount and flow of heat in soil, Soil temperature and crop growth.

Unit III

Soil physico chemical and chemical properties: Soil reaction-pH, soil acidity and alkalinity, buffering, effect of pH on nutrient availability. Electrical conductivity. Soil colloids - inorganic and organic. Silicate clays: constitution and properties, sources of charge, ion exchange, cation and anion exchange capacity and base saturation.

Unit IV

Soil organic matter: composition, properties and its influence on soil properties. Humic substances - nature and properties. Soil Biology : Soil organisms : macro and micro organisms, their beneficial and harmful effects. Soil enzymes. Soil pollution – Types and behaviour of pesticides. Inorganic contaminants. Prevention and mitigation of soil pollution.

PRACTICAL SCHEDULE

Study of soil profile in field. Study of soil sampling tools, collection of representative soil sample, its processing and storage. Study of soil forming rocks and minerals. Determination of soil density, moisture content and porosity. Determination of soil colour. Determination of soil texture by feel and Bouyoucos Methods. Studies of capillary rise phenomenon of water in soil column and water movement in soil. Demonstration of heat transfer in soil. Preparation and standardization of laboratory reagents, indicators and buffers. Determination of soil pH and electrical conductivity. Determination of cation exchange capacity of soil. Estimation of organic matter content of soil. Study of soil map.

Lecture Schedule:

1. Soil definition - Soil as a three dimensional natural body, Pedological and edaphological concepts of soil
2. Components of soil – soil a three phase system- Composition of Earth's crust.
3. Soil genesis: soil forming rocks-definition, formation, Classification of rocks- igneous, sedimentary and metamorphic rocks
4. Brief description of important rocks - mineralogical composition
5. Minerals- definition, occurrence, classification of important soil forming primary minerals - silicate and non silicate minerals, ferro and non-ferro magnesium minerals
6. Formation of secondary minerals - clay minerals and amorphous minerals
7. Weathering - Rocks and minerals - Physical, chemical and biological weathering
8. Factors of soil formation- Passive and active soil forming factors soil forming factors
9. Soil forming process- Fundamental - Simenson's four fold soil forming process -eluviation, illuviation, translocation and humification

10. Specific Soil forming processes - podzolization, laterization, salinization, alkalization, calcification, decalcification and pedoturbation
11. Soil Profile – Horizons, Master horizons and subordinate horizons, subdivisions, Lithological discontinuity.
12. Soil physical properties: Soil texture - particle size distribution - textural classes - textural triangular diagram - significance of soil texture
13. Soil structure - classification - genesis - factors influencing structural stability - significance of soil structure
14. Soil bulk density, particle density and porosity - factors influencing – significance.
15. Soil colour - causes and measurement - Munsell colour chart - factors influencing soil colour – Significance of soil colour.
16. Soil consistence - cohesion, adhesion, plasticity, Atterberg's constants - upper and lower plastic limits, plasticity number- significance of soil consistence
17. **Mid semester Examination**
18. Soil water- forms of water, units of expression and pF scale
19. Soil water potentials - gravitational, matric, osmotic- Soil moisture constants and Soil moisture measurements.
20. Movement of soil water - Saturated and unsaturated flow - infiltration, hydraulic conductivity, percolation, permeability and drainage
21. Soil air, composition, gaseous exchange – Problem and its effect on crop growth.
22. Source, amount and flow of heat in soil, soil temperature and crop growth. and crop growth.
23. Soil reaction (pH) - definition, pH scale, soil acidity and alkalinity, buffering, effect of pH on nutrient availability and factors affecting soil pH
24. Soil Electrical Conductivity - Factors affecting EC – its significance
25. Soil colloids - inorganic and organic
26. Silicate clays: constitution and classification - 1:1, 2:1 expanding and non expanding - 2:2 clay minerals, amorphous minerals and their properties
27. Sources of charge, ion exchange – positive and negative charge – isomorphous substitution, pH dependant charge.
28. Ion exchange - Cation and anion exchange capacity and base saturation
29. Soil organic matter: composition, properties and its influence on soil properties
30. Humic substances – fractionation, nature and properties, Theories of humus formation.
31. Soil Biology- Soil organisms: macro and micro organisms, their beneficial and harmful effects, Soil enzymes
32. Soil carbon sequestration and carbon trading
33. Soil pollution - behaviour of pesticides and inorganic contaminants
34. Prevention and mitigation of soil pollution

Practical schedule:

1. Study of soil sampling tools, collection of representative soil sample, its processing and storage.
2. Study of soil profile in field.
3. Study of soil forming rocks and minerals.
4. Determination of soil density and porosity.
5. Determination of soil colour and moisture content and porosity.
6. Determination of soil texture by feel and Bouyoucos Methods
7. Determination of soil texture by International pipette method.
8. Studies of capillary rise phenomenon of water in soil column and water movement in soil (Infiltration Rate)

9. Studies of capillary rise phenomenon of water in soil column and water movement in soil (Hydraulic conductivity)
10. Determination of soil temperature and demonstration of heat transfer.
11. Preparation and standardization of laboratory reagents, indicators and buffers
12. Determination of soil pH and electrical conductivity.
13. Determination of cation exchange capacity of soil - I.
14. Determination of cation exchange capacity of soil - II
15. Estimation of soil organic carbon.
16. Study of soil map (India and Tamil Nadu)

17. Final Practical Examination

References

1. Brady, N.C. and Raymond, C.Weil. 2013. The Nature and Properties of Soils (14th Edition). Pearson Education, Inc. Publishing as Prentice Hall.
2. Fundamentals of Soil Science. 2009. ISSS Publication, New Delhi.
3. Sehgal, J. 2005. Pedology concepts and applications, Kalyani Publishers, New Delhi.
4. Das, D.K. 2013. Introductory Soil science, Kalyani Publishers, New Delhi.

e-references

1. <http://www.sciencedirect.com/science/books>
2. <http://202.200.144.17/sykc/hjx/content/ckzl/6/2.pdf>
3. http://www.pedosphere.com/volume01/pdf/Section._01.pdf
4. [http://waterquality.montana.edu/docs/homeowners/Septic Drain field Soil Suitability, Presentations /6 _Soil Texture and _Structure.pdf](http://waterquality.montana.edu/docs/homeowners/Septic_Drain_field_Soil_Suitability_Presentations/6_Soil_Texture_and_Structure.pdf)
5. http://wfrec.ifas.ufl.edu/landscape_horticulture/PDFdocuments/SoilProp.pdf
6. [http://www.rootsofpeace.org/assets/Soil%20Testing%20Manual%20V6%20\(Feb%2008\).pdf](http://www.rootsofpeace.org/assets/Soil%20Testing%20Manual%20V6%20(Feb%2008).pdf)
7. <http://www.soils.wisc.edu/courses/SS325/morphology.htm>
8. <http://www.soils.wis.edu/courses/SS325/morphology.htm>
9. <http://landresources.montana.edu/>
10. [http://ftp.wcc.nrcs.usda.gov/H....soil Other/soil-USDA-textural-class.pdf](http://ftp.wcc.nrcs.usda.gov/H....soil%20Other/soil-USDA-textural-class.pdf)

FOR 111 Introduction to Forestry (1+1)

UNIT I

Forest and Forestry

Introduction - Definition of Forest and Forestry - Role of Forest (Production, Protection and Amelioration) - Classification of Forest (Regeneration, Age, Composition, ownership, object of management, growing stock) - National Forest Policy 1988.

UNIT II

Silviculture and Forest plantation

Forest regeneration - Natural regeneration- Seeds and vegetative parts (Coppice, Root suckers) - Artificial regeneration, Objectives - Nurseries - Types of nurseries, Quality seedling production techniques - Silvicultural practices for *Eucalyptus spp*, *Casuarina equisetifolia*, *Tectona grandis*, *Ailanthus excelsa*, *Melia dubia*, *Leucaena leucocephala*. Tending operations - Weeding, Cleaning, Thinning and pruning.

UNIT III

Forest Mensuration

Forest Mensuration - Objectives- Diameter measurements, instruments used in diameter measurement-Height measurement, instrumental methods of height measurement - Tree form, form factor, Volume estimation of standing and felled trees.

UNIT IV

Social forestry and Agroforestry

Social Forestry and its branches - Extension Forestry, Urban forestry - Agroforestry, definition-Importance- Agroforestry systems - Shifting Cultivation, Taungya, Alley cropping, Wind break, Shelter belt, Home garden - Tree and crop combination in Agroforestry - Tree crop interaction in Agroforestry - National Agroforestry Policy 2014.

UNIT V

Forest Utilization

Forest Utilization - Definition - Wood products - solid wood and composite wood.- Non Wood Forest Products - fibres, floss, bamboo, tan, dye, resin, oleoresin.

Practical

Identification of important farm grown trees - Identification of tree seeds and seedlings- Site selection for tree nursery and layout of nursery- Study of nursery techniques for *Casuarina equisetifolia* and *Tectona grandis* - Practicing clonal propagation in trees Practicing land preparation, stacking, pitting, planting techniques and after care operations in plantations- Height measurement in trees, diameter measurement in trees, Volume estimation in trees- Identification of wood and non- wood forest products - Visit to Agroforestry plantations

Lecture schedule:

1. Introduction about forests, Definition of Forest and Forestry, branches in forestry
2. Role of Forest - Production function, Protection function and ameliorative functions of forests
3. Classification of Forest based on mode of regeneration, age, composition, ownership, object of management and growing stock
4. National Forest Policy 1988- Objectives and salient features
5. Forest regeneration - Types of regeneration - Natural regeneration through seeds and vegetative parts including coppice and root suckers
6. Artificial regeneration, Objectives - Nurseries - Types of nurseries, Quality seedling production techniques
7. Silvicultural practices for *Eucalyptus spp*, *Casuarina equisetifolia*, *Tectona grandis*, *Ailanthus excelsa*,

8. Silvicultural practices for *Melia dubia*, *Leucaena leucocephala*. Tending operations - Weeding, Cleaning, Thinning and pruning.

9. Mid Semester Examination

10. Forest Mensuration - Objectives- Diameter measurements, instruments used in diameter measurement
11. Height measurement, instrumental methods of height measurement - Tree form, form factor, Volume estimation of standing and felled trees.
12. Social Forestry and its branches - Extension Forestry and Urban forestry.
13. Agroforestry, definition- Importance- Agroforestry systems - Shifting Cultivation, Taungya, Alley cropping, Wind break, Shelter belt, Home garden
14. Tree and crop combination in Agroforestry- Tree crop interaction in Agroforestry -
15. National Agroforestry Policy 2014 , objectives and salient features
16. Forest Utilization - Definition - Wood products - solid wood and composite wood.
17. Forest Utilization - Non Wood Forest Products - fibres , floss, bamboo, tan, dye, resin, oleoresin

Practical schedule:

1. Identification of important farm grown trees
2. Identification of tree seeds and seedlings
3. Site selection for tree nursery and layout of nursery
4. Study of nursery techniques for *Casuarina equisetifolia*
5. Study of nursery techniques for *Tectona grandis*
6. Practicing clonal propagation in trees *Eucalyptus* / *Casuarina*
7. Practicing land preparation, stacking, pitting,
8. Planting techniques in plantation
9. After care operations in plantations
10. Height measurement in trees
11. Diameter measurement in trees
12. Volume estimation in standing and felled trees
13. Identification and study of wood products
14. Identification and study non- wood forest products
15. Visit to Agroforestry plantations
16. Visit to forest based industry
17. **Final Practical Examination**

References

1. Dwivedi, A.P. 1992. Principles and Practices of Indian Silviculture. Surya publications, Dehradun. 177p
2. Gupta. R.K 1993. Multipurpose trees for Agroforestry and Wasteland utilization. Oxford and IBH Publishing Company, New Delhi. 580p.
3. Nair. P.K.R. 1993. Introduction to Agroforestry. Kluwer Academic Publishers, Dordrecht, Netherlands. 499p
4. Negi, S.S. 1986. A Hand book of Social Forestry. International Book Distributors, Dehradun. 177p
5. Puri, S and P.K. Khosla. 1993. Nursery Technology for Agroforestry - applications in Arid and Semi arid regions. Oxford and IBH Publishing Company, New Delhi. 392p
6. Khanna. L.S. 1981. Principles and Practice of Silviculture. Khanna Bandhu publications, Dehradun. 472p
7. Chaturvedi, A.N and L.S. Khanna. 1982. Forest Mensuration. International Book Distributors, Dehradun. 350p

8. Tirubhawan Mehta. 1981. Hand book of Forest Utilization. International Book Distributors, Dehradun. 208 p.

ENG 101 Comprehension and Communication Skills in English (1+1)

Theory

War Minus Shooting- The sporting Spirit. A Dilemma- A layman looks at science Raymond B. Fosdick. You and Your English – Spoken English and broken English G.B. Shaw. Reading Comprehension, Vocabulary-Antonym, Synonym, Homophones, Homonyms, often confused words. Exercises to Help the students in the enrichment of vocabulary based on TOEFL and other competitive examinations. Functional grammar: Articles, Prepositions, Verb, Subject verb Agreement, Transformation, Synthesis, Direct and Indirect Narration. Written Skills: Paragraph writing, Precise writing, Report writing and Proposal writing. The Style: Importance of professional writing. Preparation of Curriculum Vitae and Job applications. Synopsis Writing. Interviews: kinds, Importance and process.

Practical

Listening Comprehension: Listening to short talks lectures, speeches (scientific, commercial and general in nature). Oral Communication: Phonetics, stress and intonation, Conversation practice. Conversation:

rate of speech, clarity of voice, speaking and Listening, politeness & Reading skills: reading dialogues, rapid reading, intensive reading, improving reading skills. Mock Interviews: testing initiative, team spirit, leadership, intellectual ability. Group Discussions.

Lecture Schedule:

1. **War Minus Shooting** (A lesson from the Text Book, “The Sporting Spirit” by George Orwell) textual grammar pertaining to factual comprehension and inferential comprehension & referential comprehension.
2. **War Minus Shooting** (A lesson from the Text Book, “The Sporting Spirit” by George Orwell) textual grammar pertaining to global comprehension and attitudinal comprehension
3. **War Minus Shooting** (A lesson from the Text Book, “The Sporting Spirit” by George Orwell) textual grammar on synonyms – antonyms – prefix – suffix – homonyms - homophones – TOEFL & IELTS vocabulary
4. **War Minus Shooting** (A lesson from the Text Book, “The Sporting Spirit” by George Orwell) textual grammar – English articles – preposition – conjunctions and its types
5. **A Dilemma** (A lesson from the Text Book, Layman looks at Science by Raymond Fosdick) textual grammar – verbs – auxiliary verbs - modals and basic tense forms
6. **A Dilemma** (A lesson from the Text Book, Layman looks at Science by Raymond Fosdick) textual grammar – sentence pattern and sentence forms (simple, compound and complex sentences)
7. **A Dilemma** (A lesson from the Text Book, Layman looks at Science by Raymond Fosdick) textual grammar – subject – verb – agreement
8. **A Dilemma** (A lesson from the Text Book, Layman looks at Science by Raymond Fosdick) textual grammar – transformation of sentences
9. **Mid Semester Examination**
10. **You and Your English** (A lesson from the Text Book, Spoken English and Broken English by G.B. Shaw) textual grammar – synthesis of sentences – reported speech (direct and indirect speech)

11. **You and Your English** (A lesson from the Text Book, Spoken English and Broken English by G.B. Shaw) textural grammar – paragraph writing (thesis sentences, supporting statements, inferential statements)
12. **You and Your English** (A lesson from the Text Book, Spoken English and Broken English by G.B. Shaw) textural grammar – four principles of writing
13. **You and Your English** (A lesson from the Text Book, Spoken English and Broken English by G.B. Shaw) textural grammar - professional writing – summary writing and paraphrasing, synopsis writing and citation
14. Graham's flow chart on writing skills
15. Letter writing – personal and social correspondence – job application
16. precise writing – report writing and proposal writing
17. Interview skills - kinds – importance and process

Practical Schedule:

1. Listening - Introduction - Listening vs Hearing - listening modes - types of listening - Intensive and Extensive Listening – practice
2. Process of Listening - methods of enhancing listening - barriers to listening and ways to overcome them – practice
3. Oral communication - organs of speech – English phonemes (consonant table, vowel table) - practice
4. English Stress & Intonation - exercises.
5. Conversation techniques and practice
6. Rate of speech (slow pace, medium pace, rhetoric)
7. Reading - types - skimming and scanning - SQ4R - critical reading - analytical reading – exercises
8. Principles and practice of presentation skills - PowerPoint preparation and presentation
9. Handout preparation - lecture notes preparation - practice and evaluation
10. Writing skills - note taking – precise writing – abstract writing – practice
11. Mind-mapping and article writing
12. Letter writing and rejoinder writing
13. Text writing - practice on table to text conversion
14. Interview skills – types of interview (group interview – panel interview – telephone interview – behavioural interview – video-conferencing interview – mock interview)
15. Practice on speaking skills – welcome address - vote of thanks - short extemporal speech
16. Group discussion – techniques – types and practice
17. **Final Practical Examination**

References

Goodale, Malcolm, *Professional Presentations*, Cambridge University, 2005.
 Greenbaum Sidney, *Oxford English Grammar*, New Delhi, Oxford University Press.
 Peregoy, 2009.
 Jones Daniel, *English Pronouncing Dictionary*, Cambridge University Press, 2006. Lynch, Tony and Kenneth Anderson, *Study Speaking*, Cambridge University, 1992. Martin Cutts, *Oxford Guide to Plain English*, Oxford University Press, 2004. Sahaneya Wandy, et.al., *IELTS, Preparation and Practice*, Oxford University, 2005. Sundararajan, N, *Attentive Listening: How it Matters*, University News, March 19-25, 2005.

E-references:

1. www.orwell.ru/library/articles/spirit/english/e_spirit
2. www.essays.com
3. www.onestopenglish.com
4. www.bogglesworld.com
5. www.eltweb.com
6. www.reportingskills.com
7. www.writing-skills.com
8. www.negotiation.com
9. www.teachersdesk.com
10. www.flexbilelearning.net.au

HOR 111 Fundamentals of Horticulture (1+1)

Unit I

History, evolution and scope of horticulture

Origin of horticulture – history – evolution – definitions – scope and importance of horticulture – division and classification of horticultural crops – fruits, vegetables, spices and plantation crops, floriculture, landscaping, ornamental gardening, medicinal and aromatic crops – nutritive value and global and national scenario of horticultural crops.

Unit II

Sexual propagation

Sexual propagation – importance, advantages and disadvantages – methods of enhancement of seed viability – types of dormancy – seed invigoration – seed treatments

Unit III

Asexual propagation

Asexual propagation, importance, advantages and disadvantages - Asexual propagation types viz., Types of cutting, layering, grafting and budding. Use of specialized plant parts in propagation. Propagation structures and their role. Rootstock influence – stock / scion relationship in fruit crops. Scope and importance of micro propagation in horticultural crops. Direct and indirect organogenesis – media for micro propagation and hardening.

Unit IV

Planting systems and pollination

Principles of orchard establishment - Methods of planting systems including HDP and UHDP in horticultural crops – crop regulatory practices for horticultural crops – training, pruning, special operations in horticultural crops – off season production of horticultural crops. Flowering, pollination, fruit set, fruit drop, parthenocarpy, fruit ripening and senescence – Unfruitfulness and its causes.

Unit V.

Principles and types of garden

Principles and types of garden – principles and types of parks – principles of herbal garden

Practical

Features of an orchard - Identification of garden tools, implements and machineries. Identification of horticultural crops and herbarium making. Preparation of potting mixture, potting and repotting. Preparation of seed bed / nursery bed. Practice of sexual and asexual methods of propagation- cutting, layering, budding, grafting – specialized plant parts - Layout and planting of fruit trees. Training and pruning of fruit trees. Transplanting and care of vegetable seedlings. Making of herbaceous and shrubby borders. Practicing irrigation, fertilizer and manures application in different crops. Preparation and application of Plant Growth Regulators – visit to tissue culture lab - Visits to commercial nurseries / orchard / garden.

Theory Lecture schedule:

1. Origin of horticulture – history – evolution – definitions – scope and importance of horticulture
2. Division and classification of horticultural crops – fruits, vegetables, spices and plantation crops, floriculture, landscaping, ornamental gardening, medicinal and aromatic crops
3. Nutritive value and global and national scenario of horticultural crops
4. Sexual propagation – importance, advantages and disadvantages – methods of enhancement of seed viability
5. Types of dormancy – seed invigoration – seed treatments
6. Asexual propagation, importance, advantages and disadvantages - Asexual propagation types
7. Vegetative propagation – merits and demerits – cutting and layering
8. Vegetative propagation – merits and demerits – grafting and budding
9. **Mid Semester Examination**
10. Use of specialized plant parts in propagation - Propagation structures and their role.
11. Rootstock influence – stock / scion relationship in fruit crops
12. Scope and importance of micro propagation in horticultural crops- Direct and indirect organogenesis
– media for micro propagation and hardening
13. Principles of orchard establishment - Methods of planting systems including HDP and UHDP in horticultural crops
14. Crop regulatory practices for horticultural crops – training, pruning, special operations in horticultural crops – off season production of horticultural crops.
15. Flowering, pollination, fruit set, fruit drop, parthenocarpy, fruit ripening and senescence, unfruitfulness and its causes
16. Principles and types of garden
17. Principles and types of parks – principles of herbal garden

Practical schedule:

1. Visit to orchard and identifying its components
2. Identification of garden tools, implements and machineries
3. Identification of horticultural crops and herbarium making
4. Preparation of pot mixture, potting and repotting
5. Preparation of nursery beds for raising rootstocks and seedlings
6. Practicing asexual methods of propagation- cutting and layering
7. Practicing asexual methods of propagation – budding and grafting
8. Plant propagation structures and specialized plant parts for propagation
9. Layout and planting of fruit trees
10. Training and pruning of fruit trees
11. Transplanting and care of vegetable seedlings
12. Making of herbaceous and shrubby borders
13. Practicing irrigation, fertilizer and manures application in different crops
14. Preparation and application of Plant Growth Regulators
15. Visit to tissue culture lab
16. Visit to commercial nurseries / garden
17. **Final Practical Examination**

References

1. Sadhu, M.K.1989. Plant Propagation. Wiley Eastern Ltd., 4835/24, Ansari Road, Daryaganj, New Delhi- 110 002.Bose, T.K., S.K. Mitra, M. K. Sadhu and B. Mitra. 1991. Propagation of Tropical and Subtropical Horticultural Crops. Naya Prakash 206, Bidhan Sarani, Calcutta-6, IndiaHartmann, H.T., D.E. Kester, F.T. Davies and R.L. Greeneve. 1997. Plant Propagation - Principles and Practices. Prentice Hall of India Private Ltd., New Delhi.Nanda, K.K and V.K. Kochhar. 1995. Vegetative Propagation of Plants. Kalyani Publishers, Ludhiana.
2. George Acquaah, 2002. Horticulture – principles and practices. Prentice Hall of India Pvt. Ltd., New Delhi.
3. Hartman, H.T. and Kester, D.E. 1986. Plant propagation – Principles and Practices – Prentice Hall of India Ltd., New Delhi.
4. Jules Janick.1979.Horticultural Science. Surjeet Publications. New Delhi.
5. Kumar, N.2014, Introduction to Horticulture, Oxford IBH Publications, New Delhi.

e-References

- 1.[http://aggie – horticulture, tamu.edu/propagation/propagation.html](http://aggie-horticulture.tamu.edu/propagation/propagation.html)
- 2.[http://www/britannica.com/](http://www.britannica.com/)
- 3.<http://www.horticulture.com.au/export/hmac.asp.asp>
- 4.<http://www.horticultureworld.net/hort-india.htm>
- 5.<http://www.fao.org/>

MAT 111 ELEMENTARY MATHEMATICS (1+1)

Unit - I

Algebra: Permutation and Combination -meaning of nPr and nCr (simple problems). Matrices- Definition of Matrices, Addition, Subtraction, Multiplication, Transpose and Inverse up to 3^{rd} order by adjoint method, Properties of determinants up to 3^{rd} order and their evaluation.

Unit - II

Analytical Geometry: Distance formula, section formula (internal and external division), Change of axes (only origin changed), Equation of co-ordinate axes, Equation of lines parallel to axes, Slope-intercept form of equation of line, Slope-point form of equation of line, Two point form of equation of line, Intercept form of equation of line, Normal form of equation of line, General form of equation of line, Point of intersection of two straight lines, Angles between two straight lines, Parallel lines, Perpendicular lines.

Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2, y_2) .

Unit - III

Differential Calculus: Definition of function, limit and continuity, Simple problems on limit, Simple problems on continuity, Differentiation of x^n , e^x , $\sin x$ & $\cos x$ from first principle, Derivatives of sum, difference, product and quotient of two functions, Differentiation of functions of functions (Simple problem based on it), Logarithmic differentiation (Simple problem based on it), Differentiation by substitution method and simple problems based on it, Differentiation of Inverse Trigonometric functions. Partial differentiation with first and second order -Maxima and Minima of the functions of the form $y = f(x)$ and $y = f(x_1, x_2)$ (Simple problems based on it).

Unit –IV

Integral Calculus: Integration of simple functions, Integration of Product of two functions, Integration by substitution method, Definite Integral (simple problems based on it), Area under simple well-known curves (simple problems based on it).

Unit-V

Mathematical Models: Agricultural systems - Mathematical models - classification of mathematical models- Fitting of Linear, quadratic and exponential models to experimental data.

Practical

Simple problems in Permutation and Combination -meaning of nPr and nCr Problems in Algebra of matrices , Transpose and Inverse up to 3^{rd} order by adjoint method, evaluation of determinants up to 3^{rd} order. Problems in Straight lines using distance formula, section formula (internal and external division), Change of axes (only origin changed)- Equation of co-ordinate axes- Equation of lines parallel to axes. Problems in equation of a line in : Slope-intercept form, Slope-point form, two point forms,

Intercept form, Normal form , General form, Point of intersection of two straight lines. Problems in Angles between two straight lines, Parallel lines, Perpendicular lines. Problems in Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2, y_2) . Simple problems in limit and continuity. Problems in differentiation of x^n , e^x , $\sin x$ & $\cos x$, derivatives of sum, difference, product and quotient of two functions. Simple problem based on differentiation of functions of functions and Logarithmic differentiation. Simple

problems based on differentiation by substitution method. Problems in partial differentiation and Maxima and Minima of the functions of the form $y=f(x)$ and $y=f(x_1, x_2)$. Problems in integration of simple functions and product of two functions- Definite Integral. Integration by substitution method-Problems in Area under simple well-known curves. Problems in fitting linear, quadratic and Exponential models to experimental data.

Theory Lecture Schedule:

1. Permutation and Combination -meaning of nPr and nCr (Simple Problems) .
2. Matrices- Definition of Matrices- Types of Matrices- Addition, Subtraction, Multiplication, Transpose
3. Determinants-Properties of determinants -up to 3^{rd} order evaluation and inverse up to 3^{rd} order by adjoint method.
4. Straight lines - Distance formula-section formula (internal and external division) - Change of axes (only origin changed) - Equation of co-ordinate axes- Equation of lines parallel to axes.
5. Forms of equation of Line-Slope-intercept form -Slope one point form - Two point form - Intercept form.
6. Normal form of equation of line- General form of equation of line- Point of intersection of two straight lines.
7. Angles between two straight lines- Parallel lines- Perpendicular lines- Angle of bisectors between two lines.
8. Circle-Equation of circle whose centre and radius is known- General equation of a circle- Equation of circle passing through three given points- Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2, y_2) .

9. Mid Semester Examination

10. Differential Calculus - Definition of function, limit and continuity- Simple problems on limit and continuity.
11. Differentiation of x^n , e^x , $\sin x$ & $\cos x$ from first principle-Derivatives of sum, difference, product and quotient of two functions- Differentiation using functions of function rule (Simple problem based on it)
12. Logarithmic differentiation (Simple problem based on it)- Differentiation by substitution method and simple problems based on it- Differentiation of Inverse Trigonometric functions
13. Maxima and Minima of the functions of the form $y=f(x)$ and $y=f(x_1, x_2)$ (Simple problems based on it).
14. Integral Calculus - Integration of simple functions and Product of two functions- Definite Integral (simple problems based on it)
15. Integration by substitution method- Area under simple well-known curves (simple problems based on it).
16. Agricultural systems - Mathematical models - classification of mathematical models- Linear model.
17. Quadratic and Exponential models- applications of mathematical models in agriculture.

Practical Schedule:

1. Simple problems in Permutation and Combination.
2. Problems in Addition, Subtraction, Multiplication and Transpose of a matrix
3. Problems in determinants and Inverse up to 3^{rd} order by adjoint method.
4. Problems in Straight lines using distance formula, section formula (internal and external division), Change of axes (only origin changed)- Equation of co-ordinate axes- Equation of lines parallel to axes.

5. Problems in Slope-intercept form of equation of line, Slope-point form of equation of line, two point forms of equation of line, Intercept form of equation of line.
6. Problems in Normal form of equation of line, General form of equation of line, Point of intersection of two straight lines.
7. Problems in Angles between two straight lines, Parallel lines, Perpendicular lines, Angle of bisectors between two lines.
8. Problems in Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2, y_2) .
9. Simple problems in limit and continuity.
10. Problems in differentiation of x^n , e^x , $\sin x$ & $\cos x$, derivatives of sum, difference, product, quotient of two functions and differentiation of functions of functions.
11. Simple problem based on Logarithmic differentiation and differentiation by substitution method.
12. Problems in Maxima and Minima of the functions of the form $y=f(x)$ and $y=f(x_1, x_2)$
13. Problems in integration of simple functions and product of two functions using integration by parts-Definite Integral.
14. Integration by substitution method-Problems in Area under simple well-known curves
15. Problems in fitting linear models to experimental data.
16. Problems in fitting Quadratic and Exponential models to experimental data.
- 17. Final Practical Examination.**

References:

1. Mehta, B. C. and G. M. K. Madnani, 2014, Mathematics for Economists, Sultan Chand & Sons, New Delhi.
2. Kailasam.C, Pangayar Selvi. R and Vasanthi. R, 2010, Applied Mathematics, Agrobios (India), Jodhpur
3. James Stewart and Barhara Frank, Calculus, 2008, International Thomson Publishers, Singapore
4. Duraipandian, 2007, Calculus and Analytical Geometry, Emerald Publishers, Chennai.
5. Ranganathan.C.R. 2006, A First Course in Mathematical Models of Population Growth (with MATLAB programs), Associated publishing company, New Delhi
6. Manickavasagam Pillai, T. K and Natarajan, T. 2004. Calculus, Viswanathan Publications, Madras.

PBG 101 Introduction to Agricultural Botany (1+1)

Unit I:

Systems of classification and general morphological description

Bentham and Hooker's classification of plant kingdom — International code of nomenclature and its major guidelines – author citation – Agricultural classification of crops; General morphology: Life span, habit, root, stem, leaf - petiole, leaf margin, leaf apex, leaf shape, venation and phyllotaxy; Modification of roots and leaf; Floral morphology: Kinds of bracts, inflorescence; Structure of flower, androecium, gynoecium, placentation, types of fruits.

Unit II:

Botanical description and economic uses of Poaceae

List of cultivated crops, economic parts, chromosome number and family description of Poaceae: Key botanical features of Rice, Wheat, Sorghum, Maize, Pearl millet, Finger millet, list of small millets, Guinea grass, Napier grass, *Cenchrus* and Sugarcane

Unit III:

Botanical description and economic uses of Papilionaceae

List of cultivated crops, economic parts, chromosome number and family description of Papilionaceae: Key botanical features of Red gram, Bengal gram, Soybean, Black gram, Green gram, Cowpea, Lablab, Horse gram, Groundnut, Lucerne, *Stylosanthes*, Clitoria, Agathi and Sunnhemp,

Unit IV:

Botanical description and economic uses of Pedaliaceae, Asteraceae, Oleaceae, Brassicaceae, Euphorbiaceae, Arecaceae and Malvaceae

List of cultivated crops, economic parts, chromosome number and family description of the following families and Key botanical features of the crops given against them: Pedaliaceae - Gingelly; Asteraceae - Sunflower, Safflower, Chrysanthemum; Oleaceae - Jasmine; Brassicaceae - Rapeseed and Mustard, Cabbage, Cauliflower; Euphorbiaceae: Castor; Jatropha and Tapioca; Arecaceae: Coconut, Arecanut, Oilpalm, Sugarpalm; Malvaceae: Cotton, Mesta and Bhendi.

Unit V:

Botanical description and economic uses of Tiliaceae, Piperaceae, Chenopodiaceae, Solanaceae, Mimosae, Moraceae, Cucurbitaceae, Alliaceae, Musaceae, Rubiaceae, Theaceae

List of cultivated crops, economic parts, chromosome number and family description of the following families and key botanical features of the crops given against them. Tiliaceae: Jute; Piperaceae: Betelvine; Chenopodiaceae: Sugar beet; Solanaceae: Tobacco, Potato, Chilli, Tomato and Brinjal; Mimosae: Desmanthes, Subabul and Acacia; Moraceae: Mulberry; Cucurbitaceae: Cucumber, Pumpkin, Ashgourd; Alliaceae: Onion and Garlic; Musaceae: Banana, Manila hemp; Rubiaceae: Coffee; Theaceae: Tea

PRACTICAL

Family features - observation and description of habit, morphology of root, stem, leaves, inflorescence, flowers, floral diagram, floral formula and economic parts of Poaceae: Rice, Wheat, Sorghum, Maize, Pearl millet, Finger millet, Guinea grass, Napier grass, *Cenchrus* and Sugarcane; Papilionaceae: Redgram, Bengal gram, Soybean, Blackgram, Greengram, Cowpea, Lab-lab, Horse gram, Groundnut, Lucerne, *Stylosanthes*, Clitoria, Agathi and Sunnhemp; Pedaliaceae: Gingelly; Asteraceae: Sunflower, Safflower and Chrysanthemum; Oleaceae:

Jasmine; Brassicaceae: Rape and Mustard, Cabbage, Cauliflower; Euphorbiaceae: Castor, Jatropha, Tapioca; Arecaceae: Coconut, Arecanut, Oil palm and Sugar palm;
 Malvaceae: Cotton, Mesta, Bhendi; Tiliaceae: Jute; Piperaceae: Betelvine; Chenopodiaceae: Sugar beet;
 Solanaceae: Tobacco, Potato, Chilli, Tomato and Brinjal; Mimosae: Desmanthes, Subabul and Acacia;
 Moraceae: Mulberry; Cucurbitaceae: Cucumber, Pumpkin, Ashgourd; Alliaceae: Onion and Garlic;
 Musaceae: Banana, Manila hemp; Rubiaceae: Coffee; Theaceae: Tea

Theory Lecture schedule:

1. Bentham and Hooker's classification of plant kingdom —International code of nomenclature and its major guidelines – author citation – Agricultural classification of crops
2. General morphology: Life span, habit, root, stem, leaf - petiole, leaf margin, leaf apex, leaf shape, venation and phyllotaxy; Modification of roots, stem and leaf
3. Floral morphology: Kinds of bracts, inflorescence; Structure of flower, androecium, gynoecium, placentation, types of fruits.
4. List of cultivated crops, economic parts, chromosome number and family description of Poaceae; Key botanical features of Rice and Wheat.
5. Key botanical features of sorghum, maize, pearl millet and finger millet. List of small millets
6. Key botanical features of Guinea grass, Napier grass, *Cenchrus* and sugarcane.
7. List of cultivated crops, economic parts, chromosome number and family description of (Papilionaceae) Key botanical features of Red gram, Bengal gram and Soybean
8. Key botanical features of Black gram, Green gram, Cowpea, Lab lab, Horse gram and Groundnut.
9. **Mid Semester Examination**
10. Key botanical features of Lucerne, *Stylosanthes*, Clitoria, Agathi, and Sunnhemp.
11. List of cultivated crops, economic parts, chromosome number and family description of Pedaliaceae and Asteraceae; Key botanical features of Gingelly, Sunflower, Safflower, Chrysanthemum; Oleaceae: Jasmine
12. List of cultivated crops, economic parts, chromosome number and family description of Brassicaceae and Euphorbiaceae; Key botanical features of Rapeseed and Mustard, Cabbage, Cauliflower, Castor, Jatropha and Tapioca
13. List of cultivated crops, economic parts, chromosome number and family description of Arecaceae and Malvaceae; Key botanical features of Coconut, Arecanut, Oil palm, Sugarpalm, Cotton, Mesta and Bhendi.
14. List of cultivated crops, economic parts, chromosome number and family description of Tiliaceae, Piperaceae and Chenopodiaceae; Key botanical features of Jute, Betelvine, Sugar beet.
15. List of cultivated crops, economic parts, chromosome number and family description of Solanaceae, Mimosae and Moraceae; Key botanical features of Tobacco, Potato, Chilli, Tomato and Brinjal, Desmanthes, Subabul, Mulberry
16. List of cultivated crops, economic parts, chromosome number and family description of Cucurbitaceae and Alliaceae; Cucurbitaceae: Key botanical features of Cucumber, Pumpkin, Ashgourd; Alliaceae: Onion and Garlic
17. List of cultivated crops, economic parts, chromosome number and family description of Musaceae, Rubiaceae and Theaceae; Key botanical features of Banana, Manila hemp, Coffee and Tea

Practical schedule:

1. Observing general morphology of roots, stems and leaves.
2. Observing general morphology of inflorescence - flowers, stamens and pistils.
3. Family characters, Botany, Economic parts, Floral diagram and Floral formula of the following crop plants:- Poaceae: Rice and Wheat
4. Poaceae: Sorghum, Maize, Pearl millet, Finger millet.
5. Poaceae: Guinea grass, Napier grass, *Cenchrus* and Sugarcane.
6. Papilionaceae: Redgram, Bengal gram and Soybean.
7. Papilionaceae: Blackgram, Greengram, Cowpea, Lab-lab, Horse gram and Groundnut.
8. Papilionaceae: Lucerne, *Stylosanthes*, Clitoria, Agathi, Sunnhemp, and Sesbania.
9. Pedaliaceae: Gingelly; Asteraceae: Sunflower, Safflower and Chrysanthemum;
Oleaceae: Jasmine
10. Brassicaceae: Rapeseed and Mustard, Cabbage, Cauliflower.
11. Euphorbiaceae: Castor, Jatropha, Tapioca; Arecaceae: Coconut, Arecanut, Oilpalm and Sugar palm.
12. Malvaceae: Cotton, Mesta, Bhendi
13. Tiliaceae: Jute; Piperaceae: Betelvine; Chenopodiaceae: Sugar beet;
14. Solanaceae: Tobacco, Potato, Chilli, Tomato and Brinjal; Mimosae: Desmanthes, Subabul ,
Moraceae: Mulberry
15. Cucurbitaceae: Cucumber, Pumpkin, Ashgourd; Alliaceae: Onion and Garlic
16. Musaceae: Banana, Manila hemp; Rubiaceae: Coffee; Theaceae: Tea
- 17. Final Practical Examination**

References:

1. Daniel Sundararaj, D. and G. Thulasidas, 1993. Botany of field crops. MacMillan India Ltd., New Delhi.
2. Sambamurthy, V.S. and N.S. Subramanian, 1989. Text Book of Economic Botany, Wiley Eastern, New Delhi
3. Purse glow, 1988. Tropical Crops - Monocotyledons. The English Language book Society and Longman Co., Singapore
4. Purse glow. 1988. Tropical Crops - Dicotyledons. The English language book Society and Longman Co., Singapore.
5. Albert F. Hill and O.P. Sharma, 1996. Economic Botany. Tata McGraw - Hill Publishing Co. Ltd., New Delhi.
6. John Joel, A., C. Vanniarajan, T.S. Raveendran, and A. Gopalan 2006. Fundamentals of Crop Botany, Directorate of ODL, Tamil Nadu Agricultural University, Coimbatore-641 003.

E-Resources

1. www.nmsu.edu
2. www.biology200.gsu.edu

AEX 101 Rural Sociology and Educational Psychology (2+0)

Theory

UNIT I

Introduction to Sociology, Social groups, Culture and Social Values

Sociology and Rural Sociology – definitions; Society – rural and urban, characteristics, differences and relationships, important characteristics of Indian rural society; Social groups – definition, classification, role of social groups in extension; Culture – concept, cultural traits, characteristics, functions, Ethnocentrism, Acculturation, Cultural lag, Cultural diffusion, Marginal man, Ethos. Social Values – definition, values and norms, characteristics of values, functions;

UNIT II

Social Structure, Social Stratification and Migration

Structure of Rural Society – patterns of rural settlement, social institutions, social organizations, ecological entities (Region, Community, Neighbourhood, Family); Social Stratification – concept, functions, types, differences between class and caste system; Migration – concept, factors influencing migration.

UNIT III

Social Control, Social Customs

Social Control – definition; Customs – conventions, folkways, mores, rituals, taboos; Social Interaction Process – definition, basic social processes; Social Change – concept, factors influencing social change, indicators of social change; Social development :

UNIT IV

Introduction to Educational Psychology, Intelligence, Teaching-Learning Process;

Education – Psychology – Educational Psychology – Social Psychology – definitions, importance in extension; Basic principles of Human behaviour – Sensation, Attention, Cognitive, affective, psychomotor domain Perception – meaning, characteristics; Intelligence – concept, types, measurement, factors affecting intelligence; Personality – concept, types, measurement, factors influencing personality; Teaching–Learning Process – Teaching – definition, meaning, principles of teaching, steps in extension teaching; Learning – definition, meaning, principles, types of learning, learning situation.

UNIT V

Motivation, Attitude

Motivation – concept, Maslow's hierarchy of needs, intrinsic and extrinsic motivation, techniques of motivation, importance in extension; Attitude – concept, factors influencing the development of attitudes.

Theory Schedule

1. Sociology and Rural Sociology – Definitions, nature of rural sociology,
2. Importance of rural sociology in extension education.
3. Society – rural and urban, characteristics, differences and relationship, important characteristics of Indian rural society;
4. Social Groups – definitions, classification, role of social groups in extension.
5. Culture – concept, cultural traits, characteristics, functions,
6. Ethnocentrism, Acculturation, Cultural lag, Cultural diffusion, Marginal man, Ethos.
7. Structure of Rural Society – patterns of rural settlement,
8. Social institutions, Social organizations and ecological entities - Region, Community, Neighbourhood, and Family.
9. Social Stratification – concept, functions, types, differences between class and caste system;
10. Social Values – definition, values and norms, characteristics of values, functions.
11. Migration – concept, factors influencing migration.
12. Social Control – definition;
13. Customs – conventions, folkways, mores, rituals, taboos;
14. Social Interaction Process – definition, basic social processes.
15. Social Change – concept, theories, factors and indicators of social change.
16. Social development
- 17. Mid semester Examination.**
18. Education – Psychology – Educational Psychology –definitions, importance in extension.
19. Social Psychology – Definitions, importance in extension.
20. Basic principles of Human behaviour –
21. Cognitive, affective, psychomotor domain
22. Perception – meaning, characteristics.
23. Sensation, Attention
24. Intelligence – concept, types,
25. Intelligence - measurement, factors affecting intelligence;
26. Personality – concept, types,
27. Personality measurement- factors influencing personality
28. Teaching–Learning Process – Teaching – definition, meaning,
29. Principles of teaching, steps in extension teaching.
30. Learning – definition, meaning, principles,
31. Types of learning, learning situation.
32. Motivation – concept, Maslow’s hierarchy of needs (including selfless-service), intrinsic and extrinsic motivation,
- 33. Techniques of motivation, importance of motivation in extension.**
34. Attitude – concept, factors influencing the development of attitudes.

References:

1. Adivi Reddy, A. 2001. Extension Education, Sree Lakshmi Press, Bapatla, Andhra Pradesh. Chatterjee, S. 2000. Advanced Educational Psychology, Books & Allied (P) Ltd., Calcutta.
2. Chauhan, S.S. 2001. Advanced Educational Psychology, Vikas Publishing House Pvt. Ltd., New Delhi. Chitambar, J.B.1997. Introductory Rural Sociology, New Age International (P) Ltd., Publishers, New Delhi.
3. Dahama, O.P. and O.P. Bhatnagar. 2007. Education and Communication for Development, Oxford &
4. IBH Publishing Co. Pvt. Ltd., New Delhi.

5. Kundu, C.L and Tutoo, D.N. 2001. Educational Psychology, Sterling Publishers Pvt. Ltd., New Delhi.
6. Lester Crow, D and Alice Crow. 1973. Educational Psychology, Eurasia Publishing House Pvt. Ltd.,
7. New Delhi.
8. Madumita Gupta. 2011. Fundamentals of Sociology, Pacific Publications, New Delhi. Mangal, S.K. 2000. Educational Psychology, Prakash Brothers, Ludhiana.
9. Shankar Rao, C.N. 2012. Sociology – Principles of Sociology with an Introduction to Social Thought,
10. S.Chand & Co. Ltd., New Delhi.
11. Sharma, R.N. 1968. Principles of Sociology, Asia Publishing House, New Delhi.
12. Supe. S.V. 2012. Text book of Extension Education, Agrotech Publishing Academy, Udaipur. Usha Rao. 2008. Advanced Educational Psychology, Himalaya Publishing House, New Delhi.
13. Vidya Bhushan and Sachdeva, D.R. 2003. An Introduction to Sociology, Kitab Mahal, Allahabad.

E-References:

1. www.sociologyguide.com
2. eu.wikipedia.org
3. www.princeton.edu

nehf;fk;

,sepiy ntshz;ik gapYk; hzth;fSf;Fk jkpH; ,yf;fpa';fs; tHp ntshz;ik kw;Wk; ntshz;ik rhh;e;j
bjhHpy;El;g';fisa[k; bra;jpfisa[k; mwpar;-jw;fhybra;jy;ntshz; bjhHpy;El;g';fnshL
bghUj;jpg; ghh;j;jy;-ntshz;ik jtpu njhl;lf;fiy_ tdtpay;-ntshz;bghwpapay;- kidapay; rhh;e;j
fUj;Jf;fis btspf;bfbzh;jy;- ntshz;;Jiwf;F ,d;wpaikahj fiyr;brhw;fs;-bkhHpg;bgah;g;g[-
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ngr;Rg;gap-neh;r;rpfhziy vjph;bffs;Sk; tifapy; bkd;jpwd;fshd jiyikg;gz;g[-MSikg;gz;g[-
fhynkyhz;ik Mfpatw;wpy; jpwk;bgwr;bra;jy;-khzth;fspd; Ma;t[f;fl;Liu jpwid tsh;j;jy;-
ntshz;ik ,jH;fs;/ E}y;fs; Fwpj;J tpHpg;g[zh;it tH';Fjy;-fzpdp tHp jkpHpy; ntshz; bra;jpfis
gjpnw;wk;/ gjptpwf;fk; bra;a[k; Kiwfis mwpar;bra;jy; Mfpatw;iwfkfhnehbfbz;L
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El;g';fs;-gjpbdz; fPH;f;fzf;F E}y;fspy; ntshz;ikmwptpay;-gs;S ,yf;fpa';fs;/ VbuGgJ/
,yf;fpaj;jpy; ntshz; bghwpapay;- njhl;ltpay;- tdtpay; kidapay;- NHypay; ntshz;ikg;
gHbkhHpf;fs;- ,yf;fpak; fhl;Lk; thH;tpay; bewpKiw-fs;;fhy ,yf;fpa';fspy; ntshz;ikr;
rpe;jidfs;-gpiHapd;wpvGJk; Kiwfs;-ghuk;ghpaj; bjhHpy;El;';fs;-yffpaj;jpy;
bkd;jpwd;fs; - mwptpay; jkpH; tsh;r;rpepiyfs;-fiyr;brhy;yhf;fk;-bkhHpbgah;g;g[-fl;Liur;
RUf;fk; vGJjy;-fzpdpcyfy; jkpH;

bra;Kiwg; gapw;rpfs;

1. bjhy;fhg;gpak; fhl;Lk; Kjw;bghUs;/ fUg;bghUs;/ jhtutpay; mwpt[/ ntshz; khe;jh; Fw
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2. r';f ,yf;fpaj;jpy; ntshz;py;bjhHEl;g';fs;-(vl;Lj;bjhif/ gj;Jg;ghl;L)
3. gjpbdz; fPH;f;fzf;F E}y;fspy; ntshz;ikmwptpay;
4. gs;S ,yf;fpa';fs;/ VbuGgJ-CHth; thH;tpay; bewpKiwfSk; ntshz;ikj; bjhHpy;
El;g';fSk;
5. ,yf;fpaj;jpy; ntshz; bghwpapay;-njhl;ltpay;-tdtpay;- kidapay;- NHypay;
6. ntshz;ikg; gHbkhHpf;fs;-cHt[tpjmwptpay;- gUtk;- kiH - ehw;WeLjy;- vU
,Ljy;-ePh;g;ghrdk;-fisnkyhz;ik_gaph;ghJfhg;g[-mWtil-CHth; rKjhak;
7. ,yf;fpak; fhl;Lk; thH;tpay; bewpKiwfs;
8. ,f;fhy ,yf;fpa';fspy; ntshz;ikr; rpe;jidfs;-ghuip/ghuipjhrd; gilg;g[fs;-g[Jf;ftpij
9. ,ilepiyg; gUti;njh;t[
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gphpg;g[g;gpiH_ thf;fpag;gpiH_bka;g;g[j; jpUj;jk;
11. ghuk;ghpa ntshz;ikj; bjhHpy;El;g';fs;
12. ,yf;fpaj;jpy; bkd;jpwd;fs;-jiyikg;gz;g[-fhynkyhz;ik
13. MSikg;gz;g[k;ghLnk kdpj cwt[j;jpwd;fs; tsh;j;jy;
14. mwptpay; jkpH; tsh;r;rpepiyfs;/ ntshz; E}y;fs;/ ntshz;-mYtyff;,jH;fs;fojk;
15. fiyr;brhy;yhf;fk;- ntshz; fiyr; brhw;fiscUthf;Fk; _jug;gLj;Jjy;Kiw-
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16. bkhHpbgah;g;g[-Kf;fpapjpf;fs;- goepiyfs;- bkhHpbgah;ghshpd; ,d;wpaikahg;
gz;g[fs;-ntshz; bra;jpfisbkhHpbgah;j;jy;-fl;Liur; RUf;fk; vGJjy;
17. fzpdpcyfy; jkpH;- tpf;fpgPoah-ntshz; bra;jpfisg; gjpnw;wk; bra;jy;-ntshz;
bra;jpfis ,izajstHpmwpjy;

nkW;ghh;it E}y;fs;

1. fe;jrhkp.,y.br.ntshz;ika[k; gz;ghLk;/ jkpH;ehLntshz;ikg; gy;fiyf;fHfk;/
nfhak;gJj 1974
2. fe;jrhkp. ,y.br.,yf;fpaj;jpy; ntshz;ik/ jkpH;ehLntshz;ikg;gy;fiyf;fHfk;/ nfhak;gJj;
1981.
3. fe;jrhkp. ,y.br. ntshz;ikgHbkhHpfs;/ fiyr;bry;tk;jpg;gfk;;/
nfhak;gJj;J}h; 1983. FHe;ijrhkp.th.br.mwptpay; jkpH;/
ghujpgjpg;gfk;/ brd;id
4. kPdhl;rpRe;juk;. kh. kw;Wk; V.,y.tprayl;Rkp./ jfty; bjhlh;gpy; jkpH;
bkhHpg;gad;ghL nf.Mh;.v.Mg;brl; gphpz;lh;/ nfhit-2002
5. kzpnkfiy.k.jkpH; bkhHpj; jlj;jpy; ntshz;ptpaypd;mw RtLfs;/
njtpgjpg;gfk;/ jpUr;rpuhg;gs;sp/ 2002
6. ,yf;fpaKk; ntshz;ika[k;/ midj;jpe;jpamwptpay; jkpH;f; fHfk;/
j";rht{h;;/ 2006 jkpHhpd; kug[r;bry;t';fs;/ cyfj; jkpHuha;r;rpepWtdk;/
brd;id
7. re;jpunrfud;/ ,uh/ bkhHpg;ghlk;-gilg;ghf;fj;jpwd; tsh;j;jy;
8. ntshz;fiyr;brhy; ngufuhjp/ jkpH; ehLntshz;ikg; gy;fiyf;fHfk;/ nfhak;gJj;J}h;/
2008. ghnte;jd;/ ,uh/ jkpHpy; mwptpay; ,jH;fs;/ rhKnty;/ @gp#; fpwp!; gjpg;gfk;/
nfhak;gJj;J}h; lhf;lh; ,uhjhby;yg;gd;/ fiyr;brhy;yhf;fk;/ jkpH;g; gy;fiyf;fHfk;/
j";rht{h;

Unit I

Basic principles of learning

Basic principles of learning - discussion - Bloom's classification of educational objectives – cognitive, affective, psychomotor domain(s) - teaching and learning.

Unit II

Career development

Career development – growth and development, education – for – life and life – long education, motivation and morale - occupation and profession, training and education, lateral thinking and convergent thinking.

Unit III

Entrepreneurship

Entrepreneur- intrapreneur – managing an intrapreneur – motivation and entrepreneurship - development – planning, monitoring and evaluation.

Unit IV

Communication skills

Interpersonal communication – transactional communication - role – play - brainstorming – demonstration -the conduct of symposium - conferencing – the concept and presentation of a paper - scientific article writing and editing - popular article writing, editing and blogging -project proposal - project report – writing.

Unit V

Simulation exercises

Simulation - educational simulation-Interactive teaching - business simulation – company's annual report for analysis.

Lecture Schedule:

1. Basic principles of learning - binary terms viz., growth and development, education – for – life and life – long education, motivation and morale .
2. Occupation and profession, training and education, lateral thinking and convergent thinking, teaching and learning – discussion.
3. Bloom's classification of educational objectives – cognitive, affective, psychomotor domain(s)
4. Career development – opportunity for graduates of agriculture and allied sciences – discussion
5. Success story of a farmer / entrepreneur – factors involved – role – play.
6. Brainstorming – demonstration.
7. Simulation – Educational Simulation-Interactive Teaching - Business Simulation – Company's annual report for analysis
8. Interpersonal communication – Transactional communication – ice breaker
- 9. Mid Semester Examination**
10. The conduct of a symposium
11. Conferencing – the concept and presentation of a paper
12. Scientific Article Writing and Editing
13. Popular Article Writing, Editing and Blogging
14. Project proposal
15. Project Report – writing

16. Entrepreneur – intrapreneur – Managing an intrapreneur – motivation and entrepreneurship development – planning, monitoring and evaluation.
17. **Final Practical Examination**

References

1. Sudarsanam.R 1985. “Development Education” Chapter 1,2
2. Krishna Mohan and Meera Banerji, (1990). “Developing Communication Skills”, Macmillan Pub. Co., Ch.6,9,10,13 and 15.

E-References :

<http://www.e-booksdirectory.com/details.php?ebook=9481>

NSS 101 NATIONAL SERVICE SCHEME (0+1)

I Year

Orientation – NSS origin – motto – symbol – NSS administration at different levels – programme planning – Rural Projects – Urban projects – Government schemes – Career guidance – Self help groups

– Environment protection – Use of natural energy – Conventional energy resources – Soil and Water conservation – Community health programmes – Women and child welfare – Education for all – National days – Commemorative days – NSS thematic programmes – literacy & computer awareness campaigns.

II Year

Popularization of agro techniques – Self employment opportunities – Animal health, Dairy and Poultry farming – Road safety – Training on First aid and emergency cell. Popularization of small savings – communal harmony and National integration – Care of Senior citizens – Personality development – meditation, Yoga Art of living – Activities on the preservation of National monuments, cultural heritage and folklore – special camp activities – National days – commemorative days – NSS thematic programmes – literacy & computer awareness campaigns.

Practical Schedule:

I Semester

1. Orientation of NSS volunteers and programme coordinator and Programme officers.
2. Origin of NSS in India and its development
3. NSS motto, symbol and NSS awards
4. Organizational set up of NSS at Central, State University and college levels.
5. Programme planning – Theme of the year – planning implementation at PC, PO and NSS volunteer level.
6. Visit to selected village - gathering basic data on socio economic status.
7. Participatory rural appraisal – studying the needs of the target group.
8. Visit of urban slum and gathering data on socio economic status.
9. Self involvement and methods of creating rapport with the target group.
10. Awareness campaign on welfare schemes of the central and state government.
11. Formation career guidance group with NSS volunteers and students welfare unit
12. Cycle rally on environmental protection.
13. Campus development activities – clean environment campaign, formation of plastic free zones.
14. Campus development, tree planting maintenance and greening the campus cleaning.
15. **15 Final Examination.**

II Semester

1. 1–3: Motivation of rural and urban youth for formation of SHG (Self Help Groups) in collaboration with Government machineries and NGOs.
2. Campaign on ill effects of plastics in the adjoining campus areas – Villages / urban areas.
3. Campaign on *Parthenium* eradication.
4. Cycle rally on air pollution – Vehicle exhaust and other means.
5. Popularization of biogas and smokeless chulah.
6. Demonstration on the use of wind energy and solar energy.
7. Demonstration of water harvesting techniques.
8. Demonstration on soil conservation techniques wherever possible.
9. Campaign on Community health programmes of central and state Government – involving Health department officials.
10. AIDS awareness campaign ; campaign on diabetes and healthy food habits and drug abuse
11. Planning formation of blood donors club – involving NGOs.
12. Campaign on gender equality and women empowerment.
13. Campaign on child health care – immunization, food habits and child labour abolition.

III Semester

1. Conducting field days with KVK to popularize improved agro techniques.
2. Conducting seminar / workshop in a nearby village to motivate the youth on agribusiness (involving DEE, KVK, NGO and local agro-entrepreneurs).
- 3–5 Campaign on self employment opportunities like Apiculture, mushroom cultivation, Food processing and value addition, production of biocontrol agents and biofertilizers, nursery techniques, seed production, tissue culture, vermicompost, manufacture of small gadgets and agricultural implements as per local needs and feasibility.
6. Animal health care campaign – Dairy and poultry farming - Forage production techniques and silage making.
7. Training the NSS volunteers on road safety measures in involving traffic wardens and RTO.
8. Training NSS volunteers on First AID and emergency call involving NGOs and organizations like St. John's Ambulance, Red Cross, etc.,
9. Organizing road safety rally.
10. Motivating NSS Volunteers on small savings concept and conveying the message to the public through them.
12. Observation of National integration and communal harmony.
- 14– 16 : Campus development and greening activities
17. **Final Examination.**

IV Semester

1. Visit to orphanages and old age homes to look after their needs.
2. Personality development programmes – Building up self confidence in youth.
3. Teaching NSS volunteers on mediation Yoga and art of healthy living with trained teachers
4. Visit of nearby National Monument / Places of tourist importance and campaign on cleanliness and preservation.
5. Exploration of hidden talents of village youth and public on folklore, traditional art, sports, martial arts and cultural heritage . Campus improvement activities Visit to special camp village and pre camp planning.
6. **Final Examination.**

Besides the above, NSS volunteers will attend work during important occasions like Convocation, Farmers day, Sports meet and other University / College functions. NSS Volunteers will attend one special camp in the selected village for a duration of 10 days and

undertake various activities based on the need of that village.

For all out door regular activities villages / slums nearby the campus may be selected to avoid

transport cost (cycle able distance)

Special camp activity will be conducted in a village situated within a radius of 15 – 20 KM.

EVALUATION

A. Regular activities

60	=	I Semester	15 marks
marks		II Semester	15 marks
		III Semester	15 marks
		IV Semester	15 marks

(Written test 10 marks – participation in programmes and behavior-5 marks) 80% attendance is mandatory for attending special camp

B. Special camp activities

- | | |
|--|----------|
| a. Attendance in daily activities during special camp: | 30 marks |
| b. Special camp activity report : | 5 marks |
| c. Viva - voce on the 10th day of the special camp : | 5 marks |

Tota	:	40 marks
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NCC 101 National Cadet Corps (0+1)

I Year

General - Military History – Introduction to NCC – Aims of NCC – Principles of NCC, NCC organization, Duties of good citizen – system of NCC training – Foot drill – Arms drill – Guard of Honour – Ceremonial Drill – Weapon training – First aid – Rifle and Light machine gun – Map reading – Civil defence – Leadership.

II Year

Drill – Weapon drill – Weapon training and firing – Introduction to National Integration – Historical – geographical – Religions back ground of India – Health and Sanitation – Aid to Civil Authorities – Civil defence – Ecology / Nature awareness – Map reading – Social service – Adventure Activities – Leadership qualities.

ISemester

1. NCC song – Aims and Motto of NCC – Motivation of cadets
2. History of NCC and organization of NCC
3. Food drill – General and word of Command
4. Human Resource Development – Motivation – Duties of Good citizen
5. National Integration – Indian History and Culture
6. Health and Hygiene – Structure and Function of a human body, hygiene and Sanitation
7. Social Service – weaker sections of our society and their needs
8. Self Defence – Theory and practice, prevention of untoward incidence
9. Map reading – introduction to map, and lay out of map
10. Disaster Management Civil defence organization and its duties
11. Communication – Different types – media
12. Signals – introduction to radio, telephony procedures
13. Field Engineering – principles and applications, camouflage and concealment
14. Adventure training introduction, different types
15. First Aid – methods and practices
16. Environment and Ecology – conservation
17. **Final Examination.**

II Semester

1. Drill – Weapon drill – Word of Commands
2. National integration- unity in diversity
3. Guard of Honour and Ceremonial drill
4. Types of weapon, Parts, Stripping and Assembling of light gun.
5. Rifle firing and follow up activities
6. Camps, types of Camps, Preparation and participation
7. Awards, different types, Ranks of officers and Cadets
8. Map reading – judging distance, conventional signs and uses of compass.
9. Leadership traits, types, perception
10. Fire Fighting, Role of NCC during natural hazards
11. Field Engineering – section formation
12. Obstacle training
13. Health and Sanitation – preventable diseases, Fractures and types of treatments
14. Environment and Ecology-Pollution and its control.
15. Social Service – contribution of youth towards social welfare
16. First Aid – Snake bite and other common medical Emergencies.
17. **Final Examination.**

III Semester

1. Drill – Individual word of command
2. Weapon training – parts of heavy weapons
3. Stripping and assembling of heavy weapons
4. Importance of team work values, code of ethics
5. Disaster management during Earth Quake
6. Evacuation of Casualties
7. Map reading – Camposs and Service Protractor
8. Aids to civil authority
9. Section and platoon formation
10. Social service, NGO's and their contribution to the society
11. Roll of NCC cadets in civil administration
12. Traffic rules and Road signs
13. Mines and types of mine fields
14. Dressing of Wounds, physical and mental health
15. Field signals
16. Air raid warning, Fire fighting
17. **Final Examination.**

IV Semester

1. Drill – Foot drill
2. Formation of squad and squad drill
3. Man Management, Morale
4. Time Management, stress management
5. Ecology and Environment wild life conservation
6. Adventure Activities, Trekking Camp
7. Map reading – Field to Map – Map to Field – Grids and scale systems
8. Communication systems – Internet – Faxi mail – Satellites
9. Collection and Distribution of Aid material
10. Field Engineering – Mines, anti tanks, explosives
11. Opportunities for NCC cadets in Army and other services
12. Social Service, Family Planning
13. Section battle drill
14. Roll of NCC cadets in National programmes.
15. Visit to Wellington, Coonoor.
16. Self defence mechanisms
17. **Final examination.**

Besides the above schedule, NCC cadets will be involved during important occasions during convocation, Independence day, Republic day, etc.

EVALUATION:

		Sem I	Sem II	Sem III	Sem IV	Total
A.	Regular activities and Behaviour	10	10	10	10	40
B.	Participation in camps and special assignments	5	5	5	5	20
C.	Written test and viva	10	10	10	10	40
	Total	25	25	25	25	100

PED 101 Physical Education (0+1)

Practical

(17 Practical classes – 2½ hours each class – 17 classes will be converted into 40 practical hours and 2½ hours for evaluation)

I Semester (20 Hours)

Exercises for strength, agility, co-ordination, flexibility, co-operation, vital capacity endurance, speed and for various systems of our body and team spirit.

Exercise for Good Posture – Conditioning and calisthenics for various Athletic activities *i.e* (a) Before start – Arm stretch, hand stretch and cat stretch (b) Loosening up jogging, bending and twisting (c) Standing – Lateral Arc, triangle and hands to feet pose (d) Sitting – camel kneel, spinal twist and supine knee bend (e) Relaxation – The corpse pose, quick and deep relaxation. Basic gymnastic exercises – participation of athletic events – running, throwing and jumping events.

Skill development in anyone of the following games

Warming up, suitable exercise, lead up games, advance skill for all the games.

Basket Ball : Dribbling, pass, two or three men pass, pivot, lay up shot, shooting, pass break, hook pass, screening, positional play, defence and offence tactics.

Volley Ball : Fingering, under arm pass, over head pass, setting, spiking, back pass, jump pass, stunts, elementary dive, flaying dive, roll, blocking and various types of services.

Ball Badminton : Grip, service, foot work, fore hand stroke, back hand stroke, lob, smash, volley, wall practice, spin service and defence tactics.

Foot ball : Dribbling, passing, dodging, kicking, heading, screening, chest pass, throwing, dragging, goal kick, defence and offence tactics.

Hockey : Grip, bully, dribbling, hitting, drive, push strokes, scoop, flick, stopping, various types of passes, dodging, defence and offence tactics.

Kho-Kho : Quadra ped, bi-ped, how to given kho, taking a direction, recede, parallel toe method, bullet tow method, distal method, foot out, dive, ring game, chains and pursue and defence skills.

Chess : Moves, move of king, move of pawns, move of rooks, move of bishops, move of queen, move of knights, en passant, castling, check and notation.

Kabaddi : Raid, touch, cant, catch, struggle, various types of defence and offence tactics.

Cricket : Grip, bowling, spin, leg spin, off spin, medium, batting, dive, sweep, mode of delivery, fielding, rolling etc.

Tennis : Grip, forehand drive, back hand drive, stroke, backhand ground stroke, service, volley, smash, wall practice, foot work, defence and offence tactics.

Table Tennis : Grip, tossing and serving, spin serve, rally, smash, flick, defence and offence tactics.

Shuttle Badminton : Grip, foot work, service, setting, smash, volley, forehand and back hand stroke, back hand serve and defence.

Gymnastics : Balanced walk, execution, floor exercise, tumbling/acrobatics, grip, release, swinging, parallel bar exercise, horizontal bar exercise, flic-flac-walk and pyramids.

ATHLETICS

- (a) **Sprint** : Medium start, long start, bunch start, set, pick up, finish, upsweep, downsweep, placement, receiving and exchanging.
- (b) **Jumps** : Western roll, belly roll, eastern cut off, fass ferry flop, approach, take off, straddle, hitch-kick, handging, clearance, landing, strides etc.
- (c) **Throws** : Grip, momentum, pre shift, sub phase, the wind up, foot work, entry to the turn, shift, angle of release, follow throw, delivery, front cross step, rear cross step, hop step, fuck method pary obraine, discoput, rotation, carry and glide.
- (d) **Hurdles** : Finding lead leg, use of lead leg and trial leg, flight, clearing, finish.

Lead up games, advance skills and game for any one of the above games.

II Semester (20+ 2 ½ hours)

Rules and regulations of anyone of the games and athletic events.

Aims and objectiaves of yoga – asanas : ie. padmasana, pujankasana, sarvangasana, chakrasana,dhanurasana, halasana, mayurasana and savasana, asanas for ailments, back pain, arthritis, abdominal problem, stress, fatiguel, Insomnia, obsity, circulation, hypertension, varicose veins, respiration, heart, digenstion, headaches, depression, addiction and eye problems.

Mental balance and importance – development of concentration suriyanamaskar – advance skills of any one of the games which were taught in the I semester.

METHOD OF EVALUATION:

a. Attendance	60 Marks
b. Behavior	10 Marks
c. Participation in Sports and Games	20 Marks
d. Final Viva Voce	10 Marks

Marks will be awarded at the end of the IV Semester based on the above method of evaluation procedure. Final class grade chart of each student will be sent to the Dean of concerned colleges of Tamil Nadu Agricultural University.

PED 102 YOGA FOR HUMAN EXCELLENCE (0+1)

UNIT - 1:

PHYSICAL HEALTH AND REJUVENATION OF LIFE-FORCE

Significance of Value Education - Types of Education – Yoga for Human Excellence Principles and Purpose of living - Body structure – Body functions – Reasons for Diseases and Prevention - Concept of Health – Role of limit and method in five deeds for good health - Importance of Naturopathy - Objectives of physical exercises Benefits of physical exercises - Kayakalpa yoga philosophy - Youthfulness practices Enriching bio-magnetism.

UNIT - 2:

MENTAL PROSPERITY AND SOCIAL WELFARE

Mind functions – Mental frequency – Thought – Brain and Memory power – Problem solving and Decision making skills - Need and benefits for meditation - SKY Yoga types of meditation Part 1: Eye brow centre meditation - Genetic centre meditation - Spinal cord clearance - Crown centre meditation - Analysis of thoughts – Moralization of desires - Neutralization of Anger – Eradication of Worries – Benefits of blessings - Human culture and values – Five-fold culture - Time management – Personality Assessment - Environment awareness and protection - Family peace – World peace - Five duties - Harmonious friendship – Greatness of Womanhood.

UNIT - 3:

YOGA PRACTICES – I

PHASE I - Simplified Physical Exercises: Hand exercise - Leg exercise – Neuro muscular breathing exercise – Eye exercise – Kapalabathi - PHASE II – Makarasana Part 1 & 2 – Body massage - Acu-pressure – Relaxation exercise - Youthfulness practices (Kayakalpa) - SKY Yoga types of meditation Part 1: Eye Brow centre meditation - Genetic centre meditation - Spinal Clearance - Crown centre meditation.

Practical Schedule:

1. Significance of Value Education - Types of Education – Yoga for Human Excellence – Eye brow centre meditation (Agha) - Simplified Physical Exercises – Objective of physical exercises – Benefits of exercises.
2. Principles and Purpose of living - Genetic centre meditation - Explanation and initiation of Genetic centre - SPE – Hand exercises, Leg Exercises, Neuro Muscular Breathing exercises, Eye exercises, Kapalabathi and Relaxation
3. Kayakalpa yoga philosophy - Youthfulness practices - Enriching bio-magnetism - Eye brow
4. centre meditation Practice (Agha) - Kayakalpa Yoga – Explanation and Kayakalpa Practice
5. Body structure – Body functions - Genetic centre meditation Practice - Simplified Physical Exercises - Makarasana, Massage and Acupressure and Relaxation - Kayakalpa
6. Concept of Health – Role of limit and method in five deeds for good health - Spinal cord Clearance - Explanation and practice - Simplified Physical Exercises Full exercises – Kayakalpa
7. Reasons for Diseases and Prevention - Crown centre meditation- Initiation (Thuriyam)
8. Importance of crown centre meditation - Simplified Physical Exercises Full exercises - Kayakalpa

9. Importance of Naturopathy - Crown centre meditation- (Thuriyam) - Simplified Physical Exercises Full exercises - Kayakalpa
10. Mind functions – Mental frequency – Thought – Brain and Memory power - Crown centre meditation- (Thuriyam) - Simplified Physical Exercises Full exercises - Kayakalpa
11. Analysis of thought - Moralization of desire - Genetic centre meditation Practice - Simplified Physical Exercises Full exercises - Kayakalpa
12. Neutralization of Anger – Eradication of Worries – Eye brow centre meditation Practice (Agni) - Simplified Physical Exercises Full exercises - Kayakalpa
13. Benefits of blessings - Human culture and values – Crown centre meditation- (Thuriyam) - Simplified Physical Exercises Full exercises - Kayakalpa
14. Fivefold culture – Time management - Crown centre meditation- (Thuriyam) - Simplified Physical Exercises Full exercises - Kayakalpa
15. Environment awareness and protection - Genetic centre meditation Practice - Simplified Physical Exercises Full exercises - Kayakalpa
16. Family peace – World peace - Harmonious friendship – Crown centre meditation- (Thuriyam)
17. Simplified Physical Exercises Full exercises – Kayakalpa
18. Greatness of Womanhood - Five duties - Genetic centre meditation Practice - Simplified Physical Exercises Full exercises - Kayakalpa
19. Physical Exercises Full exercises - Kayakalpa
20. Personality Assessment - Crown centre meditation- (Thuriyam) - Simplified Physical Exercises Full exercises – Kayakalpa
21. Physical health and mental health – revision

II Semester

1	SWE 112	Soil and Water Conservation Engineering	1+1
2	CRP 101	Fundamentals of crop Physiology	2+1
3	AEC 102	Fundamentals of Agricultural Economics	1+1
4	AGM 101	Fundamentals of Microbiology	2+1
5	AEX 102	Fundamentals of Agricultural Extension Education	2+1
6	FSN 111	Principles of Food Science and Nutrition	1+1
7	AGR 102	Introductory Agro-meteorology & Climate Change	1+1
8	HOR 112	Production Technology for Fruit and Plantation Crops	1+1
9	RSG 101	Geo- informatics for Precision Farming	1+0
10	NSS/NCC 101	NSS/NCC	0+1*
11	PED 101	Physical Education	0+1*
		Total	12+8=20
		*Non-gradual courses compulsory courses	

THEORY**UNIT I SURVEYING**

Surveying and levelling – chain and compass – levelling – land measurement – difference in elevation.

UNIT II SOIL EROSION

Soil Erosion – causes and evil effects of soil erosion – geologic and accelerated erosion – universal soil loss equation - water erosion - causes - stages of water erosion - splash, sheet, rill and gully erosion - ravines - land slides – wind erosion - factors influencing wind erosion - mechanics of wind erosion – suspension, saltation, surface creep

UNIT III SOIL CONSERVATION AND WATERSHED MANAGEMENT

Erosion control measures for agricultural lands – biological measures – contour cultivation – strip cropping – cropping systems – vegetative barriers - windbreaks and shelterbelts - shifting cultivation - mechanical measures – contour bund – graded bund – broad beds and furrows – basin listing – random tie ridging – mechanical measures for hill slopes – contour trench – bench terrace – contour stone wall – Rain water harvesting — Runoff computation - rational formula - water harvesting – farm ponds and percolation ponds -watershed concept – integrated approach and management

UNIT IV IRRIGATION AND DRAINAGE

Irrigation - measurement of flow in open channels - velocity area method - rectangular weir - Cippoletti weir - V notch - orifices - Parshall flume - duty of water - irrigation efficiencies - conveyance of irrigation water - surface irrigation methods - borders, furrows and check basins - drip and sprinkler irrigation component– agricultural drainage - surface and sub-surface drainage systems - drainage coefficient

UNIT V WELLS AND PUMPS

Types of wells– pump types – reciprocating pumps – centrifugal pumps – turbine pumps – submersible pumps – jet pumps – airlift pumps

PRACTICAL

Study of survey instruments - chains and cross staff surveying - linear measurement - plotting and finding areas. Compass survey - observation of bearings - computation of angles- radiation, intersection. Levelling – fly levels – determination of difference in elevation – contouring. Design of contour bund and graded bund. Drip systems and Sprinkler irrigation systems. Problems on water measurement. Problems on duty of water, irrigation efficiencies. Problems on water requirement - agricultural drainage. Study of different types of wells and its selection.- Study of pumps and its selection- Visit to soil and water conservation areas

LECTURE SCHEDULE

1. Introduction - land surveying - uses in agriculture - chain and cross staff surveying.
2. Compass surveying - computation of angles.
3. Dumpy level - setting, observation and tabulation of readings - difference in elevation.
4. Soil Erosion – causes and evil effects of soil erosion – geologic and accelerated erosion – universal soil loss equations.
5. Water erosion - causes - stages of water erosion - Splash, sheet, rill and gully erosion - ravines - land slides
6. Wind erosion - factors influencing wind erosion - mechanics of wind erosion – suspension, saltation, surface creep - Effects of water and wind erosion
7. Erosion control measures for agricultural lands – biological measures – contour cultivation – strip cropping - Cropping systems – vegetative barriers - Windbreaks and shelterbelts - shifting cultivation.
8. Mechanical measures – contour bund – graded bund - Broad beds and furrows – basin listing – random tie ridging - Mechanical measures for hill slopes – contour trench – bench terrace – contour stone wall
9. Mid semester examination.
10. Rain water harvesting – runoff computation - rational formula - runoff water harvesting - farm ponds and percolation ponds.
11. Watershed concept – Integrated approach and management

12. Irrigation - measurement of flow in open channels - velocity area method - Rectangular weir - Cippoletti weir - V notch - Orifices - Parshall flume
13. Duty of water - irrigation efficiencies - conveyance of irrigation water.
14. Surface irrigation methods - borders, furrows and check basins
15. Components of drip and sprinkler irrigation system
16. Agricultural drainage – need – surface and subsurface drainage systems - drainage coefficient
17. Types of wells - Pump types – reciprocating pumps – centrifugal pumps - Turbine pumps – submersible pumps - Jet pumps – Airlift pumps

PRACTICAL SCHEDULE

1. Study of survey instruments - chains - compass - dumpy level.
2. Chains and cross staff surveying - linear measurement - plotting and finding areas.
3. Compass survey - observation of bearings - computation of angles.
4. Compass - radiation, intersection.
5. Levelling – fly levels
6. Computation of area
7. Computation determination of difference in elevation.
8. Design of contour bund and graded bund.
9. Design of drip Irrigation systems.
10. Design of sprinkler irrigation system
11. Problems on water measurement.
12. Problems on duty of water, irrigation efficiencies.
13. Problems on water requirement - agricultural drainage.
14. Study of different types of wells
15. Study of pumps and Selection of pumps.
16. Visit to soil and water conservation areas.
17. Final practical examination.

TEXT BOOKS

1. Basak, N.N. 2008. Surveying and Levelling. 25th reprint. Tata Mc-Graw Hill Publishing Company Ltd
2. Michael, A.M. and Ojha, T.P. 2008. Irrigation Theory and Practice. Second Edition. Vikas Publication House, New Delhi

E- REFERENCES

- <http://nptel.ac.in/courses/105107122/13>
- <http://soilwater.okstate.edu/courses/lectures-powerpoint>

CRP 101 Fundamentals of Crop Physiology (2+1)

Aim

To impart basic knowledge on various functions and processes related to crop production, mineral nutrition, plant growth regulators and environmental stresses.

Theory

Unit I

Introduction to Crop Physiology and importance of Crop Physiology in Agriculture – Plant cell: an overview, organelles- plasma membrane, chloroplast, mitochondria, peroxisome and vacuole, Structure and role of water, water potential and its components, diffusion and osmosis; imbibition, plasmolysis, Field Capacity and Permanent Wilting Point, Absorption of water, Mechanisms of water absorption, Pathways of water movement, Apoplast and symplast, Translocation of water, ascent of sap and its mechanisms - Transpiration and Stomatal physiology: structure of stomatal pore, mechanisms of stomatal opening and closing, guttation, antitranspirants.

Unit II

Mineral nutrition of plants: Criteria of essentiality, classification of nutrients, macro, micro, mobile, immobile and beneficial elements, Physiological functions and deficiency symptoms of nutrients, nutrient uptake mechanism; Hidden hunger, Foliar nutrition, root feeding and fertigation, sand culture, hydroponics and aeroponics.

Unit III

Photosynthesis: Light and dark reactions, Photosystems, red drop and Emerson enhancement effect, Photolysis of water and photophosphorylation, Z scheme, C₃, C₄ and CAM plants; Photosynthetic pathways of C₃, C₄ and CAM plants, difference between three pathways, Factors affecting photosynthesis, Photorespiration – pathway and its significance, Phloem transport, Munch hypothesis, Phloem loading and unloading, Source and sink strength and their manipulations. Respiration: Glycolysis, TCA cycle and electron transport chain; Oxidative phosphorylation – difference between photo and oxidative phosphorylation — energy budgeting - respiratory quotient. Fat metabolism: fatty acid synthesis and breakdown.

Unit IV

Plant growth regulators: physiological roles and agricultural uses, Hormones- classifications - Biosynthetic pathway and role of auxins, gibberellins, cytokinins, ethylene and ABA, Novel and new generation PGRs, Brassinosteroids and salicylic acid, Growth retardants, Commercial uses of PGRs. Photoperiodism - short, long and day neutral plants, Chailakhyan's theory of flowering, Forms of phytochrome, Pr and Pfr, regulation of flowering, Vernalisation - Theories of vernalisation, Lysenko theories, Seed germination - physiological and biochemical changes, seed dormancy and breaking methods, Senescence and abscission, physiological and biochemical changes, Physiology of fruit ripening, climacteric and non-climacteric fruits, factors affecting ripening, Manipulations. Physiological aspects of growth and development of major crops: growth analysis, role of physiological growth parameters in crop productivity.

Unit V

Classification of stresses - Physiological changes and adaptations to drought, flooding, high and low temperature, salinity and UV radiation – compatible osmolytes – membrane properties - compartmentalization – stress alleviation - Global warming – green house gases – physiological effects on crops - Carbon Sequestration.

Practicals

Study of plant cells, structure and distribution of stomata, imbibition, osmosis, plasmolysis, measurement of root pressure, rate of transpiration, Separation of photosynthetic pigments through paper chromatography, Rate of transpiration, photosynthesis, respiration, tissue test for mineral nutrients, estimation of relative water content, Measurement of photosynthetic CO₂ assimilation by Infra Red Gas Analyser (IRGA).

Theory lecture schedule

1. Introduction and importance of Crop Physiology in Agriculture, an over view of Plant cell.
2. Structure and role of water –water potential and its components – Diffusion – Osmosis – imbibition – Plasmolysis - Field Capacity and Permanent Wilting Point
3. Mechanisms of water absorption – Pathways of water movement – Apoplast and symplast
4. Translocation of water – ascent of sap – mechanisms of xylem transport
5. Transpiration – significance – Stomatal physiology: structure of stomata with mechanisms of stomatal opening and closing – guttation - antitranspirants
6. Mineral nutrition of plants – criteria of essentiality - classification of nutrients – macro, micro, mobile and immobile – beneficial elements, mechanism of nutrient uptake
7. Physiological functions and disorders of macronutrients, Hidden hunger
8. Physiological functions and disorders of micronutrients
9. Foliar nutrition- root feeding and fertigation – sand culture, hydroponics and aeroponics
10. Light reaction of photosynthesis – photolysis of water and photophosphorylation - Z scheme
11. Dark Reaction of photosynthesis - C₃, C₄ and CAM pathways and differences.
12. Factors affecting photosynthesis - Photorespiration – pathway and its significance
13. Phloem transport – Munch hypothesis - Phloem loading and unloading - Source and sink strength and their manipulations
14. Respiration - Glycolysis – TCA cycle.
15. Photo and oxidative phosphorylation - Electron transport chain - energy budgeting - respiratory quotient.
16. Fat metabolism: fatty acid synthesis and breakdown
17. Mid Semester Examination
18. Growth – phases of growth – factors affecting growth.
19. Hormones and plant growth regulators (PGR): physiological roles and agricultural uses - Biosynthetic pathway and role of auxins and gibberellins
20. Plant growth regulators (PGR): physiological roles and agricultural uses - Biosynthetic pathway and role of cytokinin, ethylene and ABA
21. Novel growth regulators viz., Brassinosteroids and salicylic acid – New Generation PGRs - Growth retardants and inhibitors -commercial uses of PGRs
22. Photoperiodism - short, long and day neutral plants – Chailakhyan's theory of flowering
23. Forms of phytochrome - Pr and Pfr - regulation of flowering
24. Vernalisation - theories of vernalisation – Lysenko and Hormonal theories – devernalization
25. Physiological aspects of growth and development of major crops
26. Growth analysis – role of physiological growth parameters in crop productivity
27. Seed germination - physiological and biochemical changes - seed dormancy and breaking methods
28. Senescence and abscission – physiological and biochemical changes
29. Physiology of fruit ripening- climacteric and non climacteric fruits - factors affecting ripening and manipulations
30. Drought - physiological changes - adaptation – compatible osmolytes - alleviation
31. High and low temperature stress – physiological changes - membrane properties - adaptation
32. Salt stress - physiological changes - adaptation – compartmentalization - alleviation
33. Flooding and UV radiation stresses – physiological changes - adaptation
34. Global warming – green house gases —physiological effects on crop productivity- Carbon Sequestration

Practical schedule

1. Preparation of solutions
2. Study of leaf epidermal, xylem and phloem cells
3. Determination of stomatal index and stomatal frequency
4. Measurement of plant water potential
5. Measurement of water imbibition by seed mass test

6. Estimation of photosynthetic pigments
7. Determination of photosynthetic efficiency in crops
8. Measurement of transpiration and photosynthesis by IRGA
9. Diagnosis of nutritional and physiological disorders in crops
10. Rapid tissue test for mineral nutrients
11. Estimation of relative water content
12. Measurement of osmosis and plasmolysis
13. Growth Analysis
14. Bioassay for gibberellin and cytokinin
15. Estimation of chlorophyll stability index
16. Estimation of proline content
17. Final Practical Examination

Outcome

Students will acquire basic knowledge on various functions and processes related to crop production, mineral nutrition, plant growth regulators and environmental stresses. In addition, hands on exposure to preparation of solutions, analysis of pigment composition, estimation of growth analytical parameters, diagnosis and correction of nutrient deficiencies, enzyme assays and demonstration of plant growth regulator applications.

Text books

1. Salisbury F.B. and C.W.Ross., 1992 (Fourth Edition). Plant Physiology. Publishers: Wadsworth Publishing Company, Belmont, California, USA.
2. Boominathan P., R. Sivakumar, A. Senthil, and D. Vijayalakshmi. 2014. Introduction to Plant Physiology, A.E. Publications. Coimbatore
3. Jain, V.K. 2007. Fundamentals of plant physiology, S.Chand & Company Ltd., New Delhi.
4. Taiz. L. and Zeiger. E., 2015 (Sixth edition). Plant Physiology and Development. Publishers: Sinauer Associates, Inc., Massachusetts, USA.

e- books and e-references

- <http://www.plantphys.org>
- [http://www. Biologie. Uni-hamburg. de/b-online](http://www.Biologie.Uni-hamburg.de/b-online)
- <http://6e.plantphys.net>

AEC 102 Fundamentals of Agricultural Economics (1+1)

Objective

This course aims to introduce the basic principles of economics including the problem of agricultural economics decision - making, laws of economics and macroeconomic concepts.

Theory

Unit 1: Nature and Scope of Economics

Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro and macroeconomics, positive and normative analysis. Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behavior. Basic concepts: Goods and services, classification and characteristics, desire, want - meaning and characteristics, demand, utility, cost and price, wealth, capital, income and welfare. Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development. Agricultural planning and development in the country.

Unit 2: Theory of Consumption

Demand: meaning, kinds of demand, law of demand, demand schedule and demand curve, determinants; *Utility theory* - cardinal and ordinal utility; law of diminishing marginal utility, equi-marginal utility principle, Indifference curve analysis and properties - budget line: definition, assumptions, limitations and applications - consumer's equilibrium and derivation of demand curve. Concept of consumer surplus and its importance. *Elasticity of demand*: concept and measurement of price elasticity, income elasticity and cross elasticity. Factors influencing elasticity of demand - Importance of elasticity of demand – Standard of Living: Definition, Engel's Law of Family Expenditure.

Unit 3: Theory of Production

Production: process, creation of utility, factors of production definition and characteristics - Input Output Relationship. *Laws of returns*: Law of variable proportions and Law of returns to scale. *Cost*: Cost concepts, short run and long run cost curves. *Supply*: Stock versus supply, law of supply, supply schedule, supply curve, determinants of supply, elasticity of supply.

Unit 4: Exchange and Theory of Distribution

Market structure: meaning and types of market, basic features of perfectly competitive and imperfect markets. Price determination under perfect competition; short run and long run equilibrium of firm and industry, shut down and break even points. *Distribution theory*: meaning, factor market and pricing of factors of production - Concepts of Rent and Quasi rent - *Wages*: Real wage and money wage - *Interest*: Pure interest and gross interest – *Profit*: Meaning of economic profit.

Unit 5: Macroeconomic Concepts

National income: Meaning and importance, circular flow, concepts of national income - accounting and approaches to measurement, difficulties in measurement. *Population*: Importance, Malthusian and Optimum population theories, natural and socio-economic determinants, current policies and programmes on population control. *Money*: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, money supply, general price index, inflation and deflation. *Banking*: Role in modern economy, types of banks, functions of commercial and central bank, credit creation policy. *Agricultural and public finance*: meaning, micro versus macro finance, need for agricultural finance, public revenue and public expenditure. *Tax*: meaning, direct and indirect taxes, agricultural taxation, VAT and GST. *Economic systems*: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning.

Practical

Law of Diminishing Marginal Utility - Law of Equi-Marginal Utility - Indifference Curve analysis - consumer equilibrium; Individual and market demand- Measurement of Arc and Point elasticities of demand - own price, income and cross price elasticities of demand – Estimation of Consumer surplus – Law of Diminishing Marginal Returns: Relationship among TPP, APP and MPP - Cost concepts and graphical derivation of cost curves - Estimation of total revenue and profit- Producer surplus - Supply elasticity – Exchange: Market Structure and Price determination – Computation of National Income – Study of structural changes in the economy - Estimation of Growth Rate - Money: Quantity theory of money - Measures of standard of living – Indices of human development.

References

1. Dewett, K. K. 2004. Modern Economic Theory, Syamlal Charitable Trust, New Delhi.
2. Samuelson, P. 2004. Economics, (18/e), Tata Mc-graw-Hill, New Delhi
3. Seth, M. L. 2005. Principles of Economics, Lakshmi Narain Agarwal Co., Agra. New Delhi.

Theory Schedule

1. *Economics*: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro and macroeconomics, positive and normative analysis.
2. Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behavior. Basic concepts: Goods and services - classification and characteristics, desire, want – meaning and characteristics, demand, utility, cost and price, wealth, capital, income and welfare.
3. Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development. Agricultural planning and development in the country.
4. *Demand*: meaning, kinds of demand, law of demand, demand schedule and demand curve, determinants.
5. *Utility theory* - cardinal and ordinal utility; law of diminishing marginal utility, equi-marginal utility principle, Indifference curve analysis and properties - budget line - definition, assumptions, limitations and applications.
6. Consumer's equilibrium and derivation of demand curve, concept of consumer surplus and its importance.
7. Elasticity of demand: concept and measurement of price elasticity, income elasticity and cross elasticity. Factors influencing elasticity of demand, Importance of elasticity of demand. Standard of Living: Definition, Engel's Law of Family Expenditure.
8. *Production*: process, creation of utility, factors of production definition and characteristics - Input Output Relationship.
9. **Mid Semester Examination**
10. *Laws of returns*: Law of variable proportions and law of returns to scale. *Cost*: Cost concepts, short run and long run cost curves.
11. *Supply*: Stock versus supply, law of supply, supply schedule, supply curve, determinants of supply, elasticity of supply.
12. Market structure: meaning and types of market, basic features of perfectly competitive and imperfect markets. Price determination under perfect competition; short run and long run equilibrium of firm and industry, shut down and break even points.
13. Distribution theory: meaning, factor market and pricing of factors of production. - Concepts of Rent and Quasi rent - *Wages*: Real wage and money wage - *Interest*: Pure interest and gross interest – *Profit*: Meaning of economic profit.
14. *National income*: Meaning and importance, circular flow, concepts of national income - accounting and approaches to measurement, difficulties in measurement.
15. *Population*: Importance, Malthusian and Optimum population theories, natural and socio-economic determinants, current policies and programmes on population control.
16. *Money*: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, money supply, general price index, inflation and deflation. *Banking*: Role in modern economy, types of banks, functions of commercial and central bank, credit creation policy.
17. *Agricultural and public finance*: meaning, micro versus macro finance, need for agricultural finance, public revenue and public expenditure. *Tax*: meaning, direct and indirect taxes, agricultural taxation, VAT and GST. *Economic systems*: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning.

Practical Schedule

1. Law of Diminishing Marginal Utility.
2. Law of Equi - Marginal Utility.
3. Indifference Curve analysis - Properties, budget line and consumer equilibrium.
4. Individual and market demand - Graphical derivation of individual and market demand.
5. Measurement of Arc and Point elasticities of demand - own price, income and cross price elasticities of demand.
6. Estimation of Consumer surplus.
7. Law of Diminishing Marginal Returns: Relationship among TPP, APP and MPP.
8. Cost concepts and graphical derivation of cost curves.
9. Estimation of total revenue and profit.
10. Estimation of Producer surplus.
11. Estimation of Supply elasticity.
12. Exchange: Market Structure and Price determination.

13. Computation of National Income – Analysis of Trends in National Income - Study of structural changes in the economy.
14. Estimation of Growth rate of population and Food grain production.
15. Money: Quantity theory of money.
16. Measures of standard of living – Human Development Index – Physical Quality of Life Index – Gender Development Index.
17. **Final Practical Examination.**

AGM 101 Fundamentals of Microbiology (2+1)

Theory

Unit I. Introduction

Definition and scope of Microbiology. Milestones in Microbiology; biogenesis and abiogenesis theory; contributions of Antonie Van Leeuwenhoek, Louis Pasteur, John Tyndall, Robert Koch, Edward Jenner, Joseph Lister, Alexander Fleming and Waksman. Germ theory of diseases and fermentation.

Unit II. Microbiological Techniques

General principles of light microscopy - magnification, resolving power and numerical aperture. Different types of light and electron microscopes; three dimensional imaging - Atomic force and Confocal scanning laser microscopy. Staining techniques - principle and types of stains; simple, negative, differential and structural staining. Sterilization and disinfection techniques; principles and methods of sterilization - physical methods – heat, filters and radiation; chemical methods. Isolation, enrichment and purification techniques of bacteria, yeast, molds and actinobacteria. Preservation of microbial cultures.

Unit III. Microbial World

Evolutionary relationship among the living organisms. Whittaker's five kingdom concept of living organism and Carl Woese systems. Procaryotic and eukaryotic microorganisms. Three domains of life – similarities and differences; Modern approach to the bacterial systematics. Bergey's Manual of Systematic Bacteriology. Bacteria - bacterial size, shape and arrangement; bacterial cell structure and function. Morphology of fungi and algae. General properties of viruses: different types; overview of bacteriophages; morphology of bacteriophages: Lytic and lysogenic cycles; lytic and temperate phages.

Unit IV. Microbial Growth and Metabolism

Bacterial growth- population growth- growth cycles of population; environment on growth – temperature, oxygen, pH and salts; nutritional classification – chemoautotrophy and photo autotroph. Energy generation in bacteria. Aerobic and anaerobic respiration and fermentation in bacteria.

Unit V. Microbial Genetics and Immunology

Central dogma of life. Genetic elements of bacteria; bacterial chromosomal DNA, plasmids, IS elements and transposons; Mutation - types and mutagens. Genetic recombinations; transformation, transduction and conjugation. Genetic engineering – an introduction. Basic concepts of immunology – antigen – antibody reactions and vaccines.

Practical

Microbiological safety in the laboratory; introduction to microbiology laboratory and its equipments. Microscopes- handling with microscope. Micrometry. Methods of sterilization and equipments used for sterilization. Nutritional media and their preparations. Enumeration of microbial population - bacteria, fungi and actinobacteria. Methods of purification and preservation of microbial cultures. Staining and microscopic observations; simple and differential staining - spore staining. Measurement of bacterial growth. Identification of microorganisms - morphological identification of yeasts, molds and algae. Identification- cultural, physiological and biochemical tests for bacteria and actinobacteria. Isolation of bacteriophages. Isolation of mutants employing physical or chemical mutagens.

Theory schedule

1. Definition and scope of microbiology
2. Biogenesis and a biogenesis theory. Contributions by Antonie Van Leeuwenhoek and Louis Pasteur
1. Contributions of John Tyndall, Joseph Lister, Edward Jenner, Robert Koch, Alexander Fleming and Waksman. Germ theory of fermentation and disease
4. Microscopy; principles – resolving power and magnification. Light microscopy
5. Different types of microscopes - UV, dark field, phase contrast and fluorescence
6. Electron microscopes; atomic and confocal scanning laser microscopy
7. Staining techniques - principle and types of stains - staining techniques- simple, negative, differential and structural staining methods
8. Sterilization – principle – physical agents and chemical methods
9. Isolation and enrichment culture techniques; preservation techniques
10. Evolutionary relationship - position of microbes in living world – concepts and developments in classification of microorganisms
11. Groups of microorganisms - prokaryotes and eukaryotes. Archaea – ecology; differences among archaea, eubacteria and eukaryotes
12. Bergey's manual of systematic bacteriology – outline only. Economic importance of

bacteria

13. Bacteria- size, shape, structure and arrangement of cells
14. Bacteria - external and internal structures in bacteria and their functionality
15. Morphology and classification of fungi and economic importance
16. Morphology and classification of algae and economic importance
17. Viruses and their properties; bacteriophages – lytic and lysogenic and temperate phages
18. **Mid Semester Examination**
19. Reproduction in bacteria - population growth and growth phases – generation time and specific growth rate
20. Batch and continuous culture – chemostat and turbidostat; synchronous culture. Diauxic growth curve.
21. Conditions for growth - temperature requirements - aerobes and anaerobes – other factors influencing growth; methods of assessment of growth.
22. Nutritional types of bacteria. Metabolic diversity in microbes.
23. Aerobic respiration and anaerobic respiration
24. Fermentative mode of respiration
25. Oxygenic and anoxygenic mode of photosynthesis
26. Energy generation by substrate level phosphorylation, oxidative and photo phosphorylation
27. Genetic elements in bacteria – structure and functions of bacterial chromosome and plasmid and transposons
28. Mutation in bacteria – principles and types. Mutagens – physical, chemical and biological
29. Genetic recombination – competency - transformation
30. Genetic recombination by Conjugation – concept of Hfr
31. Genetic recombination by Transduction – generalized and specialized
32. Microorganisms as tools in genetic engineering
33. Immunology – principles – specific and non-specific defense
34. Antigen – antibody reactions – vaccines - applications

Practical schedule

1. Safety in Microbiology laboratory. Microscopes – handling light microscope
2. Micrometry-measurement of microorganisms
3. Aseptic techniques – working with equipment and apparatus
4. Preparation of growth media for bacteria, yeast, molds and actinobacteria
5. Isolation of microorganisms by serial dilution and plating technique
6. Purification of bacteria and actinobacteria
7. Purification of yeasts and molds
8. Preservation of bacteria, fungi and actinobacteria
9. Staining techniques - positive and negative staining
10. Differential staining - Gram staining
11. Turbidometric assessment of growth of bacteria
12. Morphological characteristics of bacteria and actinobacteria
13. Biochemical characteristics of bacteria and actinobacteria
14. Identification of yeasts, molds and algae - morphological characterization
15. Isolation of bacteriophages
16. Isolation of bacterial mutants by UV irradiation / chemical mutagenesis

17. Practical Examination

Text Books

1. Michael T. Madigan , Kelly S. Bender Daniel H. Buckley , W. Matthew Sattley, David A. Stahl 2017. Brock Biology of Microorganisms, 15th edition
2. ebook.: Prescott, Harley and Klein, 2013. Microbiology, 9th edition, McGraw Hill Publishing
3. ebook: Michael J. Leboffee and Burton E. Pierce 2011. A photographic Atlas for the Microbiology Laboratory 4th edition, Marton Publishing Company

Reference Books

1. Hans G. Schlegel, 2012. General Microbiology, 7th edition
2. Ronald M. Atlas, 1997. Principles of Microbiology, Second edition

3. Tortora, G.J., B.R.Funke and C.L. Case, 2009. Microbiology- An Introduction, 9th edition
4. Stanier, R.Y., Adelberg, E.A. and Ingram, J.L. (1991). General Microbiology, 5th Ed., Prentice Hall of India Pvt. Ltd., New Delhi.

Web pages

<http://www.microbes.info>

<http://aem.asm.org>

<http://microbelibrary.com>

<http://www.rapidmicrobiology.com>

Objective:

This course intends to expose students to the fundamentals of extension education, extension systems in India, programme planning and rural development efforts, extension administration. To impart knowledge to the students on different extension methods and approaches used for transfer of agricultural technology. The course will also provide an opportunity to students to visit different organizations involved in extension activities and rural development work. To enable them to develop practical skills on preparation of different extension teaching methods.

Theory

Unit I Extension education and programme planning

Education- meaning, definition & types; extension education –meaning, definition, scope and process; objectives and principles of extension education. Programme planning – definition, meaning, process, principles and steps in programme development

Unit II Extension System In India

Extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development scheme, Gurgaon Experiment, etc.) Post – independence era (Etawah pilot project, Nilokheri Experiment, etc.,) Various extension/ agricultural development programmes launched by ICAR/Govt. of India(IADP, IAAP, HYVP,KVK, ORP, ND, NATP, NAIP etc.,)

Unit III Rural Development, Administration, monitoring and evaluation

Rural Development –Concept, meaning, definition: various rural development programmes launched by Govt. of India. Community development –meaning, definition, concepts and principles, physiology of community development. Rural leadership: concept and definition, types of leaders in rural context: extension administration: meaning, concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programmes

Unit IV New Trends in Agricultural Extension

New trends in agricultural extension –Privatization of extension, Cyber extension/ E-extension, (Internet, cyber cafes, video and teleconferencing, Interactive Multimedia Compact disk (IMCD), Agri portals, Information Kiosks, Kisan Call Centre (KCC), Mobile phone, Village Knowledge Centre (VKC), DEMIC, Geographical Information System (GIS), market led extension, farmer led extension, expert systems etc.,

Unit V Transfer of Technology, Diffusion of Innovations and extension methods

Transfer of technology concept, models, capacity building of extension personnel, extension teaching methods: meaning, classification, individual, group and mass contact methods, media mix strategies: communication: meaning, definition, models elements, characteristics and barriers to communication Agricultural Journalism: Agricultural journalism (Print media) - definition, principles, importance, ABC of news, types of news. Diffusion of Innovations – definition, elements; Innovation – definition, attributes; Adoption – meaning, steps in adoption process, adopter categories, factors influencing adoption of innovations; process and stages of adoption, adopter categories.

Practical

To get acquainted with university extension system, group discussion- exercise, handling and use of audio visual equipments and digital camera and LCD projector: preparation and use of AV aids, preparation of extension literature- leaflet, booklet, folder, pamphlet newstories and success stories, Presentation skills exercise: micro teaching exercise: A visit to village to understand the problems being encountered by the villagers/ farmers : to study organization and functioning of DRDA and other development departments at district level: visit to NGO and learning from their experience in rural development: understanding PRA techniques and their application in village development planning: exposure to mass media; visit to community radio and television studio for understanding the process of programme production: Script writing, writing for print and electronic media, developing script for radio and television.

Theory Schedule

1. Education- meaning, definition and types;
Extension education – meaning, definition, scope and process; objectives and principles and function of

extension education.

2. Programme planning – definition, meaning, process, principles and steps in programme planning / development
3. Extension efforts in pre-independence era (IVP, Sriniketan, Marthandam, Firka Development scheme, Sevagram, Gurgaon Experiment, Baroda Village Reconstruction Project Grow more Food Campaign, IVS, Firka Vikas Yojana etc.) Post – independence era (Etawah pilot project, Nilokheri Experiment,
4. Extension/ agricultural development programmes launched by ICAR/Govt. of India ICAR Programmes – National demonstration, ORP, Lap to Land Programme, FTC.,
5. Extension programmes of Ministry of Agriculture – Training and Visit (T&V) System, Broad Based Extension System (BBES), Agricultural Technology Management Agency (ATMA); Firstline Extension System – KVK, IVLP, ATIC, Frontline demonstrations.
6. Rural Development – meaning, definition, concept and importance. Rural Development in India. Democratic Decentralization –Meaning of Democratic Decentralization and Panchayat Raj – Three tiers of Panchayat Raj system – Powers, Functions and Organizational setup.
7. Community Development Programme (CDP), National Extension Service (NES), Intensive Agricultural District Programme (IADP), Intensive Agricultural Area Programme (IAAP) - their strengths and weaknesses
8. High Yielding Variety Programme (HYVP), Institution Village Linkage Programme (IVLP), Watershed Development Programme (WDP), Integrated Rural Development Programme (IRDP) - their strengths and weaknesses.
9. National Agricultural Technology Project (NATP), Integrated Tribal Development Agency (ITDA), Small Farmers Development Agency (SFDA), Marginal Farmers and Agricultural Labourers Development Agency (MFAL) - their strengths and weaknesses
10. National Rural Employment Programme (NREP), Rural landless Employment Guarantee Programme (RLEGP), Drought Prone Area Programme (DPAP), Command Area Development Programme (CADP), Food for Work Programme (FFW), Jawahar Rozgar Yojana (JRY), Employment Assurance Scheme (EAS),
11. Indira Awaas Yojana (IAY), Swarnajayanthi Gram Swarozgar Yojana (SGSY), Prime Minister Employment Yojana (PMEY), Swarna Jayanthi Shahari Rozgar Yojana (SJSRY), Pradhan Mantri Gram Sadak Yojana (PMGSY), ARYA - their strengths and weaknesses.
12. Sampoorna Grameen Rozgar Yojana (SGRY), Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA), Providing Urban Amenities to Rural Areas (PURA), National Agricultural Innovation Project (NAIP), NADP (RKVY) - their strengths and weaknesses
13. Community development –meaning, definition, concepts and principles, physiology of community development
14. Rural leadership: concept and definition, types of leaders in rural context and selection of leaders.
15. Extension administration: meaning, concept, scope, principles and functions.
16. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programmes, types and evaluation
17. **Mid semester Examination**
18. New trends in agricultural extension -Privatization of Agricultural extension- Meaning- definition-importance in Agricultural Extension.
19. Cyber extension/ E-extension, (Internet, cyber cafes, video and teleconferencing, web streaming and multimedia.
20. Agri portals, Information Kiosks, Kisan Call Centre (KCC), Mobile phone, Village Knowledge Centre (VKC), DEMIC, Geographical Information System (GIS),
21. Market led extension, farmer led extension : Meaning, definition, challenges and importance in agricultural extension.
22. Expert systems –meaning, definition, application in agriculture.
23. Transfer of technology concept, models, PTD, FSRE.
24. Capacity building of extension personnel- Training- definition, need for training, training process, models, strategies, steps in conducting training programmes
25. Training need assessment, building up of training programme- trainer roles: training insituteion for extension personnel- KVK, EEI, MANAGE, NAARM.
26. Extension teaching methods: meaning, classification; Individual methods- Farm and Home, Personal letter, Official call, observation and Result demonstration
27. Group Contact- Method demonstration, meeting, lecture, debate, workshop, seminar, forum, conference, symposium, panel, brain storming, buzz session, role playing and simulation games.
28. Mass contact methods- Campaign, exhibition, farmers day and field trip- purpose procedure, merit and demerits and media mix strategies
29. Communication – meaning, definition, types, elements and characteristics

- 30 Communication models (Aristotle, Shanon-Weaver, Berlo, Schramm, Leagans, Rogers & Shoemaker) – elements and their characteristics; Barriers in communication
- 31 Agricultural Journalism: Agricultural journalism (Print media) - definition, principles, importance, ABC of news, types of news.
- 32 Diffusion of Innovations – definition, elements; Innovation – definition, attributes;
- 33 Adoption – meaning, steps in adoption process, stages, adopter categories, factors influencing adoption of innovations ;Consequences of innovations

34 **Final Examination**

Practical schedule

1. Visit to State department of Agri/ Horti to understand the organizational setup, roles, functions and various schemes.
2. Study the organizational set up and functions of DRDA.
3. Visit to NGO and learning from their experience in rural development
4. Visit to KVK to study the mandated activities
5. To study the ToT system of SAUs / Agricultural colleges
6. Exercise on practicing group discussion technique and presentation skills
7. Study on Art of Photography, Video techniques and preparing multimedia presentations and handling of AV aids and LCD projectors
8. Preparation of Posters, charts, leaflet, folder, booklet and Pamphlet
9. Preparation of news stories and success stories.
10. Exercise on practicing Art of Public Speaking (micro teaching skills)
11. To visit the village and understand the socio cultural and agricultural related problems being encountered by the villagers/ farmers
12. Practicing selected PRA techniques in a village setting
13. Visit to Community Radio/ Educational Media Centre to understand the process of programme production.
14. Exercise on Script writing for Radio and TV programme
15. Visit to All India Radio Station / TV to study the various activities & programmes.
16. Visit to the News Agency /TNAU press to study the process
17. Final Practical Exam

Suggested Readings (Textbooks, Reviews, Journals)

- Ahuja, B.N. 1997. Theory and Practice of Journalism, Surjeet Publications, New Delhi.
- Benor Daniel, Q. James Harrison and Baxter Michael. 1984. Agricultural Extension – The Training and Visit System, A World Bank Publication, Washington, USA.
- Chauhan Nikulsinh. 2013. Use of ICTs in Agricultural Extension, Biotech Books.
- Gamble Dennis, Blunden, S. and Wallace, G. 2000. A Systematic Framework for understanding and Improving a Farming or Equine System, (AgPak SA 22), University of Western Sydney, Hawkesbury, NSW, Australia.
- Hough George, A. 2004. News Writing. Kanishka Publishers, New Delhi.
- Dahama, O.P and O.P. Bhatnagar. 1985. Education and Communication for Development, Oxford & IBH Publishing Co. Pvt. Ltd., New Delhi.
- Dipak de, Basavaprabhu Jirli. 2010. A Handbook of Extension Education, Agrobios, India.
- Katar Singh. 1999. Rural Development – Principles, Policies and Management, Sage Publications India Pvt. Ltd., New Delhi.
- Kelsey, L.D and C.C. Hearne. 1967. Cooperative Extension Work, Cornell University Press, New York.
- Manoharan Muthiah, P. and R. Arunachalam. 2003. Agricultural Extension, Himalaya Publishing House, Mumbai.
- Narayanasamy, N. 2009. Participatory Rural Appraisal Principles, Methods and Application, Sage Publications India Pvt. Ltd., New Delhi.
- Neela Mukherjee. 1993. Participatory Rural Appraisal: Methodology and Applications, Concept Publishing Co.
- Pandey, B.K. 2005. Rural Development, ISHA Books, New Delhi.
- Pandey, V.C. 2003. Information Communication Technology and Education (The Changing World ICT Governance), Isha Publishers.

- Ray, G.L. 1999. Extension Communication and Management, Naya Prokash, 206, Bidhan Sarani, Calcutta.
- Reddy Adivi, A. 1993. Extension Education, Shree Lakshmi Press, Bapatla, Andhra Pradesh.
- Rishipal. 2011. Training and Development Methods, S.Chand and Co. Ltd., New Delhi.
- Rogers, E.M. 1995. Diffusion of Innovations, The Free Press, New York.
- Sagar Mondal and Ray, G.L. 2007. Text book of Rural Development, Kalyani Publishers, New Delhi.
- Sandhu, A.S. 1996. Agricultural Communication: Process and Methods, Oxford & IBH Publishing Co. Pvt. Ltd, New Delhi.
- Sandhu, A.S. 1996. Extension Programme Planning, Oxford & IBH Publishing Co. Pvt. Ltd, New Delhi.
- Sanjay Prakash Sharma. 2006. Panchayat Raj, Vista International Publishing House, New Delhi.
- Singh, A.K. 2012. Agricultural Extension, Agrobios, New Delhi.
- Sivasudevaro, B and Rajannikanthu, G. 2007. Rural Development and Entrepreneurship Development, The Associated Publications, Ambala.
- Supe, S.V. 1997. An Introduction to Extension Education, Oxford & IBH Publishing Co. Pvt. Ltd., New Delhi.
- Van den Ban, A.W and H.S. Hawkins. 2002. Agricultural Extension, CBS Publishers & Distributors, New Delhi.

Journals

- International Journal of Extension Education
- Indian Journal of Extension Education
- Journal of Extension Education – Coimbatore
- Journal of Extension Education – Bhubaneshwar
- Rajasthan Journal of Extension Education
- The Journal of Agricultural Education and Extension
- Journal of Agricultural Extension Management
- Journal of Agricultural Education and Extension
- Indian Journal of Gender Studies
- Indian Research Journal of Extension Education
- Journal of Community Mobilization and Sustainable Development
- Indian Journal of Social Sciences
- Agricultural Extension Review
- Journal of Extension Education - Coimbatore
- Journal of Rural Development
- Yojana

Web resources

- rural.nic.in
- www.panchayat.gov.in
- wcd.nic.in
- moud.nic.in
- mhupa.gov.in
- www.i4d.com
- www.panasia.org
- www.joe.org

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Theory**Unit I: Principles of Food Science and Nutrition**

Food Science - definition – classification of foods – functional and nutritional classification. Food groups and food pyramid. Methods of cooking - moist, dry and microwave - principles, merits and demerits. Importance and scope of nutrition – relation of nutrition to health.

Unit -II: Carbohydrate, Protein and Fat

Carbohydrate – classification, functions, digestion and absorption, sources and Recommended Dietary allowance (RDA). Energy value of foods – determination. Protein – classification, functions digestion and absorption, sources and requirements. Protein quality of foods – supplementary value of protein. Fat - classification functions, digestion and absorption, sources and requirements. Rancidity – types of rancidity and prevention. Deficiency states of protein, carbohydrate and fat nutrition – signs and symptoms.

Unit III: Vitamin and Mineral Nutrition

Fat Soluble vitamins – A, D, E and K- functions, sources, requirements and deficiency. Water soluble vitamins – thiamine , riboflavin , niacin, pyridoxine, folic acid, cyanacobalamin, biotin, pantothenic acid ascorbic acid – functions, sources, deficiency and requirements. Minerals - calcium, iron, phosphorus, iodine, magnesium, zinc, sodium, potassium, fluorine and chlorine – functions, sources, deficiency and requirements. Importance of water – maintenance of electrolyte balance. Dietary fibre - importance, health benefits, sources and requirements.

Unit IV: Food Preservation and Processing

Introduction – preservation by sugar - processing of jam, squash, jelly, marmalade and beverages. Preservation by using salt, chemicals, dehydration technology, canning technology, preservation by low temperature and irradiation techniques. Processing of puffed, flaked and extruded products. Quality control of raw and processed products.

Unit V: Food Quality and Safety

Food packaging materials – requirements – methods – nutrition labeling. Food adulterants and their detection methods. Food laws and regulations and quality control standards - FSSAI, ISO, EU standards, FDA, HACCP and Codex Alimentarius Commission.

Practical

Determination of energy value of Foods, cooking quality tests – cereals and pulses. Estimation of moisture, protein and fat. Processing of jam, jelly, squash, ready to serve beverages (RTS). Preparation of flaked, puffed and extruded products. Visit to food industries and quality control laboratory.

Theory Schedule

1. Food Science – definition, scope and classification, food pyramid.
2. Methods, merits and demerits of moist heat, dry heat and microwave cooking of foods.
3. Importance and scope of nutrition and the relation of nutrition to health.
4. Carbohydrate – classification, functions, digestion and absorption, deficiency symptoms, sources and requirements.
5. Protein – classification, functions, digestion and absorption, deficiency symptoms, sources and requirements. Protein quality – supplementary value of protein.
6. Fat - classification, functions, digestion and absorption, deficiency symptoms, sources and requirements. Rancidity – types. Determination of energy value of foods.
7. Fat soluble vitamins – A, D, E and K – functions, deficiency symptoms, sources and requirements.
8. Water soluble vitamins - thiamine, riboflavin, niacin, pyridoxine, folic acid, cyanacobalamin, biotin, pantothenic acid, ascorbic acid – functions, deficiency symptoms, sources and requirements.
9. **Mid Semester Examination**
10. Minerals – calcium, iron, phosphorus, iodine, magnesium, zinc, sodium, potassium, fluorine and chlorine – functions, sources, requirements and deficiency diseases.
11. Importance of water and maintenance of electrolyte balance. Health benefits of fibre.
12. Preservation of food by low and high temperature and food irradiation.
13. Processing of puffed, flaked and extruded products

14. Preservation by using sugar (jam, jelly, squash and marmalade), preservation by using salt (brining and pickling) and use of preservatives in food preservation.
15. Food packaging – importance, types of packaging materials and nutrition labeling.
16. Common food adulterants and their detection.
17. Food laws and regulations and quality control standards - FSSAI, ISO, EU standards, FDA, HACCP and Codex Alimentarius Commission.

Practical Schedule

1. Cooking tests for cereals and pulses
2. Determination of energy value of food
3. Estimation of moisture
4. Estimation of protein
5. Estimation of fat
6. Estimation of ascorbic acid
7. Estimation of iron
8. Estimation of crude fibre
9. Processing of jam and jelly
10. Processing of squash and RTS
11. Puffing of pulses
12. Extrusion of cereals and millets
13. Canning of fruits and vegetables
14. Processing of dehydrated fruits and vegetables
15. Identification of common food adulterants
16. Visit to food processing unit and quality control lab
17. Final Practical Examination

TEXT BOOKS

1. Srilakshmi, B. 2005. Food Science. New Age International (P) Ltd., Publishers, New Delhi.
2. Srivastava, R.P., and Sanjeevkumar. S. 2013. Fruit and Vegetable preservation. International Book Distributing Co. Lucknow.
3. Srilakshmi .B. 2015. Nutrition Science. New Age International Pvt. Ltd. New Delhi.

JOURNALS

1. The Indian Journal of Nutrition and Dietetics
2. Journal of Food Science and Technology
3. Critical Reviews of Food Science and Nutrition

WEB RESOURCES

1. www.cellinteractive.com
2. www.nutrition.org.uk
3. www.fnics.nal.usda.gov
4. www.myfooddiary.com

AGR 102. Introductory Agro-meteorology & Climate Change (1+1)

Theory:

Unit - I: Climate and weather

Meteorology - Agricultural Meteorology - Importance and scope in crop production - Co-ordinates of India and Tamil Nadu - Atmosphere - Composition and vertical layers of atmosphere (stratification) - Climate - Weather - Factors affecting climate and weather - Climatic types - Different agricultural seasons of India and Tamil Nadu and climatic characteristics of India.

Unit - II: Solar radiation, RH and Wind

Solar radiation - Light intensity, quality, direction and duration - Air and Soil temperature - Diurnal variation - importance in crop production. Heat unit and its importance in agriculture. Relative Humidity and its importance - vapor pressure deficit and its importance - Wind and its effect on crops.

Unit - III: Atmospheric pressure and precipitation

Atmospheric pressure - cyclones, anticyclones, tornado, hurricane and storms - Wind systems of the world - Clouds - types and their classification. Precipitation - forms - monsoon - Seasons of India- rainfall variability drought, flood and their effect - Cloud seeding - Evaporation - transpiration - Evapotranspiration - PET.

Unit - IV: Agroclimatic zones and remote sensing

Agro climatic Zones of India and Tamil Nadu - Agro climatic normals - Weather forecasting - synoptic chart - crop weather calendar - Remote sensing - Impact of climate and weather on crop production and pest and diseases.

Unit - V: Climate change

Climate change- climate variability - definition and causes of climate change - Impact of climate change on Agriculture.

Practical:

Observatory - Site selection and layout. Acquiring skill in use of Pyranometers - Sunshine recorder - Maximum, Minimum, Grass minimum and Soil thermometers - Thermograph, Dry and wet bulb thermometers - Hygograph - Psychrometers - Fortein's barometer - Barograph - Altimeter; Wind vane, Anemometer - Raingauge - Ordinary and self-recording - Dew guage; Automatic weather station - Evaporimeters - Lysimeters, Automatic weather station - Preparation of synoptic charts and crop weather calendars. Rainfall probability analysis. Mapping of Agroclimatic Zones.

Theory

Lecture Schedule:

1. Meteorology - Agricultural Meteorology - Definition, their importance and scope in crop production.
2. Coordinates of India and Tamil Nadu. Atmosphere - Composition of atmosphere - Vertical layers of atmosphere based on temperature difference / lapse rate.
3. Climate and weather - Factors affecting climate and weather. Macroclimate - Meso climate - Microclimate - Definition and their importance - Different climates of India and Tamil Nadu and their characterization.
4. Solar radiation - Radiation balance - Wave length characteristics and their effect on crop production - Light - effect of intensity, quality, direction and duration on crop production.
5. Air temperature - Factors affecting temperature. Diurnal and seasonal variation in air temperature - Isotherm, Heat unit and its use - Heat and cold injuries.
6. Role of temperature in crop production. Soil temperature - Importance in crop production. Factors affecting soil temperature, diurnal and seasonal variation in soil temperature.
7. Humidity - Types - Dew point temperature - Vapour pressure deficit - Diurnal variation in Relative humidity and its effect on crop production - Wind and its role on crop production.
8. Atmospheric pressure, diurnal and seasonal variation - Isobar – cyclone, hurricane, tornado and storms.
9. **Mid Semester Examination.**
10. Wind systems of the world - wind speed in different seasons -. Clouds and their classification - Concepts of cloud seeding - present status.
11. Precipitation - Forms of precipitation - Isohyte - Monsoon - Different monsoons of India - Rainfall variability - Drought and flood - Impact on crop production.
12. Evaporation - Transpiration, evapotranspiration - Potential evapotranspiration - Definition and their importance in agricultural production. Agroclimatic zones of Tamil Nadu - Agroclimatic normals for field crops.
13. Weather forecasting - Types, importance, Agro Advisory Services - Synoptic chart - Crop weather calendar.

14. Remote sensing and its application in agriculture.
15. Effect of weather and climate on crop production, soil fertility and incidence of pest and diseases.
16. Climate change, climate variability - definition and causes of climate change.
17. Impact of climate change on Agriculture.

Practical Schedule:

1. Site selection and layout for Agromet Observatory - Calculation of local time - Time of observation of different weather elements - Reviewing agromet registers.
2. Measurements of solar radiation (pyranometers), sunshine hours (sunshine recorder) - working out weekly and monthly mean for graphical representation.
3. Measurement of air and soil temperature and grass minimum thermometers and thermographs - drawing isolines.
4. Humidity measurements - use of wet and dry bulb thermometers - Psychrometers - Hygrograph - Measurement of wind direction and wind speed and conversion (KMPH, KNOT, and M/Sec.) - Beaufort's scale.
5. Measurement of atmospheric pressure - barograph - Fortin's barometer - Isobars based on past data for different seasons.
6. Measurement of rainfall - Ordinary and self-recording rain gauges - Measurement of Dew - dew gauge - study of Automatic weather station.
7. Measurement of Evaporation - Open pan evaporimeter - application of evaporation data - Measurement of Evapotranspiration - Lysimeter.
8. Heat Unit concept - GDD, HTU, PTU for fixing time of sowing.
9. Probability analysis of rainfall for crop planning.
10. Drawing Synoptic charts for understanding weather.
11. Preparation of crop weather calendars and forecast based agro advisories.
12. Preparation pest weather calendar and pest forewarning.
13. Estimation of length of growing periods using weekly rainfall data.
14. Water balance studies.
15. Identification of efficient cropping zone - RYI, RSI.
16. Mapping of agro climatic Zones of India and Tamil Nadu and its characterization.
17. **Practical Examination.**

References

Mavi, H.S., 1996. Introduction to Agrometeorology, Oxford and IBH Publishing Co., New Delhi.

Gopalaswamy, N. 1994. Agricultural Meteorology, Rawat publications, Jaipur.

Prasad, Reddy, S.R. and Reddy, D.S. 2014. Agro meteorology. Kalyani Publishers, New Delhi.

Reddy, S.R. 2014. Introduction to Agriculture and Agrometeorology. Kalyani Publishers, New Delhi.

Patra, A.K. 2016. Principles and applications of Agricultural Meteorology. New India Publishing Agency, New Delhi.

Smith, J.W. 2013. Agricultural meteorology. Axis Books (India), Jodhpur.

Rao, P. 2008. Agricultural Meteorology. Prentice Hall of India Pvt. Ltd, New Delhi.

Murthy, R.V. 2002. Basic Principles of Agricultural Meteorology. BS Publications, Hyderabad.

Singh, J. 2014. Textbook of Agricultural meteorology. Oxford Book Company, New Delhi.

Rao, G.S.L.H.V. 2005. Agricultural Meteorology. Kerala Agricultural University Press, Thrissur.

E:References:

www.tawn.tnau.ac.in
www.usbr.gov/pn/agri.met
www.imd.gov.in

HOR 112 . Production Technology for Fruit and Plantation Crops (1+1)

Aim

To impart knowledge on the production techniques of tropical, sub tropical, temperate fruit crops and plantation crops.

Syllabus

Theory

Unit I: Production status of fruit and plantation crops

Importance and scope of fruit and plantation crop industry in India – nutritional value of fruit crops - classification of fruit crops – area, production, productivity and export potential of fruit and plantation crops.

UNIT II: Crop production techniques in tropical fruit crops

Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high density planting - cropping systems - after care - training and pruning - water, nutrient and weed management –fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest- value addition.

Fruit crops: mango, banana, papaya, guava, sapota

UNIT III: Crop production techniques in subtropical fruit crops

Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest and value addition.

Fruit crops: citrus, grape, litchi, pineapple, pomegranate, jackfruit and minor fruits

UNIT IV: Crop production techniques temperate fruit crops

Climate and soil requirements – varieties – propagation and use of rootstocks - planting density and systems of planting -High density and ultra high density planting -cropping systems - after care - training and pruning - water, nutrient and weed management –fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest and value addition.

Fruit crops: apple, pear, peach, strawberry, nut crops.

UNIT V Crop production techniques in palms and plantation crops

Climate and soil requirements - varieties - propagation - nursery management - planting and - planting systems - cropping systems - after care - water, nutrient and weed management - intercropping - multi-tier cropping system - mulching - special horticultural practices - maturity indices, harvest and yield - pests and diseases - processing - value addition

Palms: Coconut, Arecanut, Oil palm and Palmyrah

Climate and soil requirements - varieties- propagation - nursery management - planting and planting systems - cropping systems- after care- training and pruning - water, nutrient and weed management - shade management - intercropping - mulching - cover cropping - special horticultural practices - maturity indices, harvest and yield - pests and diseases - processing - value addition.

Plantation crops: Tea, Coffee, Cocoa, Cashew, Rubber

Practical

Propagation methods for fruit crops - description and identification of varieties - preparation of plant bio regulators & their uses – nutrient deficiency and disorders of fruit crops - fertilizers- application - pests and diseases- micro propagation in fruit crops- Visit to commercial orchard.

Fruit Crops: Mango, banana, papaya, guava, sapota, grapes, citrus (Mandarin and acid lime), pomegranate and jackfruit

Propagation methods for plantation crops - description and identification of plantation crops - preparation of plant bio regulators & their uses - nutritional disorders of plantation crops - fertilizers- application - pests and diseases- cost economics of plantation crops. Visit to plantations and plantation industries.

Palms and plantation Crops: Coconut, Arecanut, Cashew, Tea, Coffee, Rubber and Cocoa

Theory schedule

1. Importance and scope of fruit and plantation crop industry in India – nutritional value of fruit crops
2. Classification of fruit crops – area, production, productivity and export potential of fruit and plantation crops
3. Climate and soil – varieties - propagation methods - planting and cropping systems - after care- training and pruning- top working - water, nutrient and weed management- canopy management - plant growth regulation - important disorders – maturity indices and harvest - post harvest management of **Mango**
4. Climate and soil – varieties - propagation methods - planting and cropping systems - after care- water and nutrient management – fertigation - Weed control - Plant growth regulation - important disorders – maturity indices and harvest- post harvest management of **Banana**
5. Climate and soil – varieties - propagation methods - planting and cropping systems - after care - water, nutrient and weed management - crop regulation- important disorders – maturity indices and harvest - post harvest management of **Papaya, Guava and sapota**
6. Climate and soil – varieties - propagation methods - planting and cropping systems - after care - water, nutrient and weed management - crop regulation – nutrient deficiencies and important disorders – maturity indices and harvest- post harvest management of **Citrus (Sweet orange, Mandarin and Acid Lime)**
7. Climate and soil – varieties - propagation methods - planting and cropping systems-after care – systems of training and pruning and bud forecasting - water, nutrient and weed management - plant growth regulation - important disorders – maturity indices and harvest - post harvest management of **Grapes**
8. Climate and soil – varieties - propagation methods - planting and cropping systems - after care - water, nutrient and weed management - plant growth regulation- important disorders – maturity indices and harvest - post harvest management of **pineapple and litchi**
9. **Mid semester examination**
10. Climate and soil – varieties - propagation methods - planting and cropping systems - after care - training and pruning - water, nutrient and weed management - plant growth regulation - important disorders – maturity indices and harvest - post harvest management of **Pomegranate, jackfruit and minor fruits**
11. Climate and soil – varieties - propagation methods - planting and cropping systems - after care- training and pruning - water, nutrient and weed management - plant growth regulation - important disorders – maturity indices and harvest- post harvest management of **Apple and pear**
12. Climate and soil – varieties - propagation methods - planting and cropping systems - after care- training and pruning - water, nutrient and weed management - plant growth regulation - important disorders – maturity indices and harvest - post harvest management of **Peach and strawberry, nut crops** .
12. Climate and soil requirements - varieties - propagation - nursery management - planting systems - planting density -nutrient, water and weed management - intercropping at various ages of plantation -multitier cropping - shade management - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of **Coconut**
13. Climate and soil requirements - varieties - propagation - nursery management - planting systems - planting density - nutrient, water and weed management - intercropping at various ages of plantation - multitier cropping – shade management - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of **Arecanut and Cocoa** .
14. Climate and soil requirements - varieties - propagation - nursery management - planting and planting density - HDP - UHDP - nutrient, water and weed management - cover cropping - tapping - use of plant growth regulators - top working - maturity indices - harvest and yield , latex yield and processing - pests and diseases - grading - processing and value addition **Rubber and Cashew** .
15. Climate and soil requirements- varieties – propagation - nursery management - planting density and systems of planting - nutrient, water and weed management - mulching - cropping systems - shade regulation - training and pruning - role of growth regulators - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of **Tea** .

16. Climate and soil requirements - varieties – propagation - nursery management - planting - nutrient, water and weed management - mixed and inter cropping - shade management - training and pruning - role of growth regulators - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of **Coffee**.
17. Climate and soil requirements - varieties – propagation - nursery management - planting - nutrient, water and weed management - water conservation techniques - leaf pruning - pollination - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition **Oil palm and Palmyrah**.

Practical schedule

1. Propagation techniques, selection of planting material, varieties, important cultural practices for **Mango**
2. Propagation techniques, selection of planting material, varieties, important cultural practices for **Banana**
3. Propagation techniques, selection of planting material, varieties, important cultural practices for **Papaya**
4. Propagation techniques, selection of planting material, varieties, important cultural practices for **Guava**
5. Propagation techniques, selection of planting material, varieties, important cultural practices for **Sapota**
6. Propagation techniques, selection of planting material, varieties, important cultural practices for **Grapes**
7. Propagation techniques, selection of planting material, varieties, important cultural practices for **Citrus (Mandarin and acid lime)**
8. Propagation techniques, selection of planting material, varieties, important cultural practices for **Pomegranate**
9. Propagation techniques, selection of planting material, varieties, important cultural practices for **Jackfruit**
10. Preparation and application of PGR's for propagation.
11. Micro propagation, protocol for mass multiplication and hardening of fruit crops.
12. Identification and description of varieties - mother palm and seed nut selection - nursery practices- seedling selection – fertilizers - application - nutritional disorders - pests and diseases of **Coconut**
13. Identification and description of varieties - mother palm and seed nut selection- nursery practices- fertilizers - application - nutritional disorders - pests and diseases of **Arecanut and cocoa**
14. Identification and description of varieties - nursery practices - training and pruning - pests and diseases – processing of **Tea and coffee**
15. Identification and description of varieties, clones - bud wood nursery practices - propagation techniques - top working – preparation of plant bio regulators and its uses- pests and diseases - processing of **Rubber and cashew**
16. Visit to commercial orchard and plantation industries.
17. **Practical examination**

Outcome

- Students will be imparted with wide knowledge on major tropical, a few sub tropical and temperate fruit and plantation crops
- Hands on training on various propagation methods and important cultural practices for major fruit and plantation crops will be provided (Practical)

Reference text books

1. Kumar, N. 2014. Introduction to Horticulture. Oxford & IBH Publishing co. Pvt. Ltd.
2. Chadha, K.L and Pareek, O.P. 1996. (Eds.). Advances in Horticulture. Vols. IIIV. Malhotra Publ. House
3. Kumar, N. 2014. Introduction to Spices, Plantation, Medicinal and Aromatic crops, IBH Publishing Co. Pvt. Ltd., New Delhi.
4. Alice Kurian and Peter, K.V. 2007. Horticulture science series Vol. 08, New India Publishing Agency, New Delhi.
5. Veeraragavathatham, D and et al.,2004. Scientific fruit culture, Sun Associates, Coimbatore.
6. Henry Louis, I. 2002. Coconut- The wonder palm. Hi - Tech Coconut Corporation, Nagercoil.

E references

<http://www.jhortscib.com>

<http://journal.ashspublications.org>

<http://www.actahort.org/>

<http://www.aphorticulture.com/crops.htm>

<http://cpcri.nic.in/>

<http://indiancoffee.org>

RSG 101 Geoinformatics for Precision Farming (1+0)

Theory

Unit I

Remote sensing: Concepts - Electromagnetic radiation: principles and theories- Energy interaction: atmosphere and earth surface features- Spectral reflectance of earth features- Platforms and sensors: types and characteristics. Image Processing and Interpretation – Digital image processing: Image Classification - Optical, Microwave and Drones

Unit II

GIS: Definition, Components and functions- Raster and vector data models and non-spatial data types - Raster Data Analysis: Local, Neighborhood and Regional Operations- Vector Data Analysis: Querying, Buffering, Overlay

Unit III

Geodesy and its basic principles -Global Positioning System – components and its functions. GPS survey methods- Error sources - DEM – Sources, Generation and application.

Unit IV

Precision agriculture: concepts and techniques; their issues and concerns –STCR / VRT approach for precision agriculture - Soil moisture, Pest and Disease incidence – nutrient deficiencies – linking with VRT using Geospatial Technologies

Unit V

Crop discrimination and Yield monitoring, soil mapping; fertilizer recommendation using geospatial technologies. Introduction to crop Simulation Models and their uses for optimization of Agricultural Inputs – Drones for precision agriculture

Theory schedule

1. Remote sensing concepts and Electromagnetic radiation. Energy interaction: atmosphere and earth surface features and Spectral reflectance
2. Platforms and sensors: types and characteristics
3. Elements of visual photo/image interpretation
4. Digital image processing -Image classification – Optical, Microwave and Drones
5. GIS: Definition and Components and functions
6. Raster and vector data models and non-spatial data types
7. Raster Data Analysis: Local, Neighborhood and Regional Operations.
8. Vector Data Analysis: Querying, Buffering, Overlay and Network Analysis
- 9. Mid-semester Examination**
10. Geodesy and its basic principles
11. GPS: components and functions - GPS Survey and Error Sources
12. Precision agriculture: concepts and techniques; their issues and concerns – STCR / VRT approach
13. Soil moisture, Pest and Disease incidence – nutrient deficiencies – linking with VRT using Geospatial Technologies
14. Crop discrimination and Yield monitoring using Remote Sensing
15. Soil mapping; fertilizer recommendation using geospatial technologies
16. Introduction to Crop Simulation Models and their uses for optimization of Agricultural Inputs
17. Drones for precision agriculture

Textbooks

For Remote Sensing part of syllabus

Anji Reddy, M. 2008. Textbook of Remote Sensing and Geographic Information Systems. Third Edition. BS Publication, Hyderabad

For GIS part of syllabus

Rolf A.de By. 2001. Principles of Geographic Information Systems. ITC Educational Textbook Series I

For Application part of syllabus

Roy, P.S., R.S.Dwivedi and D.Vijayan.2010. Remote Sensing Applications. NRSC Publication. ISBN: 978-81-909460-0-1

Web resources

1. www.physicalgeography.net
2. www.gisdevelopment.net
3. www.gis.nic.in
4. www.geos.iitb.ac.in/remotesensing.html
5. www.dimensionigis.com/remote_sensing.html

III SEMESTER

S. No.	Course code	Course Title	Credit load
1	PAT 201	Fundamentals of Plant Pathology	2+1
2	AEN 201	Fundamentals of Entomology	2+1
3	SST 201	Principles of Seed Technology	2+1
4	AGR 201	Crop Production Technology – I (<i>Kharif</i> crops)	1+1
5	HOR 211	Production Technology for Vegetables and Spices	1+1
6	ENS 201	Environmental Studies & Disaster Management	2+1
7	AMP 201	Livestock and Poultry Management	2+1
8	AEC 201	Farm Management, Production & Resource Economics	1+1
9	SAC 201	Soil Resource Inventory	1+1
10	FMP 211	Farm Machinery and Power	1+1
11	AGR 202	Study tour	0+1*
12	NSS/NCC 101	NSS/NCC	0+1*
13	PED 101	Physical Education	0+1*
		Total	15+10=25
		*Non-gradual courses compulsory courses	

PAT 201 Fundamentals of Plant Pathology (2+1)

Unit I: Introduction to Plant Pathology

Plant Pathology- Definition – Importance of plant diseases, scope and objectives of Plant Pathology. History of Plant Pathology with special reference to Indian work. Terms and concepts in Plant Pathology. Causes and classification of plant diseases. Important plant pathogenic organisms, different groups: fungi, bacteria, fastidious vesicular bacteria, phytoplasmas, spiroplasmas, viruses, viroids, algae, protozoa and phanerogamic parasites

Unit II: Pathogenesis

Koch's postulates- Types of parasitism- Pathogenesis - Mode of infection – pre-penetration, penetration and post penetration - Role of enzymes and toxins on disease development-Effect of pathogen on physiological functions of the plants

Unit III: General characters and taxonomy of Protozoa, Chromista and Fungi

General characters, definition of fungus, somatic structures, types of fungal thalli, fungal tissues, modifications of thallus, reproduction (asexual and sexual). Binomial system of nomenclature and rules of nomenclature of fungi. Classification of fungi. Key to divisions, sub-divisions, orders and classes based on Ainsworth and Bisby's dictionary of fungi 10th edition compiled by Kirk *et al.*, (2008). General characters and symptoms of **Kingdom:Protozoa**, **Phylum: Plasmodiophoromycota**, *Plasmodiophora brassicae*. **Kingdom:Chromista**, **Phylum:Oomycota**-*Pythium*, *Phytophthora*, *Sclerospora*, *Plasmopara* and *Albugo* **Kingdom:Fungi** , **Phylum:Chytridiomycota**-*Synchytrium*, **Phylum: Zygomycota** -*Mucor*, *Rhizopus*

Unit IV: General characters and taxonomy of fungi - Ascomycota and Basidiomycota

Phylum: Ascomycota: Classification , symptoms and taxonomy characters of *Taphrina*, *Capnodium*, *Mycosphaerella*, *Macrophomina*, *Cochliobolus*, *Lewia*, *Venturia*, *Eurotium*, *Talaromyces*, *Sclerotinia*, *Erysiphe*, *Leveillula*, *Phyllactinia*, *Claviceps*, *Gibberella*, *Ustilaginoidea*, *Verticillium*, *Glomerella*, *Pestalotiopsis* and *Magnaporthe*

Phylum: Basidiomycota: Classification, symptoms and life cycle of *Puccinia*, *Uromyces*, *Hemileia* and *Ustilago*. Important taxonomic characters of *Ganoderma* , *Agaricus*, *Pleurotus* and *Calocybe*. Symptoms and Important taxonomic characters of *Athelium*, *Rhizoctonia* and *Exobasidium*

Unit V: Bacteria, Phytoplasma, virus, viroid, Algae, Phanerogams and abiotic disorders

General characters and symptoms- phytopathogenic bacteria, *Candidatus Phytoplasma*, *Spiroplasma*, Fastidious vascular bacteria, viruses, viroids, algae, Phanerogams –Abiotic disorders.

Practical

Acquaintance with various laboratory equipments and microscopy. Preparation of media, isolation and Koch's postulates. General characters of fungi – Types of mycelia -Types of vegetative, asexual and sexual spores- asexual and sexual fruiting bodies. Study of important taxonomic characters and symptoms produced by *Plasmodiophora*, *Pythium*, *Phytophthora*, *Sclerospora*, *Plasmopara* , *Albugo*, *Mucor*, *Rhizopus*, *Taphrina*, *Capnodium*, *Cercospora*, (*Mycosphaerella*), *Botryodiplodia* (*Botryosphaeria*), *Curvularia*, *Drechslera* (*Helminthosporium*), *Alternaria*, *Venturia*, *Erysiphe*, *Phyllactinia*, *Uncinula*, *Leveillula* , *Claviceps*, *Fusarium* (*Gibberella* ,*Nectria*), *Verticillium* ,*Colletotrichum* (*Glomerella*) *Pestalotia* (*Pestalosphaeria*), *Pyricularia*(*Magnaporthe*) *Sarocladium*, *Macrophomina* , *Puccinia*, *Uromyces* , *Hemileia*, *Ustilago* *Sphacelotheca* (*Sporisorium*), *Tolyposporium* (*Moesziomyces*), *Exobasidium*, *Sclerotium*, *Rhizoctonia* (*Thanatephorus*), *Ganoderma*, *Agaricus*, *Pleurotus* and *Calocybe*.

Symptoms of bacterial diseases, *Candidatus Phytoplasma*, Fastidious vascular bacteria, algal parasite, phanerogamic parasites and non-parasitic diseases.

Theory

1. Definition of Plant Pathology – History of Plant Pathology
2. Importance of plant diseases
3. Causes of Plant diseases – Protozoa , Chromista, , fungi, Bacteria, Fastidious vascular bacteria, Spiroplasma, *Candidatus Phytoplasma*,
4. Causes of Plant diseases -Virus, viroid, algal, phanerogamic parasites and abiotic disorders
5. Pathogenesis – stages in pathogenesis – pre-penetration, penetration and post penetration
6. Role of enzymes in pathogenesis
7. Role of toxins in pathogenesis
8. Effect of pathogen on physiological functions of the plants- Effect on Photosynthesis- Transpiration- Respiration- translocation of water and nutrients
9. General characters of fungi- Mycelia – vegetative resting structures
10. Asexual reproduction in fungi
11. Sexual reproduction in fungi
12. Parasitism in fungi- Types of parasitism – parasite, saprophyte, obligate parasite, facultative parasite, facultative saprophyte- Mode of nutrition in fungi- biotrophs, hemibiotrophs, perthotrophs/ necrotrophs and symbiosis
13. Classification of Kingdom Protozoa - important taxonomic characters , symptoms and life cycle of *Plasmodiophora brassicae* and symptoms of Protozoan diseases
14. Classification of Kingdom Chromista- General characters of Oomycetes- Symptoms and life cycle of *Pythium*, *Phytophthora* and *Albugo*
15. Symptoms and life cycle of *Sclerospora* and *Plasmopara*
16. Classification of Kingdom– *Chytridiomycota* and *Zygomycota* - important characters, symptoms and life cycles of *Synchytrium* and *Rhizopus* and *Mucor*
- 17. Mid Semester Examination**
18. Classification of Kingdom– *Ascomycota*- important characters
19. Symptoms and life cycles of *Taphrina*, *Capnodium*, *Cercospora*, (*Mycosphaerella*), *Macrophomina*, *Cochliobolus* (*Helminthosporium*), *Lewia* (*Alternaria*) and *Venturia*
20. Symptoms and life cycles of *Eurotium*, *Talaromyces*, *Erysiphe*, *Leveillula* and *Phyllactinia*,
21. Symptoms and taxonomic characters of *Claviceps*, *Fusarium* (*Gibberella*, *Nectria*) and *Verticillium*
22. Symptoms and taxonomic characters of *Colletotrichum* (*Glomerella*) *Pestalotia* (*Pestalotia*), *Pyricularia* (*Magnaporthe*), *Sarocladium* and *Ustilago*
23. Classification of Kingdom - *Basidiomycota*- important characters
24. Symptoms and life cycles of *Puccinia*, *Uromyces*, *Hemileia*
25. Symptoms and taxonomic characters of *Ustilago*, *Sphacelotheca* (*Sporisorium*), *Tolyposporium* (*Moesziomyces*), *Tilletia* and *Exobasidium*
26. Symptoms and taxonomic characters of *Athelium*, *Thanetophorus* and *Ganoderma*
27. Important taxonomic characters of *Agaricus*, *Pleurotus* and *Calocybe*
28. Classification and general characters of phytopathogenic bacteria
29. Symptoms and characters of *Xanthomonas*, *Ralstonia*, *Erwinia*, *Pantoea*, *Pectobacterium* *Agrobacterium* (*Rhizobium*), *Corynebacterium* (*Clavibacter*,) and *Streptomyces*
30. Important characters and symptoms of *Candidatus Phytoplasma* diseases – Phyllody, little leaf, yellow dwarf and sandal spike, Fastidious vascular bacteria and Spiroplasma
31. Virus - definition, nature and properties of plant virus, Single stranded, Double stranded RNA and DNA viruses and Transmission of plant viruses
32. Common symptoms of virus diseases – mosaic, chlorosis, leaf curl, stem pitting, spotted wilt, necrosis , ring spot, vein clearing, leaf crinkle, rosette and bunchy top
33. Important characters and symptoms of Viroid, Algal and Phanerogamic parasites

34. Symptoms and characters of non-parasitic diseases

Practical

1. Acquaintance with light microscope- Preparation of media for isolation and proving Koch's postulates
2. General characters of fungi – Types of mycelia -Types of vegetative, asexual and sexual spores- asexual and sexual fruiting bodies
3. Study of important taxonomic characters and symptoms produced by *Plasmodiophora*, *Pythium* and *Phytophthora*
4. Study of important taxonomic characters and symptoms produced by *Sclerospora*, *Plasmopara* and *Albugo*
5. Study of important taxonomic characters and symptoms produced by *Rhizopus*, *Taphrina*, *Capnodium*, *Cercospora*, (*Mycosphaerella*), *Botryodiplodia* (*Botryosphaeria*), *Drechslera* (*Helminthosporium*) and *Alternaria*
6. Study of important taxonomic characters and symptoms produced by *Eurotium*, *Talaromyces*, *Erysiphe*, *Leveillula*, *Phyllactinia*, *Uncinula*, *Podosphaera* and *Sphaerotheca*
7. Study of important taxonomic characters and symptoms produced by *Claviceps*, *Fusarium* (*Gibberella*, *Nectria*) and *Verticillium*
8. Study of important taxonomic characters and symptoms produced by *Colletotrichum* (*Glomerella*), *Pestalotia* (*Pestalosphaeria*), *Pyricularia* (*Magnaporthe*) *Sarocladium* and *Macrophomina*
9. Study of important taxonomic characters and symptoms produced by *Puccinia*, *Uromyces*, and *Hemileia*
10. Field visit for exposing students on different crop diseases
11. Study of important taxonomic characters and symptoms produced by *Ustilago*, *Sphacelotheca* (*Sporisorium*), *Tolyposporium* (*Moesziomyces*), and *Exobasidium*
12. Study of important taxonomic characters of *Agaricus*, *Pleurotus*, *Calocybe* and *Volvariella*
13. Study of important taxonomic characters and symptoms produced by *Athelium*, *Thanetophorus* and *Ganoderma*
14. Symptoms of bacterial diseases – leaf blight, leaf streak, canker, scab, crown gall, wilt and soft rot.
15. Symptoms and vectors of viral diseases – mosaic, chlorosis, leaf curl, stem pitting, spotted wilt, necrosis, ring spot, vein clearing, leaf crinkle, rosette and bunchy top
16. Symptoms of *Candidatus Phytoplasma*, Algae, Phanerogamic parasites and non-parasitic diseases

17. Final Practical Examination.

Note: Students should submit 50 well-preserved disease specimens

Reference text book

1. Alice D, and Jeyalakshmi C 2014. Plant Pathology. A.E Publications, Coimbatore
2. Agrios, G.N. 2005. Plant Pathology – (5th Edition). Academic Press, New York.

3. Kirk, P.M., P.F. Cannon, D.W. Minter and J.A. Stalpers, (2008). Ainsworth and Bisby's dictionary of fungi , 10th edition. CAB international Wallingford, UK.

E- references

1. Agrios, G.N. 2005. Plant Pathology – (5th Edition). Academic Press, New York.
2. Richard N. Strange. 2003. Introduction of Plant Pathology - John Wiley & Sons Ltd, London
3. John Webster and Ronald Weber, 2007. Introduction to fungi by Cambridge University Press, UK

AEN 201 FUNDAMENTALS OF ENTOMOLOGY (2+1)

Theory

Unit I: History and importance of Entomology; Insect morphology

History of Entomology in India; Position of insects in the animal kingdom and their relationship with other classes of Arthropoda; Reasons for insect dominance. General organisation of insect body wall - structure and function, cuticular appendages, moulting; Body regions - insect head, thorax and abdomen, their structure and appendages.

Unit II:

Anatomy and physiology (Part – I) : Digestive, excretory, respiratory, circulatory and nervous systems in insects.

Unit III:

Anatomy and physiology (Part – II) : Reproductive systems in insects, sense organs and their functions, exocrine and endocrine glands; Embryonic and post embryonic development.

Unit IV:

Taxonomy of Apterygota and Exopterygota

Insect systematics; Distinguishing characters of agriculturally important orders and families of Hexapoda. Apterygota (Thysanura, Diplura, Protura and Collembola); Exopterygota (Ephemeroptera, Odonata, Orthoptera, Phasmida, Dictyoptera, Embioptera, Dermaptera, Hemiptera, Isoptera, Psocoptera, Mallophaga, Thysanoptera and Siphunculata).

Unit V:

Taxonomy of Endopterygota

Distinguishing characters of agriculturally important families of Lepidoptera, Coleoptera, Diptera, Hymenoptera, Siphonaptera, Neuroptera and Strepsiptera.

Practical

Observations on external features of grasshopper / cockroach, Methods of insect collection, preservation – Preparation of Riker mount. Types of insect head, antenna, mouth parts – Structure of thorax. Types of insect legs, wings and their modifications – wing coupling. Structure of abdomen, and its modifications. Metamorphosis in insects – immature stages in insects. Study of digestive and reproductive systems of grasshopper / cockroach – Observing the characters of agriculturally important orders and families.

Theory lecture schedule:

9. History of Entomology in India; Position of insects in the animal kingdom - relationship with other members of Arthropoda
10. Structural, morphological and physiological factors responsible for dominance
11. Insect body wall - its structure and function; cuticular appendages
12. Moulting process in insects
13. Structure of insect head and its appendages
14. Structure of insect thorax and its appendages
15. Structure of insect abdomen and its appendages

35. Structure of alimentary canal and its modifications; Digestive enzymes, digestion and absorption of nutrients
36. Excretory system - Malpighian tubules - accessory excretory organs and physiology of excretion
37. Respiratory system – types - structure of trachea - tracheoles - types of spiracles - respiration in aquatic and endoparasitic insects
38. Circulatory system - haemocoel and dorsal vessel - circulation of blood -composition of haemolymph - haemocytes and their functions
39. Nervous system - Structure of neuron – types of nervous systems
40. Axonic and synaptic transmissions of nerve impulses
41. Male and female reproductive systems in insects – structure and modifications - Spermatogenesis and Oogenesis
42. Oviparous, viviparous, paedogenesis, polyembryony, ovoviporous and parthenogenesis
43. Embryogenesis; Types of metamorphosis – Immature stages of insects
44. **Mid-semester examination**
45. Structure of sense organs - types of sensilla – photoreceptors, chemoreceptors and mechanoreceptors
46. Exocrine and endocrine glands and their function - effect on metamorphosis and reproduction
47. Tropism and Biocommunication in insects — Sound and light production
48. Systematics - principles and procedures of classification and nomenclature of insects
49. Distinguishing characters of insect orders — Apterygota (Thysanura, Diplura, Protura and Collembola), Exopterygota — (Ephemeroptera, Odonata and Phasmida)
23. Orthoptera (Ensifera - Tettigonidae, Gryllidae and Gryllotalpidae; Caelifera - Acrididae and Tetrigidae), Dictyoptera, Dermaptera and Embioptera
17. Isoptera — social life in termites
18. Thysanoptera, Psocoptera, Mallophaga and Siphunculata.
26. Hemiptera – Homoptera (Delphacidae, Flatidae, Cercopidae, Cicadidae, Membracidae, Cicadellidae, Psyllidae, Aleyrodidae, Aphididae, Margarodidae, Kerridae, Pseudococcidae, Coccidae, Asterolecaniidae and Diaspididae)
3. Hemiptera - Heteroptera (Tingidae, Reduviidae, Cimicidae, Anthracoridae, Miridae, Lygaeidae, Pyrrhocoridae, Coreidae, Scutellaridae, Pentatomidae, Veliidae, Gerridae, Naucoridae, Belostomatidae, Nepidae, Notonectidae and Corixidae)
4. Endopterygota — Classification of Lepidoptera – suborders; butterfly families (Nymphalidae, Lycaenidae, Pieridae, Papilionidae, Satyriidae and Hesperidae)
29. Moth families (Psychidae, Gelechiidae, Metarbellidae, Cochliidae, Pyralidae, Crambidae, Pterophoridae, Geometridae, Bombycidae, Saturniidae, Sphingidae, Arctiidae, Noctuidae and Lymantriidae)
7. Classification of Coleoptera – suborders; Adephaga (Carabidae, Cicindellidae, Dytiscidae, Gyrinidae)
8. Polyphaga (Hydrophilidae, Staphylinidae, Passalidae, Lucanidae, Scarabaeidae, Dynastidae, Melolonthidae, Cetonidae, Buprestidae, Elateridae, Lampyriidae, Cantharidae, Dermestidae, Anobiidae, Bostrychidae, Coccinellidae, Tenebrionidae, Meloidae, Cerambycidae, Bruchidae, Chrysomelidae, Apionidae and Curculionidae)
9. Diptera – Suborders; Nematocera (Tipulidae, Psychodidae, Culicidae, Bibionidae, and Cecidomyiidae), Brachycera (Tabanidae, Asilidae and Bombyliidae), Cyclorhapha (Syrphidae, Drosophilidae, Muscidae, Calliphoridae, Tachinidae, Hippoboscidae, Micropezidae, Agromyzidae, Chloropidae and Tephritidae)

9. Hymenoptera—Suborders; Symphyta (Tenthredinidae) Apocrita (Ichneumonidae, Braconidae, Evaniidae, Agaonidae, Chalcididae, Encyrtidae, Eulophidae, Trichogrammatidae, Bethyidae, Chrysididae, Scolidae, Mutillidae, Formicidae, Vespidae, Sphecidae, Megachilidae, Anthophoridae, Xylocopidae and Apidae)
10. Neuroptera (Mantispidae, Chrysopidae, Myrmeleontidae and Ascalaphidae); Siphonaptera and Strepsiptera

Practical schedule:

10. Observations on external features of grasshopper / cockroach and other members of phylum Arthropoda
11. Methods of insect collection, preservation, display and storage
12. Types of insect head and antenna
13. Mouth parts of cockroach, modifications in the mouth parts in plant bug, female mosquito, honeybee, thrips, antlion grub, house fly, moths and butterflies
14. Structure of thorax and their appendages —modifications in insect legs and wings — wing venation, regions and angles — wing coupling.
15. Structure of abdomen and their appendages
16. Types of immature stages of insects
17. Study of digestive system, male and female reproductive systems
18. Observing the characters of Apterygota - Collembola and Thysanura and Exopterygota -Odonata and Ephemeroptera and Phasmida
19. Observing the characters of Dictyoptera, Dermaptera, Embioptera, Orthoptera (Ensifera - Tettigonidae, Gryllidae and Gryllotalpidae; Caelifera - Acrididae and Tetrigidae), Mallophaga and Siphunculata
20. Observing the characters of Exopterygota —Isoptera and Hemiptera — Homoptera (Delphacidae, Flatidae, Cercopidae, Cicadidae, Membracidae, Cicadellidae, Psyllidae, Aleyrodidae, Aphididae, Margarodidae, Kerridae, Pseudococcidae, Coccidae, Asterolecaniidae and Diaspididae) Heteroptera (Tingidae, Reduviidae, Cimicidae, Anthocoridae, Miridae, Lygaeidae, Pyrrhocoridae, Coreidae, Scutellaridae, Pentatomidae, Veliidae, Gerridae, Naucoridae, Belastomatidae, Nepidae, Notonectidae and Corixidae)
14. Observing the characters of orders Thysanoptera and Diptera- Nematocera (Tipulidae, Psychodidae, Culicidae, Bibionidae, and Cecidomyiidae), Brachycera (Tabanidae, Asilidae and Bombyliidae,), Cyclorhapha (Syrphidae, Drosophilidae, Muscidae, Calliphoridae, Tachinidae, Hippoboscidae, Micropezidae, Agromyzidae, Chloropidae and Tephritidae)
15. Observing the characters of Hymenoptera-Symphyta (Tenthredinidae) Apocrita (Ichneumonidae, Braconidae, Evaniidae, Agaonidae, Chalcididae, Encyrtidae, Eulophidae, Trichogrammatidae, Bethyidae, Chrysididae, Scolidae, Mutillidae, Formicidae, Vespidae, Sphecidae, Megachilidae, Anthophoridae, Xylocopidae and Apidae)
16. Observing the characters of Coleoptera - Adephaga (Carabidae, Cicindellidae, Dytiscidae, Gyrinidae) Polyphaga (Hydrophilidae, Staphylinidae, Passalidae, Lucanidae, Scarabaeidae, Dynastidae, Melolonthidae, Cetonidae, Buprestidae, Elateridae, Lampyriidae, Cantharidae, Dermestidae, Anobiidae, Bostrychidae, Coccinellidae, Tenebrionidae, Meloidae, Cerambycidae, Bruchidae, Chrysomelidae, Apionidae and Curculionidae)
17. Observing the characters of Lepidoptera - Butterfly families (Nymphalidae, Lycaenidae, Pieridae, Papilionidae, Satyriidae and Hesperidae), Moth families (Psychidae, Gelechiidae, Metarbellidae, Cochliidae, Pyralidae, Crambidae, Pterophoridae, Geometridae, Bombycidae, Saturniidae, Sphingidae, Arctiidae, Noctuidae and Lymantriidae)
35. Observing the characters of Neuroptera (Mantispidae, Chrysopidae, Myrmeleontidae and Ascalaphidae), Siphonoptera. Identification and naming of collected insects based on characters — order and family

36. Final Practical examination

References:

Richards O.W. and R.G. Davies. 1977. *Imm's General Text Book of Entomology*. Vol.I and II. Chapman and Hall Publication, London. 1354p. {ISBN 0412 15220 7}

Chapman, R.F. 1998. *The Insects: Structure and Function*. Fourth Edition. Cambridge University Press. 770p. {ISBN 0 521 78732 7}

Snodgrass, R.E. 1994. *Principles of Insect Morphology*. CBS publishers and distributors, New Delhi. 667p.

David, B.V. and V.V. Ramamurthy. 2011. *Elements of Economic Entomology*, Namrutha Publications, Chennai, 386 p. {ISBN: 978-81-921477-0-3}

Srivastava, P. D. and R. P. Singh. 1997. *An Introduction to Entomology*. Concept Publishing Company, New Delhi.

E- References:

11. <http://www.itis.usda.gov/it is/>

12. www.zin.ru/animalia

13. <https://courses.cit.cornell.edu/ent201/content/anatomy2.pdf>

14. www.insectsexplained.com/03external.htm

15. www.earthlife.net/insects/anatomy.html

16. www.insectidentification.org/orders_insect.asp

SST 201 Principles of Seed Technology (2+1)

Theory

Unit I –

Seed : Importance and biology

Seed and seed technology: definition -importance -Characters of good quality seed -Seed development and maturation - Germination - phases of seed germination - Dormancy - types of seed dormancy - Different classes of seed - generation system of seed multiplication in supply chain - Seed replacement rate and varietal replacement - Seed Multiplication Ratio -Seed renewal period. Varietal deterioration of crops - causes and maintenance.

Unit II –

Seed Production

Principles of seed production- Foundation and certified seed production of varieties and hybrids - Cereals - rice, maize, sorghum and bajra - Pulses - greengram, blackgram and redgram - Oilseeds - groundnut, sesame, sunflower and castor - Cotton, Forage crops - Cenchrus sp and lucerne - Vegetables - tomato, brinjal, chillies, bhendi, onion and gourds - bittergourd, ashgourd, snakegourd, ribbedgourd, bottlegourd and pumpkin. Principles of GM crop and organic seed production.

Unit III –

Post harvest handling of seeds

Post harvest handling of seeds - threshing methods - drying - methods of seed drying - Seed processing - seed cleaning and grading - Processing equipments -cleaner cum grader -Upgrading equipments - specific gravity separator, colour sorter, indented cylinder separator, spiral separator, magnetic separator, needle separator - working principles - Seed quality enhancement techniques - importance - seed fortification, seed priming, seed coating, seed pelleting.

Unit IV –

Seed Quality Control and Seed Testing

Seed certification - phases of certification, procedure for seed certification, field inspection, field counts, field and seed standards. Post harvest inspection - processing, bagging and tagging. Seed Act and Rules - Seed law enforcement. Duty and powers of seed inspector, offences and penalties. Seeds Control Order 1983 - Salient features of PPV&FRA 2001 - National Seed Policy 2002 - Seed Bill 2004. Seed testing for quality assessment - importance - Varietal identification through grow out test, molecular and biochemical test. Detection of genetically modified crops.

Unit V - Seed Storage and marketing

Seed storage – principles- factors affecting seed longevity during storage – Seed treatments and packaging materials - measures for pest and disease control during storage and godown sanitation. Seed marketing - structure and organization - sales promotional activities. factors affecting seed marketing and demand - International seed movement - role of international organizations, WTO and OECD in seed trade.

Practical

Study on seed structure of major cereals - rice, wheat, maize, sorghum and bajra - Pulses - greengram, blackgram, redgram, bengalgram and field pea - Oilseeds - groundnut, sesame, sunflower, castor, soybean and mustard - Cotton - Forage crops - Vegetable crops. Seed production techniques - identification of physical and genetic contaminants - supplementary pollination in hybrid rice - detasselling techniques in hybrid maize - emasculation and dusting technique in cotton - supplementary pollination in sunflower - pre-germinative technique and enhancing female flowers in cucurbits - assessment of physiological maturity indices - seed extraction methods in vegetables. Visit to seed production farms - Seed enhancement techniques - seed coating, seed priming and seed pelleting. Visit to seed processing plant. Seed certification - field inspection and counting procedure - Seed sampling and testing - seed moisture content, physical purity, seed germination, viability - Seed and seedling vigour test - Seed health test- Genetic purity test -grow out test and electrophoresis - Seed production planning - economics - Visit to seed testing laboratory.

Theory lecture schedule:

7. Seed and seed technology - definition -importance -Characters of good quality seed.
8. Seed development and maturation.
9. Germination - phases of seed germination - Dormancy - types of seed dormancy.
10. Different classes of seed - generation system of seed multiplication in supply chain - Seed replacement rate and varietal replacement - Seed Multiplication Ratio -Seed renewal period.
11. Varietal deterioration of crops - causes and maintenance.
12. Principles of seed production - factors affecting seed production - physical and genetic contaminants.
13. Seed production techniques in varieties and hybrids of rice.
14. Seed production techniques in varieties and hybrids of maize.
15. Seed production techniques in varieties and hybrids of sorghum and bajra.
16. Seed production techniques in greengram and blackgram varieties.
17. Seed production techniques in varieties and hybrids of redgram.
18. Seed production techniques in varieties and hybrids of sunflower and groundnut varieties.
19. Seed production techniques in varieties and hybrids of castor and sesame varieties.
20. Seed production techniques in varieties and hybrids of cotton.
21. Seed production techniques in Cenchrus species and lucerne.
22. Seed production techniques in varieties and hybrids of tomato, brinjal and chillies.
23. Seed production techniques in varieties and hybrids of bhendi and onion.
- 24. Mid semester examination.**
25. Seed production techniques in varieties and hybrids of gourds - bittergourd, ashgourd, snakegourd, ribbedgourd, bottlegourd and pumpkin.
26. Principles of GM crop and organic seed production.
27. Post harvest handling of seeds - threshing methods - drying - methods of seed drying.
22. Seed processing - principle - importance - seed processing sequence for different crops - equipments.
18. Principles and mechanism of seed cleaning and grading - processing equipments - cleaner cum grader - specific gravity separator.
19. Principles and mechanism of upgrading equipments - colour sorter - indented cylinder separator - spiral separator - magnetic separator - needle separator
20. Seed quality enhancement techniques - importance - seed fortification - seed priming - seed coating - seed pelleting.
21. Seed certification - phases of certification, procedure for seed certification, field inspection, field counts, field and seed standards - post harvest inspection - processing - bagging and tagging.
22. Seed Act and Rules - Seed law enforcement - Duties and powers of seed inspector - offences and penalties - Seeds Control Order 1983.
18. Salient features of PPV&FRA, 2001 - National Seed Policy, 2002 - Seed Bill, 2004.
19. Seed testing for quality assessment - importance - methods.

20. Varietal Identification - grow out test - molecular and biochemical test - Detection of genetically modified crops.
21. Seed storage - principles - factors affecting seed longevity during storage.
22. Seed treatment and packaging materials - measures for pest and disease control during storage and godown sanitation.
23. Seed marketing - structure and organization - sales promotional activities - factors affecting seed marketing and demand.
24. International seed movement - role of international organizations - WTO and OECD in seed trade.

Practical schedule:

9. Study on external and internal seed structure and identification of major cereals - pulses - oilseeds - cotton - forage crops and vegetable crops.
10. Practicing supplementary pollination techniques in hybrid rice and detasselling techniques in hybrid maize.
11. Practicing emasculation and dusting technique in cotton and supplementary pollination in sunflower.
12. Practicing pre-germinative technique and female flowers production enhancement techniques in cucurbits.
13. Assessment of physiological maturity indices in various crops and seed extraction methods in vegetables.
14. Visit to seed production farms.
15. Seed enhancement techniques - Seed coating - seed priming and seed pelleting.
16. Seed certification - field inspection and counting procedure - identification of physical and genetic contaminants in seed production plots and roguing.
17. Seed testing - seed sampling - mixing - dividing - equipments.
18. Estimation of seed moisture content and physical purity.
19. Seed germination testing - tetrazolium test for viability - evaluation.
20. Genetic purity test - grow out test - electrophoresis.
21. Seed health testing - methods.
22. Seed and seedling vigour test - brick gravel test, paper piercing test - cool and cold test - accelerated ageing test.
23. Seed production planning - economics.
24. Visit to seed processing plant and seed testing laboratory.
- 25. Final practical examination.**

References

12. Agrawal, R.L. 1996. Seed Technology, Oxford & IBH Publishing Co., New Delhi.
13. Bhaskaran, M., A.Bharathi and K.Vanangamudi. 2013. Text Book on Principles of seed production and quality control. Kalyani Publishers, New Delhi.
14. Indian Minimum Seed Certification Standards. 2014. Published by GOI, MOA, New Delhi.
15. Seed legislations. 2014. Published by GOI, MOA, New Delhi.

E-References

1. www.fao.org
2. www.seednet.gov.in
3. www.agricoop.nic.in
4. www.online.library.wiley.com
5. www.sciencedirect.com

6. Seed Science Research (www.jgateplus.com)
7. Seed Science and Technology (www.jgateplus.com)

AGR 201 Crop production technology – I (Kharif crops) (1+1)

Theory :

Unit - I:

Cereals

Rice, Maize, - Origin, geographic distribution, economic importance, soil and climatic requirements, varieties, cultural practices (from land preparation to harvest) and yield.

Unit - II:

Millets

Sorghum, Pearl millet, Small millets - Finger millet, Foxtail millet, little millet, Kodo millet, Barnyard millet and Proso millet - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.

Unit - III:

Pulses

Redgram, Blackgram, Greengram, , Cowpea, - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.

Unit - IV:

Oilseeds (Kharif)

Groundnut, sesame, Soybean- Origin, and geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Unit - V:

Fibre and forage

Cotton, jute, fodder sorghum, cumbu napier- Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Cereals	Rice, maize
Millets	Sorghum, pearl millet, finger millet and minor millets
Pulses	Pigeonpea, green gram, black gram, cowpea,
Oilseeds	Groundnut, sesame, soybean
Fibre & Forage	Cotton, jute, fodder sorghum, Cumbu napier

Theory Lecture Schedule:

1. Importance and area, production and productivity of major cereals and millets of India and Tamil Nadu.
2. Importance and area, production and productivity of pulses and oilseeds crops of India and Tamil Nadu.
3. Rice - Origin - geographic distribution - economic importance - varieties - soil and climatic requirement.
4. Rice - cultural practices - yield - economic benefits.
5. Special type of Rice cultivation – SRI - and Hybrid rice cultivation.
6. Maize - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.
7. Sorghum and Pearl millet - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.
8. Finger millet and Minor millets - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.
9. **Mid semester Examination.**
10. Pigeonpea - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield.

2. Greengram, Blackgram and Cowpea - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield - Agronomy of rice fallow pulses.
3. Groundnut - Origin, geographical distribution, economic importance, soil and climatic requirements - varieties, cultural practices yield and economics.
4. Sesame and Soybean - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield.
5. Cotton - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield.
6. Jute- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield.
7. Fodder sorghum- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield.
8. Cumbu napier- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield.

Practical Schedule:

18. Identification of cereals, millets, pulses and oilseed crops in the crop cafeteria.
19. Practicing various nursery types and main field preparation for rice crop.
20. Nursery and main field preparation for important millets, pulses and oilseeds.
21. Acquiring skill in different seed treatment techniques in important kharif crops.
22. Estimation of plant population per unit area for important kharif crops.
23. Acquiring skill in field preparation, sowing and manuring of crops under pure and intercropping situations for cereals and millets.
24. Acquiring skill in field preparation, sowing and manuring of crops under pure and intercropping situations for pulses and oilseeds.
25. Acquiring skill in using seed drill for sowing operations.
26. Acquiring skill in foliar nutrition for important field crops.
27. Observations on growth parameters of cereals and millets.
28. Observations on growth parameters of pulses and oilseeds.
29. Study of yield parameters and estimation of yield in cereals and millets.
30. Study of yield parameters and estimation of yield in pulses and oilseeds.
31. Working out cost and returns of important cereals, millets, pulses and oilseeds crops.
32. Visit to farmers field / research stations to study the cultivation techniques of cereal, millets, pulses , cotton and oilseeds.
33. Visit to nearby Agricultural Research Station / Farmer's field.
34. **Practical Examination.**

References:

- Ahlawat, I.P.S., Om Prakash and G.S. Saini. 1998. Scientific Crop Production in India. Rama publishing House, Meerut.
- Chidha Singh. 2010. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd, New Delhi.
- Singh. S.S. 2015. Crop management under irrigated and rainfed conditions. Kalyani Publishers, New Delhi.
- Reddy,S.R. 2012. Agronomy of field crops. Kalyani publishers, New Delhi.
- Joshi, M. 2015. Textbook of Field crops. PHI Learning Pvt. Ltd., New Delhi
- ICAR 2015. Hand book of Agriculture. Indian Council of Agriculture, New Delhi
- Crop production Guide 2012. Directorate of Agriculture, Chennai.

E-References:

1. www.crida.org
2. www.cgiar.org
3. www.tnau.ac.in/agriportal

HOR 211 Production technology of vegetables and spices (1+1)

Theory

Vegetables

Unit I: Scope, Importance and classification of vegetables

Importance of vegetable growing –area and production of vegetables in India and Tamil Nadu- National economy- nutritive value of vegetables and human nutrition.

Unit II: Production technology of tropical vegetable crops

Origin - climate and soil – varieties and hybrids – seeds and sowing – transplanting – water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulators - physiological disorders - maturity indices – harvest – pest and diseases – seed production

Crops: Tomato, chilli, brinjal, bhendi, gourds (ash gourd, pumpkin, bitter gourd, ridge gourd, bottle gourd, snake gourd and watermelon) onion, cassava, amaranthus and moringa.

Unit III: Production technology of temperate vegetable crops

Origin -climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrient and plant growth regulators- physiological disorders- maturity indices – harvest – pest and diseases – seed production

(Crops: Cabbage, cauliflower, potato, carrot, radish, beetroot, peas and french beans, Protected cultivation of vegetables (tomato, capsicum and cucumber).

SPICES

Unit IV: Crop production techniques of major spices

Spices- scope and importance - classification of spices - origin, area and production - role of commodity boards- export potential of spices.

Climate and soil - varieties - propagation - nursery management and planting – cropping systems- training practices - nutrient, water and weed management - shade regulation - maturity indices - harvest and yield - pests and diseases - processing - value addition.

Black pepper, Cardamom, Turmeric, Ginger and Garlic

Unit V: Crop production techniques in seed spices, tree spices and other spices

Climate and soil- varieties - propagation, nursery management and planting- training , pruning canopy management- weed and water management- shade regulation- nutrient management including drip and fertigation – harvest and yield – pests and diseases – processing – value addition.

Coriander, Fenugreek, Cumin, Fennel, Clove, Nutmeg, Cinnamon, Curry leaf, Tamarind and Herbal spices

Practical

Vegetables

Layout of kitchen garden – seed sowing – nursery management – grafting in vegetables water and nutrient management – fertigation – weed management – practices in use of plant growth regulators - Special horticultural practices in vegetable production - study of maturity indices - Identification of physiological disorders - protected cultivation - visit to vegetable nursery unit/ protected cultivation unit.

Spices

Identification of spices - description of varieties - Propagation methods - rapid multiplication techniques

18. seed collection and extraction - raising of nurseries - seed sowing - seed treatment - fertilizer application - harvesting – pests and diseases - processing - cost economics - visit to spice gardens

Black pepper, Cardamom, Turmeric, Ginger, Coriander, Fenugreek, Curry leaf, Clove, Nutmeg and Cinnamon

Theory lecture schedule

2. Importance of vegetable growing –area and production of vegetables in India and Tamil Nadu- National economy- nutritive value of vegetables and human nutrition .
3. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of tomato, chilli and brinjal
4. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of bhendi and onion.
5. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield– pest and diseases – seed production of gourds (ash gourd, pumpkin, bitter gourd, ridge gourd, bottle gourd, snake gourd and watermelon)
6. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator- physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of cassava and potato
7. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of moringa and amaranthus.
8. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of cabbage and cauliflower.
9. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of carrot , beetroot and radish.
10. **Mid -semester examination**
11. Climate and soil – varieties and hybrids – seeds and sowing – transplanting - water and nutrient management – fertigation – weed management – use of micronutrients and plant growth regulator - physiological disorders - maturity indices – harvest and yield – pest and diseases – seed production of french beans and peas .
12. Protected cultivation of vegetables (tomato, capsicum and cucumber)
6. Spices- scope and importance - classification of spices - origin, area and production - role of commodity boards- export potential of spices. **Black pepper** - climate and soil- varieties – propagation – rapid multiplication techniques - nursery management and planting- nutrient, water and weed management - special horticultural practices - role of growth regulators - shade regulation- maturity indices - harvest and yield - pests and diseases –post harvest practices-processing and value addition.
7. **Cardamom** - climate and soil - varieties - propagation - nutrient, water and weed management-shade regulation- mulching - maturity indices - harvest and yield - pests and diseases –post harvest practices-processing and value addition.

8. **Turmeric , Ginger and Garlic** - Climate and soil- varieties - propagation, nursery management and planting- nutrient, water and weed management- inter cropping- physiological disorders -maturity indices - harvest and yield - pests and diseases –post harvest practices- processing and value addition.
15. **Seed spices** - climate and soil- varieties - seed treatment/ sowing – nutrient, water and weed management- intercropping - maturity indices - harvest and yield - pests and diseases –post harvest practices- processing and value addition.(**Coriander, Fenugreek, Cumin and Fennel**)
16. **Tree spices** - climate and soil - varieties - propagation, nursery management and planting - nutrient, water and weed management- training and pruning practices- cropping system- special horticultural practices maturity indices - harvest and yield - pests and diseases –post harvest practices- processing and value addition.(**Clove, Nutmeg and Cinnamon**)
17. **Tamarind, Curry leaf and herbal spices** - climate and soil- varieties - propagation, nursery management and planting- nutrient, water and weed management- canopy management - maturity indices - harvest and yield - pests and diseases –post harvest practices- processing and value addition.

Practical schedule

1. Layout of kitchen garden.
2. Seed treatment and sowing practices in direct sown vegetables
3. Nursery management of transplanted ,bulb and tuber vegetable crops
4. Grafting in vegetable crops
5. Water and nutrient management – fertigation in vegetable crops
6. Practices in use of plant growth regulators in vegetable crops
7. Special horticultural practices in vegetable production
8. Identification of physiological disorders in vegetable crops
9. Study of maturity standards and harvesting of vegetables
10. Practices in protected cultivation of vegetable crops
11. Visit to vegetable nurseries/protected vegetable cultivation unit
12. Black pepper and cardamom- identification and description of varieties – seed propagation and vegetative propagation – fertilizers application - preparation of plant bio regulators and application – pests and diseases- harvest and post harvest practices.
13. Turmeric and ginger- identification and description of varieties- propagation, fertilizers application - preparation of plant bio regulators and application – pests and diseases- harvest and post harvest practices.
14. Coriander and Fenugreek - identification and description of varieties - seed treatment, sowing fertilizer application - pests and diseases- harvest and post harvest practices.
18. Clove, Nutmeg and Cinnamon - identification and description of varieties – seed collection and extraction - propagation – fertilizer application – training and pruning – pests and diseases- harvest and post harvest practices.
19. Tamarind and curry leaf - identification and description of varieties – seed collection and extraction - propagation – fertilizer - application – canopy management – pests and diseases- harvest and post harvest practices.
20. Visit to spice gardens or commodity boards and working out cost economics of spice crops.

REFERENCES

Gopalakrishnan, T.R. 2007. Vegetable Crops. Horticultural Science Series (Series Editor K.V.Peter). New India Publishing Agency.

Mandal, R.C. 2006. Tropical root and tuber crops. Agrobios (India) Peter.K.V.

2000. Genetics and Breeding of Vegetables, ICAR, Publication.

Singh, P.K., S.K. Dasgupta and S.K. Tripathi, 2006. Hybrid Vegetable Development. International Book Distributing Co.

Mini, C. and Krishnakumar, K. 2004. Leaf Vegetables. Agro tech Publishing Academy, Udaipur Thamburaj, S. and Narendra Singh .2001. Vegetables, Tuber crops and Spices, Directorate of information and publications of agriculture, ICAR, New Delhi.

Veeraragavaththam ,D., M.Jawaharlal and Seemanthini Ramadas 2000 “ Vegetable Culture”

Kumar, N. 2014. Introduction to Spices, Plantation, Medicinal and Aromatic crops, IBH Publishing Co. Pvt. Ltd., New Delhi.

Alice Kurian and Peter, K.V. 2007. Horticulture science series Vol. 08, New India Publishing Agency, New Delhi.

Veeraragavaththam, D and et al.,2004. Scientific fruit culture, Sun Associates, Coimbatore.

Henry Louis, I. 2002. Coconut- The wonder palm. Hi - Tech Coconut Corporation, Nagercoil.

E-References

1. <http://www.jhortscib.com>
2. <http://journal.ashspublications.org>
3. <http://www.actahort.org/>
4. <http://www.aphorticulture.com/crops.htm>
5. <http://cpcri.nic.in/> <http://indiancoffee.org>

Theory

Unit 1: Multidisciplinary nature of environmental studies Definition, scope and importance

Unit 2: Natural Resources: Renewable and non-renewable resources Natural resources and associated problems. a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people. b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources. Case studies. f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification. • Role of an individual in conservation of natural resources. • Equitable use of resources for sustainable lifestyles.

Unit 3: Ecosystems • Concept of an ecosystem. • Structure and function of an ecosystem. • Producers, consumers and decomposers. • Energy flow in the ecosystem. • Ecological succession. • Food chains, food webs and ecological pyramids. • Introduction, types, characteristic features, structure and function of the following ecosystem. a. Forest ecosystem. b. Grassland ecosystem. c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Unit 4: Biodiversity and its conservation:- Introduction, definition, genetic, species & ecosystem diversity and biogeographical classification of India. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. Biodiversity at global, National and local levels, India as a mega-diversity nation. Hot-spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India. Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

Unit 5 : Environmental Pollution: Definition, cause, effects and control measures of : a. Air pollution. b. Water pollution. c. Soil pollution. d. Marine pollution. e. Noise pollution. f. Thermal pollution. g. Nuclear hazards. Solid Waste Management: causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Pollution case studies.

Unit 6: Social Issues and the Environment: From Unsustainable to Sustainable development. Urban problems related to energy. Water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Wasteland reclamation. Consumerism and waste products. Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation. Public awareness.

Unit 7: Human Population and the Environment: population growth, variation among nations, population explosion, Family Welfare Programme. Environment and human health: Human Rights, Value

Education, HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health. Case Studies.

DISASTER MANAGEMENT

Unit 8: Natural Disasters- Meaning and nature of natural disasters, their types and effects. Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves, Climatic change: global warming, Sea level rise, ozone depletion.

Unit 9 : Man Made Disasters- Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, air pollution, water pollution, deforestation, industrial waste water pollution, road accidents, rail accidents, air accidents, sea accidents.

Unit 10 : Disaster Management- Effect to migrate natural disaster at national and global levels. International strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community - based organizations and media. Central, state, district and local administration; Armed forces in disaster response; Disaster response; Police and other organizations.

Practical

Field work: Visit to a local area to document environmental assets river/forest/grassland/hill/mountain, visit to a local polluted site -Urban/Rural/Industrial/Agricultural, study of common plants, insects, birds and study of simple ecosystems-pond, river, hill slopes, etc.

Lecture Schedule

5. Multidisciplinary nature of environmental studies - Definition, scope and importance - Natural Resources: Renewable and non-renewable resources - Natural resources and associated problems
6. Forest resources: Use and over-exploitation, deforestation, case studies - Timber extraction, mining, dams and their effects on forest and tribal people
7. Water resources: Use and over-utilization of surface and ground water - Floods, drought, conflicts over water, dams - benefits and problems
8. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies - Food resources: World food problems, changes caused by agriculture and overgrazing
9. Effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies - Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources. Case studies.
10. Land resources: Land as a resource, land degradation, man induced landslides - Soil erosion and desertification - Role of an individual in conservation of natural resources - Equitable use of resources for sustainable lifestyles
11. Ecosystems - Concept of an ecosystem - Structure and function of an ecosystem - Producers, consumers and decomposers - Energy flow in the ecosystem - Ecological succession - Food chains, food webs and ecological pyramids
7. Introduction, types, characteristic features, structure and function of Forest ecosystem, Grassland ecosystem and Desert ecosystem
8. Introduction, types, characteristic features, structure and function of Aquatic ecosystems : ponds, streams, lakes - Rivers, oceans, estuaries
9. Biodiversity and its conservation - Introduction, definition, genetic, species & ecosystem diversity and biogeographical classification of India- Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values
10. Biodiversity at global, National and local levels - India as a mega-diversity nation - Hotspots of biodiversity - Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts
11. Endangered and endemic species of India - Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

12. Environmental Pollution - Definition, cause, effects and control measures of Air pollution and Noise pollution
13. Definition, cause, effects and control measures of Water pollution and Soil pollution
14. Definition, cause, effects and control measures of Marine pollution, Thermal pollution and Nuclear hazards
15. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes
- 16. Mid Semester Examination**
17. Role of an individual in prevention of pollution - Pollution case studies - Social Issues and the Environment - From Unsustainable to Sustainable development - Urban problems related to energy
18. Water conservation, rain water harvesting, watershed management - Environmental ethics: Issues and possible solutions, climate change, global warming
19. Acid rain, ozone layer depletion, Nuclear accidents and holocaust - Wasteland reclamation-Consumerism and waste products
20. Environment Protection Act - Air (Prevention and Control of Pollution) Act - Water (Prevention and control of Pollution) Act - Wildlife Protection Act - Forest Conservation Act
21. Issues involved in enforcement of environmental legislation - Public awareness - Human Population and the Environment: Population growth, variation among nations, population explosion, Family Welfare Programme
22. Environment and human health: Human Rights, Value Education, HIV/AIDS - Women and Child Welfare - Role of Information Technology in Environment and human health - Case Studies
23. **Disaster Management** - Natural Disasters - Meaning and nature of natural disasters, their types and effects - Floods, drought
24. Cyclone, earthquakes, Landslides, avalanches
25. Volcanic eruptions, Heat and cold waves, Climatic change: global warming, Sea level rise, ozone depletion
26. Man Made Disasters - Nuclear disasters, chemical disasters, biological disasters, Building fire, coal fire, forest fire, oil fire
27. Air pollution, water pollution, deforestation, industrial waste water pollution
28. Road accidents, rail accidents, Air accidents, sea accidents
29. Disaster Management - Effect to migrate natural disaster at national and global levels
30. International strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements
31. Role of NGOs, community - based organizations and media in disaster management
32. Central, state, district and local administration in disaster management
33. Armed forces in disaster response - Disaster response; Police and other organizations.

Practical schedule

18. Visit to a local area to document environmental assets river/forest/grassland/hill/mountain
19. Energy: Biogas production from organic wastes
20. Visit to wind mill / hydro power / solar power generation units
21. Biodiversity assessment in farming system
22. Floral and faunal diversity assessment in polluted and un polluted system
23. Visit to local polluted site-Urban/Rural/Industrial/Agricultural to study of common plants, insects and birds
24. Environmental sampling and preservation
25. Water quality analysis: pH, EC and TDS
26. Estimation of Acidity, Alkalinity
27. Estimation of water hardness
28. Estimation of DO and BOD in water samples
29. Estimation of COD in water samples
30. Enumeration of *E. coli* in water sample
31. Assessment of Suspended Particulate Matter (SPM)

32. Study of simple ecosystem – pond/river/hills
33. Visit to areas affected by natural disaster

34. Practical Examination

References

4. Erach Bharucha, Text book for Environmental studies. University Grants Commission, New Delhi
5. Tyler Miller and Scot Spoolman. 2009. Living in the Environment (*Concepts, Connections, and Solutions*). Brooks/cole, Cengage learning publication, Belmont, USA
6. P.D. Sharma, 2009, Ecology and Environment, Rastogi Publications, Meerat, India
7. De. A.K., 2010. Environmental chemistry. Published by New Age International Publishers, New Delhi. ISBN:13–978 81 224 2617 5. 384 pp

E-References

1. Dhar Chakrabarti. P.G., 2011. Disaster management - India's risk management policy frameworks and key challenges. Published by Centre for Social Markets (India), Bangalore. 36 pp.
18. Proceedings of 2nd India disaster management congress, New Delhi. Organized by National Institute of Disaster Management, New Delhi during 4 – 6, November 2009.

AMP 201 Livestock and Poultry Management (2+1)

Theory

Unit I: Introduction to Livestock Management

Significance of Livestock and Poultry in Indian Economy – Livestock and Poultry census – Different livestock development programs of Government of India and Tamil Nadu- Various systems of livestock production- extensive – semi intensive - intensive- mixed- Integrated and specialized farms.

Unit II: Dairy Cattle Management

Important White and Black cattle breeds-classification-indigenous and exotic – Breed characteristics – Breeding - Cross breeding- Upgrading - Economic traits of cattle –Culling - Estrus Cycle – Artificial Insemination – Introduction to Embryo transfer – Housing – Space requirement calf and adult stock – System and types of housing - Feeding and Management of Calf, Heifer, Pregnant, Milch animal and working animals – Nutrition – Ration – Balanced Ration - Characteristics of ration and classification of feed and fodder –Total Mixed Ration – composition of concentrate mixture for different stage - Milking methods - Clean milk production – Factors affecting milk composition – Common diseases of cattle – classification – symptoms - preventing and control measures.

Unit III: Sheep and Goat Management

Breeds - Sheep and goat classification — Economic traits - system of rearing - Housing Management – Floor space requirement - Care and Management of young and adult stock – Nutrition – Feed and fodders of Small ruminants – Flushing - Common diseases – prevention and control.

Unit IV: Management of Swine

Classification of breeds – Economic traits - Housing - Nutrition – creep feeding - Care and Management of Adult and Young Stock - Common disease- prevention and control.

Unit V: Poultry Management

Classification of breeds - Commercial Strains of broilers and layers – Housing – brooding – deep litter and cage system – care and Management of broilers and layers -Nutrition of Chick, grower, Layer and broiler – Incubation and Hatching of Eggs - Common Diseases - Control and prevention.

Practical

Study of external parts of Livestock - Identification of livestock and poultry-Tattooing-ear tags-wing and leg bands-Common restraining methods-Disbudding (or) Dehorning-Different methods of castration-Dentition- Study of type design of animal and poultry houses-Selection of dairy cow and work bullock-Determination of specific gravity, fat percentage and total solids of milk- Demonstration of cream separation, - Identification of feeds and fodder- Economics Dairy, Goat and Swine farming - Study of external parts of Fowl - Preparation of Brooder House - Brooder management-Identification of layer and non layer- Debeaking, delousing and deworming of poultry-Vaccination schedule for broiler and layer-Dressing of broiler chicken - Economics of Broiler and Layer Farming - Visit to a modern Dairy and commercial layer and broiler farms - Demonstration of incubator and setter.

Lecture schedule:

1. Significance of livestock and poultry in Indian economy-livestock and poultry census. Different livestock development programmes of Government of India and Tamil Nadu - www.indiastat.com, Livestock census 2012, Dairying in Tamil Nadu 2014 by NDDB
2. Various systems of livestock production-extensive – semi intensive, intensive- mixed- integrated and specialized farms. -357- 396 Handbook of Animal Husbandry – ICAR
3. Definition of breed-classification of indigenous white and black cattle-breed characteristics of Tamil Nadu cattle breeds and Indian breeds -Sindhi, Gir and Sahiwal. - 1-53- Handbook of Animal Husbandry – ICAR
4. Breed-characteristics of exotic cattle -Jersey and Holstein Friesian – Indian Buffaloes- Murrah, Surti and Toda. - 1-53- Handbook of Animal Husbandry - ICAR
5. Breeding-cross breeding-upgrading-economic traits of cattle-culling importance and methods - 1-53- Handbook of Animal Husbandry – ICAR.
6. Estrous cycle – signs of estrous - artificial insemination-merits and demerits-Principles and outline of embryo transfer -722-723 Handbook of Animal Husbandry - ICAR

7. Housing management-farm site selection and floor space requirement for calves, heifer, milch animal and work bullocks. - 364-379 Handbook of Animal Husbandry – ICAR
8. Systems of housing-single row system-double row system- head to head and tail to tail-merits and demerits - Type design of house. - 364-379 Handbook of Animal Husbandry – ICAR
9. Care and management of new born calf and heifers -358-362 Handbook of Animal Husbandry – ICAR
10. Care and management of pregnant animal and lactating animals. - 362-363 Handbook of Animal Husbandry – ICAR
11. Care and management of dry cows and work bullock. - 756-757 Handbook of Animal Husbandry - ICAR
12. Nutrition-definition-ration-balanced ration-desirable characteristics of a ration. Classification of feed stuffs-concentrate and roughage-comparison, Total Mixed Ration - 395-447 Handbook of Animal Husbandry – ICAR
13. Model composition of concentrate mixture of young and adult stock-age wise feed and fodder requirement-Importance of green fodder. - 395-447 - Handbook of Animal Husbandry – ICAR
14. Milking methods-clean milk production-factors affecting milk yield and composition - 363 Handbook of Animal Husbandry – ICAR
15. Diseases-classification-viral, bacterial and metabolic-general control and preventive measures. - 448-551 Handbook of Animal Husbandry – ICAR
16. Viral diseases-foot and mouth disease, bacterial diseases, anthrax, hemorrhagic septicemia- black quarter - metabolic- tympanites, acidosis, ketosis and milk fever - 448-551 Handbook of Animal Husbandry – ICAR
17. **Mid semester examination**
18. Sheep and goat farming-classification of breeds of Indian and exotic origin – economic traits. - 54-120 Handbook of Animal Husbandry – ICAR
19. Systems of rearing-housing management - type design- floor diagram-space requirement for adult and young stock. - 101 Handbook of Animal Husbandry – ICAR
20. Care and management of ram, ewe and lamb-nutrition- feeds and fodder for small ruminants. - 99-101 Handbook of Animal Husbandry – ICAR
21. Care and management of buck, doe and kid- nutrition- flushing. -102 Handbook of Animal Husbandry – ICAR
22. Common ailments of sheep and goat-sheep pox-foot and mouth-blue tongue- PPR- enterotoxaemia-Ecto and endo parasites. - 448-551 Handbook of Animal Husbandry – ICAR
23. Swine husbandry –Common breeds of exotic origin-Large White Yorkshire, Landrace and Duroc - economic traits- housing of Swine. -256-271Handbook of Animal Husbandry – ICAR
24. Care and management of sow, boar and piglets-nutrition- creep feeding. - 256-271Handbook of Animal Husbandry – ICAR
25. Disease prevention and control of swine diseases –hog cholera, foot and mouth, ecto and endo parasites. - 448-551 Handbook of Animal Husbandry – ICAR
26. Classification of breeds - commercial strains of layer and broiler. - 206-255 Handbook of Animal Husbandry – ICAR
27. Care and management of Chicks-brooder management. - 206-255 Handbook of Animal Husbandry – ICAR
28. Systems of housing- deep litter and cage system- floor space requirement-common litter material-litter management-merits and demerits. - 206-255 Handbook of Animal Husbandry – ICAR
29. Care and management of Grower and Layers- vaccination schedule. - 206-255 Handbook of Animal Husbandry – ICAR
30. Care and management of broilers-vaccination schedule. - 206-255 Handbook of Animal Husbandry – ICAR
31. Incubation and hatching of eggs. - 206-255 Handbook of Animal Husbandry – ICAR
32. Nutrition-feed formulation-composition of chick, grower, layer broiler- starter and Finisher mash-Feed Conversion Ratio /dozen egg or kg of meat production. - 206-255 Handbook of Animal Husbandry – ICAR

33. Classification of disease –viral – bacterial - protozoan- causative organisms, symptoms and prevention – viral diseases- Ranikhet – IBD-avian flu - 448-551 Handbook of Animal Husbandry – ICAR
34. Bacterial disease-E.coli-coryza-salmonellosis-protozoan–coccidiosis-casulative organism, symptoms and preventive measures. Management of dead birds and manure - 448-551 Handbook of Animal Husbandry - ICAR

Practical Schedule

35. Study of external parts of livestock
36. Identification of livestock and poultry
37. Common restraining methods of livestock
38. Disbudding, Dehorning, Castration and Dentition of livestock
39. Study of type design of animal and poultry houses
40. Selection of dairy cow and work bullock
41. Determination of specific gravity, fat %, total solids, solids not fat
42. Demonstration of cream separation
43. Identification of feed & fodder
44. Economics of dairy, goat and swine Farming
45. Study of external parts of fowl. Preparation of brooder house
46. Identification of layer and non- layer
47. Debeaking, delousing, deworming of poultry Vaccination schedule for broiler and layer
48. Demonstration of dressing of broiler. Economics of layer and broiler farming
49. Visit to a modern dairy and commercial layer and broiler farms
50. Demonstration of incubator and setter
51. **Practical examination**

Reference:

ICAR (2002) Hand of Animal Husbandry, ICAR, New Delhi.

E- Eeference:

1. <http://www.elearnvet.net/>
2. http://agridr.in/expert_system/cattlebuffalo/Breeding%20management%20of%20cattle%20and%20buffaloes-2.html

AEC 201 Farm Management, Production and Resource Economics (1+1)

Theory

Unit 1: Production Economics and Farm Management - Nature and Scope

Meaning and concept of farm management, objectives and relationship with other sciences. Meaning and definition of farms, its types and characteristics, factors determining types and size of farms. Types of farming:

Specialized, Diversified, and Mixed farming – Systems of farming: Peasant Farming, State Farming, Capitalistic, Collective and Co – operative Farming.

Unit 2: Factor – Product, Factor – Factor and Product – Product Relationships

Principles of farm management: concept of production function and its characteristics and its type, use of production function in decision-making on a farm. Factor-Product relationship. Meaning, Definition – Laws of Returns. Meaning and concept of cost, types of costs, cost curves - and their inter-relationship - shut down and break-even points, importance of cost in managing farm business and estimation of gross farm income, net farm income, family labor income and farm business income. Economies of Scale – Economies of Size - Determination of Optimum Input and Output – Physical and Economic Optimum. Factor – Factor relationship: Least Cost Combination of inputs; Product – Product relationship: Optimum Combination of Products – Principle of Equi – Marginal Returns – Principle of Opportunity Cost and Minimum Loss Principle. Law of Comparative Advantage.

Unit 3: Farm Planning and Budgeting

Farm business analysis: meaning and concept of farm income and profitability, technical and economic efficiency measures in crop and livestock enterprises. Importance of farm records and accounts in managing a farm, various types of farm records needed to maintain on farm, farm inventory, balance sheet, profit and loss accounts. Meaning and importance of farm planning and budgeting, partial and complete budgeting, steps in farm planning and budgeting - linear programming, appraisal of farm resources, selection of crops and livestock's enterprises.

Unit 4: Risk and Uncertainty in Agriculture Production

Concept of risk and uncertainty occurrences in agriculture production, nature and sources of risks and their management strategies, Crop / livestock / machinery insurance. Weather based crop insurance - Features and determinants of compensations.

Unit 5: Resource Economics

Resource Economics: Concepts, Classification, differences between Natural Resource Economics (NRE) and agricultural economics, unique properties of natural resources. Natural Resources - Issues – Scarcity of resources – Factors mitigating scarcity – Property Rights: Common Property Resources (CPRs): meaning and characteristics of CPRs – Externalities: meaning and types - positive and negative externalities in agriculture, Inefficiency and welfare loss, solutions; Important issues in economics and management of common property resources of land, water, pasture and forest resources.

Practical

Preparation of farm layout. Determination of cost of fencing of a farm. Computation of depreciation cost of farm assets. Application of equi-marginal returns / opportunity cost principle in allocation of farm resources. Determination of most profitable level of inputs use in a farm production process. Determination of least cost combination of inputs. Selection of most profitable enterprise combination. Application of cost principles including CACP concepts in the estimation of cost of crops – Estimation of costs and returns of livestock products. Preparation of farm plan and budget, farm records and accounts and profit and loss accounts. Break – even analysis- Graphical solution to Linear Programming problem. Collection and analysis of data on various resources in India.

References

- Sankayan, P.L. 1983. Introduction to Farm Management. Tata McGraw Hill Publishing Company Ltd. New Delhi.
- Johl, S.S & Kapoor, T.R. 1973. Fundamentals of Farm Business Management. Kalyani Publishers. Ludhiana.
- Kahlon, A.S and Singh K. 1992. Economics of Farm Management in India. Allied Publishers. New Delhi.

- Doll, J.P. and F. Orazem. 1983. Theory of Production Economics with Applications to Agriculture. John Wiley, New York.
- Debertin, D.L. 1986. Agricultural Production Economics. Macmillan. New York.
- Heady, E.O. and H.R. Jensen. 1954. Farm Management Economics. Prentice – Hall. Englewood Cliffs.
- Kay, Ronald D., and William M. Edwards, and Patricia Duffy. 2004. Farm Management. Fifth Edition. McGraw–Hill Inc. New York.
- Panda, S.C. 2007. Farm Management and Agricultural Marketing. Kalyani Publishers. Ludhiana. India.

Theory lecture schedule

1. Meaning and concept of farm management, objectives and relationship with other sciences. Meaning and definition of farms, its types and characteristics, factors determining types and size of farms.
2. Types of farming: Specialized, Diversified, and Mixed farming – Systems of farming: Peasant Farming, State Farming, Capitalistic, Collective and Co – operative Farming.
3. Principles of farm management: concept of production function and its characteristics and its type, use of production function in decision-making on a farm.
4. Factor - Product relationship: Meaning, Definition – Laws of Returns: Classical production function and its characteristics.
5. Meaning and concept of cost, types of costs, cost curves - and their inter-relationship -shut down and break even points, importance of cost in managing farm business and estimation of gross farm income, net farm income, family labor income and farm business income.
6. Economies of Scale – Economies of Size - Determination of Optimum Input and Output – Physical and Economic Optimum.
7. Factor – Factor relationship: Least Cost Combination of inputs.
8. Product – Product relationship: Optimum Combination of Products – Principle of Equi –Marginal Returns – Principle of Opportunity Cost and Minimum Loss Principle. Law of Comparative Advantage.
9. **Mid Semester Examination.**
10. Farm business analysis: meaning and concept of farm income and profitability, technical and economic efficiency measures in crop and livestock enterprises.
11. Importance of farm records and accounts in managing a farm, various types of farm records needed to maintain on farm, farm inventory, balance sheet, profit and loss accounts.
12. Meaning and importance of farm planning and budgeting, partial and complete budgeting, steps in farm planning and budgeting - linear programming, appraisal of farm resources, selection of crops and livestock's enterprises.
13. Concept of risk and uncertainty occurs in agriculture production, nature and sources of risks and its management strategies.
14. Crop / livestock / machinery insurance. Weather based crop insurance - Features and determinants of compensations.
15. Resource Economics: Concepts, Classification, differences between Natural Resource Economics (NRE) and agricultural economics, unique properties of natural resources.
16. Natural Resources Issues – Scarcity of resources – Factors mitigating scarcity – Property Rights – Common Property Resources (CPRs): meaning and characteristics of CPRs – Externalities: meaning and types - positive and negative externalities in agriculture,
17. Inefficiency and welfare loss, solutions, Important issues in economics and management of common property resources of land, water, pasture and forest resources.

Practical Schedule

1. Preparation of farm layout. Determination of cost of fencing of a farm.
2. Computation of depreciation and cost of farm assets: Valuation of assets by different methods.
3. Application of equi - marginal returns / opportunity cost principle in allocation of farm resources.

4. Determination of most profitable level of inputs use in a farm production process.
5. Determination of least cost combination of inputs.
6. Selection of most profitable enterprise combination.
7. Application of cost principles including CACP concepts in the estimation of cost of cultivation and cost of production of agricultural crops.
8. Estimation of cost of cultivation and cost of production of perennial crops / horticultural crops.
9. Estimation of cost of returns of livestock products.
10. Preparation of farm plan and budget.
11. Farm records and accounts: Usefulness, types of farm records: farm production records and farm financial records.
12. Preparation of Cash flow statement
13. Preparation and Analysis of Net worth Statement and Profit and Loss statement
14. Estimation of Break – even analysis.
15. Graphical solution to Linear Programming problem.
16. Collection and analysis of data on various resources in India.
17. **Final Practical Examination.**

SAC 201 Soil Resource Inventory (1+1)

Theory

Unit-I

Soil Survey: Importance-Definition-Purpose of soil survey- Standard soil survey - Scope and objectives - Fundamental and Applied. Soil systematics- pedon and polypedon, control section and three dimensional nature of soil body. Soil mapping units: Soil series, soil association, soil complex, variants, inclusions and miscellaneous land types.

Unit II

Methods of soil survey: Base maps, Traversing: Grid survey and Free survey. Types of soil survey:

Detailed, Reconnaissance, Detailed- Reconnaissance and Semi–Detailed soil survey. Soil mapping.

Unit-III

Soil Classification -Purpose -Early and modern systems -USDA system of soil classification and its advantages. Soil taxonomy - Diagnostic horizons: surface and sub surface-structure and differentiating characteristics and limitations. Soil orders – description. Distribution of Soil orders in world. Soils of India and Tamil Nadu.

Unit-IV

Soil survey reports - Soil Survey Interpretations - Land Capability Classification - Soil and Land Irrigability Classification - Storie Index Rating - Productivity potential - Fertility Capability Classification - Crop suitability: Field crops, horticultural crops and forest trees. Delineation of soils for fertility – Nutrient indexing. Land Use Planning concepts and objectives.

Practical

Site characteristics and Soil Profile description - Morphological study of soil profiles - Estimation of CEC in soil - Estimation of cations and free CaCO_3 - Study of base maps- Interpretation of soil survey data and maps - Interpretation of soil data for land capability,

crop suitability - Interpretation of soil data for fertility capability classes - Interpretation of soil data for productivity rating - Interpretation of soil data for Nutrient Indexing.

Lecture schedule

1. Soil Survey: Importance-Definition-Purpose of soil survey.
2. Standard soil survey - Its scope and objectives. Fundamental and Applied.
3. Soil systematics - pedon and polypedon, control section and three dimensional nature of soil body.
4. Soil mapping units: Soil series, soil association, soil complex, variants, inclusions and miscellaneous land types.
5. Methods of soil survey: Base maps,Traversing: Grid survey and Free survey.
6. Types of soil survey: Detailed, Reconnaissance, Detailed- Reconnaissance and Semi–Detailed soil survey. Soil mapping.
7. Soil Taxonomy – Purpose. Early and modern systems. USDA system of soil classification and its advantages.

18. Mid semester examination.

19. Diagnostic horizons: surface and sub surface.
20. Soil taxonomy – Structure and differentiating characters and limitations.
21. Soil orders, characteristics and their distribution in world.
22. Soils of India and Tamil Nadu.
23. Soil maps, kinds of soil maps and their preparation.
24. Interpretative groupings of soils. Land capability classification and Fertility Capability Classification.
25. Land irrigability classification, Storie index and productivity potential.
26. Land suitability classification for field crops, horticultural crops and forest trees.
27. Land Use Planning - Concepts and objectives - Tropical, subtropical and temperate regions.

Practical schedule

1. Site characteristics and Soil Profile description
2. Morphological study of soil profile 1 (Red soil)
3. Morphological study of soil profile 2 (Black soil)
4. Morphological study of soil profile 3 (Alluvial / Laterite soil)
5. Estimation of CEC in soil
6. Estimation of cations and free CaCO_3
7. Study of base maps- Topo sheets and cadastral maps.
8. Study of base maps- Aerial photographs or satellite imageries.

9. Interpretation of soil survey data and maps.
10. Interpretation of soil data for land capability.
11. Interpretation of soil data for crop suitability for field crops.
12. Interpretation of soil data for crop suitability for horticultural crops.
13. Interpretation of soil data for crop suitability for forest trees.
14. Interpretation of soil data for fertility capability classes.
15. Interpretation of soil data for productivity rating.
16. Interpretation of soil data for Nutrient Indexing.

17. Practical Examination

References

1. Buol, S.W., Hole, F.D., McCracken, R.J., (1973). Soil genesis and classification. Oxford and IBH publishing Co., New Delhi.
2. Lillesand, T.M. and Kiefer, R.W., 1987. Remote sensing and image interpretation, John Wiley and sons, inc, New York.
3. Sehgal, J. 2005. Pedology concepts and applications, Kalyani Publishers, New Delhi.
4. Soil Survey Division Staff 1999. Soil Survey Manual, USDA publication.
5. Steven, M.D. and Clark, J.A. 1990. Applications of Remote Sensing in Agriculture, Cambridge University, UK.
6. USDA 1954. Diagnosis and improvements of Saline and alkali soils. (Ed) L.A. Richards. Handbook No.60. USDA Washington DC.
7. Anji Reddy, M., 2002. Remote sensing and geographical information systems, BS publication, Hyderabad.
8. Somani, L.L. and K.L. Totawat 1993. Management of Salt Affected Soils and Water.

E-references

10. <ftp://ftp-fc.sc.egov.usda.gov/NSSC/NCSS/Conferences/scanned/>
11. ftp://ftp-fc.sc.egov.usda.gov/NSSC/Lab_References/SSIR_51.pdf
12. tp://ftp-fc.sc.egov.usda.gov/NSSC/Lab_References/SSIR_51.pdf
13. www.iuss.org/Bulletins/00000096.pdf
14. www.oosa.unvienna.org/pdf/sap/centres/rscurrE.pdf -
15. en.wikipedia.org/wiki/Geographic_information_system
16. www.annauniv.edu/cia/Curric%20Syllabi/M.../Remote%20Sensing.pdf
17. www.csre.iitb.ac.in/~dd/detail.html
18. www.dvsinstitute.org/forms/pg/M.Sc.%20-%20RS%20&%20GIS-350.pdf
19. inkinghub.elsevier.com/retrieve/pii/S0166248197800335
20. www.scribd.com/doc/40246764/Description-Pedon-Copy -
21. [www.angrau.net/BSc\(Ag\)CourseCurriculum.htm](http://www.angrau.net/BSc(Ag)CourseCurriculum.htm)
22. www.springerlink.com/index/BJG00EL8FLNTFUNL.pdf
23. www.eurojournals.com/ejsr_42_2_10.pdf
24. inkinghub.elsevier.com/retrieve/pii/S0166248197800335
25. www.springerlink.com/index/R177R744722222UN.pdf - Similar
26. content.alterra.wur.nl/Internet/webdocs/ilri-publicaties/.../Bib10.pdf
27. www-wds.worldbank.org/external/.../INDEX/multi_page.txt - Cached
28. openaccess.icrisat.org/.../Proceedings-integrated-watershed-management-for-land-Asia.pdf
29. www.springerlink.com/index/jlu87tk58363.pdf
30. www.buc.edu.in/sde_book/msc_soil.pdf

FMP 211 FARM MACHINERY AND POWER (1+1)

THEORY

Unit I: Farm Power & IC engines

Status of Farm Power in India, Sources of Farm Power, I.C. engines, working principles of IC engines, comparison of two stroke and four stroke cycle engines , Study of different components of IC engine, I.C. engine terminology and solved problems, Familiarization with different systems of I.C. engines: Air cleaning, cooling, lubrication, fuel supply.

Unit-II : Tractor & functional components

Hydraulic control system of a tractor, Familiarization with Power transmission system clutch, gear box, differential and final drive of a tractor ,Tractor types, Cost analysis of tractor power and attached implement,

Unit –III: Tillage implements

Familiarization with Primary and Secondary Tillage implement, Implement for hill agriculture.

Unit-IV: Sowing & Intercultural implements

Familiarization with sowing and planting equipment, calibration of a seed drill and solved examples, implement for intercultural operations.

Unit-V: Plant Protection and Harvesting equipments

Familiarization with Plant Protection equipment, Familiarization with harvesting and threshing equipment.

PRACTICALS

Study of different components of I.C. engine - To study air cleaning and cooling system of engine - Familiarization with clutch – Transmission - Differential and final drive of a tractor - Familiarization with lubrication and fuel supply system of engine - Familiarization with brake – Steering - Hydraulic control system of engine - Learning of tractor driving - Familiarization with operation of power tiller - Implements for hill agriculture - Familiarization with different types of primary and secondary tillage implements - Mould board plough - Disc plough and disc harrow -Familiarization with seed-cum- fertilizer drills their seed metering mechanism and calibration - Planters and transplanter - Familiarization with different types of sprayers and dusters – Familiarization with different inter-cultivation equipment - Familiarization with harvesting and threshing machinery.

LECTURE SCHEDULE:

1. Farm power in India - sources of farm power and their use in agriculture
2. Working principles of IC Engines-Two stroke and Four stroke engines - applications – comparison-Engine terminology.
3. Components of IC engine and systems of IC engine – air cleaning, cooling, lubricating and fuel supply systems.
4. Tractors- types - transmission system- clutch, gearbox, differential and final drive - hydraulic system.
5. Cost analysis of tractor with attached implement.
6. Tillage, objectives, types - ploughing methods. Primary tillage- mould board plough, disc plough, chisel plough and subsoil plough - components and functions, types, advantages and disadvantages
7. Secondary tillage equipment – cultivators, harrows, levelers, land forming equipment – rotovators – puddlers - manure trammers and cage wheels, Implements for Hill agriculture.
8. Sowing methods - seed drills and planters- seed cum fertilizer drills - components and functions-Calibration.
9. Mid semester examination
10. Paddy transplanters, types, working principle, field and nursery requirements
11. Implements for intercultural operations – cultivators, sweep, junior hoe, manual weeders and power operated weeders for wet land and garden land
12. Sprayers and their functions, classification, manually operated sprayers, terminology, Nozzle types.
13. Power operated sprayers – Tractor operated boom sprayer, Knapsack mist blower cum duster – Tall tree sprayer-dusters, types and uses.
14. Tools for horticultural crops – propagation tools, planters and harvesting tools and machinery
15. Threshing of crop, thresher and its principles of operation - threshing losses.
16. Harvesting equipment – reapers - mowers and combine harvesters – types, construction and operation-Balers.
17. Harvesting machinery for groundnut, tuber crops, Cotton and sugarcane

PRACTICAL SCHEDULE:

1. Study of working of two and four stroke IC engines and their systems with solved problems.

2. Study of Tractor clutch, gearbox, differential and final drive. Study of brake steering, and hydraulic control.
3. Learning driving of tractor and power tiller
4. Study of tractors and power tillers – their operation and maintenance
5. Study of mould board plough, - methods of ploughing- with solved problems.
6. Disc plough and subsoiler and their components- Hitching and adjustment of plough - field operation of different tractor drawn primary tillage machinery.
7. Study of cultivator, disc harrows, Rotavator, bund former, ridger, leveller and puddling implements and their operation.
8. Study of seed drills, planters and seed-cum-fertilizer drills and their components and metering mechanisms - calibration- simple problems on calibration.
9. Study and operation of machinery for rice cultivation - puddling implements- rotary puddlers and cage wheels, tray seeder for rice nursery, transplanters- types operation and maintenance- Drum seeder, conoweeder, power weeder and finger type weeder.
10. Study of different inter-cultivation equipment for uplands - manual, animal drawn, power operated - tractor and power tiller operated - field operation
11. Study of plant protection equipment – manually operated sprayers and dusters, knapsack mist blower cum duster, tractor operated sprayers- their operation, adjustment, calibration and safety requirements
12. Study of tools for Hill agriculture and horticultural crops – propagation tools, vegetable transplanter, harvesting tools -lawn mower, hole diggers, tree climber, shredders for crop residue.
13. Threshing machinery for paddy and identification of its components- different threshing drums - calculation of efficiency and losses.
14. Study of paddy reaper and paddy combine- their systems, method of operation and adjustment.
15. Study of harvesters for root crops - turmeric and tapioca and groundnut diggers
16. Problems on cost of operation of tractor operated machinery.
17. Final practical examination

References:

1. Jagadishwar Sahay, 2010 - **Elements of Agricultural Engineering**. Standard Publishers Distributors, Delhi. ISBN 978-8180140440
2. Ojha, T. P. and Michael, A.M. **Principles of Agricultural Engineering**. Vol. I, Jain Brothers, 16/893, East Park Road, Karol Bagh, New Delhi – 110005
3. S.C.Jain and C.R.Rai. **Farm Tractor – Maintenance and Repair**. Standard Publishers, 1705-B, Nai Sarak, Delhi – 110006
4. Senthilkumar, T., R. Kavitha and V.M.Duraisamy 2015. **A Text Book of Farm Machinery**, Thannambikkai Publications, Coimbatore . ISBN: 978-9381102305

E- RESOURCES:

www.agricoop.nic.in/dacdivision/Machinery1/directory.htm
www.farmmachineryshow.org
<http://www.hillagric.ac.in/edu/coa/agengg/lecture/243/agriengg-243.htm>
http://www.digitalbookindex.org/subject_search/search010agricultureequipmenta
<http://ecoursesonline.iasri.res.in/course/view.php?id=540>

The students will undertake the short tour during third semester for seven days covering KVK's, Research stations, Sister campuses and ICAR institutes in the southern part of Tamil Nadu. The study tour will provide an exposure to the students to know about the soil, climatic conditions and cropping patterns in the respective agro-climatic zones. The students will also have first-hand information on latest technologies on various crops and allied activities.

IV SEMESTER

S. No.	Course code	Course Title	Credit load
1	PBG 201	Fundamentals of Genetics	2+1
2	AEX 201	Communication Skills and Personality Development	1+1
3	STA 211	Statistical Methods	1+1
4	PAT 202	Principles of integrated plant disease management	1+1
5	AEN 202	Management of beneficial and harmful insects	2+1
6	AGR 203	Crop Production Technology – II (<i>Rabi</i> crops)	1+1
7	AGR 204	Farming System & Sustainable Agriculture	1+1
8	SAC 202	Problematic soils and their management	2+0
9	HOR 212	Production Technology for Ornamental Crops, MAP and Landscaping	1+1
10	ANM 201	Introductory Nematology	0+1
11	NST 201	Fundamentals and Applications of nanotechnology	1+0
12	ERG 211	Renewable Energy and green technology	1+1
12	NSS/NCC 101	NSS/NCC	0+1*
13	PED 101	Physical Education	0+1*
		Total	14+10=24
		*Non-gradual courses compulsory courses	

PBG 201. Fundamentals of Genetics (2+1)

THEORY

Unit I: Cytology

Definition of genetics, heredity, inheritance, cytology, cytogenetics; Brief history of developments in genetics and cytogenetics. Physical basis of heredity. Structure and function of cell and cell organelles – Differences between Prokaryotes and Eukaryotes. Cell division – mitosis- meiosis and their significance - Gametogenesis and syngamy in Plants- identical and fraternal twins. Chromosome structure, chemical composition, nucleosome, centromere, telomere, euchromatin, heterochromatin, NOR, satellite chromosome, karyotype, ideogram. Types of chromosomes based on position of centromere, based on structure and function: normal and special chromosomes - polytene, lampbrush, B chromosomes, ring and isochromosomes. Chromosomal aberration: Variation in chromosome structure – deletion, duplication, inversion and translocation – genetic and cytological implications. Chromosomal aberration: Variation in chromosome number – euploid, aneuploid, types of aneuploids and their origin; Klinefelter syndrome and Turner syndrome; Polyploid - auto and allopolyploids, their characters; meaning of genome; evolution of wheat, triticale, cotton, tobacco, *Brassica*

Unit II: Mendelian laws and modifications of Mendelian laws

Pre-Mendelian ideas about heredity – Vapour and fluid theory, Magnetic power theory, Preformation theory, Lamarck's theory, Darwin's theory, Germplasm theory and Mutation theory. Mendel's experiments and laws of inheritance. Rediscovery of Mendel's work. Terminologies: gene, allele, locus, homozygous, heterozygous, hemizygous, genotype, phenotype, monohybrid, dihybrid, trihybrid, polyhybrid. Chromosomal theory of inheritance. Allelic interactions – Dominance vs recessive, complete dominance, codominance, incomplete dominance, threshold characters. Deviation from Mendelian inheritance – Non allelic interaction without modification in Mendelian ratio – Bateson and Punnett's experiment on fowl comb shape. Non allelic interaction with modification in Mendelian ratio – i.) Dominant epistasis (12:3:1). ii.) Recessive epistasis (9:3:4) iii.) Duplicate and additive epistasis (9:6:1). iv.) Duplicate dominant epistasis (15:1). v) Duplicate recessive epistasis (9:7) vi.) Dominant and recessive epistasis (13:3); Summary of epistatic ratios (i) to (vi). Lethal genes, Pleiotrophy, penetrance and expressivity, Multiple alleles, blood group in human, coat colour in rabbits, self incompatibility in plants; pseudo alleles, isoalleles.

Unit III: Quantitative inheritance, Linkage and Crossing over

Quantitative inheritance – Multiple factor hypothesis – Nilsson Ehle experiment on wheat kernel colour. Polygenes – transgressive segregation, comparison of quantitatively and qualitatively inherited characters; modifiers; Linkage - coupling and repulsion; Experiment on Bateson and Punnett. Chromosomal theory of linkage of Morgan – Complete and incomplete linkage- Linkage group. Crossing over – significance of crossing over; cytological proof for crossing over - Stern's experiment - Factors controlling crossing over. Strength of linkage and recombination; Two point and three point test cross. Double cross over, interference and coincidence; genetic map, physical map.

Unit IV: Sex determination, sex linkage and cytoplasmic inheritance

Sex determination: Autosomes and sex chromosomes - chromosomal theory of sex determination - different types – sex determination in human, fowl, butterfly, grasshopper, honey bee, fumea; Sex determination in plants – *Melandrium*, papaya, maize. Genic balance theory of Bridges – Gynandromorphs. Sex linked inheritance – criss cross inheritance – reciprocal difference; holandric genes; sex influenced and sex limited inheritance - Genetic disorders. Cytoplasmic inheritance and maternal effects – features of cytoplasmic inheritance, chloroplast, mitochondrial - plastid colour in *Mirabilis jalapa* - cytoplasmic male sterility in maize, kappa particles of paramecium

Unit V: Modern concept of genetics and mutation

DNA, the genetic material – Griffith's experiment, Avery, McCleod and McCarthy Experiment – confirmation by Hershey and Chase; RNA as genetic material – Frankel, Conrat and Singer experiment. Structure of DNA – Watson and Crick model. Proof for semi conservative method of DNA replication; Models of DNA replication; steps involved in DNA replication. RNA types - mRNA, tRNA, rRNA. Protein synthesis - Regulation of gene expression – Operon model of Jacob and Monod – Lac and Trp operons. Cistron, muton and recon. Mutation – characteristics of mutation – micro and macro mutation – CLB technique - molecular basis of mutation- Transition and transversion; major physical and chemical mutagens.

PRACTICAL

Study of microscopes – Preparation of fixatives and stains – pre treatment of materials for mitosis and meiosis – study of mitosis and meiosis. Study of genetic ratios of – monohybrid, dihybrid – incomplete dominance. Gene interaction - multiple alleles and multiple factors. Study of linkage, Estimation of strength of linkage and recombination frequency in three point test cross data and F₂ data – Drawing of genetic map – interference and coincidence. Studies on sex linked inheritance in Humans and *Drosophila*

Theory schedule

- Definition of genetics, heredity, inheritance, cytology, cytogenetics; Brief history of developments in genetics and cytogenetics.
 - Physical basis of heredity: Structure and function of cell and cell organelles –
 - Differences between Prokaryotes and Eukaryotes. Cell division – mitosis
 - Cell division - meiosis and their significance
 - Gametogenesis and syngamy in Plants- identical and fraternal twins
 - Chromosome structure, chemical composition, nucleosome, centromere, telomere, euchromatin, heterochromatin, NOR, satellite chromosome, karyotype, ideogram
 - Types of chromosomes based on position of centromere, based on structure and function: normal and special chromosomes - polytene, lampbrush, B chromosomes, ring and isochromosomes.
 - Chromosomal aberration: Variation in chromosome structure – deletion, duplication, inversion and translocation – genetic and cytological implications.
 - Chromosomal aberration: Variation in chromosome number – euploid, aneuploid, types of aneuploids and their origin; Klinefelter syndrome and Turner syndrome;
 - Polyploid - auto and allopolyploids, their characters; meaning of genome; evolution of wheat, Triticale, cotton, tobacco, *Brassica*
 - Pre-Mendelian ideas about heredity – Vapour and fluid theory, Magnetic power theory, Preformation theory, Lamarck's theory, Darwin's theory, Germplasm theory and Mutation theory.
 - Mendel's experiments and laws of inheritance. Rediscovery of Mendel's work
7. Terminologies: gene, allele, locus, homozygous, heterozygous, hemizygous, genotype, phenotype, monohybrid, dihybrid, trihybrid, polyhybrid.
 8. Chromosomal theory of inheritance. Allelic interactions – Dominance vs recessive, complete dominance, codominance, incomplete dominance, threshold characters

9. Deviation from Mendelian inheritance – Non allelic interaction without modification in Mendelian ratio – Bateson and Punnett's experiment on fowl comb shape. Non allelic interaction with modification in Mendelian ratio – i.) Dominant epistasis (12:3:1)
10. ii.) Recessive epistasis(9:3:4) iii.) Duplicate and additive epistasis(9:6:1). iv.) Duplicate dominant epistasis(15:1)
11. **Mid Semester Examination**
12. v) Duplicate recessive epistasis (9:7) vi.) Dominant and recessive epistasis (13:3); Summary of epistatic ratios (i)to (vi).
13. Lethal genes, Pleiotrophy, penetrance and expressivity, Multiple alleles, blood group in humans, coat colour in rabbits, self incompatibility in plants; pseudo alleles, isoalleles.
14. Quantitative inheritance – Multiple factor hypothesis – Nilsson Ehle experiment on wheat kernel colour.
15. Polygenes – transgressive segregation, comparison of quantitatively and qualitatively inherited characters; modifiers;
16. Linkage - coupling and repulsion; Experiment on Bateson and Punnet
17. Chromosomal theory of linkage of Morgan – Complete and incomplete linkage, Linkage group.
18. Crossing over – significance of crossing over; cytological proof for crossing over - Stern's experiment; Factors controlling crossing over.
19. Strength of linkage and recombination; Two point and three point test cross.
20. Double cross over, interference and coincidence; genetic map, physical map.
21. Sex determination: Autosomes and sex chromosomes - chromosomal theory of sex determination - different types – sex determination in human, fowl, butterfly, grasshopper, honey bee, fumea; Sex determination in plants – *Melandrium*, papaya, maize.
22. Genic balance theory of Bridges - Gynandromorphs
23. Sex linked inheritance – criss cross inheritance – reciprocal difference; holandric genes; sex influenced and sex limited inheritance - Genetic disorders
24. Cytoplasmic inheritance and maternal effects – features of cytoplasmic inheritance, chloroplast, mitochondrial - plastid colour in *Mirabilis jalapa* - cytoplasmic male sterility in maize, kappa particles of paramecium
25. DNA, the genetic material – Griffith's experiment, experiment of Avery, McCleod and McCarthy – confirmation by Hershey and Chase; RNA as genetic material – Frankel, Conrat and Singer experiment.
26. Structure of DNA – Watson and Crick model Proof for semi conservative method of DNA replication; Models of DNA replication; steps involved in DNA replication.RNA types - mRNA, tRNA, rRNA – Protein synthesis
27. Regulation of gene expression – Operon model of Jacob and Monad – Lac and Trp operons. Cistron,muton and recon.
28. Mutation – characteristics of mutation – micro and macro mutation – CIB technique - molecular basis of mutation- Transition and transversion; major physical and chemical mutagens.

Practical Schedule

18. Use of microscopes
19. Principles of killing and fixing; preparation of stains and preservatives.
20. Study of behavior of chromosomes in mitosis.
21. Study of mitotic phases in root tips of onion / *Aloe sp.*
22. Procedure for fixing and observing different meiotic phases in the inflorescence of rice, maize
23. Procedure for fixing and observing different meiotic phases in the inflorescence in pearl millet, sorghum
24. Repetition of meiotic studies in maize/ sorghum/ pearl millet and making temporary and permanent slides.

25. Principles of dominance, recessive, back cross, test cross, incomplete dominance, codominance and lethal factor; Chi square test; Monohybrid genetic ratio with dominance, with incomplete dominance and test cross.
26. Dihybrid ratio with dominance, with incomplete dominance and test cross
27. Simple interaction of genes-comb character in fowls; Dominant epistasis.
28. Recessive epistasis, Duplicate and additive epistasis.
29. Duplicate dominant epistasis, Duplicate recessive epistasis, Dominant and recessive epistasis.
30. Multiple alleles and polygenic inheritance
31. Estimation of linkage with F₂ and test cross data; Coupling and repulsion.
32. Problems on three point test cross; Working out interference, coincidence and drawing genetic maps.
33. Studies on sex linked inheritance in Humans and Drosophila
34. **Final Practical examination.**

References

1. Gupta P.K., 1997. Cytogenetics. Rastogi Publications, Meerut
2. Verma, P.S. and V.K. Agarwal. 2007. Genetics. S.Chand and Company Ltd./ New Delhi.
3. Stansfield, W.D. 1990. Theory and problems of genetics. Mc-Graw Hill Book Co., New York
4. Pundhansingh. 2014. Elements of Genetics. Kalyani Publishers
5. Benjamin Lewin. 2005. Genes IX Oxford University Press, Oxford.
6. Russel, P.J. 2000. Fundamentals of genetics. Addison Wesley Longman Publishers, USA
7. Daniel Sundararaj, G. Thulasidas and M. Stephen Dorairaj, 1997. Introduction to Cytogenetics and Plant Breeding. Popular Book Depot, Chennai –15.
8. Strickberger. M.W. 1996. Genetics. Prentice-Hall of India Pvt. Ltd. New Delhi.
9. Singh, B.D. 2004. Fundamentals of Genetics, Kalyani Publishers, Chennai.

50. E- References

1. www.nmsu.edu,
2. www.biology200.gsu.edu

AEX 201 Communication Skills and Personality Development (1+1)

Theory

Communication Skills: meaning and process of communication, verbal and nonverbal communication; listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures. Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting; individual and group presentations, impromptu presentation, public speaking; Group discussion. Organizing seminars and conferences.

Unit I Communication Skills: meaning and process of communication, verbal and nonverbal communication

Unit II Listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures.

Unit III Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting

Unit IV Individual and group presentations, impromptu presentation, public speaking **Unit V** Group discussion. Organizing seminars and conferences.

Practical

Listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures. Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting; individual and group presentations

Theory Schedule

19. Communication – meaning and process –Functions and Types of communication
 20. Communication models - Aristotle, Shannon and Weaver, Schramm, Berlo Westly and Maclean, Leagan, Rogers and Shoemaker, Littererls model and Dance’s Helical Model – Elements of communication – communication barriers
 21. Verbal and Non verbal communication – Verbal communication – definition and meaning – Verbal vs Oral communication – Types – Styles - Barriers to effective verbal communication ;
 22. Non verbal communication – definition and meaning – Proxemics, Chronemics, Movement and body position, Posture, Facial Expression, Gestures and Eye Contact – importance of non verbal communication
 23. Listening – Definition – Listening vs Hearing – Active listening – Types of listening –Guidelines for effective listening – Developing listening skills - Barriers to listening – Listening misconceptions
 24. Writing skill – Importance – Effective writing - Components of writing : Introduction , Audience and format ,Composition and style, Structure, Grammatical errors , Proofing and Conclusion – Ways to improve writing skills – Technical writing
 25. Oral presentation skills – Basics of effective oral presentation : Planning , preparing (Introduction, Body and conclusion), Delivery, Body language and Handling anxiety – Strategies for giving oral presentation
 26. Field diary – Definition – Components to be included – Parts of field diary – Field diary in social sciences
-
5. Lab record : Definition –Importance of keeping a lab record - Features of a lab record - Contents of lab record – Guidelines for keeping a lab record
 6. **Mid semester examination**
 7. Indexing – Definition – Importance – Types of indexing with advantages and limitations
 8. Footnote and Bibliographic procedure : Footnote system of citation ; Bibliographic procedures : Citation in Text, Citation in Journal, Citation from Book(One author / Multiple authors), Citation from an Edited Book, Citation of Seminar/Conference Proceedings, Citation from Institutional Publication, Citing Government Publications, Abbreviations for Names of Journals, Paraphrasing, Abbreviations in citations (Art of publication)

9. Reading skills – Definition – Kinds of reading skills – Critical reading skills – Reading readiness skills- Guidelines for effective reading- Extensive reading- Intensive reading. Comprehension : Definition and meaning – Comprehension skills-Readability Index
10. Precise writing – Derivation and Meaning – Skills required – Method or procedure – Guidelines; Summarising – Meaning- Steps to write a summary
11. Abstracting : Definition - Purpose of abstract – Types of abstract - Abstract Styles – Steps for Writing Effective Abstracts- Some Do's Don'ts in preparing abstracts
12. Individual presentation - Meaning –Steps for individual presentation; Group presentation – Meaning – Stages of group presentation ; Impromptu presentation
13. Public speaking : meaning – Points to be considered in public speaking – Effective public speaking: Group Discussion: Meaning –Procedure – Advantages – Limitations ;Seminar Conferences : Definition and meaning – Steps in organizing seminar / conferences / symposium / workshop

Practical Schedule

10. Practicing active listening
11. Exercise on note taking methods
12. Exercise on technical writing and practicing proof correction
13. Practicing oral presentation
14. Exercise on writing field diary and Lab record
15. Visit to library and learn indexing
16. Exercise on preparing foot notes and citations
17. Practice on effective reading skills
18. Comprehension of technical article
19. Comprehension of general article
20. Exercise on precise writing
21. Practice on summarizing articles
22. Practice on preparing abstracts
- 14&15 Developing skill on individual presentation
11. Developing skill on group presentation
12. **Practical Examination**

Referecnce

- G.L. Ray and Sagar mondal. 2010. Journalism –Farm journalism and communication skills. Kalyani publishers.
- Sagar Mondal. 2016. Agricultural extension , Kalyani publishers
- G. L. Ray 2007 Extension Communication and Management , Kalyani publishers
- Communication and Instructional Technology, By: Indu Grover, Shusma Kaushik, Lali Yadav, Deepak Grover & Shashikanta Verma
- Indu Grover, Lali Yadav & Deepak Grover Extension Management, Agrotech
- Everett Rogers, and Floyd Shoemaker, Communication of Innovation – a Cross Cultural Approach, New York Free Press.
- Knapp, Mark L., & Hall, Judith A .(2007) Nonverbal Communication in Human Interaction. (8th ed.) Wadsworth: Thomas Learning.
- Kathleen M. German, Bruce E Gronbeck Principles of Public Speaking

e- Referecnce

1. www.managementstudyguide.com
2. www.ajms.co.in
3. www.mindtools.com

STA 201 Statistical Methods (1+1)

Theory

Unit I: Descriptive Statistics

Basic concepts – statistics – variable – types and sources of data – classification and tabulation of data. Diagrammatic and graphical representation of data – simple, multiple, component and percentage bar diagrams, pie diagram – frequency polygon, frequency curve and histogram. Construction of frequency distribution tables.

Measures of central tendency: arithmetic mean, geometric mean, harmonic mean, median and mode – merits and demerits. Measures of dispersion: range, quartile deviation, mean deviation, standard deviation, and coefficient of variation – skewness and kurtosis – merits and demerits.

Unit II: Probability Distributions and Sampling Theory

Probability – basic concepts – additive and multiplicative laws (without proof). Probability distributions – Discrete distributions: Binomial and Poisson. Continuous distribution: Normal distribution – definitions and properties.

Sampling theory – population – sample – parameter and statistic – sampling distribution – sampling vs complete enumeration – Types of sampling – simple random sampling – selection of simple random sample using random number tables.

Unit III: Testing of hypotheses

Null and alternative hypothesis – types of errors – critical region and level of significance – degrees of freedom. Large sample test – single proportion and difference between two proportions – single mean and difference between two means.

Small sample tests – F-test – t-test for testing the significance of single mean – independent t test and paired t test – chi square test for goodness of fit – chi square test for testing the association of attributes by m x n contingency table – 2 x 2 contingency table – Yates' correction for continuity.

Unit IV: Correlation and Regression

Correlation – Scatter diagram – Karl Pearson's correlation coefficient definition – computation – types of correlation and properties. Regression – simple linear regression – fitting of simple linear regression equation – properties of regression coefficient.

Unit V: Analysis of Variance and Experimental Designs

Analysis of Variance (ANOVA) – assumptions – one way and two way classifications. Basic principles of experimental designs – Completely Randomized Design (CRD) – Randomized Block Design (RBD) – Latin Square Design (LSD) – lay out, analysis, merits and demerits of the above mentioned designs.

Practical

Formation of frequency distribution tables – Diagrammatic and graphical representation. Computation of different measures of central tendency and computation of various measures of dispersion for raw and grouped data – calculation of coefficient of variation (CV) – measures of skewness and kurtosis. Simple problems in Binomial distribution, Poisson and Normal distribution – Selection of simple random sampling. Large sample test for single proportion and difference between two proportions and Large sample test for single mean and difference between two means. t-test for single mean – t-test for testing the significance of two means for independent and paired samples – chi square test for goodness of fit and test for independence of two attributes in a contingency table – Yates correction for continuity

– calculation of the correlation coefficient – fitting of simple linear regression equation – One way and two way ANOVA – completely randomized design (CRD) – randomized block design (RBD) – Latin square design (LSD).

Theory Lecture Schedule

18. Basic concepts – statistics – variable – types and sources of data – classification and tabulation of data. Diagrammatic and graphical representation of data – simple, multiple, component and percentage bar diagrams, pie diagram – frequency polygon, frequency curve and histogram. Construction of frequency distribution tables.
19. Measures of Central Tendency – meaning – limitations – properties – mean, median mode geometric mean and harmonic mean for ungrouped and grouped data.
20. Measures of Dispersion – meaning – limitations – properties – range and mean deviation, Quartile deviation, standard deviation, variance and coefficient of variation for ungrouped and grouped data. Skewness and kurtosis – types – uses.
21. Probability – basic concepts – axioms – mathematical and statistical probabilities – additive and multiplicative laws (without proof). Theoretical discrete distributions – Binomial and Poisson distribution and its applications.
22. Theoretical continuous distribution – Normal distribution and its properties and importance – standard normal distribution.
23. Sampling theory – population – sample – sampling vs complete enumeration – parameter and statistic – need for sampling – sampling distribution – standard error.
24. Sampling methods – probability sampling method – simple random sampling – Selection using random number tables and lottery method.
25. Tests of significance – basic concepts – null and alternative hypotheses – critical region – level of significance – degrees of freedom.
26. **Mid Semester Examination**
5. Large sample test – single proportion and difference between two proportions – single mean and difference between two means
17. Small sample tests – F-test – t-test for testing the significance of single mean independent t test and paired t test
18. Chi square test for goodness of fit – chi square test for testing the association of m x n contingency table – 2 x 2 contingency table – Yates' correction for continuity
28. Correlation – Scatter diagram – Karl Pearson's correlation coefficient definition – computation – types of correlation and properties.
29. Regression – simple linear regression – fitting of simple linear regression equation – properties of regression coefficient.
30. Analysis of Variance (ANOVA) – assumptions – one way and two way classifications. Basic principles of experimental designs – randomization, replication and local control.
31. Completely Randomized Design (CRD) – Randomized Block Design (RBD).
32. Latin Square Design (LSD).

Practical schedule

23. Construction of frequency distribution tables.

24. Diagrammatic representation – simple, multiple, component and percentage bar diagrams, pie diagram. Graphical representation – frequency polygon, frequency curve and histogram.
25. Computation of arithmetic mean, geometric mean, harmonic mean, median and mode for ungrouped and grouped data.
26. Computation of range, standard deviation, variance, coefficient of variation for ungrouped and grouped data. Computation skewness and kurtosis for ungrouped and grouped data.
27. Simple problems in Binomial distribution and Poisson distribution.
28. Simple problems in Normal distribution.
29. Selection of simple random sample using simple random sampling method.
30. Large sample test – test for single proportion and difference between two proportions.
31. Large sample test – test for single mean and difference between two means.
25. Small samples test – t-test for single mean – independent t test for difference between two sample means (equal variances only) – Paired t-test.
26. Chi square test for goodness of fit – Chi square test for testing the association of attributes.
27. Computation of Karl Pearson's correlation coefficient.
28. Fitting of simple linear regression equation y on x .
29. One way ANOVA – analysis of experimental data using Completely Randomised Design (CRD) (for equal replications only).
30. Two way ANOVA – analysis of experimental data using Randomised Block Design (RBD).
31. Analysis of experimental data using Latin Square Design (LSD).

32. Final Practical Examination

References

26. Rangaswamy, R. 2000, A Text book of Agricultural Statistics, Wiley Eastern Limited, New Delhi.
27. K.P. Dhamu and K. Ramamoorthy, 2007, Statistical Methods, Agrobios (India), Jodhpur.
28. R. Gangai Selvi and C. Kailasam, 2017, Applied Statistics, Kalyani Publishers, New Delhi.
29. K. M. Palaniswamy and Usha Palaniswamy, 2006, Handbook of Statistics for Teaching and Research in Plant and Crop Science, , IBDC Publishers, , Lucknow.

References

16. G. Nageshwara Rao , 2007, Statistics for Agricultural Sciences, BS Publications, Andhra Pradesh.
17. Gupta. S.P., 2007, Statistical Methods, Sultan Chand and Sons, New Delhi.
18. S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 2003, Sultan Chand and Sons, New Delhi.
19. S.C. Gupta and V.K. Kapoor, Fundamentals of Applied Statistics, 2003, Sultan Chand and Sons, New Delhi.
20. Gomez, K.A. and Gomez, A.A., 1984, Statistical Procedures for Agricultural Research, John Wiley and Sons, New York.
21. Panse, V.G. and Sukhatme, P.V. 1961, Statistical methods for Agricultural Workers, ICAR, New Delhi.
22. Zar. J.H., 1974, Bio-statistical analysis, Prentice - Hall, Inc., New Jersey.
23. Cochran, W.G. and Cox, G.M. 1962. Experimental Design, Asia Publishing House, New Delhi.
24. M.N. Das and Narayan C. Giri, Design and Analysis of Experiments, New Age Publishers, New
18. D.C. Sanchetti and V.K. Kapoor, 2007, Statistics (Theory, Methods and Application), Sultan Chand and Sons, New Delhi.

E- References

2. <http://www.statistics.com/resources/glossary/>
3. www.statsoft.com
4. http://www.iasri.res.in/ebook/EB_SMAR/index.htm
5. [http://www.iasri.res.in/design/analysis of data/analysis of data.html](http://www.iasri.res.in/design/analysis%20of%20data/analysis%20of%20data.html)
6. www.stats.gla.ac.uk/steps/glossary/index.html
7. <http://davidmlane.com/hyperstat/>
8. <http://www.stattrek.com/>
9. [http://www.businessbookmall.com/Statistics Internet Library.htm](http://www.businessbookmall.com/Statistics%20Internet%20Library.htm)
10. <http://www.stat-help.com/>
18. www.statsci.org/jourlist.html

PAT 202 Principles of Integrated Plant Disease Management - 1+1

Theory:

UNIT I: Epidemiology and Diagnosis of Plant Diseases

Classification of plant diseases - Disease triangle/ Disease Pyramid - Epidemiology of plant diseases- role of weather factors in disease development - Disease surveillance, assessment and forecasting- Diagnosis of plant diseases- Seed health tests- chemodiagnosis, serodiagnosis and Molecular diagnosis

UNIT II: Principles - Avoidance & Exclusion

Avoidance- Role of cultural practices in plant disease management. Exclusion- Plant quarantine – domestic, International and Embargo - Phytosanitary certificate- Quarantine in India- Post Entry Quarantine- Exotic diseases introduced into India- Pest risk analysis.

UNIT III: Eradication

Eradication of pathogens from seed and Planting materials – Eradication of diseased plants- Surgery and Rouging – Eradication of Alternate and Collateral host- different methods of eradication- Mechanical, physical , chemical and Biological methods.

UNIT IV: Protection

Protection of crops from air borne, seed borne, soil borne and vector borne plant diseases-Physical methods- soil solarization, Hot water treatment, Incineration. Chemical control of plant diseases- fungicides- Different group of fungicides and antibiotics in plant disease management- Biological control of plant diseases - Plant products, Plant activators and Antiviral principles- method of application- plant protection appliances.

Unit V: Immunization, Biotechnological approaches and IPM module

Immunization - cross protection and host plant resistance – Types of resistance - vertical and horizontal resistance – resistance breeding and Resistant varieties. Mechanism of resistance- structural and bio chemical resistance in plants -Biotechnological approaches for crop disease management. Implementation and impact of IPM. IPM module for Soil borne, air borne, Seed borne and vector borne plant disease management

Practical

Survey and Assessment of important plant diseases- Diagnosis of Plant diseases- Classification and grouping of fungicides- Preparation of Bordeaux mixture (1%) and Bordeaux paste (10%), Calculation of fungicides quantity and methods of application of fungicides – Special methods of application. Mass multiplication of *Trichoderma asperellum*, *Pseudomons fluorescens* and *Bacillus subtilis* and method of application-Preparation of leaf extracts, oil emulsion of neem and antiviral principles. Cross protection-Tissue culture –meristem tip culture technique. Visit to commercial biocontrol production unit/seed Testing Laboratory and pesticide testing laboratory.

Theory

1. Plant diseases – abiotic , biotic diseases, classification based on mode of infection, multiplication of inoculum , spread, symptoms, occurrence & distribution
2. Epidemiology – disease triangle/ disease Pyramid - role of weather factors in plant disease development. Boom and bust cycle in disease outbreak
3. Disease surveillance –different methods- surveillance report-disease surveillance programme in Tamil Nadu.
4. Assessment of plant diseases- different methods- measurement of disease growth rate by Area Under Disease Progressive Curve (AUDPC)
5. Forecasting of plant diseases- forecasting models in plant disease management
6. Diagnosis of plant diseases-seed health tests, chemodiagnosis, serodiagnosis and molecular diagnosis

7. Avoidance and Exclusion- plant quarantine – domestic, international and embargo - phytosanitary certificate- Quarantine in India. Post entry quarantine in India. Exotic diseases introduced into India

8. Eradication: Role of cultural practices in plant disease management- different methods of eradication of plant diseases

9. Mid semester examination

10. Protection –physical methods of protection- chemical fungicides – ideal characters- formulations and adjuvants- safety measures to be followed while handling fungicides

11. Sulphur and copper fungicides,- classification -phytotoxicity, mode of action and uses

12. Mercury fungicides, Heterocyclic nitrogen compounds , Organo tin, Quinone, Benzene and Miscellaneous compounds , mode of action and uses

13. Systemic fungicides including antibiotics – classification – mode of action - uses.

New generation fungicides, plant activators/ SAR inducing chemicals in plant disease management. Methods of application of fungicides: seed treatment, foliar spray, soil drenching and special methods of application.

14. Biological control – definition - mechanism of action – mass production of *Trichoderma asperellum* , *Pseudomonas fluorescens* & *Bacillus subtilis* - methods of application - plant products – antiviral principles – preparation – methods of application

15. Disease resistance- types- resistant varieties. methods of developing resistant varieties- Mechanisms of resistance- structural and bio chemical resistance in plants- cross protection

16. Biotechnological approaches in plant diseases management: Tissue culture techniques- meristem tip culture, somoclonal variation and transgenic plant production by genetic engineering.

17. IPM module for soil borne , Airborne , Seed borne and vector borne plant diseases

Practical Schedule

1. Survey and assessment of important plant diseases

2. Diagnosis of plant diseases: Tetrazolium test, Iodine test , ELISA test and ooze test, paraquat test

3. Seed health tests for diagnosis of seed borne pathogens - dry seed examination, seed washing, blotter tests

4. Classification and grouping of fungicides.

5. Preparation of Bordeaux mixture (1%) and Bordeaux paste (10%)

6. Calculation of spray fluid and methods of application of fungicides – seed (wet and dry) soil, foliar and post harvest dipping

7. Special methods of application: swabbing, acid delinting, pseudostem injection, capsule application

8. Special methods of application: corm injection, paring and prolinage, root feeding and trunk injection.

9. *In vitro* assay of fungicides against fungal pathogens

10. *In vitro* assay of biocontrol agents and their compatibility with agrochemicals

11. *Trichoderma viride* -mass production and methods of application

12. *Pseudomonas fluorescens* and *Bacillus subtilis* -mass production & methods of application

13. Visit to commercial biocontrol production unit /seed and pesticide testing laboratories

14. Preparation of leaf extracts, oil emulsion of neem and antiviral principles.
15. Cross protection: production of pre immunized citrus seedlings against tristeza virus.
16. Tissue culture – Production of virus free plants through meristem tip culture technique.
17. Practical Examination

Reference Books

1. Arjunan.G. Karthikeyan, G, Dinakaran ,D. Raguchander,T. 1999 Diseases of Horticultural Crops, AE Publications, Coimbatore.
2. Rangasawmi ,G and Mahadevan, A. 1998. Diseases of crop Plants in India, Prentice Hall of India Pvt. Ltd., New Delhi
3. Prakasam, V., Valluvaparidasan, V., Raguchander, T. and K.Prabakar. 1997. Field crop diseases, AE Publication, Coimbatore.

E- Books

1. Agrios, G.N. 2008. Plant Pathology, Academic Press, New York
2. Rangaswami, G. 2005. Diseases of Crop plants in India. Prentice Hall of India Pvt. Ltd., New Delhi
3. Thakur, B.R. 2006. Diseases of field crops and their management

AEN 202 MANAGEMENT OF BENEFICIAL AND HARMFUL INSECTS (2+1)

Theory

Unit I: Classification of insects based on economic importance - [Apiculture - Bee species – comparison- castes of bees, bee behaviour and bee dance; Apiary management practices – bee pasturage, foraging, seasonal variations; Bee products – properties and uses; Effect of agricultural inputs on bee activity – pesticide poisoning.](#)

Unit II: [Moriculture; Silkworm rearing;](#) Lac insect- biology-strains-natural enemies of lac insect and lac products; Weed killers, pollinators, scavengers and soil builders; Balance of life in nature – population dynamics – role of abiotic and biotic factors. Life table – interspecific and intraspecific relationships

Unit III: Pests – definition and categories – pest outbreak – factors governing pest outbreak– pest monitoring, surveillance and forecasting. Economic Threshold Level – Economic Injury Level- [Integrated Pest Management – history, principles and strategies – requirements for successful pest management programme;](#) Cultural, physical, mechanical, ecological engineering methods and host plant resistance in pest management

Unit IV: Parasitoids, predators and microbial agents in pest management. [Legal methods – definition – pest introductions – quarantine – phytosanitary certificate – pest legislation.](#) Pesticides – history,

classification – mode of action of insecticides. Pesticides compatibility, safety and hazards in the use of pesticides – pesticide poisoning - impact of pesticides in agro-ecosystem.

Unit V: Insecticide act. Insecticides residues and resistance. Semiochemicals – allomones – kairomones – pheromones- semiochemicals in pest management. Sterile male technique – chemosterilants, insect growth regulators – moult inhibitors – Juvenile Hormone mimics – antifeedants and repellents. Natural pesticides. Biotechnology in pest management. Bio safety of transgenic plants. Impact of global warming on pests. [Bio-intensive/Bio-rational/ Eco-friendly Integrated Pest Management – Indigenous/traditional technologies in Integrated Pest Management](#)

Practical

Identification, morphology and structural adaptations in honey bees. Bee keeping appliances, bee enemies and diseases. Sericulture. Lac insect-life history, hosts and culturing of lac, natural enemies and lac products. Study of useful insects- Pollinators, weed killers, scavengers and soil builders. Symptoms and types of damage caused by insect pests. Assessment of insect population and their damage in field crops. Cultural, mechanical and physical control of insects. Identification and mass culturing of different types of parasitoids, predators and entomopathogens. Behavioral approaches in pest management – Pheromone traps, light traps, sticky traps and others. Pesticide formulations and toxicity parameters. Pesticide application techniques. Preparation of spray fluids and botanicals for field application. Plant protection appliances.

Theory lecture schedule:

1. Economic classification of insects
2. [Bee species – comparison – castes of bees – bee behaviour and bee dance](#)
3. [Apiary management practices – bee pasturage – foraging – seasonal variations.](#)
4. [Bee products – their properties and uses](#)
5. [Effect of agricultural inputs on bee activity – pesticide poisoning](#)
6. Ecological requirements for mulberry cultivation – soil type – mulberry varieties – Methods of propagation – merits and demerits – selection of semi hard wood cuttings

7. Pests and diseases of mulberry
8. Types of silkworm - Mulberry silkworm – origin – classification based on voltinism, multivoltinism, geographical distribution and genetic nature – Characters of multivoltine races, bivoltine races, cross breeds and bivoltine hybrids – double hybrids– suitability for rearing in different seasons.
9. Morphology and biology of silkworm – sexual dimorphism in immature and adult stages – silkworm genetics – chromosome number – sex limited characters in egg, larva and cocoon for grainage use.
10. Lac insect- biology-strains-Natural enemies of lac insect and lac products
11. Weed killers, pollinators, scavengers and soil builders
12. [Insect ecology – definition – balance of life in nature – reproductive potential and environmental resistance](#)
13. [Population dynamics – role of biotic factors – competition – parasitoids and predators. Life table – Interspecific and intraspecific relationship](#)
14. [Abiotic factors – physical, nutritional and host plant associated factors on insect population.](#)
15. [Pests – definition, categories and causes for outbreak of pests. Losses caused by pests](#)
16. [Pest monitoring – pest surveillance and forecasting – objectives, survey, sampling techniques and decision making. Economic Threshold Level and Economic Injury Level. Factors influencing Economic Injury Level and Economic Threshold Level](#)
17. [Midsemester examination](#)
18. [Integrated Pest Management – history, principles and strategies – requirements for successful pest management programme. Components of pest management](#)
19. [Cultural methods – definition – characteristics, requisites – farm level practices and community level practices, advantages and disadvantages- Ecological Engineering in pest management](#)
20. [Physical methods – definition – use of heat, moisture, light, electromagnetic energy and sound energy – Mechanical methods – definition – mechanical destruction and exclusion – merits and demerits](#)
21. Host plant resistance – types and mechanisms of resistance and role of host plant resistance in pest management
22. Biological control – definition, parasitoids and predators and their role in pest management
23. Microbial control – viruses, bacteria, fungi, protozoa and nematodes and their role in pest management
24. [Legal methods – definition – pest introductions – quarantine – phytosanitary certificate – pest legislation](#)
25. [Chemical control – definition – history of insecticide development – toxicity parameters – ideal qualities of an insecticide](#)
26. [Classification of insecticides based on mode of entry, mode of action and chemical nature](#)
27. [Mode of action of organophosphates, carbamates, synthetic pyrethroids, neonicotinoids, diamides and avermectins](#)
28. [Pesticide compatibility, safety and hazards – pesticide poisoning - antidotes – safe handling – impact of pesticides on agroecosystems](#)
29. [Insecticides Act 1968 – insecticide residues and waiting periods, role of pesticides in pest management, insecticide resistance management](#)
30. [Semiochemicals – definition – intraspecific semiochemicals – allomone, kairomone, synomone and apneumone - Interspecific semiochemicals – pheromone, sex pheromone, alarm and trail marking pheromone. Pheromones in Integrated Pest Management](#)
31. [Sterility methods – definition – principles – methods – requirements and limitations](#)
32. [Insect growth regulators – moult inhibitors – Juvenile Hormone mimics – mode of action and uses. Insect antifeedants and repellents – mode of action, groups and uses](#)
33. Botanicals and Biotechnological approaches in pest management – bio safety of transgenic plants
34. [Impact of global warming on pests. Integrated Pest Management : Issues and options. Bio-intensive/Bio-rational/ Eco-friendly Integrated Pest Management – Indigenous/traditional technologies in Integrated Pest Management](#)

Practical schedule:

1. Identification, morphology and structural adaptations in honey bees
2. Bee keeping appliances, bee enemies and diseases
3. Mulberry nursery bed preparation – methods of planting - Pruning methods – leaf / shoot harvest– preservation of leaves.
4. Identification of damage symptoms of insects, diseases and nematodes of mulberry
5. Chawki rearing and shoot rearing
6. Lac insect-life history, hosts and culturing of lac, natural enemies and lac products
7. Study of useful insects-Pollinators, weed killers, scavengers and soil builders
8. Symptoms and types of damage caused by insect pests , Assessment of insect population and their damage in rice, cotton and brinjal
9. Cultural, mechanical and physical control of insects
10. Identification and mass culturing of different types of parasitoids
11. Identification and mass culturing of different types of predators
12. Identification and mass production of entomopathogens
13. Behavioral approaches in pest management – Pheromone traps, light traps, sticky traps and others
14. Pesticide formulations and toxicity parameters
15. Pesticide application techniques, Preparation of spray fluids and botanicals for field application
16. Plant protection appliances
17. Final Practical examination

References:

1. David, B.V. and V.V. Ramamurthy. 2011. *Elements of Economic Entomology*, Namrutha Publications, Chennai, 386 p. {ISBN: 978-81-921477-0-3}
1. Pedigo, L.P. and M.E.Rice.1996. *Entomology and Pest Management*. Prentice-Hall of India Pvt Ltd, New Delhi. 812p. {ISBN-978-8120338869}
2. Dhaliwal, G.S. and R.Arora. 2001. *Integrated Pest Management – Concepts and approaches*. Kalyani publishers, New Delhi. 427p. {ISBN: 81-7663-904-4}
3. Dandin, S.B., J.Jayaswal and K. Giridhar.2003. *Hand book of Sericulture Technologies*. Central Silk Board, Bangalore, 287 p.

e resources

1. <http://www.sristi.org/hbnew>
2. <http://www.ncipm.org.in/recent-publications.htm>
3. <http://www.ipmnet.org>
4. www.silkbase.org
5. www.papilo.ab.a.u.tokyo.ac.jp

AGR 203 Crop Production Technology- II (Rabi crops) (1+1)

Theory

Unit I : Cereals

Wheat, barley, Oats - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Unit II : Pulses

Chickpea, lentil, peas - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Unit – III Oilseeds

Rapeseed, mustard and sunflower- Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Unit -IV: Sugar Crops

Sugarcane - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.

Unit V: Forage crops

Berseem, Lucerne , Fodder maize : Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices.

Practical:

Identification of rabi cereals, pulses, oilseeds, sugarcane, and forage crops - nursery preparation and management for sugarcane - main field preparation; Seed treatment techniques - Sowing and manuring
35. Seeding equipment's - Estimation of population - After cultivation practices - Study of growth and yield parameters and yield estimation, harvesting of above crops; Fodder preservation techniques - Silage and hay making, Cost and returns - Visit to institutes and industries - Farmers' fields

Lecture Schedule:

8. Wheat- Origin, geographic distribution, economic importance, soil and climatic requirement,
9. Wheat - varieties, cultural practices and yield.
10. Barley and oats - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
11. Chickpea- Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
12. Lentil and peas - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
13. Peas - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
14. Rapeseed and Mustard - Origin, geographic distribution, economic importance, Classification , soil and climatic requirement, varieties
15. Rapeseed and mustard - cultural practices, yield.

16. Mid semester examination

17. Sunflower- Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
19. Sugarcane - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties,
20. Sugarcane - cultural practices and yield.
21. Sugarcane- package of practices for SSI
22. Sugarcane - Crop logging, maturity and ripening
23. Sugarcane - Gur manufacture , Value addition and byproduct utilization.

24. Berseem and Lucerne - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
25. Fodder maize - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices, yield.
2. Identification of rabi crops and recording their importance in the crop cafeteria.
3. Acquiring skill in field preparation, sowing and manuring of rabi crops under pure and intercropping situations.
4. Acquiring skill in different seed treatment techniques and foliar nutrition of rabi crops.
5. Estimation of plant population per unit area for rabi crops.
6. Nursery preparation for Sugarcane.
7. Acquiring skill in after - cultivation practices in sugarcane - detrashing, and Propping.
8. Study on growth parameters of sugarcane.
9. Study on yield parameters and estimation of yield in sugarcane.
10. Study on yield parameters and estimation of yield in rabi crops.
11. Estimating Cost and returns of important rabi crops.
12. Visit to Sugarcane Breeding Institute/ Research Station to study cultivation of sugarcane and its by products.
13. Visit to - nearby sugar mill, for observing juice extraction, quality assessment, sugar manufacture and by products.
14. Silage making.
15. Practicing field preparation and sowing Lucerne.
16. Practicing field preparation and sowing for fodder maize.
17. Visit to Wheat research station, Wellington to study rabi crops – wheat, barley, rye, oats.
18. **Practical Examination.**

References:

- Rajendra Prasad. 2012. Text Book on Field Crop Production, Indian Council of Agrl. Research, New Delhi.
- Ahlawat,I.P.S., Om Prakash and G.S. Saini. 2010. Scientific Crop Production in India. Rama publishing House, Meerut
- Chidda Singh, Prem Singh and Rajbir Singh. 2011. Modern Techniques of Raising Field Crops. Oxford and IBH Pub. Co. Pvt. Ltd., New Delhi.
- Reddy,S.R. 2012. Agronomy of field crops. Kalyani publishers, New Delhi.
- Crop production guide.2012. Directorate of Agriculture, Chennai
- ICAR 2015. Hand book of Agriculture. Indian Council of Agricuture, New Delhi

E.References:

1. <http://sugarcane.tn.nic.in>
2. <http://fibre.tn.nic.in> ,
3. www.tnau.ac.in/agriportal

AGR 204 Farming System and Sustainable Agriculture (1+1)

Theory :

Unit - I: Cropping System

Cropping systems - Definition - Principles - Concepts - Classification - mono cropping - intensive cropping - multiple cropping systems of India and Tamil Nadu - Interaction between different cropping systems - Cropping system management - Resource management - land, nutrient, water and weed.

Unit - II: Evaluation of Cropping System

Index for evaluation of cropping systems - Land use - yield advantages - Economic evaluation - sustainability.

Unit - III: Farming System

Farming systems - Definition - Principles - Concepts - Enterprises selection and management - interaction between different enterprises with cropping - scope and advantages of Integrated Farming system - Integrated farming system models for different agro eco-systems - interaction between enterprises.

Unit - IV: Evaluation of Farming System

Resource recycling in IFS - Evaluation indicators of integrated farming system - LEISA & HEIA - concepts and principles - Conservation agriculture - principles, concept and scope.

Unit - V: Resource and labour management in farming system

Resource management under constraint situation - Cost reduction strategies in crop production - Non-monetary inputs and low cost technologies - Labour management - farming system and environment.

Practical:

Preparation of cropping scheme - working out input requirements for crops, cropping systems - preparation of calendar of operations for wetland, irrigated upland and dry land cropping system - visit to cropping system experiments - working out indices for evaluation of cropping systems - visit to different units: dairy, goat, poultry, fishery. Mushroom, sericulture and biogas - study on evaluation indicators on farming system - preparation of integrated farming system models for different eco-systems - on farm field visit - analysis of farming system models.

Lecture Schedule

15. Cropping system: Definition, Principles and basic concepts.
16. Classification of cropping system - Mono cropping, intensive cropping, multiple cropping, mixed cropping.
- 3 Major cropping systems prevailing in India and Tamil Nadu for different agro eco systems.
25. Complementary and competitive interaction in different cropping system - light, nutrient, water and weed.
26. Cropping system management: agronomic requirement for crops and cropping system selection of crops and varieties, tillage and land shaping, plant population and crop geometry.
27. Cropping system management: agronomic requirement for crops and cropping system - water management, soil fertility management and plant protection.
28. Indices for evaluation of cropping system - land use, yield advantage and economics.
29. Farming system: definition, principles and concepts and factors influencing choice and size of enterprises
- 35. Mid Semester Examination.**
36. Scope and advantages of integrated farming system.
37. Allied enterprises for wetland, irrigated upland and dryland - selection and management and their interaction.
38. Resource recycling in integrated farming system.

39. Integrated Farming System evaluation indicators.
40. Integrated farming system - models for wetland, irrigated upland and dryland eco system.
41. LEISA and HEIA - principles and concepts and Labour management in integrated farming system.
42. Conservation agriculture and environmental impact of integrated farming system.
43. Cost reduction technologies and non monetary inputs in integrated farming system.

Practical Schedule:

52. Visit to cropping system experiments in wetland.
53. Visit to cropping system experiments in irrigated upland and dryland.
54. Preparation of cropping scheme for wetland and working out input requirement.
55. Preparation of cropping scheme for irrigated upland and working out input requirement.
56. Calendar of operations for wet land and irrigated upland cropping system.
57. Working out indices for evaluating the cropping system - land use, yield advantage.
58. Working out indices for evaluating the cropping system - Economics, sustainability.
59. Visit to dairy, goat and poultry units.
60. Visit to mushroom unit.
61. Visit to sericulture and biogas unit.
62. Preparation of integrated farming system models : wetland eco-system.
63. Preparation of integrated farming system models : irrigated upland and dryland eco systems.
64. Resource recycling in integrated farming system models of different eco systems.
65. Evaluation of integrated farming system models : wetland eco-system.
66. Evaluation of integrated farming system models : irrigated upland and dryland eco systems.
67. On-farm visit to cropping fields and integrated farming system units.
68. **Practical examination.**

References:

1. Palaniappan, SP and K. Sivaraman.1996. Cropping systems in the tropics Principles and management.
2. New Age International (P) Ltd., New Delhi.
3. Jayanthi, C. Devasenapathy, P and C. Vennila. 2007. Farming Systems. Principles and practices. Satish Serial Publishing House.Delhi.
4. S.C. Panda. 2003. Cropping and Farming Systems. Agrobios Publishers. Jodhpur. Jana, B.L. 2014. Farming Systems. Agrotech Publishing Academy, Udaipur Shagufta. 2015. Cropping and Farming Systems. APH Publishing Corporation

E references:

1. www.agriinfo.in
2. www.fao.org
3. www.agritech.tnau.ac.in

SAC 202 Problematic Soils and their Management (2+0)

Theory

Unit-I

Soil quality and health, Distribution of Waste land and problem soils in India and Tamil Nadu. Categorization of waste lands based on properties.

Unit-II

Characteristics, reclamation and management of soil physical and chemical constraints - Eroded and Compacted soils, Flooded soils, Saline and sodic soils, Acid soils, Acid Sulphate soils, degraded alkali soils and Polluted soils. Effect of salts on soil and plants.

Unit-III

Remote sensing and GIS in assessment and management of problem soils. Irrigation water – quality and standards. Utilization of saline water in agriculture.

Unit-IV

Multipurpose tree species, bio remediation of soils through MPTs, land capability classification, land suitability classification. Problematic soils under different Agro ecosystems - Soil fertility improvement through carbon build up.

Lecture Schedule:

10. Soil health - Definition - Soil Quality Indices – Physical indicators
11. Soil Quality Indices - Chemical and biological indicators
12. Distribution of waste lands and problem soils in India and Tamil Nadu
13. Categorization of waste lands based on properties
14. Soil physical constraints – slow permeable, excessively permeable soils and fluffy paddy soils - Characteristics and management
15. Soil crusting, soil compaction, sub soil hard pan, sand dunes and shallow soils – characteristics and management
16. Eroded soil – Genesis, types and characteristics: water- sheet, rill, gully, ravines, wind – Aeolian, loess, saltation, suspension , soil creep
18. Universal soil loss equation and erosion control measures
19. Flooded soils – Formation, characteristics and management
20. Acid soil and acid sulphate soil – Genesis and characteristics.
18. Lime requirement of acid soil, liming materials, reclamation and management of acid soil
19. Formation and classification of Saline, Sodic and saline sodic soils
20. Effects of Salts on soils- Physical: Clay swelling and Dispersion, permeability, Infiltration, Crust, Water transmission. Chemical: pH and EC. Biological : Microbial activity.
21. Effects of Salts on plants – Plants response to saline and sodic conditions, Factors affecting salt tolerance, crop response to salinity, ratings of crop salt tolerance.
22. Salts and plant mineral nutrition- Salinity and nutritional effects: Salinity and N, P, K, Ca, Mg, S, and Micronutrients. Alkalinity and nutritional effects.
23. Saline, Sodic, saline sodic, and degraded alkali soils- characteristics and their management
8. Saline soil-reclamation – Leaching requirement. Sodic soil – reclamation -gypsum requirement – calculations.
9. **Mid semester examination**
10. Polluted soils- industrial effluent s- Characteristics, reclamation and management
11. Polluted soils- mine spoils- Characteristics, reclamation and management
12. Irrigation water – quality and standards - EC, SAR, RSC, RSBC SSP, PSI and PS
13. Irrigation water – quality and standards -USDA system and specific ion toxicity-USSL system
14. Factors affecting suitability of irrigation water and Management of poor quality water in agriculture

15. Remote sensing and GIS in assessment of wastelands and problem soils
16. Remote sensing and GIS in monitoring and management of wastelands and problem soils
17. Multipurpose tree species for waste lands and problem soils
18. Bio remediation through MPTs of soils
19. MPTs - Nutrient cycling under waste lands and problem soils
20. Land capability and classification
21. Land suitability classification
22. Problematic soils under different Agro ecosystems- coastal salinity, inland salinity
23. Problematic soils under different Agro ecosystems- marshy, swampy soils, red sand dunes (Theri soils) Tsunami affected soils.
24. Agricultural Ecosystem services- Soil fertility improvement in problem soils
25. Potential of agro forestry systems in management of problem soils
26. Carbon sequestration and its role in problem soil management

REFERENCES

28. Sehgal, J. 2005. Pedology concepts and applications, Kalyani Publishers, New Delhi.
29. Gupta, S.K. and I.C. Gupta 2014. Salt affected soils: Reclamation and Management. Scientific Publishers.
30. Richards, L.A. 2012. Diagnosis and improvement of saline and alkali soils. Scientific Publishers.
31. Soil Survey Staff. 2006. Keys to Soil Taxonomy. United States Department of Agriculture, Natural Resources Conservation Service.
32. Maliwal, G.L. and L.L. Somani. 2010. Nature, Properties and Management of saline and alkali soils. Agrotech publishing academy, Udaipur.

E-REFERENCES

18. <ftp://ftp-fc.sc.egov.usda.gov/NSSC/NCSS/Conferences/scanned/>
19. <ftp://ftp-fc.sc.egov.usda.gov/NSSC/Lab References/SSIR 51.pdf>
20. <ftp://ftp-fc.sc.egov.usda.gov/NSSC/Lab References/SSIR 51 .pdf>
21. www.iuss.org/Bulletins/00000096.pdf
22. www.oosa.unvienna.org/pdf/sap/centres/rscurrE.pdf
23. www.csre.iitb.ac.in/~dd/detail.html
24. www.dvsinstitute.org/forms/pg/M.Sc.%20-%20RS%20&%20GIS-350.pdf
25. inkinghub.elsevier.com/retrieve/pii/S0166248197800335
26. www.scribd.com/doc/40246764/Description-Pedon-Copy-
9. [www.angrau.net/BSc\(Aq\)CourseCurriculum.htm](http://www.angrau.net/BSc(Aq)CourseCurriculum.htm)
10. www.eurojournals.com/ejsr 42 2 10.pdf
11. www.springerlink.com/index/R177R744722222UN.pdf-Similar
12. [content.alterra.wur.nl/Internet/webdoc\\$/ilri-publicaties/.../Bib10.pdf](http://content.alterra.wur.nl/Internet/webdoc$/ilri-publicaties/.../Bib10.pdf)
13. www-wds.worldbank.org/external/.../INDEX/multi_..page.txt-Cached
14. openaccess.icrisat.org/.../Proceedings-integrated-watershed-management-for-land-Asia.pdf

HOR 212 Production Technology for Ornamental Crops, MAPs and Landscaping (1+1)

Theory

Unit I: Landscaping

Importance and scope of ornamental crops landscaping. Principles of landscaping. Landscape uses of trees, shrubs and climbers.

Unit II: Production technology of cut flower crops under protected conditions

Production technology of important cut flowers like rose, gerbera, carnation, lily and orchids under protected conditions

Unit III: Production technology of flowers under open conditions and value addition in ornamental crops

Production technology of important cut flowers like gladiolus, tuberose, chrysanthemum under open conditions. Package of practices for loose flowers like marigold and jasmine under open conditions. Processing and value addition in ornamental crops.

Unit IV: Production technology of medicinal crops

Medicinal crops- importance and scope – current status - soil and climate – varieties – propagation– planting methods – nutrient, irrigation and organic practices – harvest – post-harvest handling – storage, packaging of Periwinkle, Asparagus, Aloe, Costus, Isabgol, Glory lily, extraction and value addition of medicinal crops.

Unit V: Production technology of aromatic crops

Aromatic crops - importance and scope – current status -- soil and climate – varieties – propagation– planting methods – nutrient, irrigation and organic practices – harvest – post-harvest handling – storage, packaging of Ocimum, Mint, Geranium, Citronella, Lemon grass, Palmarosa and Vetiver – Distillation of oil and value addition.

Practical

Identification of Ornamental plants. Nursery bed preparation and seed sowing. Training and pruning of Ornamental plants. Planning and layout of garden. Protected structures – care and maintenance. Intercultural operations in flowers. Harvesting and post harvest handling of cut and loose flowers. Visit to commercial flower unit.

Medicinal and Aromatic Plants

Identification of Medicinal and Aromatic Plants- varieties-propagation-special practices - nutrient management, extraction and distillation of essential oil - Periwinkle, Asparagus, Aloe, Costus, Isabgol, Glory lily, Ocimum, Mint, Geranium, Citronella, Lemon grass, Palmarosa and Vetiver – visit to commercial medicinal and aromatic plants fields and processing units

Theory lecture schedule

31. Importance and scope of ornamental crops and landscaping.
32. Principles of landscaping
33. Landscape uses of trees, shrubs and climbers.
34. Production technology of cut rose under protected conditions
- 2 Production technology of gerbera and carnation under protected conditions
- 3 Production technology of lily and orchids under protected conditions
- 4 Production technology of gladiolus and tuberose under open conditions
- 5 Production technology of chrysanthemum and marigold under open conditions
- 6 **Mid Semester Examination.**

- 7 Production technology of jasmine under open conditions.
- 8 Processing and value addition in ornamental crops.
- 9 Scope and Importance of medicinal & aromatic crops– current status - conservation methods
- 10 Periwinkle, Asparagus and Aloe - varieties -- soil and climate – propagation- sowing and planting, nutrient, water management – harvest and processing
- 11 Costus, Isabgol and Glory lily - Propagation- soil and climate – propagation and planting- standards - pollination-nutrient, irrigation management – harvest, yield and processing
- 12 Ocimum, Mint, Geranium - varieties – soil and climate- propagation - planting - nutrient, water management – harvest - distillation of essential oil
- 13 Citronella, Lemon grass, Palmarosa and Vetiver - varieties – soil and climate- propagation - planting – nutrient- water and weed management – harvest- distillation of essential oil.
- 14 Processing and value addition in medicinal and aromatic plants.

Practical schedule

18. Identification, planting, care and maintenance of trees, shrubs and climbers used in garden
19. Identification of varieties in cut flowers under protected conditions.
20. Identification of varieties in flowers under open conditions.
21. Practices of nursery bed preparation, seed sowing in ornamental plants.
22. Training and pruning and intercultural operations in Ornamental plants
23. Planning and layout of garden.
24. Protected structures – care and maintenance.
25. Harvesting and post harvest handling of cut and loose flowers.
26. Identification of medicinal and aromatic plants –economic parts
27. Propagation techniques, planting, cultural operations in Periwinkle, Asparagus and Aloe.
28. Propagation techniques, planting, cultural operations in Costus, Isabgol and poppy.
29. Propagation techniques, planting, cultural operations in Ocimum, Mint, Geranium
30. Propagation techniques, planting, cultural operations in lemon grass, palmarosa, vetiver and citronella
31. Extraction and distillation of medicinal & Aromatic crops.
32. Visit to commercial floriculture and floral oil extraction units
33. Visit to commercial medicinal and aromatic crops field and extraction unit.

References

- 15 Bhattacharjee, S.K and De L.C (2003) Advanced Commercial Floriculture Vol. (1) Aavishkar publishers, Distributors, Jaipur.
- 16 Bhattacharjee, S.K and De L.C (2005) Medicinal Herbs & Flowers, Aarishkar, Jaipur.
- 17 Bhattacharjee, S.K., 2004. Hand book of medicinal plants, Pointer publications, Jaipur.
- 18 Bose, T.K., Yadav, L.P., Pal. P., Parthasarathy, V.A., Das. P., 2003. Commercial flowers. Vol. I and II. Naya udyog, Kolkata-6.
- 19 Ravindrasharma (2004) Agro techniques of Medicinal plants. Daya publishing, New Delhi.
3. Trivedi, P.C. (2004) Medicinal Plants: Utilization and Conservation, Aavishkar Publisher, Distributors, Jaipur.
4. Allan M. Armitage and Judy M. Laushman “Speciality Cut Flowers” , Second Edition, Published by Timber press 2003, ISBN - 0881925799
5. Atal. C. K. and B. M. Kapur. 1992. Cultivation and utilisation of medicinal plants RRL. CSIR, Jammu – Tawi.
6. Bose, T.K., Yadav, L.P., Pal. P., Das. P. and Parthasarathy, V.A., (2002) Commercial Flowers. Vol.1, Naya Prakash, Calcutta.
7. Chadha, K.L.1994. Advances in Horticulture, Vol.10. Malhotra Publishing house, New Delhi.
8. Chadha, K.L.1994. Advances in Horticulture, Vol.11. Malhotra Publishing house, New Delhi.

9. Farooqi, M., M. M. Khan and M. Vasundhara. 2004. Production technology of medicinal and aromatic crops. Natural Remedies Pvt. Ltd., Bangalore – 561229.
10. Surendraprasad and Updesh Kumar (1998), Commercial floriculture, Agrobotanica, Bikaner.
11. Kumar, N. Introduction to Horticulture. 2010. Oxford and IBH Publications, New Delhi.
12. Kumar, N. Introduction to Spices, Plantation, Medicinal and Aromatic crops. 1995. Oxford and IBH Publications, New Delhi.

E- References

1. <http://www.theflowerexport.com>
2. <http://www.intuxford.tripod.com>
3. <http://www.webct.uark.edu>
4. <http://www.pubmed.com>
5. <http://www.bestgarden.net/>
6. <http://www.indiaagronet.com/>
7. <http://www.intuxford.tripod.com/>
8. <http://www.lawngrasses.com/>
9. <http://www.frlht.org>
10. www.herbs.org

ANM 201 INTRODUCTORY NEMATOLOGY (0+1)
SYLLABUS

PRACTICAL

Usage and handling of microscopes (binocular, trinocular, zoom and compound microscopes) -Soil and root sampling – Extraction of active nematodes and cysts from soil and roots (Cobb's sieving and decanting technique, Baermann funnel technique, conical flask technique, Sugar floatation technique, Fenwick can method, Incubation and Blender technique) – Nematode processing techniques (preservation, slow and rapid method of processing, making semi permanent and permanent slides) – Morphology of orders *Tylenchida* (*Hoplolaimus*), and *Dorylaimida* (*Xiphinema*) – Identification of important nematodes (*Tylenchorhynchus*, *Helicotylenchus*, *Pratylenchus*, *Hirschmanniella*, *Hemicriconemoides* / *Criconea* *Heterodera* / *Globodera*, *Tylenchulus*, and *Aphelenchoides*) – Life stages of sedentary and migratory endoparasites – symptoms of important nematode diseases – Nematicides and their application – Biocontrol agents-bacteria and fungi.

PRACTICAL

1. Soil and root sampling. Extraction of nematodes by Cobb's sieving method; Baermann funnel Technique and modified Baermann funnel technique.
2. Extraction of nematodes by sugar flotation technique; Extraction of cysts by conical flask technique and fenwick can method.
3. Extraction of nematodes from roots and staining of roots infested with endoparasitic and semi – endoparasitic nematodes.
4. Preservation of nematodes and preparation of temporary and permanent slides.
5. Observing morphology of the order Tylenchida (*Hoplolaimus*) and Dorylaimida (*Xiphinema*, *Longidorus*).
6. Identification of nematodes – *Tylenchorhynchus*, *Helicotylenchus*.
7. Identification of nematodes – *Pratylenchus*, *Hirschmanniella*.
8. Identification of nematodes – *Hemicriconemoides* – *Criconea*, *Heterodera* – *Globodera*.
9. Identification of nematodes – *Tylenchulus*, *Aphelenchoides*.
10. Study of life stages of *Meloidogyne*, *Rotylenchulus*.
11. Study of life stages of *Radopholus*.
12. Study of Entomopathogenic nematodes
13. Study of life stages of Nematodes diseases of rice (White tip and rice root nematode)
14. Damage caused by root – knot and reniform nematodes indifferent crops.
15. Symptoms of damage caused by citrus nematode; the lesion nematode and the burrowing nematode of banana.
16. Study of types of nematicides, application methods and calculation of dosages; study of biocontrol agents.
17. Practical examination.

NST 201 Fundamentals and Applications of Nanotechnology (1+0)

Theory

Unit I - Principles of Nanoscience (4 Lecture) : History, definition, terminologies in nanoscience - Importance of Moore's law- Introduction to nanomaterials – Semiconductor – Diode – Quantum Dots- Buckyball - CNT - Polymers- types – PLGA – coreshell nanoparticles - micelle - Introduction to nanobiosensor- types- properties and applications

Unit II - Synthesis of Nanomaterials (3 Lectures): Top-down and bottom-up approaches - Physical, Mechanical, Chemical and Biological synthesis of nanomaterials

Unit III - Properties and Characterization of Nanomaterials (4 Lectures): Physical, Mechanical, optical, magnetic, thermal and electrical properties – Characterization – SEM, TEM, AFM, FT-IR, XRD

Unit IV - Application of Nanotechnology (2 Lectures) : Agriculture and Food Systems

Unit V - Application of Nanotechnology (3 Lectures): Energy, Environment, Health – Social, Economic and Ethical issues – Nanotoxicology

Lecture schedule

Unit 1 Principles of Nanoscience (4 lectures)

1. History, definition, terminology in nanoscience and importance of Moore's law.
2. Nanomaterials – Semiconductor – Diode – Quantum Dots - Buckyball - CNT – - characteristics – Applications
3. Polymers - Types – PLGA – Coreshell nanoparticles - Micelles - characteristics – Applications
4. Biosensors – Principle, Components, Types, Applications

Unit 2 Synthesis of Nanomaterials (3 lectures)

5. Top down and Bottom up approaches - Physical method, Physical Vapour Deposition (PVD), Etching - Molecular Beam Epitaxy – Sputtering – Lithography - Mechanical synthesis - Ball milling – Types - Mechanical alloying
6. Chemical synthesis – Sol-gel Method – Chemical Vapour Deposition (CVD) – electro-deposition- thin film
7. Biological synthesis using Microorganisms and Plants

Unit 3 Properties and Characterization of Nanomaterials (4 lectures)

8. Mechanical, magnetic and thermal properties of nanomaterials
9. Optical and electrical properties of nanomaterials
10. *Principle, components and application of nanotechnology equipments*: Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM)
11. *Principle, components and application of nanotechnology equipments*: X-ray Diffraction (XRD) – Fourier Transform Infra Red Spectroscopy (FT-IR) – Atomic Force Microscope (AFM)

Unit 4 Applications of Nanotechnology in Agriculture & Food Systems (2 Lectures)

12. Agriculture – Nano fertilizers – Nano-herbicides – Nano-pesticides – Seed technology
13. Nanotechnology in Food Systems – Nano foods, Nano-encapsulation of functional foods, Nano-packaging, Quality assessment.

Unit 5 Applications of Nanotechnology in Energy, Environment, Health (3 Lectures)

14. Nanotechnology applications in Energy and Environment

15. Applications in Health Sciences and Nanotoxicology

16. Social, Economic and Ethical Issues in Nanotechnology

References:

3. Nano: The essentials understanding nanoscience and Nano- T.Pradeep - 2009 - Mc Graw Hill.
4. Nano materials - B.Viswanathan - 2009 -Narosa.
5. Introduction to nanotechnology - Charles P. Poole; Frank J. Owens – 2008 – Wiley.
6. Fundamentals of biomems and medical microdevices - Steven S.Saliterman – 2006 - Wiley Interscience.
7. Instrumental methods of analysis - Hobart H. Willam; Lynne L. Merrit – 2006 -CBS.
8. Fundamentals of physics - David Halliday; Robert Resnick – 2007 – Willey.
9. Chemistry Raymond Chang – 2009 - Tata Mcgraw Hill.
10. Nanomaterial chemistry - C.N. Rao, A. K . Chettam, A. Muller – 2007 – Wiley – VCH.
11. Nanotechnology Applications in Agriculture – C.R. Chinnamuthu, B.Chandrasekaran and C. Ramasamy – 2008.

THEORY**Unit I- Introduction to Renewable energy Sources**

Energy crisis – classification of energy sources – renewable energy – significance – potential - achievements in India. Biomass – methods of energy conversion.

Unit-II Biochemical Energy Conversion

Biofuels – importance – biodiesel and bioethanol production method – flowchart – by products utilization. Biogas technology – classification - types - factors affecting biogas plants- alternate feedstocks – applications - biodigested slurry and enrichment.

Unit III – Thermochemical Energy Conversion

Briquetting –methods- advantages and disadvantages -combustion –definition- Improved chulhas – types – construction features - applications. Pyrolysis – methods for charcoal /biochar production- comparison of slow and fast pyrolysis. Gasification – chemistry – types – updraft gasifier -downdraft gasifier – working principles – operation and applications.

Unit IV – Solar Energy Conversion

Solar Energy – characteristics - types of radiation – solar constant-solar thermal devices – solar water heater – solar cooker – solar pond – solar distillation – working principles and applications. Solar PV systems – principle – solar lantern - water pumping. Solar driers – natural and forced convection types – solar tunnel drier – working principles and operation.

Unit V- Wind and other Renewable Energy Sources

Wind – formations - Wind mills – types – horizontal and vertical axis – components – working principles – applications. Geothermal energy – wave energy – tidal energy – ocean energy – principle and operation - types – advantages and disadvantages

PRACTICAL

Familiarization with renewable energy gadgets. To study biogas plants, To study gasifier, To study the production process of biodiesel, To study briquetting machine, To study the production process of bio-fuels. Familiarization with different solar energy gadgets. To study solar photovoltaic system: solar light, solar pumping, solar fencing. To study solar cooker, To study solar drying system. To study solar distillation and solar pond.

Lecture Schedule

- | | | |
|-----|--|-------------------------------|
| 1. | Energy crisis – renewable energy sources – significance – potential and achievements in India – energy requirements of agricultural and horticultural crops. | TB-1: 1-10 |
| 2. | Biomass – methods of energy conversion – biochemical conversion methods – thermochemical conversion methods. | TB-1: 12-26 |
| 3. | Biofuels – importance – biodiesel and bioethanol production method – flowchart – by products utilization | TB-1: 164-177; 182-183 |
| 4. | Biogas technology – classification - types of biogas plants – KVIC and Deenabandhu model biogas plants – factors affecting biogas plants. | TB-1: 30-43 |
| 5. | Alternate feedstocks for biogas production – applications of biogas cooking, lighting and engine operations - biodigested slurry and enrichment. | TB-1: 45-49 |
| 6. | Briquetting – MED – VED – methods – need for briquetting - benefits of biomass briquettes. | TB-1: 92-99 |
| 7. | Combustion – improved chulha – single pot – double pot – conventional chulha – biomass gas stove – constructional features – principles and applications. | TB-1: 52-57; 64-67 |
| 8. | Pyrolysis – methods for charcoal production –biochar production– comparison between slow and fast pyrolysis. | TB-1: 67-73 |
| 9. | Mid semester examination | |
| 10. | Gasification – chemistry – types – updraft gasifier – working principles | TB-2: 395-411 |

- operations – application
11. Downdraft gasifier – working principles – operation and applications.
 12. Solar energy – characteristics of solar radiation - types of radiation – solar constant **TB-1:** 101-105
 13. Solar thermal devices – solar water heater – solar cooker – solar pond – solar distillation – working principles and applications. **TB-1:** 105-114
TB-2: 138-142, 195-197
 14. Solar PV systems – principle – solar lantern - water pumping applications. **TB-1:** 117-123
 15. Solar driers – natural and forced convection types – solar tunnel drier – working principles and operation. **TB-1:** 115-117
 16. Wind mills – types – horizontal and vertical axis – components – working principles – applications. **TB-1:** 136 - 144
 17. Energy from ocean, waves, tides. Geothermal energy sources – principles and operation. **TB-1:** 189-205

Practical schedule

1. Basic principles of working of renewable energy gadgets
2. Experiments on biodiesel production
3. Experiments on bioethanol production process
4. Construction and working principle of KVIC biogas plant
5. Construction and working principle of deenbandhu biogas plant
6. Experiments on biogas applications
7. Experiments on briquetting technology
8. Performance evaluation of improved chulha
9. Evaluation of biochar production systems
10. Experiments on biooil production method
11. Performance evaluation of producer gas production system
12. Performance evaluation of solar dryers
13. Experiments on solar cookers and distillation systems
14. Performance evaluation of solar water heaters
15. Experiments on solar water pumping system
16. Performance assessment of solar street light and fencing
17. Final practical examination

References:

- S. Pugalendhi, R. Shalini, J. Gitanjali and P. Subramanian. 2017. Introduction to Renewable Sources of Energy. TNAU, Coimbatore
- G.D. Rai. 2012. Nonconventional Energy Sources. Khanna Publishers, New Delhi.
- C.S. Solanki, 2009. Renewable Energy Technologies : A Practical Guide for Beginners. PHI Learning Pvt. Ltd., New Delhi.
- S. Rao and B.B. Parulekar. 2007. Energy Technology: Non-Conventional, Renewable and Conventional. Khanna Publishers, Naisarak, Delhi.
- G.D. Rai. 1993. Solar Energy Utilisation. Khanna Publishers, New Delhi.
- J. F. Manwell, J. G. McGowan and A. L. Rogers. 2009. Wind Energy Explained: Theory, Design and Application. Wiley & Sons Ltd.,

VI SEMESTER

S. No.	Course code	Course Title	Credit load
1	AEC 302	Agricultural Finance and Co-Operation	2+1
2	PAT 302	Diseases of Field and Horticultural crops and their management -II	2+1
3	COM 311	Agro Informatics	1+1
4	ENS 301	Environmental Pollution and Management	1+1
5	AEN 301	Pests of Crops and Stored grain and their Management	2+1
6	AGR 303	Practical Crop Production - II (<i>Rabi</i> crops)	0+2
7	AGR 304	Principles of organic Farming	1+1
8	ABT 301	Plant Bio technology	2+1
9	PBG 302	Crop Improvement	2+1
10	OPT 301	Optional course	1+1
11	NCC 101	NCC*	
		Total	13+11=24
		*Non-gradual courses compulsory courses	

AEC 302 Agricultural Finance and Co - operation (2+1)

Theory

Unit 1: Agricultural Finance – Nature and Scope : Agricultural Finance- meaning, scope and significance, credit needs and its role in Indian agriculture. Agricultural credit: meaning, definition, need, classification. Sources of credit - advantages and disadvantages - Rural indebtedness- History and Development of rural credit in India.

Unit 2: Financial Institutions : Sources of agricultural finance: institutional and non-institutional sources and their roles, commercial banks - social control and nationalization of commercial banks. Micro financing including KCC, Micro finance – SHG Models, Lead Bank Scheme, RRBs, Scale of finance and unit cost. Cost of credit. An introduction to higher financing institutions – RBI, NABARD, ADB, IMF, world bank, Insurance and Credit Guarantee Corporation of India. Recent development in agricultural credit: Rural credit policies of Government – Subsidized farm credit - Differential Interest Rate (DIR) Scheme – Loan relief measures

Unit 3: Farm Financial Analysis: Credit analysis: 4 R's, 7 P's and 3C's of credit. Preparation of bankable projects / Farm credit proposals – Feasibility; Appraisal - Time value of money: Compounding and Discounting - Undiscounted and Discounted measures. Preparation and analysis of financial statements – Balance Sheet, Income Statement and Cash Flow Statement. Basic guidelines for preparation of project reports - Bank norms – SWOT analysis.

Unit 4: Co-operation: Agricultural Cooperation in India – Meaning, brief history of cooperative development in India - Pre and Post - Independence periods and Co-operation in different plan periods, objectives, principles of cooperation, significance of cooperatives in Indian agriculture. Co-operating credit structure: short term and long term. Agricultural Cooperation - credit, marketing, consumer and multi-purpose cooperatives, farmers' service cooperative societies, processing cooperatives, farming cooperatives, cooperative warehousing; role of ICA, NCUI, NCD, NAFED. Strength and weakness of co-operative credit system, Policies for revitalizing co-operative credit.

Unit 5: Banking and Insurance: Negotiable Instruments: Meaning, Importance and Types - Central bank: RBI – functions - Credit control – Objectives and Methods: CRR, SLR and Repo rate - Credit rationing - Dear money and cheap money - Financial Inclusion and Exclusion: credit widening and credit deepening monetary policies. Credit gap: Factors influencing credit gap. Non - Banking Financial Institutions (NBFI). NPA – Causes, consequences and mitigation. Crop Insurance: Schemes, Coverage, Advantages and Limitations in Implementation - Estimation of Crop Yields - Assessment of crop losses, Determination of compensation - Weather based crop insurance, features, determinants of compensation. Livestock Insurance Schemes Agricultural Insurance Company of India Ltd (AIC): Objectives and functions.

Practical

Determination of most profitable level of capital use. Optimum allocation of limited amount of capital among different enterprise. Analysis of progress and performance of cooperatives using published data. Analysis of progress and performance of commercial banks and RRBs using published data. Visit to a commercial bank, cooperative bank / cooperative society to acquire first - hand knowledge of their management, schemes and procedures. Visit to District Central Co-operative Bank (DCCB) to study its role, functions and procedures for availing loan – Fixation of Scale of Finance. Estimation of credit requirement of farm business – A case study. Preparation and analysis of Balance Sheet, and Cash Flow Statement – A case study. Exercise on Financial Ratio Analysis. Appraisal of farm credit proposals – A case study. Preparation and analysis of income statement – A case study. Preparation of Bankable

projects / Farm Credit Proposals and appraisal - Undiscounted methods and Discounted methods. Techno-economic parameters for preparation of projects for various agricultural products and its value added products. Seminar on selected topics. Analysis of Different Crop Insurance Products / Visit to crop insurance implementing agency.

Theory Schedule

35. Agricultural Finance - meaning, scope and significance, credit needs and its role in Indian agriculture.
36. Agricultural credit: meaning, definition, need and classification.
37. Sources of credit - advantages and disadvantages.
38. Rural indebtedness - History and Development of rural credit in India.
39. Sources of agricultural finance: institutional and non-institutional sources - their roles.
40. Commercial banks - social control and nationalization of commercial banks.
41. Micro financing including KCC, Micro finance – SHG Models, Lead bank scheme.
42. RRBs, Scale of finance and unit cost. Cost of credit.
43. An introduction to higher financing institutions—RBI, NABARD, ADB, IMF and World Bank.
44. Role of Insurance and Credit Guarantee Corporation of India.
45. Recent developments in agricultural credit.
46. Rural credit policies of Government: Subsidized farm credit- Differential Interest Rate (DIR) Scheme – Loan relief measures
47. Credit analysis: 4 R's, 7 P's and 3C's of credit.
48. Preparation of bankable projects / Farm credit proposals – Feasibility.
49. Appraisal: Time value of money: Compounding and Discounting - Undiscounted and Discounted measures.
50. Preparation and analysis of financial statements – Balance Sheet, Income Statement and Cash Flow Statement.

51. Mid Semester Examination

18. Basic guidelines for preparation of project reports- Bank norms – SWOT analysis.
19. Agricultural Cooperation in India – Meaning, brief history of cooperative development in India.
20. Pre and Post - Independence periods and Co-operation in different plan periods, objectives, principles of cooperation, significance of cooperatives in Indian agriculture.
21. Co-operating credit structure: short term and long term. Agricultural Cooperation - credit, marketing, consumer and multi-purpose cooperatives, farmers' service cooperative societies, processing cooperatives, farming cooperatives, cooperative warehousing;
22. Role of ICA, NCUI, NCDC and NAFED.
23. Strength and weakness of co-operative credit system, Policies for revitalizing co-operative credit.
24. Negotiable Instruments: Meaning, Importance and Types.
25. Central bank: RBI – functions, Credit control – Objectives and Methods: CRR, SLR and Repo rate.
26. Credit rationing - Dear money and cheap money.
27. Financial Inclusion and Exclusion: credit widening and credit deepening monetary policies.
28. Credit gap: Factors influencing credit gap.
29. Non - Banking Financial Institutions (NBFI). NPA – Causes, consequences and mitigation.
30. Crop Insurance: Schemes, Coverage, Advantages and Limitations in Implementation.
31. Estimation of Crop Yields - Assessment of crop losses, Determination of compensation.
32. Weather based crop insurance, features, determinants of compensation.
33. Livestock Insurance Schemes
34. Agricultural Insurance Company of India Ltd (AIC): Objectives and functions.

Practical Schedule

1. Determination of most profitable level of capital use.
2. Optimum allocation of limited amount of capital among different enterprise.
3. Analysis of progress and performance of cooperatives using published data.
4. Analysis of progress and performance of commercial banks and RRBs using published data.
5. Visit to a commercial bank, cooperative bank / cooperative society to acquire first - hand knowledge of their management, schemes and procedures.
6. Visit to District Central Co-operative Bank (DCCB) to study its role, functions and procedures for availing loan – Fixation of Scale of Finance.
7. Guest lecture on Role and functions of Commercial Bank and Lead Bank / NABARD and its Role and Functions.
8. Estimation of credit requirement of farm business – A case study.
9. Preparation and analysis of Balance Sheet and Cash Flow Statement – A case study.
10. Exercise on Financial Ratio Analysis. Appraisal of farm credit proposals – A case study.
11. Preparation and analysis of income statement – A case study.
12. Preparation of Bankable projects / Farm Credit Proposals and appraisal.
13. Undiscounted methods and Discounted methods.
14. Techno-economic parameters for preparation of projects for various agricultural products and its value added products.
15. Analysis of Different Crop Insurance Products / Visit to crop insurance implementing agency.
16. Seminar on selected topics.
17. **Practical Examination.**

References

29. Muniraj, R. 1987. Farm Finance for Development. Oxford & IBH. New Delhi.
30. Subba Reddy, S and P. Raghu Ram. 2011. Agricultural Finance and Management. Oxford & IBH. New Delhi.
31. Lee, W.F., M.D. Boehlje, A.G. Nelson and W.G. Murray. 1998. Agricultural Finance. Kalyani Publishers. New Delhi.
32. Mammoria, C.B. and R.D. Saxena. 1973. Cooperation in India. Kitab Mahal. Allahabad. Patnaik, V.E. and A.K. Roy. 1988. Cooperation and Cooperative Management. Kalyani Publishers. Ludhiana.

1. PAT 302 Diseases of Field and Horticultural crops and their management-II (2+1)

Theory

Etiology, symptoms, mode of spread, survival, epidemiology and integrated management of important diseases

Unit-I Diseases of cereals: Wheat

Unit- II Diseases of Pulses, Oilseeds and Cash crops

Pulses: chick pea and lentil; **Oilseeds:** sunflower and mustard; **Cash crops:** sugarcane and cotton

Unit- III Diseases of Fruits and vegetables crops

Fruits: mango, citrus, grapevine, sapota, jackfruit, pineapple, ber, apple, peach plum and strawberry; **Vegetables:** cucurbits, peas, potato, beet root, radish, cassava, colacasia and yam

Unit- IV Diseases of Spices, Plantation and Flower crops

Spices: chillies, turmeric, ginger, onion, garlic, coriander, cardamom; **Plantation crops:** black pepper and vanilla; **Flower crops:** rose, Jasmine, marigold, crossandra, chrysanthemum, tube rose, carnation, lillium and orchids

Unit- V Diseases of medicinal crops and mushroom cultivation

Medicinal crops: gloriosa, coleus, stevia and aloe; **Mushroom cultivation:** Importance of mushroom and cultivation of button mushroom, oyster mushroom, milky mushroom and paddy straw mushroom- pest and diseases of mushroom

PRACTICAL

Study of symptoms and host parasite relationship of the important diseases of wheat, chick pea, lentil, sunflower, mustard, cotton, sugarcane, mango, citrus, grapevine, sapota, jackfruit, pineapple, ber, apple, peach, plum, strawberry, cucurbits, potato, peas, beet root, radish, cassava, colacasia, yam, chillies, turmeric, ginger, onion, garlic, coriander, cardamom, black pepper, vanilla, rose, Jasmine, marigold, crossandra, chrysanthemum, tube rose, carnation, lillium, orchids, gloriosa, coleus, stevia and aloe and cultivation of button mushroom, oyster mushroom, milky mushroom and paddy straw mushroom.

THEORY

Etiology, symptoms, mode of spread, survival, epidemiology and integrated management of

1. Diseases of wheat
2. Diseases of chickpea and lentil
3. Diseases of sunflower and mustard
4. Diseases of cotton
5. Diseases of sugarcane
6. Diseases of mango
7. Diseases of citrus
8. Diseases of grapevine
9. Diseases of sapota, jack fruit, pineapple and ber
10. Diseases of apple
11. Diseases of peach, plum and strawberry
12. Diseases of cucurbits
13. Diseases of Potato-I (Fungal diseases)
14. Diseases of Potato-II (bacterial and viral diseases)
15. Diseases of peas, beet root and radish

16. Mid semester examination

17. Diseases of cassava, colacasia and yam
18. Diseases of chillies
19. Diseases of turmeric and ginger
20. Diseases of onion and garlic
21. Diseases of cardamom and coriander
22. Diseases of black pepper, betel vine and vanilla
23. Diseases of rose and jasmine
24. Diseases of marigold, crossandra and chrysanthemum
25. Diseases of tube rose and carnation
26. Diseases of lillium and orchids
27. Diseases of gloriosa and coleus
28. Diseases of stevia and aloe
29. Diseases of stored grains and their management
30. Post harvest diseases of fruit and vegetables
31. Mushroom-edible and poisonous mushroom- importance of mushroom
32. Cultivation of button mushroom and oyster mushroom
33. Cultivation of milky mushroom and paddy straw mushroom
34. Pest and diseases of mushroom

PRACTICAL

Study of symptoms and host-parasite relationship of:

1. Diseases of wheat
2. Diseases of chick pea, lentil, sunflower and mustard
3. Diseases of cotton and sugarcane
4. Diseases of mango and sapota
5. Diseases of citrus and grapevine
6. Diseases of jackfruit , pineapple, ber, apple , peach, plum , strawberry
7. Diseases of cucurbits
8. Diseases of potato , peas , beet root and radish
9. Diseases of cassava, colacasia and yam
10. Field visit/ exposure visit to hilly fruits , vegetables and plantation crops / mushroom unit
11. Diseases of chillies, turmeric and ginger
12. Diseases of coriander, cardamom, black pepper and vanilla,
13. Diseases of rose, Jasmine, marigold and crossandra
14. Diseases of tube rose , carnation, lillium and orchids,
15. Diseases of gloriosa, coleus, stevia and aloe
16. Cultivation of oyster , milky and paddy straw mushroom cultivation
17. Final practical examination

Reference Books

1. Arjunan.G. Karthikeyan, G, Dinakaran ,D. Raguchander,T. 1999 Diseases of Horticultural Crops, AE Publications, Coimbatore.

2. Rangasawmi ,G and Mahadevan, A. 1998. Diseases of crop Plants in India, Prentice Hall of India Pvt. Ltd., New Delhi
3. Prakasam, V., Valluvaparidasan, V., Raguchander, T. and K.Prabakar. 1997. Field crop diseases, AE Publication, Coimbatore.

E- Books

1. Agrios, G.N. 2008. Plant Pathology, Academic Press, New York
2. Rangaswami, G. 2005. Diseases of Crop plants in India. Prentice Hall of India Pvt. Ltd., New Delhi
3. Thakur, B.R. 2006. Diseases of field crops and their management

COM 311 Agro- Informatics (1+1)

Theory

Unit I: Information and Communication Technology (ICT)

ICT and its importance – Computer Fundamentals - Basic anatomy of the computer system: Input devices, CPU, Output devices, Memory: Primary and secondary - Software – Types: System software, Application software and Utility software – Software terminologies: Firmware, Liveware, Freeware, Shareware, Commercial software, Proprietary software, Semi-free software - Internet - World Wide Web – URL – Domain names - Protocols: HTTP, HTTPS - Internet Applications: Email, File sharing web apps, Social Networks, Online shopping, Video Conferencing – HTML: Introduction, Editor, HTML Documents – Tags: <head>, <body>, <title>, <heading>, <paragraph>, </br>, <table>, , , <href>, , <hr> and <marquee>.

Unit II: Spreadsheet and Database

Electronic spreadsheet – Microsoft Excel - Worksheet manipulation: insert, delete, move, copy and hide worksheet – Cell manipulation: copy, edit and format cell data – Charts - Create Bar and Pie charts - PIVOT table - DBMS: Database terms: Data, Database, DBMS, RDBMS, Row, Column, Table – Database Architecture – Data types: char, varchar(), int, float() – Use of databases in agriculture.

Unit III: C Programming

Introduction to Computer Programming – Programming languages - Translators: Compilers and Interpreters - Algorithm – Flowchart - Introduction to C – Structure of C program - Data types, Variables, Constants, Operators: Arithmetic, Relational, Logical, Assignment - Input/Output: scanf(), printf() - Control statements: if, if else – Loop: while, do while, for.

Unit IV: Agroinformatics

Agroinformatics – Needs and objectives - e-Agriculture : Concept, Meaning, Terminologies and Importance - e-Agriculture – National and International scenario - ICT for Data collection, formation of development programmes, monitoring and evaluation of programmes - Decision support systems: Taxonomy, Components, Framework, Classification and applications in Agriculture -Expert systems - Concepts and Importance – Components – User Interface – Knowledge Base – Inference Mechanism – Inference Rule - Designing an Expert Systems - Advantages and disadvantages of Expert Systems - Information systems for supporting farm decisions.

Unit V: Models and Computer Controlled Devices

Introduction to computer based agricultural models: Model, Simulation, Systems analysis models, Subsystems, Types: Mechanistic process models, Operational models, Statistical models and dynamic simulation models - List of agricultural models - Computer controlled devices – Sensor – Drones – Robots – Internet of Things (IoT) and Cloud Computing for Agriculture.

Practical

Innards of computer – Booting and shutdown – Practice of DOS commands: dir, cd, mkdir, rmdir, del, cls, attrib, ren, copy, move, ipconfig, ping - Software practices – Installation / Uninstallation – Windows apps: Sticky Notes, Steps Recorder, Snipping Tool – Pin and unpin the programs – System tray customization – Shortcut keys - Microsoft Excel - Entering a formula in a cell, Built-in functions: SUM, AVERAGE, MIN, MAX, COUNT, COUNTIF, IF – Import and export data - Charts - Create Bar and Pie charts – PIVOT table - MS-ACCESS: Creating agriculture database – Entering, editing, deleting data – Creating Forms – Query wizard: select, update, delete – Reports - Internet Applications: Email, File sharing web apps: Dropbox, Google drive - Social Networks, Online shopping, Video Conferencing – Creating a web page: HTML editor – Tags: <head>, <body>, <title>, <heading>, <paragraph>, </br>, <table>, , , <href>, , <hr> and <marquee> - Develop algorithms and represent the same in the flowchart for

the following problems -To calculate Leaf Area Index (LAI) -To calculate the Crop Growth Rate (CGR) - To find the greatest average seed sales of two districts during samba season - Familiarizing with the Integrated Development Environment of C Editor for coding, saving, compiling, debugging and executing – C Programs: Display TNAU motto "Till, Toil, Triumph" – Calculate Leaf Area Index (LAI) – Calculate the Crop Growth Rate (CGR) - Find the greatest average seed sales of two districts during samba season - e-Agriculture – Leveraging social media in agriculture (Social networks) - ICT in agriculture – Paperless data collection using google survey tools - Online photo and video editing tools - Simulating crop yield: InfoCrop - Base file creation for rice and maize (Weather, Varietal characters, Agronomy practices, Soil data) – Interpretation - InfoCrop – Climate change impact studies on rice and maize - Smartphone mobile apps in Agriculture for farm advices, crop protection, market price, postharvest management - Decision support systems - Expert systems - Information systems for supporting farm decisions - Crop calendar – Crop planning tool for farmers.

Lecture Schedule

6. Introduction to Computers - Basic anatomy of the computer system: Input devices, CPU, Output devices, Memory: Primary and secondary.
7. Software – Types: System software, Application software and Utility software – Software terminologies: Firmware, Liveware, Freeware, Shareware, Commercial software, Proprietary software, Semi-free software.
3. Internet - World Wide Web – URL – Domain names - Protocols: HTTP, HTTPS - Internet Applications: Email, File sharing web apps, Social Networks, Online shopping, Video Conferencing – HTML: Introduction, Editor, HTML Documents – Tags: <head>, <body>, <title>, <heading>, <paragraph>, </br>, <table>, , , <href>, , <hr> and <marquee>.
27. Electronic spreadsheet – Microsoft Excel - Worksheet manipulation: insert, delete, move, copy and hide worksheet – Cell manipulation: copy, edit and format cell data – Charts - Create Bar and Pie charts - PIVOT table.
28. DBMS: Database terms: Data, Database, DBMS, RDBMS, Row, Column, Table – Database Architecture – Data types: char, varchar(), int, float() – Use of databases in agriculture.
29. Introduction to Computer Programming – Programming languages - Translators: Compilers and Interpreters - Algorithm – Flowchart.
30. Introduction to C – Structure of C program - Data types, Variables, Constants, Operators: Arithmetic, Relational, Logical, Assignment - Input/Output: scanf(), printf().
31. Control statements: if, if else – Loop: while, do while, for.

32. Mid-Semester Examination

6. Agriinformatics – Needs and objectives - e-Agriculture : Concept, Meaning, Terminologies and Importance
7. e-Agriculture – National and International scenario
8. ICT for Data collection, formation of development programmes, monitoring and evaluation of programmes.
9. Decision support systems: Taxonomy, Components, Framework, Classification and applications in Agriculture.
10. Expert systems - Concepts and Importance – Components – User Interface – Knowledge Base – Inference Mechanism – Inference Rule - Designing an Expert Systems - Advantages and disadvantages of Expert Systems - Information systems for supporting farm decisions.
11. Introduction to computer based agricultural models: Model, Simulation, Systems analysis models, Subsystems, Types: Mechanistic process models, Operational models, Statistical models and dynamic simulation models - List of agricultural models.
12. Computer controlled devices – Sensor – Drones – Robots.
13. Internet of Things (IoT) and Cloud Computing for Agriculture.

Practical Schedule

19. Innards of computer – Boot and shutdown – Windows apps: Sticky Notes, Steps Recorder, Snipping Tool – Pin and unpin the programs – System tray customization – Shortcut keys.
20. Software practices – Installation / Uninstallation – Practice of DOS commands: dir, cd, mkdir, rmdir, del, cls, attrib, ren, copy, move, ipconfig, ping.
21. Microsoft Excel - Entering a formula in a cell, Built-in functions: SUM, AVERAGE, MIN, MAX, COUNT, COUNTIF, IF – Import and export data - Charts - Create Bar and Pie charts – PIVOT table.
22. MS-ACCESS: Creating agriculture database – Entering, editing, deleting data – Creating Forms – Query wizard: select, update, delete – Reports.
23. Internet Applications: Email, File sharing web apps: Dropbox, Google drive - Social Networks, Online shopping, Video Conferencing – Creating a web page: HTML editor – Tags: <head>, <body>, <title>, <heading>, <paragraph>, </br>, <table>, , , <href>, , <hr> and <marquee>.
24. Develop algorithms and represent the same in the flowchart for the following problems -To calculate Leaf Area Index (LAI) -To calculate the Crop Growth Rate (CGR) - To find the greatest average seed sales of two districts during samba season.
25. Familiarizing with the Integrated Development Environment of C Editor for coding, saving, compiling, debugging and executing – C Programs: Display TNAU motto "Till, Toil, Triumph" – Calculate Leaf Area Index (LAI) – Calculate the Crop Growth Rate (CGR) - Find the greatest average seed sales of two districts during samba season.
26. Looping statements: Calculate the average yield of last 10 years Rice yield of our District - Write a C program to find total, maximum, minimum and average rain fall of last five years in our District.
27. e-Agriculture – Leveraging social media in agriculture (Social networks).
28. ICT in agriculture – Paperless data collection using google survey tools - Online photo and video editing tools.
29. Simulating crop yield: InfoCrop - Base file creation for rice and maize (Weather, Varietal characters, Agronomy practices, Soil data) – Interpretation.
30. InfoCrop – Climate change impact studies on rice and maize.
31. Smartphone mobile apps in Agriculture for farm advices, crop protection, market price, postharvest management.
33. Decision support systems
34. Expert systems - Information systems for supporting farm decisions.
35. Crop calendar – Crop planning tool for farmers.

36. Final Practical Examination

Reference:

32. Introduction to Information Technology, 2012, Second Edition, ITL Education Solutions Limited, PEARSON Education.
33. Firuza Aibara, HTML 5 for Beginners, 2012, Shroff Publications.
34. John Walkenbach, Excel 2010 Bible, Wiley publishing, Inc
35. Balagurusamy, E., Programming in ANSI C, 2017, Seventh Edition, McGraw Hill Education.
36. Saravanan, R., Kathiresan, C and Indra Devi, T., 2011. Information & communication technology for agriculture and rural development. New India Publ. Agency.
37. Aggarwal, P.K., Naveen Kalra and Subhash Chander, Infocrop: A generic simulation model for annual crops in tropical environments, Indian Agricultural Research Institute, New Delhi.
38. Malcolm J. Blackie, Information Systems for Agriculture, 2012, Springer Netherlands.
39. Smart Sensing Technology for Agriculture and Environmental Monitoring, 2012, Editors: Mukhopadhyay and Subhas Chandra (Ed.), Springer
40. John Billingsley, Arto Visala and Mark Dunn, 2008, Robotics in Agriculture and Forestry – 46th Chapter from book Springer Handbook of Robotics.

33. Introduction to Expert Systems, 3rd Edition by Peter Jackson
34. Introduction to Artificial Intelligence and Expert Systems, 2007 by Dan W. Patterson.
35. Balagurusamy, E., Computing Fundamentals & C Programming, Second Edition, 2017, McGraw Hill Education.
36. A.S. Sandhu, 2004. Text book on Agricultural Communication Process and Methods. Oxford & TBH.
37. C.J. Date: Data Base Design, Addison Wesley.

E-Reference

25. <https://www.scribd.com/document/249057939/InfoCrop-Help>
26. InfoCrop: A dynamic simulation model for the assessment of crop yields, losses due to pests, and environmental impact of agro-ecosystems in tropical environments. P.K. Aggarwal et al., Agricultural Systems 89 (2006) 47–67.
27. <http://www.sciencedirect.com/science/article/pii/S0168169916303623>
28. Web-based crop model: Web InfoCrop – Wheat to simulate the growth and yield of wheat
29. <http://excelpro.ir/wp-content/uploads/2015/10/Excel-2010-Bible.pdf>
30. https://www.researchgate.net/publication/226105300_Decision_Support_Systems_Concepts_Progress_and_Issues_-_A_Review
31. <https://www.hindawi.com/journals/js/2015/195308/> *Applications of Smartphone-Based Sensors in Agriculture: A Systematic Review of Research*
32. <http://ncert.nic.in/ncerts/l/kect214.pdf>
33. <http://teacherlink.ed.usu.edu/tlresources/training2/Google/GoogleForms.pdf>
19. <http://www.fao.org/publications/card/en/c/24f624ea-7891-45e8-9b24-66cbf13f004d/>
20. http://indiagovernance.gov.in/files/ict_in_agriculture.pdf
21. www.manage.gov.in/studymaterial/AKM-E.pdf
22. https://www.researchgate.net/publication/233910963_Application_of_Cloud_Computing_in_Agricultural_Sectors_for_Economic_Development

ENS 301 – Environmental Pollution and Management (1+1)

Theory

Unit-I-Pollution in Environment-Introduction-Pollution- Pollutants – Contaminants – Source and types of pollution in Soil-Water-Air-Impact on environment-Pollution Status in India

Unit– II Waste water Management: Waste water – Different types of waste water-pollutants and contaminants-Impact of waste water on ecosystem –Eutrophication – Biomagnification – Water borne diseases –Wastewater treatment methods – Physical, chemical and Biological – General water treatments-Wastewater recycling – Constructed wetlands-Reed bed system -Legislation and standards

Unit-III-Management of polluted soils: Soil pollutants – Sources – Urban and Industrial – Heavy metal – Pesticides – PAH's and PCB's-E-Waste-Fate of pollutants in Soil - Management of soil pollution – Bio and phyto remediation of polluted soil

Unit-IV - Air Pollution and its Management: Air pollutants from industrial and domestic sources – Fate of air pollutants-Air pollution indicators - Monitoring and Control measures – Role of plants in controlling air pollutants-Legislation and Air quality standards - – Noise Pollution – Sources, Effect and Control Measures-Indoor air pollution and control measures

Unit-V- Solid waste management: Solid waste –Sources – Sludge from Industry and farm waste-Characteristics – Environmental problems – Management of sludge and farm wastes – Disposal methods

– Sanitary land fills – Incineration – Pyrolysis - Recycling –Energy recovery –Composting – Vermicomposting – Maturity indices assessment-Standards and Legislation

Unit-VI-Environmental standards, Regulation and EIA - Environmental standards-CPCB Norms for discharging industrial effluents to public sewers- CDM and Carbon foot print-Environmental Impact Assessment: Stages of EIA -Monitoring and Auditing – Environmental clearance procedure in India

Lecture Schedule:

11. Introduction-Pollution- Pollutants – Contaminants – Source and types of pollution in Soil-Water-Air-Impact on environment-Pollution Status in India
12. Waste water – Different types of waste water-pollutants and contaminants-Impact of waste water on ecosystem –Eutrophication – Biomagnification – Water borne diseases –
9. Wastewater treatment methods – Physical, chemical and Biological – General water treatments-
10. Wastewater recycling – Constructed wetlands-Reed bed system -Legislation and standards
11. Soil pollutants – Sources – Urban and Industrial – Heavy metal – Pesticides – PAH's and PCB's-E-Waste
12. Fate of pollutants in Soil - Management of soil pollution – Bio and phyto remediation of polluted soil
13. Air pollutants from industrial and domestic sources – Fate of air pollutants-Air pollution indicators – Air pollution episodes-Monitoring and Control measures–
14. Role of plants in controlling air pollutants- Legislation and Air quality standards,
- 15. Mid Semester Examination**
- 16.** Noise Pollution, Sources, Effect and Control Measures, Indoor air pollutants and control mesures
- 17. Solid waste** –Sources – Sludge from Industry and farm waste-Characteristics – Environmental problems
18. Management of solid waste, Disposal methods, Sanitary land fills, Incineration, Pyrolysis
19. Recycling –Energy recovery –Composting – Vermicomposting – Maturity indices assessment-Standards and Legislation
20. Environmental standards-CPCB Norms for discharging industrial effluents to public sewers

21. Environment Impact Assessment, Introduction, Stages of EIA, -Monitoring and Auditing
22. CDM and Carbon foot print
23. Environmental clearance procedure in India

Practical Schedule

35. Sample collection and preservation from contaminated sites
36. Waste water treatment by physical (column study with vermiculite and activated charcoal) and chemical (Alum treatment)
37. Waste water treatment through constructed wetland system and characterisation
38. Estimation of Chlorides, Phosphates in waste water
39. Analysis of Nitrogen in industrial effluent and sludge
40. Collection of PAH's contaminated soils and analysis by GC-MS
41. Biosorption of heavy metal (Cr) by using Water hyacinth and analysis through AAS
42. Pesticide Residue analysis in contaminated water
43. Analysis of SPM in air, Methane and CO₂ in Municipal dumping site
44. Assessing the efficiency of plants to control Indoor air pollutants
45. Analysis of Organic carbon in Sludge and Organic manure
46. Composting and Vermicomposting of farm wastes
47. Energy recovery from wastes
48. Maturity indices of compost- C:N ratio and Phytotoxicity test
49. Maturity indices of compost: starch iodine test and sulphide test
50. Visit to water treatment plant

51. Final practical examination

Reference:

19. Tyler Miller and Scot Spoolman. 2009. Living in the Environment (*Concepts, Connections, and Solutions*). Brooks/Cole, Cengage learning publication, Belmont, USA
20. P.D. Sharma, 2009, Ecology and Environment, Rastogi Publications, Meerut, India **E-Books:** Chiras D.D., 2016. Environmental Science, Tenth Edition. Jones & Bartlett Learning, Burlington, MA. ISBN: 978-1-284-05705-8, 708 Pages

AEN 301 Pests of Crops and Stored Produce and their Management (2+1)

Theory

Unit I: Distribution, bionomics, symptoms of damage and management strategies for insects and non-insect pests of rice, wheat, maize, sorghum, cumbu, ragi, tenai, redgram, green gram, black gram, bengal gram, cowpea, groundnut, castor, gingelly, sunflower, safflower, jatropha, soybean and mustard.

Unit II: Distribution, bionomics, symptoms of damage and management strategies of insects and non-insect pests of cotton and sugarcane, green manures (Sunnhemp, Sesbania, Daincha. Glyricidia), forage crops (Lucerne and Subabul)

Unit III: Distribution, bionomics, symptoms of damage and management strategies of insect and non-insect pests of Brinjal, Bhendi, Tomato, Chillies, Onion, Garlic, Moringa, Amaranthus, Crucifers, Cucurbits, Mango, Citrus, Banana, Guava, Grapevine and Sapota

Unit IV: Distribution, bionomics, symptoms of damage and management strategies of insect and non-insect pests of Pomegranate, Papaya, Aonla, Apple, Pine apple, Custard apple and Jack, Potato, Sweet potato, Tapioca, Yam, Colocasia, Coconut, Arecanut, Tea, Coffee, Cashew, Cocoa, Betelvine, Ginger, Turmeric, Coriander, Cardamom, Pepper, Curry leaf and Tamarind

Unit V: Distribution, bionomics, symptoms of damage and management strategies of insect and non-insect pests of Rose, Jasmine, Crossandra, Chrysanthemum, Tuberose, Cut flowers, Glory lily, Coleus, Stonebreaker, Aswagantha, Senna, Periwinkle and Lawn. Distribution, bionomics, symptoms of damage and management strategies of pests of and stored products. Rodents and birds of agricultural importance and their management. Locusts and their management.

Practical

Identification of symptoms of damage and life stages of important pests of different field crops *vi.*, cereals, millets, pulses, oilseeds, cotton, sugarcane and green manure crops and horticultural crops *viz.*, vegetables, fruits, spices, tubers, plantation crops, flower crops, medicinal plants, lawn and stored products.

Lecture schedule:

Distribution, bionomics, symptoms of damage and management strategies for insects and non-insect pests of

18. Rice – Sucking pests
19. Rice – Borers and defoliators
20. Maize, sorghum and cumbu
21. Wheat, ragi and tenai
22. Redgram, bengalgram, blackgram, greengram and cowpea
23. Groundnut, gingelly and sunflower
24. Castor, soybean, safflower, jatropha and mustard
25. Cotton – Sucking pests
26. Cotton – Bollworms, borers and defoliators
12. Sugarcane
13. Green manures and forage crops - sunnhemp, sesbania, daincha, lucerne, subabul and gliricidia
14. Brinjal, bhendi and tomato
15. Chillies, onion, garlic, moringa and amaranthus
16. Crucifers
17. Cucurbits

18. Mango
19. **Mid semester examination**
 - Citrus and banana
 - Guava, grapevine and sapota
 - Pomegranate, papaya and aonla Apple, pine apple, custard apple and jack
 - Potato, sweet potato, tapioca, yam and colocasia Coconut and arecanut
 - Tea and coffee
 - Cashew, cocoa and betelvine Ginger, turmeric and coriander, Cardamom, pepper, curry leaf and tamarind
 - Rose, jasmine, crossandra, chrysanthemum, tuberose and cut flowers Gloriv lily, coleus, stone breaker, aswagantha, senna, periwinkle and lawn
 - Role of physical, biological, mechanical and chemical factors in deterioration of grain Stored product pests
 - Methods of grain storage and various methods of stored product pest management
 - Mites, slugs and snails, rodents and bird pests
 - Locusts and their management

Practical schedule: Identification of symptoms of damage and life stages of pests of

34. Pests of rice
35. Pests of maize, sorghum , cumbu, ragi and tenai
36. Pests of pulses
37. Pests of groundnut, gingelly sunflower and castor
38. Pests of cotton
39. Pests of sugarcane
40. Pests of green manures and forage crops -sunnhemp, sesbania, daincha, lucerne, subabul and gliricidia
41. Pests of brinjal, bhendi and tomato
42. Pests of chillies, onion, garlic, moringa and amaranthus
43. Pests of crucifers and cucurbits
44. Pests of mango, citrus, sapota, banana, grapevine and guava
45. Pests of pomegranate, aonla, papaya, jack, pine apple, custard apple, ber and apple
46. Pests of potato, sweet potato and tapioca
47. Pests of coconut , cashew, cocoa , betelvine , coffee and tea
48. Pests of turmeric, ginger, coriander, cardamom, pepper and curry leaf
49. Pests of rose, jasmine, crossandra, chrysanthemum and tuberose
50. Pests of stored products

References:

1. Manisegaran, S. and R.P.Soundararajan. 2010. *Pest Management in Field Crops- Principles and Practices*. Agrobios, Jodhpur, India. 316p. {ISBN (10): 81-7754-321-0}
2. David, B.V. and V.V. Ramamurthy. 2011. *Elements of Economic Entomology*, Namrutha Publications, Chennai. 386 p. {ISBN: 978-81-921477-0-3}
3. Muthukrishnan, N., N.Ganapathy, R.Nalini and R.Rajendran. 2005. *Pest Management in Horticultural Crops*. New Madura Publishers, Madurai. 325p. {ISBN: 81-902832-0-0}
4. Awasthi, V.B. 2007. *Agricultural Insect Pests and their Control*, Scientific publishers (India), Jodhpur, 267p. {ISBN 81-7233-491-5}

5. Dhaliwal, G.S. and Ramesh Arora. 2004. *Integrated pest management Concepts and Approaches*, Kalyani Publishers, Ludhiana, 427p. {ISBN: 81-7663-904-4}
6. Regupathy, A. and R.Ayyasamy. 2013. *A Guide on Crop Pests*. Namrutha Publications, Chennai, 368 p. {ISBN: 978-81-921477-1-0}
7. Srivastava, K.P. and G.S. Dhaliwal. 2011. *A text book of Applied Entomology*. Vol. II, Kalyani Publishers, Ludhiana. 368p. {ISBN: 978-81-272-6752-0}
8. Nair, M.R.G.K. 1986. *Insects and mites of crops in India*. Publications and Information Division, ICAR, New Delhi. 408p.
9. Parvatha Reddy. 2010. *Insect, Mite and Vertebrate Pests and their Management in Horticultural Crops*. Scientific Publishers, Jodhpur. 384p. {ISBN: 978-81-7233-628-8}
10. Butani, D.K. and M.G. Jotwani. 2013. *Insects in Vegetables*. Daya Publishing House, New Delhi. 356p.
11. Regupathy, A. and R.Ayyasamy. 2013. *A Guide on Crop Pests*. Namrutha Publications, Chennai. 368p. {ISBN: 978-81-921477-1-0}
12. Nair, M.R.G.K. 1995. *Insects and Mites of Crops in India*. Indian Council of Agricultural Research, New Delhi, 408p.
13. Ayyar, T.V.R. 1963. *Hand Book of Economics Entomology for South India*. Govt. Press Madras. 21.
21. Sivasubramanian, P., K. Samiayyan, N. Ganapathy, K. Bhuvaneshwari and S. Jayaprabhavathi. 2012. *A treatise on Integrated Pest Management*. Associated Publishing Company, New Delhi. 287 p.
14. Srivastava, K.P. and D.K. Butani. 2009. *Pest Management in Vegetables* (Vol. I & II). Studium Press (India) Pvt. Ltd., New Delhi. 777p. {ISBN: 978-81-907577-3-7}
15. Sathe, T.V. 2012. *Pests of Ornamental Plants*. Daya Publishing House, New Delhi. 199p. {ISBN: 978-81-7035-757-5}

E- References:

17. <http://www.ncipm.org.in>
18. <http://agritech.tnau.ac.in/>
19. <http://www.nbaii.res.in/>
20. <http://www.nrcg.res.in/>
21. ipm.illinois.edu

AGR 303 Practical crop Production – II (Rabi crop) (0+2)

30. Each student will be allotted a minimum land area of 100/200 m² and he / she will do all field operations in the allotted land from field preparation to harvest and processing.
31. Field preparation, seed, treatment, nursery raising, sowing, nutrient, water and weed management and management of insect-pests diseases of crops, harvesting, threshing, drying winnowing, storage and marketing of produce.
32. Any irrigated dry crop (maize / sorghum / pearl millet / finger millet / cotton / groundnut / sunflower / sesame)

Practical Schedule for Irrigated dry crop (Eg. Maize):

33. Ecosystem - Climate and weather - Seasons and varieties of Tamil Nadu
34. Selection of field - Main field preparation - seed treatment - Application of manures and fertilizers - Sowing - Weed management and practicing pre- emergence application of herbicides - Thinning and gap filling - Estimation of plant population - Top dressing - Weed management - Water management - Pest management - Observation on nutrient and weeds - Recording growth, yield attributes and yield
35. Harvesting, threshing and cleaning the produce - Cost of cultivation and economics

- 1 & 2 .Study of ecosystems, climate, weather, seasons and varieties of Tamil Nadu
- 3 & 4. Selection of field for maize cultivation
- 5 & 6. Acquiring skill in seed treatment practices
- 7 & 8. Study and Practice of main field preparation for maize
- 9 & 10. Practicing of application of manures and fertilizers for maize
- 11 & 12. Practicing sowing of maize
- 13 & 14. Acquiring skill in pre-emergence application of herbicides
- 15 & 16. Estimation of plant population and acquiring skill in gap filling and thinning
- 17 & 18. Observation on nutritional deficiency symptoms and corrective measures
- 19 & 20. Study of weeds and weed management in maize
- 20 & 21. Recording growth parameters and assessing dry matter production
- 22 & 23 Study of water management practices for maize
- 24 & 25. Observation of insect pests and diseases and their management
- 26 & 27. Estimation of yield and yield parameters in maize
- 28 & 29. Harvesting, threshing and cleaning of the produce
- 30 & 31. Harvesting, threshing and cleaning of the produce
- 32 & 33. Working out cost of cultivation and economics
44. **Practical**

References:

- Ahlawat, I.P.S., Om Prakash and G.S.Saini.2010. Scientific Crop Production in India. Rama Publishing House, Meerut.
- Crop Production Guide. 2012. Directorate of Agriculture, Chennai and Tamil Nadu Agricultural University, Coimbatore.
- Rajendra Prasad. 2012. Text Book on Field Crop Production, Indian Council of Agrl. Research, New Delhi. Reddy,S.R. 2012. Agronomy of field crops. Kalyani publishers, New Delhi.
- Chidra Singh.1997. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi. Crop Production Guide. 2012. Directorate of Agriculture, Chennai and Tamil Nadu Agricultural University, Coimbatore.

E-References:

www.cimmyt.org

AGR 304 Principles of Organic Farming (1+1)

Theory:

Unit - I: Components and Principles of Organic Cotton

Organic farming: Definition - Scope - principles and concepts - history of organic farming - global scenario - biodiversity: importance and measure to preserve biodiversity - pre requisites for Organic farming:- Soil organic carbon: status and improvement strategies.

Unit - II: Organic sources of Nutrients

Organic sources of nutrients - manures and other inputs - on farm and off farm sources - organic waste recycling - methods - Soil and crop management - inter cropping, crop rotation, green manures, cover crops, mulching - bio fertilizers.

Unit - III: Non - Chemical weed and Pest disease management

Non-chemical weed management methods: preventive, physical, cultural, mechanical and biological measures - Bio-intensive pest and disease management.

Unit - IV: Indigenous Technical Knowledge (ITK)

Indigenous Technical Knowledge (ITK) in organic agriculture - scientific rationale - soil, nutrient, weed, water, management - prospects and problems in organic farming.

Unit - V: Certification of label

Organic certification - NPOP guidelines - Certification agencies in India - crop production standards - Quality considerations - labeling and accreditation process - marketing and export opportunities.

Lecture Schedule:

69. Organic farming; definition - prospects - principles and concepts - History and genesis of organic farming in World and India: Present status in World, India and Tamil Nadu.
70. Introduction to bio - diversity; importance and measures to preserve bio - diversity.
71. Pre-requisites and basic steps for organic farming; conversation to organic farming - planning and processes in practices - IFS approach - Integration of animal components.
72. Organic carbon; status and improvement strategies - conservative tillage systems.
5. Sources of organic manures - plant, animal and microbial origin - on - farm resources; FYM, green manures, crop residues, poultry manure, sheep and goat manures, biogas slurry and vermicompost.
17. Off-farm resources; coir pith, press mud, oilcakes, flyash, bio compost, minerals, bone meal, bio fertilizers, traditional preparations.
18. Organic waste recycling methods and techniques - composting, vermicomposting, *in situ* composting - system approach.
19. Soil and crop management in organic farming; Inter cropping and companion planting, crop rotation green manures and cover crops, mulching.
- 20. Mid semester examination**
21. Weeds - Ecology - habitat management of weeds - Non - chemical weed management methods; preventive, physical, cultural, use of tools and implements and biological measures - good crop husbandry practices.
22. Integrated pest and diseases management - bio control agents, bio rational pesticides; minerals, botanicals, soaps, trap crops, bird perches, and traditional preparations - sanitation.
23. Indigenous technical knowledge (ITK) in organic agriculture - rationale and principles - general, indigenous practices for soil, nutrient, weed, water pest and disease management in farming - ITK's in farmers practice.
24. Benefits and problems in organic farming.

25. Organic farming; Promotional activities; role of government and NGO's - action plan - policy considerations.
26. Economic evaluation of organic production systems - cost - benefit analysis and comparison with conventional systems.
27. Organic certification - procedures - certification agencies in India - labeling, marketing and export opportunities.
28. Crop production standards - NPOP guidelines - principles, recommendations and standards - Quality considerations - assessment methods - premium and export opportunities.

Practical Schedule:

24. Resource inventory of organic farm- Soil sampling and analysis for organic carbon and pesticide residues / contaminants.
25. Raising of green manures (Sunnhemp / Daincha / Fodder cowpea).
26. Incorporation of green manure - seed treatment and raising of field crop (Rice / Maize / Cowpea / Cotton / Gingelly).
27. Hands on practice on preparatory cultivation; soil and water conservation methods.
28. Hands on experience on recycling techniques; bio-composting and vermicomposting.
29. Quantification of nutrients from organic sources and application of manures and bio- fertilizers.
30. Exposure visit to an organic farm to learn ITK based preparations.
31. Organic crop production and weed management.
32. Skill development in composting farm residues.
33. Organic crop production and pest management.
34. Exposure visit to bio-control agent (*Pseudomonas*, *Trichoderma* etc.,) production units.
35. Organic crop production and diseases management.
36. Skill development in vermicompost preparation.
37. Hands on training on grading, packaging and post-harvest management.
38. Exposure visit to organic market out lets.
39. Exposure visit to organic certification agencies / Directorate of Organic Certification, Tamil Nadu.

40. Practical Examination

References:

Dahama, A.K.2009. Organic farming for sustainable agriculture, Agrobros publishers.
 SP. Palaniappan and K Annadurai. 2008. Organic Farming: Theory and Practice. 2008. Scientific Publishers.
 Panda, S.C. 2012. Principles and Practices of Organic Farming. Agribios (India), Jodhpur.
 Gehlot, D. 2010. Organic Farming- Components and Management. Agribios (India), Jodhpur.
 Dushyant Gehlot . 2010. Organic farming: Components and management.Agrobios (India), Jodhpur.
 Ranjan Kumar Biswas.2014. Organic farming in India. N.D.Publishers.New Delhi.

E:References:

1. www.ifoam.org
2. www.apeda.org
3. www.cowindia.org
4. www.ncof.org
5. www.earthfooda.co.uk,
6. www.newfarm.org/training

ABT 301 Plant Biotechnology (2+1)

Theory

Unit I Basics of Plant Tissue Culture

Plant tissue culture: Concepts, history and scope - Media and Culture Conditions - Sterilization techniques- Regeneration methods - morphogenesis, organogenesis and embryogenesis - culture types - callus culture and cell suspension culture; shoot tip and meristem tip culture; anther and pollen culture; ovule and embryo culture

Unit II Applied Plant Tissue Culture

Micropropagation - banana and ornamental plants; National certification and Quality management of TC plants- Applications of organ culture - Meristem tip culture (virus free plants) and anther culture (doubled haploids)- Protoplast isolation and fusion- somaclonal variation- synthetic seeds - secondary metabolite production- invitro germplasm conservation

Unit III Basic Molecular Biology

Genome organization- prokaryotes vs eukaryotes- Central dogma of life - Structure of nucleic acids - DNA replication, aminoacids and their classification- genetic codes- transcription, translation and protein synthesis- Structure of a gene, regulation of gene expression, Operon concept- basic techniques in molecular biology-Blotting techniques- Polymerase chain reaction- DNA sequencing methods.

Unit IV Recombinant DNA Technology and Genetic Transformation

DNA manipulation enzymes: Polymerases, restriction endonucleases and ligases - Different types of vectors: plasmids, phagemids, cosmids, BAC - Construction of recombinant DNA molecules- Bacterial transformation - Direct and indirect gene transfer methods in plants: microinjection, electroporation, particle bombardment, *Agrobacterium* mediated method - Tissue specific promoters, selectable and scorable markers, reporter genes- Molecular analysis of transgenic plants – Transgenic plants: herbicide, pest and disease resistant, abiotic stress resistant, nutritional enhancement and traits for improved quality- Detection of GMOs – regulations and biosafety.

Unit V Molecular Marker Technology and Molecular Breeding

DNA markers - hybridization based markers (RFLP) - PCR based markers: RAPD, SSR, AFLP, and SNPs - DNA fingerprinting of crop varieties – Development of mapping populations- linkage and QTL analysis- principles, methods and applications of Marker Assisted Selection in crop improvement- Applications of Plant Genomics and genome databases

Practicals

Biotech Laboratory organization, safety regulations – basics of reagents and solution preparation- Plant tissue culture media preparation- shoot tip culture (rose) - Meristem culture (tapioca)- Micro propagation of banana - Callus culture – Culturing of *E. coli* and determination of growth curve-Isolation of bacterial plasmid DNA- Restriction Digestion and Ligation- Competent cell preparation and Bacterial transformation – confirmation of transformation through colony screening - DNA extraction from plants- Quantification of DNA and quality check through Agarose gel electrophoresis - Molecular marker analysis- DNA fingerprinting using RAPD/SSR markers - NTSys- analysis of diversity in crop plants-Visit to tissue culture units /biotech labs in seed industry/Bt cotton field/tissue culture banana fields

Lecture Schedule

34. Plant tissue culture: Concepts, history and scope
35. Media and Culture Conditions and Sterilization techniques
36. Regeneration methods - morphogenesis, organogenesis and embryogenesis
37. Culture types - callus culture and cell suspension culture; shoot tip and meristem tip culture
38. Anther and pollen culture; ovule and embryo culture
39. Micropropagation - banana and ornamental plants
40. National certification and Quality management of TC plants
41. Meristem tip culture (virus free plants) and anther culture (doubled haploids)
42. Protoplast isolation and fusion- somaclonal variation-synthetic seeds
43. Secondary metabolite production, *invitro* germplasm conservation
44. Genome organization- prokaryotes vs eukaryotes
45. Central dogma of life - Structure of nucleic acids
46. DNA replication- Mechanism
47. Transcription and Post transcriptional processing - RNA splicing
48. Translation - Amino acids and their classification, genetic codes and protein synthesis
49. Concept and structure of a gene- classical and modern concept

50. Mid semester Examination

51. Regulation of gene expression, Operon concept
52. Blotting techniques and Polymerase chain reaction
53. DNA sequencing methods
54. DNA manipulation enzymes: Polymerases, restriction endonucleases and ligases
55. Different types of vectors: plasmids, phagemids, cosmids, BAC
56. Construction of recombinant DNA molecules- Bacterial transformation
57. Direct and indirect gene transfer methods in plants: microinjection, electroporation, particle bombardment, *Agrobacterium* mediated method
58. Tissue specific promoters, selectable and scorable markers, reporter genes, Molecular analysis of transgenic plants
59. Transgenic plants: herbicide, pest and disease resistant, abiotic stress resistant,
60. Transgenic plants: nutritional enhancement and traits for improved quality
61. Detection of GMOs – regulations and biosafety.
62. DNA markers - hybridization based markers (RFLP) - PCR based markers: RAPD, SSR, AFLP, and SNPs
63. DNA fingerprinting of crop varieties
64. Development of mapping populations
65. Linkage and QTL analysis
66. Principles, methods and applications of Marker Assisted Selection in crop improvement
67. Applications of Plant Genomics and genome databases

Practical schedule

- 20 Biotech Laboratory: Organization and Safety Regulations
- 21 Basics of Reagents and Solution Preparation
- 22 Plant Tissue Culture Media Preparation
- 23 Shoot Tip Culture of Rose
- 24 Meristem Tip Culture of Tapioca
- 25 Micropropagation of Banana
13. Callus Culture
14. Isolation of Bacterial Plasmid DNA
15. Restriction Digestion and Ligation
16. Competent Cell Preparation and Bacterial Transformation
17. Confirmation of Transformation through Colony Screening

18. Genomic DNA Extraction from Plants
19. Quantification of DNA and Quality Check through Agarose Gel Electrophoresis
20. DNA Fingerprinting using PCR
21. NTSys- Analysis of Diversity in Crop Plants
22. Visit to Tissue Culture Units /Biotech Lab in Seed Industry/Bt Cotton Field – Lateral Flow Strip Assay
23. **Final Practical Examination**

Reference

- Chawla, H S. 2009. **Introduction to Plant Biotechnology (3/e)**. CRC Press , London. 730 P
ISBN 9781578086368
- Boopathi, N. M. 2013. Genetic Mapping and Marker Assisted Selection - Basics, Practice and Benefits. Springer Publications
- Brown, T. A. 2010. **Gene Cloning and DNA Analysis: An Introduction**, 6th Edition, Wiley-Blackwell- Companion site (Chapters 1 to 12 and 15)
- Chahal, G. S. and Gosal, S. S. 2003. *Principles and Procedures of Plant Approaches Breeding Biotechnological and Conventional*. Narosa Publishing House, New Delhi
- Dubey, R. C. 2014. A textbook of Biotechnology. 5th revised Edn. S. Chand Publications. New Delhi
- George, E. F, Hall M. A. and Geert-Jan De Klerk. 2009. **Plant Propagation by Tissue Culture**, 3rd Edition, Springer, The Netherlands. 501p.
- Gupta, P. K. , 2015. *Elements of Biotechnology* 2nd Edn. Rastogi and Co. , Meerut.
- Neal Stewart, Jr. C. 2008. Plant Biotechnology and Genetics: Principles, Techniques and Applications John Wiley & Sons, Inc ISBN: 978-0-470-04381-3
- Nelson, D. S. and M. M. Cox. 2012. **Lehninger's Principles of Biochemistry. Sixth edition. Chapters- 1,3,8,9,25,26,28** (weblinks, tutorials and lecture companion art) W. H. Freeman and
- Razdan M K, 2014. *Introduction to plant Tissue Culture* 2nd Edn. Science Publishers, inc. USA.
- Singh, B. D. 2012. *Plant Biotechnology*. Kalyani publishers, Ludhiana
- Tomar, R. S. , Parakhia, M. V. , Patel, S. V. and Golakia, B. A. , 2010. *Molecular markers and Plant Biotechnology*, New Publishers, New Delhi.
- Xu, Y 2010. **Molecular Plant Breeding**. International Maize and Wheat Improvement Centre (CIMMYT). 752 Pages

E- Reference

1. <http://www.isaaa.org/india/-> Briefs
2. <http://www.nal.usda.gov>.
3. <http://www.agbiotechnet.com>.
4. <http://www.agbioworld.org>
5. <http://www.cropgen.org>.
6. <http://www.agbiosafety.unl.edu/>.

PBG 302 Crop Improvement (2+1)

THEORY

Unit I: Cereals

Place of origin – putative parents – related wild species – breeding objectives–breeding methods–conventional and innovative methods-heterosis breeding and important varieties in following cereals: Rice, Wheat, Maize, Sorghum, Pearl millet, Finger millet

Unit II: Pulses and Oilseeds

Place of origin – putative parents – related wild species – breeding objectives–breeding methods–conventional and innovative methods-heterosis breeding and important varieties in following crops
Pulses: Redgram, Bengal gram, Greengram, Blackgram, Cowpea, Soybean. Oilseeds: Groundnut, Sunflower, Gingelly, Castor, Rape and Mustard.

Unit III: Cash crops, Fodder and Horticultural crops

Place of origin – putative parents – related wild species – breeding objectives–breeding methods–conventional and innovative methods-heterosis breeding and important varieties in following crops

Fibres: Cotton; Sugars: Sugarcane; Starch: Potato; Fumitories: Tobacco, Fodder: Guinea grass, Napier, Cumbu – Napier, Lucerne, *Stylosanthes*; Horticultural crops: Bhendi, Tomato, Brinjal, Papaya, Banana

Unit IV: Breeding for Biotic and Abiotic stresses and Quality

Breeding for insect resistance – mechanisms, basis, genetics of insect resistance - suitable breeding methods- merits and demerits of resistance breeding; Breeding for disease resistance – horizontal and vertical resistance- Gene for gene hypothesis – mechanisms, genetics of disease resistance; Suitable breeding methods for disease resistance- exploitation of vertical resistance in plant breeding- multilines, gene pyramiding, gene deployment.

Breeding for Abiotic stress – drought – mechanisms, basis, genetics of drought resistance - suitable breeding methods -limitations of drought resistance breeding; Breeding for Abiotic stress – salinity and alkalinity;

Breeding for quality traits- Important quality traits in different crops- nutritional quality of cereals and pulses-Genetics of nutritional traits-breeding methods- Breeding for low toxic substances- limitations of breeding for enhanced nutritional quality

Unit V: Hybrid seed production techniques and ideotype breeding

Hybrid seed production techniques in rice, maize and redgram

Ideotype breeding- main features-difference between traditional and ideotype breeding- - crop ideotypes in rice, wheat, cotton- steps in ideotype breeding- merits and demerits of ideotype breeding

PRACTICAL

Observation on floral biology – anthesis and pollination – selfing – crossing techniques – observation on cultivated germplasm, wild species – Experimental design – handling segregating generations- Yield trials in following crops- Rice, Maize and Sorghum, Pearl millet and Finger millet, Redgram, Bengal gram, Green gram, Black gram, Cowpea and Soybean, Groundnut and Sunflower, Sesame and Castor, Cotton, Sugarcane, Guinea grass, Cumbu – Napier hybrids, Lucerne and *Stylosanthes*, Bhendi, Brinjal, Tomato, Papaya and Banana, Study of quality characters in rice, Study of donor parents for different characters,

General seed production techniques in field crops, Visit to AICRP and seed production plots of different field crops

Lecture schedule

Place of origin – putative parents – related wild species – breeding objectives–breeding methods– conventional and innovative methods-heterosis breeding and important varieties in following crops:

27. Cereals: Rice.
28. Cereals: Rice.
29. Cereals: Rice.
30. Cereals: Wheat
31. Cereals : Maize
32. Cereals: Sorghum
33. Cereals: Pearl millet, Finger millet,
34. Pulses: Redgram
35. Pulses: Greengram, Blackgram,
36. Pulses: Soybean, Bengal gram
37. Pulses: Cowpea
38. Oilseeds: Groundnut
39. Oilseeds: Gingelly, Rapeseed and Mustard
40. Oilseeds: Castor and Sunflower
41. Fibres: Cotton
42. Sugars: Sugarcane
43. **Mid Semester Examination.**
44. Starch: Potato
45. Fumitories: Tobacco
46. Forage grasses and legumes: Guinea grass, Napier, Cumbunapierhybrid, Lucerne, *Stylosanthes*
47. Breeding for sexually propagated horticultural crops- Bhendi, Tomato
48. Breeding for sexually propagated horticultural crops- Brinjal, Papaya
49. Breeding for clonally propagated horticultural crops- Banana
50. Breeding for insect resistance – mechanisms, basis, genetics of insect resistance- suitable breeding methods- merits and demerits of resistance breeding
51. Breeding for disease resistance –horizontal and vertical resistance- Gene for gene hypothesis – mechanisms, genetics of disease resistance
52. Suitable breeding methods for disease resistance- exploitation of vertical resistance in plant breeding- multilines, gene pyramiding, gene deployment.
53. Breeding for Abiotic stress – drought – mechanisms, basis, genetics of drought resistance - suitable breeding methods-limitations of drought resistance breeding
33. Breeding for Abiotic stress – salinity and alkalinity
34. Breeding for quality traits- Important quality traits in different crops- nutritional quality of cereals and pulses-
35. Genetics of nutritional traits-breeding methods- Breeding for low toxic substances-limitations of breeding for enhanced nutritional quality
36. Hybrid seed production techniques in rice
37. Hybrid seed production techniques in maize
38. Hybrid seed production techniques in redgram

39. Ideotype breeding- main features-difference between traditional and ideotype breeding- - crop ideotypes in rice, wheat, cotton- steps in ideotype breeding- merits and demerits of ideotype breeding

Practical schedule

Observation on floral biology – anthesis and pollination – selfing – crossing techniques – observation on cultivated germplasm, wild species – Experimental design – handling segregating generations- Yield trials in following crops.

27. Rice
28. Maize and Sorghum
29. Pearl millet and Finger millet
30. Red gram, Bengal gram and Soybean
31. Green gram, Black gram and Cowpea
32. Groundnut and Sunflower.
33. Sesame and Castor
34. Cotton
35. Sugarcane
36. Guinea grass, Cumbu – Napier hybrids Lucerne and *Stylosanthes*
37. Bhendi, Brinjal, Tomato
38. Papaya and Banana
39. Study of quality characters in rice
40. Study of donor parents for different characters
41. General seed production techniques in field crops
42. Visit to AICRP and seed production plots of different field crops

43. Final Practical Examination

References

- ❖ Singh, B.D. 2007. Plant breeding - Principles and methods.
- ❖ Phundan Singh. 2015. Essentials of Plant Breeding. Kalyani Publishers, New Delhi
- ❖ Harihar Ram and HariGovind Singh, 1994. Crop breeding and Genetics. Kalyani Publishers, New Delhi.
- ❖ D.N.Bharadwaj.2012. Breeding Field Crops. Agrobios (India), Jodhpur - 342002
- ❖ HariHar Ram, 2011. Vegetable Breeding– Principles and Practice, Kalyani Publishers, New Delhi.
- ❖ N.Kumar.2006. Breeding of horticultural crops- Principles and Practices. New India Publishing Agency. New Delhi
- ❖ D.A.Sleper and J.M.Poehlman. 2007. Breeding Field Crops. Blackwell Publishing Professional (USA)
- ❖ H.H.Ram. 2011. Crop Breeding and Biotechnology. Kalyani Publishers (India)
- ❖ Chopra, V.L. 1990. Plant Breeding. Theory and Practice. Oxford and IBH Publishing Co., New Delhi.
- ❖ Daniel Sundararaj, D., G.Thulasidas, and M. Stephan Dorairaj. 1997. Introduction to Cytogenetics and Crop improvement. Popular Book Depot, Chennai - 15.
- ❖ Sharma, J.R. 1994. Principles and practice of Plant Breeding. Tata McGraw - Hill Publishing Co. Ltd., New Delhi.



Singh, R.B., R.M. Singh and B.D. Singh, 1984. Advances in Cytogenetics and crop improvement. Kalyani Publishers, New Delhi.

E- References

1. www.cimmyt.org
2. www.nbpgri.nic.in
3. www.irri.org
4. www.icrisat.org

Optional courses for 2 credits			
S. N.	Code No.	Courses	Credit Hours
1.	CRP 351	Physiological Techniques in crop production	2 (1+1)
2.	SAC351	Designer fertilizer Production	2 (1+1)
3.	SAC352	Rejuvenation of Deteriorated lands	2 (1+1)
4.	SAC353	Soilless crop production	2 (1+1)
5.	SAC354	Instrumental methods of analysis	2 (1+1)
6.	SST 351	Seed entrepreneurship skill development and management	2 (1+1)
7.	AGR 351	Weed and water management	2 (1+1)
8.	PGP 351	Plant Genetic Resources Collection, Conservation and Utilization	2 (1+1)
9.	NEM 351	Commercial Production of Nematode Antagonistic bio-agents	2 (1+1)
10.	AGM351	Downstream Processing for Industrially Important Microbial Products	2 (1+1)
11.	AGM 352	Microbial Enzymes	2 (1+1)
12.	AGM 353	Microbial Quality and Safety of Foods	2 (1+1)
13.	AGM 354	Plant –Microbe Interaction	2 (1+1)
14.	AGM 355	Quality Control of Bio-inoculants	2 (1+1)
15.	SAC 355	Crop and Pesticide Chemistry	2 (1+1)

CRP 351 Physiological Techniques in crop production (1+1)

Unit I Stress physiology: Abiotic Stress: Water deficit – impact of drought on crop productivity - characteristic features of drought tolerant plants, Drought resistance and tolerance mechanisms, Osmotic adjustment and osmoregulation, stress proteins, Water use efficiency – carbon isotope discrimination-Physiological traits associated with drought and Mitigation techniques. Flooding: Physiological mechanism of adaptation, Mitigation techniques, Temperature stress: High and Low Temperature, Tolerance mechanisms - Functions of HSPs and CSPs, Oxidative stress: Reactive Oxygen Species (ROS). Role of scavenging systems – Enzymatic and Non-Enzymatic, Physiological traits associated with high and low temperature, Mitigation techniques. Salt stress: Physiological basis of tolerance mechanisms, Physiological traits associated with salt stress, Mitigation techniques. UV stress and climate change: Physiological adaptation of crops to UV radiation and tolerance mechanisms.

Unit II Nutriophysiology: Diagnosis and correction measures for nutritional disorders including Macro, Micro and secondary nutrients in Cereals, Pulses, Oilseeds, Fibre and Sugar crops; Fruits, Vegetables, Flowers, Spices, Plantation and Aromatic crops. Impact of heavy metals on physiology and productivity of crops, Phytoremediation, Importance of beneficial elements – Na, Si, Se, Co.

Unit III Hormonal physiology: Role of hormones in plant growth and yield enhancement, stress management and quality improvement - Auxins, Gibberellins, Cytokinins, Absciscic acid, Ethylene and Brassinosteroids. Role of other phytohormones in crop production-triacontanol, polyamines, jasmonates and salicylic acid, New generation PGRs - 1- MCP, Triazoles, strigalactone, pro-hexadione Ca.

Unit IV Production Physiology: Physiological limitations of crop productivity, Physiological and genetic basis of crop environment interaction, Plant architecture – Ideotype concept, Crop photosynthetic efficiency – C_3 , C_4 and CAM Strategies to improve the crop photosynthesis, Source- sink balance and Harvest Index.

Unit V Postharvest physiology: Environmental factors influencing senescence, ripening and postharvest life of fruits, flowers, vegetables and seeds. Physiological and biochemical aspects of senescence and fruit ripening. Regulatory role of ethylene in senescence and ripening, Pre and post harvest measures to influence shelf life.

Theory Lecture schedule

- 26 Classification of abiotic stresses - Drought – types-Drought resistance and tolerance mechanisms- adaptations-Physiological traits associated with drought -osmotic adjustment.
- 27 Reactive Oxygen Species- scavenging enzymes- stress proteins-water use efficiency – carbon isotope discrimination- concept -mitigation techniques.
- 28 Flooding - Physiological mechanism of adaptation- physiological traits associated with flooding- Role of ethylene.
24. Temperature stress-High and Low Temperature -Tolerance mechanisms-Functions of HSPs and CSPs - Physiological traits associated with high and low temperature.
25. Salt stress - Physiological basis of tolerance mechanisms –adaptations- Physiological traits associated with salt stress.
26. Physiological adaptation of crops to high and low light and UV radiation.
27. Impact of heavy metals on physiology and productivity of crops –Phytoremediation.
- 28. Mid Semester examination**
29. Diagnosis and correction measures for nutritional disorders in Cereals, Millets, Pulses, Oilseeds.

30. Diagnosis and correction measures for nutritional disorders in Fibre, Sugar crops, Fruits and Vegetable, Flowers, Spices, Plantation, Medicinal and Aromatic crops.
31. Importance of beneficial elements – Na, Si, Se, Co- Effect of crop specific application of beneficial elements.
32. Role of hormones in plant growth and yield enhancement, stress management and quality improvement – Auxins, Gibberellins and Cytokinins.
33. Role of hormones in plant growth and yield enhancement, stress management and quality improvement –Absciscic acid, Ethylene and Brassinosteroids.
34. Role of other phytohormones -triacontanol, polyamines, jasmonates and salicylic acid. New generation PGRs - 1- MCP, Triazoles, strigalactone, pro-hexadione Ca.
35. Physiological limitations of crop productivity, Physiological and genetic basis of crop environment interaction, Plant architecture – Ideotype concept.
36. Crop photosynthetic efficiency – C₃, C₄ and CAM Strategies to improve the crop photosynthesis, Source- sink balance and harvest index.
37. Environmental factors influencing senescence and ripening-Physiological and biochemical aspects of senescence and fruit ripening-Factors affecting post harvest life of fruits- measures for enhancing the shelf life of fruits, vegetables and flowers.

Practical schedule

1. Determination of osmotic potential
2. Gas Exchange measurements using Infra Red Gas Analyzer
3. Stress induction response techniques
4. Water Use Efficiency or Transpiration Efficiency of crops
5. Estimation of anti oxidant enzymes activity – Catalase and peroxidase
6. Estimation of Membrane thermal stability
7. Bioassay for Cytokinins
8. Effect of PGRs on flowering
9. Effect of PGRs on fruit ripening
10. Influence of ABA on stomatal regulation
11. Diagnosis of nutritional disorders and their amelioration measures
12. Nutrient application techniques
13. Estimation of Macro nutrients in plant samples
14. Estimation of micro and secondary nutrients using AAS
15. Field Visit I- Diagnosis of nutrient disorders
16. Field Visit II-Postharvest storage facilities
- 17. Final practical Examination**

References

11. Barker AB & Pilbeam DJ. 2007. Handbook of Plant Nutrition. 2nd Edition, CRC Press, Taylor & Francis Group. Pp 1-773.
12. Bewley, J. D. and Black, M. 1985. Seed Physiology of Development and Germination. Plenum Publishing Corp. New York, NY. pp 70-73.
13. Bleasdale, J. K. A. 1984. Plant Physiology in Relation to Horticulture. 2nd Ed. MacMillan & Avi Publishing Company, USA.
14. Leopold, A. C. and Kriedemann, P. E. 1985. Plant Growth and Development. 3rd Ed. McGraw-Hill, New York, pp 545.
15. [Madhava Rao KV](#), [Raghavendra, AS](#) and [Janardhan Reddy K](#). 2006. Physiology and Molecular Biology of Stress Tolerance in Plants. Springer publishers, Netherland.
16. Taiz. L. and Zeiger. E. , 2015 (Sixth edition). Plant Physiology and Development. Publishers: Sinauer Associates, Inc. , Massachusetts, USA.

17. Wilkins MB. 1969. Physiology of Plant Growth and Development. Tata McGraw-Hill, New York, pp 127-162.

E-REFERENCES

1. <http://www.plantphys.net>
2. <http://www.plantstress.com>
3. <http://www.ipni.net>
4. <http://www.edis.ifas.ufl.edu>
5. <http://www.greenair.com/plantlnk.htm>
6. <http://www.tvdsb.on.ca>

AGR 351 WEED AND WATER MANAGEMENT (1+1)

Unit - I : Weeds: Introduction, Definitions; harmful and beneficial effects, classification,; crop weed competition and allelopathy. Methods of weed control: physical, cultural, chemical and biological methods. Integrated weed management.

Unit – II: Herbicides - Classification, characteristics, formulations, methods of application; advantages- Weed management in major field crops - aquatic and problematic weeds and their control.

Unit – III: Role of water in plant growth - Importance of irrigation- Soil - water - plant relationship - Soil Plant Atmospheric Continuum (SPAC) - Hydrological cycle - Soil water movement - soil moisture constants - Moisture extraction pattern - Absorption of water.

Unit IV: Crop water requirement- Factors affecting water requirement- Factors affecting water requirement. Scheduling of irrigation – Water use efficiency

Unit V: Methods of irrigation: surface, sub-surface sprinkler and drip irrigation – Micro irrigation: layout, suitability, merits and scope.

Practical

Identification of weeds in wetlands, gardenland and drylands – Practicing different methods of weed control -Calculations on weed control efficiency and weed index; Classification and characteristics of herbicides - Computation of herbicide doses- Study of herbicide application equipment and calibration; Demonstration of methods of herbicide application;

Estimation of soil moisture - Measurement of irrigation water through water measuring devices (flumes, weirs and water meter) - Calculation of irrigation water requirement (problems)- Acquiring skill in land shaping for different surface irrigation methods - Operation and economics of drip and sprinkler irrigation systems - Estimation of crop water requirement - Irrigation efficiency (problems) –

Lecture Schedule :

5. Weeds - Definition, classification
6. Characteristics of weeds, harmful and beneficial effect of weeds.
7. Crop weed interactions - Critical crop weed competition, competitive and allelopathic effects of weeds and crops.
8. Methods of weed control: physical, cultural
9. Methods of weed control: chemical and biological methods. Integrated weed management.
10. Herbicides - Classification, characteristics, formulations,
11. Methods of herbicide application
12. Weed management in major field crops - aquatic and problematic weeds and their control.
- 13. Mid semester examination**
14. Role of water in plants - Importance of irrigation
15. Soil - Plant -water relationship - Soil-plant-atmospheric continuum - Hydrologic cycle - absorption of water and evapotranspiration.
16. Soil water movement - saturated and unsaturated flow and vapour movement - soil moisture constants and their importance in irrigation.
17. Available soil moisture - definition and importance - moisture extraction pattern
8. Crop water requirement - factors affecting crop water requirement -Critical stages for irrigation - water requirement for different field crops.
9. Scheduling of irrigation - Different approaches- Water use efficiency
10. Methods of irrigation - surface (flooding, beds and channels, border strip, ridges and furrows, broad bed and furrows, surge irrigation) and sub-surface methods.

11. Micro irrigation system (drip and sprinkler irrigation) - suitability, components, layout, operation, advantage and disadvantage.

Practical schedule:

12. Identification, classification and characterization of wetland weeds.
13. Identification, classification and characterization of gardenland and dryland weeds.
14. Practicing skill development on cultural and non chemical weed management.
15. Identification, classification and characterization of herbicides.
16. Practicing skill development on herbicide application techniques.
17. Practicing Skill development on spray equipment's and spray fluid calibration.
18. Calculation of herbicide quantity and recommendation for different eco systems
19. Working out weed index, weed control efficiency and economics in weed management studies.
20. Estimation of soil moisture by gravimetric method and tensiometer.
21. Estimation of soil moisture by resistance blocks and neutron probe and other improved devices.
22. Measurement of irrigation water with flumes and weirs.
23. Calculation of irrigation water based on source, water flow, soil moisture status and depth of irrigation.
24. Land leveling and land shaping - Beds and channels - ridges and furrows.
25. Layout, operation and maintenance of drip and sprinkler irrigation systems.
26. Estimation of crop water requirement by direct and indirect methods.
27. Calculations on irrigation efficiency parameters.
- 28. Practical examination**

References:

- Gupta, O. P. 1998. Modern Weed Management. Agro Botanica Bikaner, India.
- Subramanian, S. A. Mohammed Ali and R. Jayakumar. 1991. All about Weed Control. Kalyani Publishers, New Delhi.
- Jaganathan R. , and R. Jayakumar. 2003. Weed Science Principles, Kalyani Publishers, New Delhi.
- Das, P. C. 2015. Weed Science. New India Publishing Agency, Jaipur
- Michael, A. M. 1997. Irrigation: Theory and Practice Vikas Publishers.
- Sankara Reddy, G. H. and T. Yellamanda Reddy. 1997. Efficient use of irrigation water. Kalyani Publishers
- Reddy, S. R. 2012. Irrigation Agronomy. Kalyani publishers, NewDelhi
- Nirmal Tripathy. 2014. Irrigation, Watershed and Drainage. Agrotech Press, Jaipur
- Gupta, A. K. 2014. Irrigation and Drainage. Oxford Book Company, Jaipur
- Israelsen, O. W. 2012. Irrigation Principles and Practices. Axis Books (India), Jodhpur.
- Michael, A. M. 2009. Irrigation- Theory and Practice, Vikas Publishing House, Noida

E-References:

1. www.tnau.ac.in
2. www.fao.org
3. www.tnau.ac.in/agriportal www.irri.org
4. www.wcc.nrcs.usda.gov/nrcsirrig www.wcc.nrcs.sda.gov/irrig.info.html www.croinfo.net/irrigschedule.html

AGM 351 Downstream Processing for Industrially Important Microbial Products (1+1)

Unit I - Introduction to Bio separation Processes

Role and importance of bioseparations in biotechnological processes. Problems and requirements of bioproduct purification. Cost-cutting strategies. Characteristics of biological mixtures. Classification of bioproducts - Biological activity, Analysis of purity-Process economics- Capital and operating cost analysis

Unit II - Physical Methods of Separation

Centrifugation and filtration. Cell disruption methods; enrichment operations: precipitation methods-with salts, organic solvents, and polymers; extractive separations - aqueous two-phase extraction, supercritical extraction; adsorption methods; membrane based separation theory - types of membranes, types of membrane processes - dialysis; ultrafiltration; microfiltration and reverse osmosis.

Unit III Physical and chemical methods - Isolation of products

Physico-chemical basis of bio-separation processes. Removal of particulate matter, biomass and insolubles: flocculation; sedimentation; adsorption: isotherm, batch, continuous and scale-up of adsorption; extraction: solvent separation, equipment and modes of extraction. Aqueous-two-phase extraction process, supercritical fluid extraction. Precipitation of proteins-methods and scale-up.

Unit IV Isolation of Products - Chromatographic methods of purification

Chromatography - principles, instruments and performance parameters. Paper, TLC, adsorption, gel filtration, reversephase, ion-exchange, hydrophobic interaction, bioaffinity, pseudo affinity chromatographic techniques; GC, HPLC, HPTLC, FPLC, parafusion chromatography and membrane based chromatographic techniques and sample preparation. Electrophoretic separations.

Unit – V Finishing Operations

Products polishing: Crystallization and drying; Purification of antibiotics, amino acids, enzymes and organic acids, proteins, pigments, plant growth hormones, monoclonal antibodies, human growth hormones – Insulin etc.

Practical

Kinetics of a bacterium / yeast in batch culture- doubling time, specific growth rate and growth curve. Batch cultivation of microbes for product formulation; calibration of pH probe, medium preparation, sterilization, and calibration of dO₂ probe. Inoculation of a bacterial / yeast strain in fermentor for batch production of an enzyme / biomass/ ; estimation of kinetic parameters of product formation, biomass production and substrate utilization. Laboratory centrifugation for separation of biomass from fermentation broth. Filtration and microfiltration for solid-liquid separation. Extraction of crude enzyme from microbial biomass and enzyme assay. Production of microbial secondary metabolites and product recovery. Mechanical cell disruption by homogenization: determination of product release kinetics. Sonication of microbial cell suspensions and determination of product release kinetics. Salting out of proteins by precipitation with ammonium sulphate. Precipitation of proteins with acids or bases: determination of isoelectric point. Batch adsorption of an acid on activated charcoal. Solvent extraction and aqueous two-phase extraction of proteins. Gel-filtration chromatography of a protein - salt mixture. Ion-exchange chromatography of a positively charged protein. Gas chromatography for quantitative estimation of volatiles- ethanol. SDS-PAGE of proteins and determination of molecular weight.

Lecture Schedule

12. Role and importance of bio separations in biotechnological processes. Problems and requirements of bioproduct purification
13. Cost- cutting strategies. Characteristics of biological mixtures. Classification of bio products
14. Biological activity, Analysis of purity. Process economics. Capital and operating cost analysis
15. Centrifugation and filtration-Cell disruption methods
16. Enrichment Operations: precipitation methods(with salts; organic solvents and polymers)
17. Extractive separations; aqueous two-phase extraction; supercritical extraction and adsorption methods
18. Membrane based separation theory - types of membranes; types of membrane processes – dialysis, ultrafiltration, microfiltration and reverse osmosis
19. Physico-chemical basis of bio-separation processes. Removal of particulate matter, biomass and insolubles: flocculation; sedimentation
- 20. Mid semester evaluation**
21. Adsorption – Isotherm, batch, continuous and scale-up of adsorption
22. Extraction: solvent separation, equipment and modes of extraction. Aqueous two-phase extraction process, supercritical fluid extraction.
23. Precipitation of proteins: methods and scale-up. Chromatography - principles, instruments and practice
24. Paper; TLC, adsorption, gel filtration; reverse phase, ion-exchange, hydrophobic interaction, bioaffinity ; pseudo affinity chromatographic techniques
25. GC, HPLC, FPLC HPTLC, parafusion chromatography and membrane based chromatographic techniques and sample preparation. and electrophoretic separation
26. Products polishing. Crystallization and drying. Purification of antibiotics
27. Purification of amino acids, organic acids, enzymes and proteins
28. Monoclonal antibodies; human growth hormones – Insulin etc

Practical Schedule

11. Kinetics of a bacterium / yeast in batch culture: calculation of doubling time, specific growth rate, and plotting of growth curve.
12. Preparation of fermentor and accessories for batch cultivation of microbes: calibration of pH probe, medium preparation, sterilization, and calibration of dO₂ probe.
13. Inoculation of a bacterial / yeast strain in fermentor for batch production of an enzyme / biomass: estimation of kinetic parameters of product formation, biomass production and substrate utilization.
14. Laboratory centrifugation for separation of biomass from fermentation broths. Filtration and microfiltration for solid-liquid separation.
15. Extraction of crude enzyme and assay of enzyme
13. Production of secondary metabolites from microbes and recovery
14. Mechanical cell disruption by homogenization: determination of product release kinetics.
15. Sonication of microbial cell suspensions and determination of product release kinetics.
16. Salting out of proteins by precipitation with ammonium sulphate.
17. Precipitation of proteins with acids or bases: determination of isoelectric point.
18. Batch adsorption of an acid on activated charcoal.
19. Solvent extraction and aqueous two-phase extraction of proteins.
20. Gel-filtration chromatography of a protein-salt mixture.
21. Ion-exchange chromatography of a positively charged protein.
22. Gas chromatography for quantitative estimation of volatiles: ethanol.
23. SDS-PAGE of proteins and determination of molecular weight
- 24. Final Practical Examination**

Reference

1. Sivasankar B. , 2010. Bioseparations: Principles and Techniques, PHI, New Delhi .
2. Stanbury, P. F. , Whitaker, A. and Hall, S. J. 2016. Principles of Fermentation Technology, BH Elsevier Publications, Third Edition.
3. Harrison, R. G. Todd, P. , Rudge, S. R. and Petrides, D. P. , 2003. Bioseparation Science and Engineering, Oxford University Press.
4. Costa, C. A. and Cabral J. S. 1991. Chromatographic and Membrane Processes in Biotechnology, Publisher: Kluwer Academic Publishers, The Netherlands.
5. Harrison et al. 2006. Bioseparation Science and Engineering. Oxford Univ. Press.
6. Nooralabettu Krishna Prasad, 2010. Downstream Process Technology: A New Horizon in Biotechnology, PHI, New Delhi.
7. Jenkins, R. O. 1992. Product Recovery in Bioprocess Technology, Biotechnology by Open Learning Series, Butterworth- Heinemann, London, Second Edition.

AGM 352 MICROBIAL ENZYMES (1+1)

Unit I: Introduction: Introduction and Scope, General distinctive features and industrial applications. Enzymes: Historical perspectives, Nomenclature and classification of enzymes. Isozymes, biological roles, activation energy, chemical nature of enzymes, characteristics of enzymes, 3'D' structure of enzymes, active site, factors affecting enzyme activity, modifiers of enzyme activity, enzyme activators, enzyme inhibitors and allosteric enzymes.

Unit II: Enzyme kinetics: Kinetics of single substrate reactions, Estimation of *Michaelis - Menten* parameters, multi substrate reactions, mechanisms and kinetics, turnover number, types of inhibition, kinetic models: substrate and product inhibition, allosteric regulation of enzymes, *Monod ChangeuxWyman* model, pH and temperature effect on enzymes and deactivation kinetics. Vitamins and their co-enzymes: structure and functions with suitable examples. Metallo enzymes and metal ions as co-factors and enzyme activators.

Unit III: Extraction and purification of microbial enzymes: Importance of enzyme purification, different sources of enzymes. Extracellular and intracellular enzymes. Physical and chemical methods used for cell disintegration. Enzyme fractionation by precipitation - using temperature, salt, solvent, pH, etc. , Liquid-liquid extraction -ultra filtration, ionic exchange, gel chromatography, affinity chromatography and other special purification methods. Enzyme crystallization techniques. Criteria of purity of enzymes.

Unit IV: Industrial applications of enzyme and enzyme engineering: Industrial applications: Microbial enzymes in textile, leather, wood industries and detergents. Enzymes in clinical diagnostics. Enzyme sensors for clinical processes and environmental analyses. Enzymes as therapeutic agents. Enzyme engineering: Chemical modification and site-directed mutagenesis to study the structure-function relationship of industrially important enzymes. Cloning- strain improvement.

Unit V: Enzyme formulation: Physical and Chemical techniques for enzyme immobilization – adsorption - matrix entrapment encapsulation - cross-linking - covalent binding - examples; advantages and disadvantages of different immobilization techniques. Freeze drying and spray drying of immobilized enzymes.

Practical

Amylase production using *Bacillus amyloliquefaciens* and its assay. Protease production by using *Bacillus* isolate and its quantification. Production of cellulase by solid state fermentation (SSF) of rice straw through lignocellulolytic fungi: (a) Estimation of filter paper lyase activity (b) Estimation of carboxy methyl cellulase activity. Production and estimation of xylanase from rice straw through submerged fermentation. Immobilization of microbial cells for enzyme production. Protease production from *Bacillus subtilis* using soybean meal. Purification of fungal α - amylase or bacterial protease by fractionation, chromatographic techniques and electrophoretic separation. Studies on enzyme kinetics of alpha amylase / protease- optimization of parameters viz. , substrate, enzyme concentration, reaction temperature, reaction pH, K_m , V_{max} and metal ions as activators and inhibitors. Enzyme extraction- concentration- ultrafiltration- chromatography- microencapsulation.

Lecture schedule

1. Introduction and scope, general distinctive features and industrial applications.
2. Enzymes: historical perspectives, nomenclature and classification .
3. Isozymes, biological roles, activation energy, chemical nature of enzymes.
4. Characteristics of enzymes, 3'D' structure of enzymes, active site, factors affecting enzyme activity.
5. Modifiers of enzyme activity, enzyme activators, enzyme inhibitors; allosteric enzymes.

6. Kinetics of single substrate reactions, estimation of *Michaelis - Menten* parameters, multi substrate reactions, mechanisms and kinetics; turnover number.
7. Types of inhibition; Kinetic models: substrate and product inhibition; Allosteric regulation of enzymes.
8. *Monod Changeux Wyman* model, pH and temperature effect on enzymes and deactivation kinetics.

9. **Mid Semester Examination**

10. Vitamins and their co-enzymes: structure and functions with suitable examples. Metallo enzymes and metal ions as co-factors and enzyme activators.
11. Importance of enzyme purification, different sources of enzymes. Extracellular and intracellular enzymes.
12. Physical and chemical methods used for cell disintegration. Enzyme fractionation by precipitation - using temperature, salt, solvent, pH, *etc.* , liquid-liquid extraction.
13. Ionic exchange, gel chromatography, affinity chromatography and other special purification methods. Enzyme crystallization techniques. Criteria of purity of enzymes.
14. Industrial applications: Microbial enzymes in textile, leather, wood industries and detergents. Enzymes in clinical diagnostics. Enzyme sensors for clinical processes and environmental analyses. Enzymes as therapeutic agents.
15. Enzyme engineering: Chemical modification and site-directed mutagenesis to study the structure-function relationship of industrially important enzymes. Cloning- strain improvement.
16. Physical and Chemical techniques for enzyme immobilization – adsorption - matrix entrapment encapsulation - cross-linking - covalent binding - examples;
17. Advantages and disadvantages of different Immobilization techniques - overview of applications of immobilized enzyme systems

1. &2. Amylase production using *Bacillus amyloliquefaciens* in submerged and solid state fermentation and its assay.
3. & 4. Protease production using *Bacillus* isolate and its quantification.
5. Production of cellulase by solid state fermentation (SSF) of rice straw through lignocellulolytic fungi - Estimation of filter paper lyase activity.
6. Production of cellulase by SSF of rice straw through lignocellulolytic fungi- Estimation of carboxy methyl cellulase activity.
7. Production and estimation of xylanase from rice straw through submerged fermentation.
8. Immobilization of microbial cells for enzyme production.
9. Protease production from *Bacillus subtilis* using soybean meal.
10. -12. Purification of fungal alpha-amylase or bacterial protease by fractionation, chromatographic techniques and electrophoretic separation.
- 13-14. Studies on enzyme kinetics of alpha amylase / protease - optimization of parameters *viz.* , substrate, enzyme concentration, reaction temperature, reaction pH, K_m , V_{max} and metal ions as activators and inhibitors.
- 15-16. Enzyme extraction- concentration-ultrafiltration-chromatography- microencapsulation

J **Final Practical**

Reference

1. Alexander N. Glazer, Hiroshi Nikaido. 2007. Microbial Biotechnology, Second Edition, ISBN 9780521842105, Cambridge University Press.
- JJ Trevor Palmer and Philip L. Bonner, 2004. Enzymes: Biochemistry, Biotechnology, Clinical Chemistry, East- West Press.
- B Shuler, M. L. and F. Kargi, 2002. Bioprocess Engineering : Basic Concepts, Second Edition, Pearson.
- C Blanch, H. W and D. S. Clark, 1997. Biochemical Engineering, Marcel & Dekker, Inc. ,.
3. Bailey, J. E and D. F. Ollis, 1986. Biochemical Engineering Fundamentals, Second Edition, McGraw-Hill.
- B Nicholas C. Price and Lewis Stevens, 1982. Fundamentals of Enzymology, Oxford University Press.
- C Alan Wiseman, 1999. Handbook of Enzyme Biotechnology, Third Edition, Ellis Harwood Publications.

E-Reference: James Lee, M. , Biochemical Engineering, PHI, USA, e-Book Version 2. 1, 2002.

AGM 353 MICROBIAL QUALITY AND SAFETY OF FOODS (1+1)

Unit I Introduction: Food safety risks and assessment– characteristic features of spoilage - significance of spoilage of different groups of foods - cereal and cereal products, vegetables and fruits, meat, poultry, sea foods, milk and milk products, packed and canned foods.

Unit II Food borne infections and intoxications: Food borne pathogens- food infections and intoxications of food borne diseases – bacteria, fungal, protozoa and viral. Investigation and management of food borne diseases.

Unit III Techniques for detection of pathogens and toxins : Advanced laboratory techniques for food-borne pathogens: principle, working and application of GC-MS, HPLC, LC/MS, inductively coupled Plasma Mass Spectroscopy, TOF and PCR-real time PCR. DGGE. Metagenomics, proteomics and immunological methods

Unit IV Food standards: Food standards – HACCP concepts, principles – EU, FDA and WHO standards - CODEX – Food Law & regulations - HACCP principles & applications – GM foods – SSOP – GMP & GAP – Food traceability- Food Audit - functions, duties and responsibilities of food safety regulators.

Practical

Food sampling procedures – Preparation & plan - Examination of microorganisms of by aerobic plate count -Assessing *Bacillus cereus*, coliforms, *Campylobacter*, *Salmonella* and *Staphylococcus* in various food samples. Microbiological examination of canned foods. Rapid detection of pathogens and toxins in foods– HACCP of fruits and vegetables - visit to food processing industry.

Lecture Schedule

18. Food Safety Risks - biological, chemical, physical risks; risk assessment.
19. Physical and chemical changes during food spoilage.
20. Significance of microbial spoilage of different groups of foods - cereals, vegetables and fruits, packed and canned foods.
21. Significance of microbial spoilage of different groups of foods-meat, poultry, sea foods, milk and milk products.
22. Food borne pathogens – food infection – intoxication - bacterial food borne diseases – viral and protozoa.
23. Mycotoxicoses in foods – occurrence and economic significance and food control measures.
24. Analytical techniques and their working principles for the detection of toxin in foods – GC-MS, HPLC and HPTLC.
25. Analytical techniques and their working principles for the detection of toxin in foods -LC/MS, ICP – MS and TOF.

26. Mid Semester Examination

27. Molecular Detection of Food borne Pathogens - PCR, real time PCR, DGGE, metagenomics, proteomics and immunological methods
- (e) Food standards – India, EU, FDA and WHO Standards of Food Safety
- (f) Food laws & Regulations - CODEX
- (g) HACCP concepts, principles and applications
- (h) GM foods and current guidelines for production and labeling
- (i) Food Safety – Standard Sanitation Operating Procedures; GMP, GAP for food safety
- (j) Food traceability– significance - Food safety Audit
- (k) Functions, duties and responsibilities of food safety regulators

Practical schedule

12. Different sampling plan in food and preparation for various foods
13. Determination of aerobic plate counts in food
14. Enumeration of yeasts and molds in food
15. Detection and confirmation of *Bacillus cereus* in food
16. Microbiological hazard analysis in processed fruit product
17. Microbiological hazard analysis in water - water quality control
18. Microbiological hazard analysis in fresh vegetables – *Campylobacter* and *Salmonella*
19. Microbiological hazard analysis in beverages - Detection of coliforms
20. Microbiological techniques for cereal based food analysis
21. Microbiological hazard analysis in meat products - *Staphylococcus aureus*
22. Rapid detection of toxin producing *Escherichia coli* (STEC) in food products –PCR method
23. Detection of aflatoxin in groundnut and maize kernels
24. Hazard Analysis and Critical Control Point (HACCP) of Fruits / vegetables
25. Hazard Analysis and Critical Control Point (HACCP) of processed foods
26. Visit to Food Processing industry
27. Microbiological examination of canned foods
28. **Final Practical Examination**

Reference Books

1. Adams, M. R. and Moss, M. O. 1995. Food Microbiology. The Royal Society of Chemistry, Cambridge.
2. Frazier, W. C. and Westhoff, D. C. 1988. Food Microbiology. TATA McGraw Hill Publishing Company Ltd. , New Delhi.
3. Jay, J. M. 1987. Modern Food Microbiology. CBS Publishers and distributors, New Delhi.
4. Banwart, G. J. 1989. Basic Food Microbiology. Chapman & Hall New York.
5. Board, R. C. 1983. A Modern Introduction to Food Microbiology. Blackwell Scientific Publications, Oxford.
6. Robinson, R. K. 1990. Dairy Microbiology Elsevier Applied Science, London.
7. Hobbs, B. C. and Roberts, D. 1993. Food Poisoning and Food Hygiene. Edward Arnold.
8. Lund B. M. , Baird Parker A. C. , and Gould G. W. 2000. The Microbiological Safety and Quality of Foods. Vol. 1-2, ASPEN Publication, Gaithersberg, MD.
9. Gould G. W. 1995. New Methods of Food Preservation. Blackie Academic and Professional, London.

AGM 354 PLANT-MICROBE INTERACTIONS (1+1)

Unit I Introduction: Introduction to plant - microbe interactions; types of interaction -positive and negative. Plants as microbial habitat. Spermosphere, phyllosphere and endophyticmicroorganisms. Bacterial secretion systems, gene regulation and quorum sensing in bacterial - plant interactions. Role of plant- microbial interactions in soil health and plant growth promotion. Signalling - effect of microbial signalling on plant productivity and plant signalling on microbial diversity and activity in the soil.

Unit II Plant- microbe interaction –I : Beneficial bacteria – *Arthrobacter*, *Azospirillum*, *Azoarcus* *Bacillus*, *Burkholderia*, *Frankia*, *Gluconacetobacter*, *Herbaspirillum*, *Paenibacillus*, *Pseudomonas*, *Rhizobium*, *Streptomyces* and *Xanthomonas* - host interactions and plant growth promotion. *Agrobacterium* induced tumorigenesis and rhizogenesis. *Azolla* - *Anabaena* symbiosis.

Unit III Plant– microbe interaction –II: Mutualistic fungal symbionts. Ectomycorrhizal and endomycorrhizal fungi. Infection processes. Interactions with host plants and other soil microbes. Infection processes- fungi and toxins. Induced resistance. Secondary product responses and fungal virulence. Lichens, algal and bacterial interaction.

Unit IV Applications – Agriculture and industries: Plant growth promoting bacteria-types and mechanism of plant growth promotion. Biocontrol agents- types and mechanism of action. Induced resistance: Protein defenseresponses and systemic responses. Phytoremediation. Industrial application and medicinal applications – antiviral, anti-cancerous, immunosuppressive and antioxidants.

Unit V Techniques to study plant–microbe interactions: Techniques to study plant- microbe interactions - Phytotron, Rhizotron and Confocal Laser Scanning Microscope.

Practical

Collection and assay of root exudates. Characterization of root exudates. Studying the effect of root exudates on selected bacterial population. Isolation of ecto and endorhizosphere microorganisms. Isolation of spermospheremicorganisms from germinating seeds. Isolation and purification of endophytic and phyllosphere microorganisms from rice. Visit to plantations and collection of ectomycorrhizal fruiting bodies, orchidaceous and ericoid mycorrhizal root samples and lichens. Examination of endomycorrhizal infection in orchids and ericaceous plants. Examination of lichens associated with trees. Collection and examination of *Anabaena azollae* associated with *Azolla*. Testing rhizogenesis by *A. rhizogenes* in laboratory conditions.

Lecture Schedule

1. Introduction to plant - microbe interactions; types of interaction - positive and negative. Plants as microbial habitat.
2. Role of microbial diversity in soil health and plant growth promotion.
3. Signalling – effect of microbial signalling on plant productivity.
4. Effect of plant signalling on microbial diversity and activity in the soil.
5. Spermosphere and phyllospheremicroroganisms; endophytic microorganisms.
6. Bacterial secretion systems, gene regulation and quorum sensing in bacterial - plant interaction
7. &8. Beneficial bacteria – *Arthrobacter*, *Azospirillum*, *Azoarcus*, *Bacillus*, *Burkholderia*, *Frankia*, *Gluconacetobacter*, *Herbaspirillum*, *Paenibacillus*, *Pseudomonas*, *Rhizobium*, *Streptomyces* and *Xanthomonas* - host interactions and plant growth promotion
9. **Mid Semester Examination**
10. *Agrobacterium* induced tumour formation and root proliferation; exploitation of tumorigenesis and rhizogenesis. Mutualistic fungal symbionts

11. Ectomycorrhizal and endomycorrhizal fungi. Infection processes. Interactions with host plants and other soil microbes
12. Infection processes - fungi and toxins. Secondary product and fungal virulence
13. Azolla - *Anabaena* symbiosis. Association of lichens with trees
14. Plant growth promoting bacteria- types and mechanism of plant growth promotion
15. Biocontrol agents- types and mechanism of action
16. Induce Resistance-Induced Systemic Resistance (ISR) and Systemic Acquired Resistance (SAR)
17. Protein defense responses and systemic responses

1. Collection of root exudates
2. Assay of root exudates – growth regulator/sugars
3. Studying the effect of root exudates on selected bacterial population – well diffusion assay
4. Isolation and purification of ectorhizosphere – rhizosphere and rhizoplane microorganisms
5. Isolation and purification of endophytic microorganisms from rice
6. Isolation and purification of phyllosphere microorganisms from rice
7. Isolation and purification of *Rhizobium*
8. Isolation and purification of *Frankia* from *Casuarina equisetifolia*
9. Visit to plantations and collection of ectomycorrhizal fruiting bodies, lichens, orchid and ericaceous mycorrhizae infected root samples
10. Examination of ectomycorrhizal fruiting bodies and isolation of ectomycorrhizal fungi
11. Examination of lichens associated with trees; orchidaceous and ericoid mycorrhizae fungal infection in plants
12. Examination of AM fungal infection in plants
13. Collection and examination of AM spores from soil
14. Testing *Agrobacterium tumefaciens* induced tumour formation in dicotyledonous plants in laboratory conditions
15. Testing *A. rhizogenes* induced root proliferation in laboratory conditions
16. Collection and examination of endosymbiont associated with Azolla
17. **Final Practical Examination**

Reference Books

1. Kamal B. , Normand B. and Fouad D. 2009. Plant-microbe interactions.
2. Chrispeels M. J. and D. E. Sadava. 2003. Plants, Genes, and Crop Biotechnology. Jones & Bartlett Publishers, Boston.
3. Susan I. 1992. Fungal-Plant Interactions, Chapman Hall,
4. George A. 2005. Plant Pathology, Academic Press, Fifth Edition.

E - Reference

1. Nautiyal C. S. and Patrice, D. 2008. Molecular mechanism of Plant and Microbe Coexistence, Springer-Verlag Berlin Heidelberg
2. Lugtenberg, B. 2015. Principles of Plant-Microbe Interactions. Microbes for Sustainable Agriculture, Springer International Publishing Switzerland

AGM 355 QUALITY CONTROL OF BIOINOCULANTS (1+1)

UNIT I: Overview of bioinoculant production and quality control. Nitrogen fixers-types. P solubilizers and mobilizers. Potassium releasing bacteria. Microbial transformation of micronutrients zinc, sulphur, iron etc. PPFM – PGPR. Mass production methods. Introduction to quality control standards. FCO standards.

UNIT II: Quality control of *Rhizobium* inoculants-Purity checking of mother culture-morphological, biochemical and cultural characteristics of *Rhizobium*-Cross inoculation groups-recent classification-methods for testing nodulation-Estimation of nitrogen fixation – direct & indirect methods - Carrier based and liquid inoculants - FCO standards-scope for new formulations-quality control at different stages of production.

UNIT III: Quality control of *Azospirillum* and *Azotobacter* inoculants-purity checking of mother culture-morphological, biochemical and cultural characteristics-different species-Nitrogen fixation-ARA and microkjeldhal method-Quality control at different stages of production-Quality control of PSB- Purity checking of mother culture-Morphological, biochemical and cultural characteristics-Quantitative and qualitative assay for P solubilization-FCO standards.

UNIT IV: Endophytic nitrogen fixation in sugarcane-*Gluconacetobacter diazotrophicus*- Potash releasing bacteria-Characteristics-Mechanism of K release – PPFM - PGPR and bioinoculants for micronutrients-Quality control of Mycorrhizae-Ecto and Endo mycorrhiza-AMF-Morphological and cultural characteristics-root infection test and spore count-IP by MPN technique-mechanism of P mobilization-Hairy root organ culture.

UNIT V: Cyanobacterial biofertilizers-types-nitrogen fixing cyanobacteria-Heterocysts-role of akinetes in survival-soil based composite culture-new formulations-quality testing-population by MPN technique-*Azolla-Anabaena* symbiosis-spore inoculum production-rapid methods for quality control of biofertilizers-molecular and immunological methods. Quality of biocontrol agents-*Pseudomonas* and *Trichoderma*.

Practical

Quality control of *Rhizobium*- *Azospirillum*- *Azotobacter*- morphological and biochemical characterization-Nodulation by Roll paper towel technique-N₂ fixation by ARA- sampling methods-population estimation in broth and inoculants by SPC/MPN methods- P solubilizing bacteria- Available P estimation in Pikosviskya's broth - Organic acid production by titrable acidity-Acid and alkaline phosphatase activity - Quantitative estimation- enumeration of population in hydroxy appetite medium-AM fungi- spore count by wet sieving method- root infection studies by staining with trypan blue-IP estimation by MPN- hairy root culture Potassium releasing bacteria-quantification of K release-population estimation- PPFM, PGPR, SOB, Zn solubilizing bacteria-qualitative and quantitative assays-

Cyanobacteria- Composite culture-enumeration of population by haemocytometer count and MPN technique-*Azolla*- determination of heterocyst frequency-sporocarps-spore inoculum production- rapid methods for quality control- Molecular methods –immunological methods.

Lecture schedule

1. Overview of bioinoculant production and quality control
2. Nitrogen fixing microbes-symbiotic , associative symbiotic, nonsymbiotic and endophytic nitrogen fixation-Mechanism of nitrogen fixation
3. Phosphate solubilizing and mobilizing microbes-Mechanism of action

4. Potash releasing bacteria -PPFM-PGPR- Zinc solubilizers- Sulphur oxidizers-mechanism
5. Mass production methods of bacterial, fungal and algal biofertilizers
6. Introduction to quality control standards-BIS-FCO standards - Sampling methods
7. Purity checking of *Rhizobium* mother culture-Morphological, biochemical and cultural characteristics-cross inoculation groups-recent classification
8. *In vitro* methods for testing nodulation- Roll paper towel technique-MPN counts, growth pouches/tubular pots -N₂ fixation-direct & indirect methods –N estimation by Microkjeldahl method, ¹⁵N technique & ARA-principles & methods-enumeration of population at different stages of production

9. Mid Semester Examination

10. Criteria for selection of carrier material-FCO standards (2011) for carrier based and liquid
 11. Purity checking of *Azospirillum* and *Azotobacter* mother culture- morphological, biochemical and cultural characteristics-different species
 12. Phosphobacteria-organisms involved-purity checking of mother culture- morphological, biochemical and cultural characteristics-quantification-available P estimation in broth-organic acid production-acid and alkaline phosphatase
 13. FCO standards for *Azospirillum*, *Azotobacter* and phosphobacteria-quality control of inoculants at different stages of production- Rapid methods for quality control-molecular and immunological methods. Quality of biocontrol agents-*Pseudomonas* and *Trichoderma*.
 14. Potassium releasing bacteria- organisms involved-morphological, biochemical and cultural Characteristics-mechanism and quantification of K release
 15. Endophytic nitrogen fixation-*Glucanacetobacter diazotrophicus*-PPFM, PGPR- bioinoculants for micronutrients viz. , sulphur, zinc and iron
 16. Ecto and Endo mycorrhizae-AM fungi-morphological and cultural characteristics-mechanism of P mobilization-quality control-root infection-spore count and IP by MPN
 17. Nitrogen fixing cyanobacteria-heterocysts-role of akinetes in survival-soil based composite culture-quality checking by MPN technique-*Azolla*-*Anabaena* symbiosis- heterocyst frequency-sporocarps-spore inoculum production-storage and shelf life

Practical schedule

1. Sampling methods and study of different inoculants-carrier based and liquid inoculants
2. Morphological, cultural and biochemical characterization of *Rhizobium*
3. Purity checking of *Rhizobium* strains-Leonard Jar, germination paper roll and tubular pot methods-MPN counts
4. Morphological, cultural and biochemical characterization of *Azotobacter*
5. Estimation of N₂ fixation- Acetylene Reduction Assay of nodules/cultures
6. Estimation of N₂ fixation in broth under *in vitro* conditions-Microkjeldahl method
7. Quantitative and qualitative assay for phosphobacteria-Available P estimation in Pikosviskya's broth by Olsen's method and organic acid production by titrable acidity
8. Assay of acid and alkaline phosphatase activity of phosphobacterial cultures
9. Enumeration of population of *Rhizobium* at different stages of production with Congo red YEMA by SPC method
10. Enumeration of population of *Azospirillum* different stages of production with N free bromothymol blue (Nfb) semi solid medium by MPN method
11. Enumeration of population of *Azotobacter* at different stages of production with Waksman No. 77 medium by SPC method
12. Enumeration of population of phosphobacteria at different stages of production with Sperber' shydroxy appetite medium by SPC method

13. Quality control of AM fungi-Spore count by wet sieving, root infection by Tryphan blue staining and Infective propagules by MPN
14. Potassium releasing bacteria- quantification of K release
15. Cyanobacterial inoculants-microscopic examination, determination of heterocyst frequency& population estimation by haemocytometer count and MPN method
16. **Final Practical Examination**

Reference Books

1. Maheshwari, D. K. and R. C. Dubey, 2008. Potential Microorganisms for Sustainable Agriculture - A techno-Commercial Perspective. I. K. International Publishing House Pvt. Ltd. , New Delhi and Bangalore
2. Rai, M. K. 2006. Hand book of microbial biofertilizers. CRC press.
3. Bagyaraj, D. J. and A. Manjunath. 1990. Mycorrhizal symbiosis and plant growth, Univ. of Agricultural Sciences, Bangalore, India.
4. Kannaiyan S. 2002. Biotechnology of Biofertilisers. Kluwer Academic publishers & Narosa Publishing House.
5. Motsara M. R. , Bhattacharya P. , and Srivastava B. 1995. In: Biofertilizer Technology, Marketing and Uses -A Source Book cum Glossary. Fertilizer Development and Consultancy Organization, New Delhi.
6. Somasegaran, P. and H. J. hoben. 1985. Methods in Legume-*Rhizobium* Technology, NifTAL Project and MIRCEN, University of Hawaii, Paia, USA
7. SubbaRao, N. S. 1993. Biofertilizers in Agriculture and Forestry. Oxford and IBH Publishing Co. Ltd. , New Delhi.

E- Reference

1. Biofertilizers and Organic Fertilizers in Fertilizer (Control) Order, 1985
2. FNCA biofertilizer project group, 2006. Biofertilizer Manual, Japan Atomic Industrial Forum, MEXT, Tokyo, Japan.

SAC 351DESIGNER FERTILIZER PRODUCTION (1+1)

Theory- Syllabus

UNIT I

Designer Fertilizers - Definitions - Concepts - Historical development - Scope and Need - Scenario of Multi nutrient deficiencies in soils and plants.

UNIT II

Designer Fertilizers – Classification- Production and Characterisation - Speciality / Customized fertilizer mixtures - Fortified fertilizers - Pelleted fertilizers - Multi nutrient liquid formulations.

UNIT III

Foliar Formulations - Leaf nutrient analysis -Value added fertilizers- enriched with organics/chelates- methods and guidelines for preparing designer Fertilizers- Filler materials- Industries and approved formulations.

UNIT IV

Quality of Designer fertilizers- Compatibility of fertilizer materials - issues in storability, hygroscopicity, clogging, etc. - Toxicity - Advantages and Disadvantages - Key Challenges - Crop response to designer fertilizers - Agricultural, Horticultural, high value crops- yield and quality - Soil health - Nutrient use efficiencies

UNIT V

Feasibility of using designer fertilizers for drip fertigation- Poly houses - roof gardening- Quality Standards- Specifications - Guidelines for Patenting, Licensing and Registration of newer products

Lecture Schedule

1. Designer Fertilizers - Definitions - Concepts -Historical development
2. Scope and need for Designer Fertilizes –Multi nutrient deficiencies in soils and plants - Critical limits - current scenario of multi nutrient disorders
3. Classification-Types- Speciality / Customised, Fortified and Pelleted fertilizers, Multi-nutrient liquid formulations
4. Speciality / Customised Fertilisers- Definitions- Production-characteristics- sources - suitability for crops -Merits and Demerits
5. Fortified fertilizers-Definitions- Production-characteristics-sources-suitability for crops-Merits and Demerits
6. Pelleted fertilizers - Definitions- Production- characteristics- sources- suitability for crops -Merits and Demerits
7. Multi nutrient liquid formulations - Definitions- Production- characteristics- sources - suitability for crops -Merits and Demerits
8. Foliar Formulations - Leaf nutrient analysis - organic and synthetic chelates
9. Mid semester examination
10. Value added fertilizers- enriched with organics/chelates
11. Methods and guidelines for preparing designer Fertilizers - Filler materials
12. Industries and Approved formulations and mixtures - Advantages and Disadvantages - Key Challenges
13. Quality of designer fertilizers - Compatibility of fertilizer materials - issues in storability, hygroscopicity, clogging, etc - Toxicity
14. Crop response to designer fertilizers - Agricultural and Horticultural crops- Yield and Quality- Soil health and Nutrient use efficiencies
15. Feasibility of using designer fertilizers for drip fertigation - Poly houses - roof gardening

16. Quality Standards-Specifications for designer fertilisers
17. Guidelines for Patenting, Licensing and Registration of newer products

Practical Syllabus

Preparation of Designer Fertilizer Mixtures for major agricultural, Horticultural and High value crops- Preparation of multi nutrient liquid formulations for drip fertigation, poly houses, roof gardening - Preparation of pelletised fertilizer mixtures for high value crops and roof gardening- Preparation of fortified fertilizer mixtures for major agricultural and horticultural crops- Preparation of value added fertilisers - Assessing the storability of the formulations and mixtures -Machineries in designer fertilizer production - Computation of cost effectiveness of the designer fertilizers - Visit to Designer Fertilizer manufacturing Unit- Protocols for establishing a Designer Fertilizer Production Unit - Procedures for Licensing, registration and Patenting.

Practical schedule

1. Preparation of Designer Fertilizer Mixtures for major agricultural crops: Rice/ Pulse
2. Preparation of Designer Fertilizer Mixtures for major horticultural crops : Banana/ Tapioca
3. Preparation of Designer Fertilizer Mixtures for high value crops : Turmeric/Cotton
4. Preparation of multi nutrient liquid formulations for drip fertigation : Sugarcane/Turmeric
5. Preparation of multi nutrient liquid formulations for poly houses: Tomato/ Cucumber/Capsicum
6. Preparation of pelletised fertilizer mixtures for high value crops : Maize/ Turmeric/Cotton
7. Preparation of pelletised fertilizer mixtures for roof gardening : Chillies/ Brinjal/ Greens
8. Preparation of fortified fertilizer mixtures for major agricultural crops: Maize, Groundnut
9. Preparation of fortified fertilizer mixtures for major horticultural crops: Onion, Bhendi
10. Preparation of value added fertilizers
11. Assessing the storability of the mixtures and formulations
12. Machineries in designer fertilizer production
13. Computation of cost effectiveness of the designer fertilizers
14. Visit to Designer Fertilizer manufacturing Unit
15. Protocols for establishing a Designer Fertilizer Production Unit - Guidelines and budget
16. Procedures for Licensing, registration and Patenting
- 17. Final practical examination**

References

1. Hagin, J. and B. Tucker (2012). Fertilization of Dry land and Irrigated Soils, Springer Science & Business Media, 06-Dec-2012 - Technology & Engineering, 190 pages
2. John Havlin, Samuel L. Tisdale, James D. Beaton, Werner L. Nelson (2013) Soil Fertility and Fertilizers: An Introduction to Nutrient Management, Pearson, 2013 - Technology & Engineering - 516 pages
3. Hari Lal Singh Tandon (2012) Fertilizer Management: Balance-efficiency-profitability Fertilizer Development and Consultation Organization, Arid regions agriculture - 187 pages
4. Anac,D. and P. Martin Pravel (1999). Improved crop quality by nutrient management , Kluwer Academic Publications, London Springer Science & Business Media, 30-Sep,1999 - Science - 310 pages
5. Gustafson,A. F. (2013) Handbook of Fertilizers - Their Sources, Make-Up, Effects, And Use,Read Books Ltd, 16-Apr-2013 - Technology & Engineering - 170 pages
6. Kolay,A. K. (2007) Manures and Fertilizers Atlantic Publishers & Dist, Fertilizers - 173 pages
7. UN Industrial Development Organization(1998), Fertilizer Manual, Int'l Fertilizer Development Center, Springer Science & Business Media, 31-Mar-1998 - Nature - 616 pages
8. Casper, M. S. (1973). Liquid fertilizers, Noyes Data Corp. , 1973 - Technology & Engineering - 268 pages

SAC 353 Soilless Crop Production (1+1)

Unit I : Protected agriculture overview and scope - Introduction to soilless cultivation of plants - History of solution culture - Present status of hydroponics - Nutrient requirements - Absorption of nutrients by the roots and interaction between the roots and the soil solution - Nutrient ratios - Effects of pH, EC and nutrient ratios on plant growth, yield and quality

Unit II : Containers - Grow bag / container media – formulations - their properties - Systems with aggregates as substrate - bag culture, container culture, trough culture, thin layer systems, other alternative systems - effect of volume and shape of container. Hydroponic systems - Systems involving solely water as a substrate - deep water culture, floating hydroponics, Nutrient Film Technique, plant plane hydroponics, aeroponics.

Unit III : Container media - Description of substrates -sand, gravel, rockwool, expanded minerals, pumice, zeolite, pyroclastic materials, peat, coir, tree bark, sawdust, wood fibres, etc. - Physical properties – impact of physical properties on irrigation management. Chemical properties - Container media analyses - Total and available nutrients - Microbiology and phytosanitation in container media

Unit IV : Composition of nutrient solution: Calculation of nutrient solutions for open systems/ closed systems: concept of drainage solution plus fresh water - Management of nutrient solution - Nutrient solution recycling - Irrigation control - characteristics of irrigation systems : capacity, uniformity - Delivery Systems : overhead systems, drip irrigation, sub-irrigation. Irrigation scheduling : preset schedule, sensor-based schedule, transpiration-based schedule

Unit V : Equipments in Soilless culture - Automated delivery of nutrient solution - sensor based monitoring - moisture, nutrient, temperature and humidity sensors - Integrated system development for electronic control of equipments for irrigation and nutrient solution recycling. Nutrient solution disinfection - heating, UV-irradiation, chemical treatments by means of ozone, hydrogen peroxide, chlorine, iodine, etc. - membrane filtration - slow sand filtration

Lectures

1. Importance and scope of protected agriculture; Factors affecting crop growth under protected cultivation – temperature, light intensity, CO₂ and humidity
2. Introduction to soilless cultivation of plants; Various systems of soil-less crop production - bag culture, container culture, trough culture, thin layer systems, other alternative systems. .
3. History of solution culture and Present methods of hydroponics - deep water culture, floating hydroponics, Nutrient Film Technique, plant plane hydroponics, aeroponics. - Basic needs and suitable crops for hydroponics.
4. Calculation of nutrient solution – mixing of nutrients, concentration and method, dose and time of application; Management of nutrient solution.
5. Fertilization - Nutrient requirements by crops, absorption of nutrients, nutrient ratios and its effect on crop growth and yield; Sensor based nutrient management.
6. Containers – kinds of substrates - sand, gravel, rockwool, expanded minerals, pumice, zeolite, pyroclastic materials, peat, coir, tree bark, sawdust, wood fibres, etc. – resources and methods of preparation.
7. Physical properties of substrates - air to water ratios, bulk density, particle size distribution, porosity, water release curves, hydraulic conductivity
8. Chemical properties of substrates - pH, electrical conductivity, ion sorption, ion exchange, concentration and composition of ions, cation exchange capacity.

9. Mid semester examination

10. Standardization of soil-less media and solution culture
11. Irrigation systems – Drip irrigation, sub-irrigation, matric suction irrigation and irrigation management.
12. Automated irrigation system – sensor based, transpiration based schedule
13. Equipments used in soil-less cultivation – various sensors used
14. Sterilization of substrate in grow bag media and disinfection of nutrient solution in hydroponics.
15. Day to day maintenance of soil-less system of crop production.
16. Suitability of crops for growing under green house cultivation; specific technology for raising vegetable crops under protected cultivation.
17. Economics and Business opportunities in soil-less system of crop production.

Practical

Propagation of plants for culture by hydroponics - testing seeds or cutting using media for adaptation to soilless culture - Growing crops in solution culture observation on growth and maturity phases - Growing crops in water culture and identification of plant nutrient deficiencies / nutritional stresses - Control of acidity of solutions by regulation of pH of nutrient solutions or by control of the sources of N - Formulation of nutrient solutions - Factors governing stability of Nutrient solutions - solubility of salts, purity of constituents, buffering of acidity, quality of water - Regulation of salinity in nutrient solutions – record of change in concentration of soluble salts in nutrient solutions - Culture of plants by nutrient film technique – growing plants using thin layers of water which flow by roots in a designed chamber - Evaluation of solid media in solution culture – growth of plants in solid media - sand, gravel, sawdust, perlite, etc. - Visit to commercial hydroponics greenhouse

Lecture Schedule

1. Preparation of soilless media using different substrates for grow bag method and sowing crop.
2. Preparation of solution culture for hydroponics and sowing crop.
3. Estimation of physical characteristics of grow bag media (cocopeat, vermiculite)
4. Determination of chemical properties like pH, EC, CN ratio of grow bag media.
5. Determination of water soluble and exchangeable nutrients in grow bag media.
6. Evaluation of porosity of medium based on moisture characteristic/ water retention curve.
7. Computation of container capacity, air filled porosity of grow bag media
8. Estimation of moisture constants by pressure plate apparatus and computation of available water and water holding capacity.
9. Preparation of common nutrient solution for hydroponics and drip system.
10. Preparation of fertilizer pellet packs for crops under matric suction irrigation.
11. Nutrient monitoring study using sensors in grow bag media.
12. Nutrient monitoring study using sensors in continuous recycling solution culture.
13. Assembling water lines and measurement of water consumption under drip / matrix suction irrigation.
14. Recording of operations involved in devices – peristaltic pump, filter pump, injectors, solenoid valves
15. Study of circuitry for sensor based nutrient monitoring system with wireless controls.
16. Visit to successful greenhouse cultivation system
17. **Practical examination**

References

- Burt, C. , K. O'Connor, and T. Ruehr. 1998. Fertigation. ITRC, California Polytechnic State University, San Luis Obispo, CA. FAO. 1990. Soilless Culture for Horticultural Crop Production.
- Jones, J. Benton. 1983. A Guide for the Hydroponic and Soilless Culture Grower. Timber Press.
- Jones, J. Benton. 2005. Hydroponics: A Practical Guide for the Soilless Grower. 2nd ed. CRC Press. Boca Raton.
- Kipp, J. A. , G. Wever and C. de Kreij. 2000. International Substrate Manual. Elsevier, the Netherlands.
- Marschner, H. 1995. Mineral Nutrition of Higher Plants (2nd ed.). Academic Press, London. Mengel, K. and Kirkby, E. A. 2001. Principles of Plant Nutrition (5th ed). Kluwer Academic Publishers, Dordrecht.
- Raviv, M and Lieth, J. H. (eds.) 2008. Soilless Culture, Theory and Practice. Elsevier, London.
- Schwartz, M. 1995. Soilless Culture Management. Springer-Verlag, Berlin.
- Proceedings of the World Congress on Soilless Culture: Agriculture in the coming millennium. Editors, A. Bar-Tal, Z. Plau. 2001.

SAC 354 Instrumental Methods of Analysis (1+1)

Unit I: Principles of instrumentation- classification of instrumental methods – selection of instruments - Principles involved in digestion methods – dry ashing and wet digestion – open vs closed digestion - Block digester , microwave digester and IR digestion systems - components -operation - special consideration

Unit II : Automated methods – Principle and applications - Total N analyser, CN analyser

Unit III: Optical methods – spectrophotometry – visible, ultraviolet and infrared spectrometry - Principle - Instrumentation – sample handling and measurement - method development and validation - accuracy

Unit IV : Emission and absorption Spectroscopy - principles and applications - flame photometry, atomic absorption spectrophotometry, inductively coupled plasma emission spectrometry - instrumentation - features and operation of components - sample handling - errors - fault finding - trouble shooting

Unit V : Chromatography techniques – classification - paper chromatography, TLC - Gas chromatography- HPLC , GC – MS - principles - Instrumentation - sample preparation and handling - errors – trouble shooting

Lecture schedule

1. Basic principles in instrumental method of analysis
2. Principle and practice of digestion methods
3. Principle and practice of N analyser and CN analyser
4. Spectrophotometry : Types, Principle and instrumentation
5. Spectrophotometry: Sample handling and measurement, method development and validation and checking for accuracy
6. Emission spectroscopy (Flame Photometer) : Instrumentation, interferences, trouble shooting and maintenance
7. Absorption spectroscopy (Atomic Absorption Spectrophotometer) : Principle, instrumentation, features and operation of components
8. Absorption spectroscopy (Atomic Absorption Spectrophotometer): Sample handling and measurement, errors due to molecular and ionic species, matrix effect and other interferences, trouble shooting and maintenance
9. Mid semester examination
10. Absorption spectroscopy (Inductively Coupled Plasma Emission Spectrometer) : Concepts and instrumentation
11. Absorption spectroscopy (Inductively Coupled Plasma Emission Spectrometer) : Preparation of samples and standards, interferences, trouble shooting and maintenance
12. Principle and practice of paper chromatography, Thin layer chromatography
13. Gas chromatography : Principle, Types and instrumentation
14. Gas chromatography : Operation ,sample handling, maintenance and trouble shooting and applications

15. High Performance Liquid Chromatography : Principle, Instrumentation and operation
16. High Performance Liquid Chromatography : Sample preparation , method development, maintenance and troubleshooting
17. GC – MS : Principle , instrumentation, Sample preparation , method development, maintenance and troubleshooting

PRACTICAL

Collection and processing of samples - Digestion of samples – block digester and microwave digester methods - N analyser – Spectrophotometry - UV –Vis Spectrophotometer -- Emission spectroscopy - Flame photometer – Absorption spectroscopy – Atomic Absorption Spectrophotometer

(AAS) and Inductively Coupled Plasma Emission Spectrometer (ICP) – Chromatography - Gas Chromatography and High Performance Liquid Chromatography- Procedures for establishing a analytical laboratory

PRACTICAL SCHEDULE

1. Collection and processing of samples (Soil, plant, water, manure and fertilizer)
2. Digestion of samples by block digester/microwave digester and sample preparation for different analysis
3. N analyser : Calibration , sample estimation and results interpretation
4. UV -Vis spectrophotometer : Getting acquainted with parts of UV -Vis spectrophotometer and preparation of standards
5. UV -Vis spectrophotometer : Calibration, sample estimation (P/S/B) and results interpretation
6. Flame photometer : Getting acquainted with components of flame photometer and preparation of standards, calibration, sample estimation (P/S/B) and results interpretation
7. Getting acquainted with components of AAS and standards (micronutrients and heavy metal) preparation
8. Calibration and sample estimation for micronutrients and heavy metals by AAS
9. Getting acquainted with components of ICP
10. Calibration , sample estimation for elements by ICP and results interpretation
11. Analyte extraction from sample and separation for GC
12. Calibration , sample introduction and interpretation of results in GC
13. Analyte extraction from sample and separation for HPLC
14. Calibration, sample introduction and interpretation of results in HPLC
15. Visit to a analytical laboratory (Government/Private)
16. Procedures for establishing a analytical laboratory – Guidelines and budget
- 17. Practical examination**

References

- Chatwal, G. and Anand. 2005. *Instrumental Methods of Chemical Analysis*. Academic Press. Muralidhar aRao, D. , A. V. N Swamy and D. Dharaneeswara Reddy. 2013. *Instrumental Methods of Analysis*. CBS Publishers & Distributors Pvt. Ltd. , New Delhi.
- Skoog, A. , Holler, F and Nieman, A. 2007. *Principles of Instrumental Analysis* (6th Ed.). Thomson Brooks/Cole publishing. ISBN: 9780030020780.
- Willard M, Merritt LL & Dean JA. 1989. *Instrumental Methods of Analysis*. (7th Ed). Wadsworth Publishing Company, Boston, USA. ISBN :9780534001423.

E-References

1. [http://blogs. rediff. com/ticconcvibi1985/2013/05/12/download-ebook-modern-hplc-for-practicing-scientists-free/](http://blogs.rediff.com/ticconcvibi1985/2013/05/12/download-ebook-modern-hplc-for-practicing-scientists-free/)
2. [http://cdn. intechopen. com/pdfs-wm/26275. pdf](http://cdn.intechopen.com/pdfs-wm/26275.pdf)
3. [http://faculty. ksu. edu. sa/Dr. almajed/Books/practical%20HPLC. pdf](http://faculty.ksu.edu.sa/Dr.almajed/Books/practical%20HPLC.pdf)
4. [http://web. uni-plovdiv. bg/plamenpenchev/mag/books/spectroscopy/PRIMER. PDF](http://web.uni-plovdiv.bg/plamenpenchev/mag/books/spectroscopy/PRIMER.PDF)

SAC 352 Rejuvenation of Deteriorated Lands (1+1)

Theory syllabus

Unit I : Type, factors and processes of soil / land deterioration and its impact on soil productivity, including soil fauna, bio deterioration and environment. Land rejuvenization and conservation / management techniques; afforestation and silviculture and soil carbon restoration.

Unit II: Causes, reclamation and management of soil physical deterioration - surface and sub surface hard pans, shallow, slowly permeable and highly permeable and fluffy paddy soils; soil erosion, ravine and sand dune, coastal and seasonally inundated soils and mined land.

Unit III : Causes, reclamation and management of salt-affected soils – saline, sodic and saline sodic soils; acid and acid sulphate soils; laterite soils.

Unit IV : Extent, diagnosis and mapping of land deterioration by conventional and modern RS-GIS tools. Monitoring land deterioration by fast assessment and modern tools.

Unit V : Land use policy, incentives and participatory approach for reversing land deterioration; global issues for twenty first century.

Practical

Determination of gypsum requirement in sodic soils, lime requirement in acid soils and soil erodibility indices – field reclamation of saline, saline-sodic, sodic and acid soils. . Characterization of soil physical constraints - surface and sub surface hard pans, shallow, slowly permeable soils, and fluffy paddy soils - characterization of mined out and ravine lands, laterite, sand dune, coastal and seasonally inundated soils.

Lecture schedule

1. Type, factors and processes of soil / land deterioration.
2. Soil / land deterioration impact on soil productivity, including soil fauna, bio deterioration and environment.
3. Land rejuvenization and conservation techniques; Land configuration techniques; Surface / vertical mulching.
4. Afforestation and silviculture methods; Soil carbon restoration – use of industrial C- rich by products.
5. Causes, reclamation and management of soil physical deterioration - surface and sub surface hard pans, shallow, slowly permeable and highly permeable and fluffy paddy soils.
6. Causes, management of soil erosion.
7. Causes, reclamation and management of mined and ravine lands.
8. Causes, reclamation and management of sand dunes, coastal and seasonally inundated soils.
9. Mid semester examination
10. Causes, reclamation and management of saline and saline sodic soils
11. Causes, reclamation and management of sodic soils
12. Causes, reclamation and management of acid and acid sulphate soils.
13. Causes, reclamation and management of laterite soils.
14. Extent, diagnosis and mapping of land deterioration by conventional and modern RS-GIS tools.
15. Monitoring land deterioration by fast assessment and modern tools.
16. Land use policy, incentives and participatory approach for reversing land deterioration.

17. Global issues for twenty first century.

Practical schedule

1. Practicing field reclamation of saline soils and assessing its impact and cost benefit.
2. Practicing field reclamation of saline-sodic soils and assessing its impact and cost benefit.
3. Practicing field reclamation of sodic soils by gypsum application and assessing its impact and cost benefit.
4. Practicing field reclamation of sodic soils by press mud application and assessing its impact and cost benefit.
5. Practicing field reclamation of sodic soils by spent wash application and assessing its impact and cost benefit.
6. Practicing field reclamation of acid soils and assessing its impact and cost benefit.
7. Practicing management of surface and sub surface hard pans and assessing its impact and cost benefit.
8. Practicing management of slowly permeable and highly permeable and fluffy paddy soils and assessing its impact and cost benefit.
- 9 to 13. Field visit and characterization of eroded, ravine and mined out lands, sand dune, coastal, seasonally inundated and laterite soils and assessing its management impact and cost benefit.
14. Field visit to agro forestry and silviculture farms and assessing its impact on soil conservation.
15. Assessing the suitability of industrial byproducts for eco friendly recycling through soil conservation and rejuvenization.
 16. Diagnosis and mapping of land degradation using RS and GIS tools.
 17. **Final Practical examination.**

References

- Biswas TD and Narayanasamy G. (Eds.). 1996. Soil Management in Relation to Land Degradation and Environment. Bull. Indian Soc. Soil Sci. 17, New Delhi.
- Doran JW and Jones AJ. 1996. Methods of Assessing Soil Quality. Soil Science Society of America, Madison.
- Greenland DJ and Szabolcs I. 1994. Soil Resilience and Sustainable Land Use. CABI.
- Lal R, Blum WEH, Vailentine C and Stewart BA. 1997. Methods for Assessment of Soil Degradation.
- Sehgal J & Abrol IP. 1994. *Soil Degradation in India - Status and Impact*. Oxford & IBH.

SAC 355 Crop and Pesticide Chemistry (1+1)

Unit-I : Chemistry of Agricultural Crops: Chemical constituents of plants - Proximate and ultimate constituents - Chemical composition and nutritional quality of cereals, pulses and forage crops. Chemical composition and nutritional quality of oilseeds and sugarcane. Post harvest changes in Sugarcane.

Unit-II : Chemistry of Horticultural Crops, alkaloids and Essential oils: Chemical composition and nutritional quality of fruits, vegetables, spices, condiments, tuber crops and beverages. Post harvest changes in fruits. Chemistry of medicinal and aromatic plants.

Unit-III : Pesticides and its Formulations: Pesticides - Definition - Classification-Trends in pesticide use. Pesticide formulations -dusts, wettable powders, emulsifiable concentrate, granules. Insecticides - classification-. Characteristics, Mode of action and use of Organophosphates - Carbamates - Pyrethroids Botanicals, Insect Growth Regulators and Newer insecticides.

Unit-IV : Fungicides and Herbicides : Fungicides - classification of fungicides -properties, mode of action of inorganic, organic and systemic fungicides - Rhodenticides- Zinc phosphide - Aluminium phosphide - Bromodiolone Herbicides - classification - properties - mode of action of inorganic and organic herbicides like phenoxy compounds, substituted ureas, amides, thiocarbamates, triazines, pyridines, imidazolines and sulphonyl ureas.

Unit-V : Pesticides and Environment : Insecticide Act and Insecticide Rules - Fate of pesticides in soil- Impact of pesticides on environment

Practical

Estimation of moisture, ash, crude protein, P, K and crude fibre in plant samples - Determination of reducing and non-reducing sugars in jaggery — Oil content in Groundnut- Estimation of total solids, ascorbic acid, titratable acidity in fruits- Phenols/ Mucilages in Vegetables - HCN content in Tapioca/ Sorghum - Analysis of pesticides - Physical tests - Bulk density, wettability, suspensibility, Emulsion stability -. Estimation of pesticide residues in soil, water , vegetables, fruits and pesticidal calculations. Visit to Pesticide Testing Laboratory.

Lecture Schedule

1. Chemical composition and nutritional quality of cereals and pulses - Rice, wheat, maize, minor millets, Red gram, blackgram, and soybean. Starch synthesis and protein synthesis
2. Chemical composition and nutritional quality of oil seed crops - Groundnut, sesame, sunflower, castor, coconut and palm.
3. Chemical composition and nutritional quality of sugarcane -Sucrose synthesis - Post harvest changes in sugarcane. Nutritional quality of forage crops.
4. Chemical composition and nutritional quality of fruits - Mango, banana, papaya, grapes, guava, apple and pomegranate. Chemistry of post harvest changes in fruits.
5. Chemical composition and nutritional quality of vegetables- Tomato, bhendi, brinjal, moringa, greens, cauliflower, radish and peas.
6. Chemical composition of spices and condiments (Turmeric, chillies, pepper, ginger, onion, garlic and Beverages (tea and coffee). Tuber crops- Potato & Tapioca
7. Alkaloids in medicinal plants (Cinchona, Gloriosa, Coleus and Aloe vera)

8. Pesticides - Definition - Classification-Trends in pesticide use
9. **Mid semester Examination**
10. Pesticide formulations - dusts - wettable powders, flowables, sprays –Emulsion concentrates – water soluble liquids - granules, fumigants and aerosols - characteristics and uses.
11. Insecticides classification -Characteristics, Mode of action and use of Organophosphates(Chlorpyrifos, Phorate, Dimethoate, Quinalphos and Profenophos)
12. Characteristics, Mode of action and use of Carbamates (Carbaryl, carbofuran, carbosulfan, aldicarb) and synthetic pyrethroids (Deltamethrin, Fenvalerate, Cypermethrin and Lambda-cyhalothrin)
13. Characteristics, Mode of action and use of Botanicals (nicotine and neem), Insect Growth Regulators (Novaluron, Buprobasin and GABA inhibitors) and *and newer insecticides (Neonicotinoids - Imidacloprid, Thiacloprid, Acetamiprid, Flupendiamide, Fipronil, Emamectin, Thiomethoxam, Indoxacarb, Chlorantraniliprole)*
14. Fungicides - Classification – Inorganics (sulfur) and Organic fungicides (Chlorobenzene and Chlorothalnil) - Characteristics, mode of action and use
15. Characteristics, mode of action and use of Systemic fungicides (Benomyl, Carbendazim, Metalaxyl, Quinones, Diclonex, Dicarboximides –vincozolin).
16. Herbicides - Classification of herbicides - Characteristics, Mode of action and use of 2, 4-D, Sulfonyl ureas - Metsulfuron, Pyrosulfuron, Imidazoline, Alachlor, Butachlor, Oxyfluorfen, Fulchloralin, Pendimethalin, Atrazine, Paraquat and Glyphosate. Bisperipac sodium.
17. Fate of pesticides in soil-Impact of pesticides on the environment , Highlights of Insecticide Act - 1968 and Insecticide Rules -1971

Practical schedule

1. Sampling, processing and storage of plant materials for chemical analysis -Estimation of moisture and ash content
2. Preparation of tri acid extracts of plant samples -Estimation of P and K in triple acid extract
3. Estimation of crude protein
4. Estimation of crude fibre
5. Estimation of reducing and non-reducing sugars in jaggery
6. Estimation of oil content in groundnut
7. Estimation of total solids, ascorbic acid and titrable acidity in fruit samples
8. Estimation of phenols in vegetables / Mucilages in Bhendi
9. Determination of HCN content in Tapioca/ forage sorghum
10. Determination of bulk density in dust formulation, wettability and suspensibility test in wettable powder formulations
11. Estimation of emulsion stability in EC formulation
12. Estimation of pesticide residues in soil using GC/HPLC
13. Estimation of pesticide residues in Water / Soft drinks using GC/HPLC
14. Estimation of pesticide residues in Vegetables using GC/HPLC
15. Estimation of pesticide residues in Fruits using GC/HPLC & Pesticide requirement calculations
16. Visit to Pesticide Testing Laboratory
17. **Practical Examination**

References

1. Brijesh Tiwari and Narpinder Singh. 2012. Pulse Chemistry and Technology. Scientific Publishers, Jodhpur, India.
2. Hand book of Agriculture, 2009. Published by Indian Council of Agricultural Research, New Delhi 110 012. Pp. 1583
3. Paul H. Moore and Frederik C. Botha. 2014. Sugarcane : Physiology, Biochemistry and Functional Biology (World Agriculture Series). Amazon Publishers, India.
4. Petra Marschner. 2012. Marschner's mineral nutrition of higher plants. ^{3rd} Edition. ISBN: 978-0-12-384905-2. Elsevier publications.
5. Yash P. Kalra, 1998, Handbook of Reference Methods for Plant Analysis, Taylor & Francis Group, LLC, New york, London
6. Dodia, D. A. , I. S. Patel and G. M. Patel. 2010. Botanical Pesticides for Pest Management. Amazon Publishers, India.
7. Gupta,A. 2006. Pesticide Residue in Food commodities. Agrobios Publishers, Jodhpur.
8. Hassall, K. A. 2013. The Chemistry of pesticides, their metabolism, mode of action and uses in crop production. Scientific Publishers, Jodhpur, India.
9. Koul, O. , G. S. Dhaliwal, S. Khohar and R. Singh. 2014. Biopesticides in Sustainable Agriculture. Progress and Potential. Amazon Publishers, India.
10. Mac Bean, C. 2013. The Pesticide Manual: A World Compendium. Amazon Publishers, India.
11. Sree Ramulu, U. S. 1979. Chemistry of Insecticides and Fungicides - Oxford and IBM Publishing Co, New Delhi.

e-references

1. www.apo-tokyo.org/OOe-books/AG-12_Leg
2. www.researchgate.net/. . . Chemical_composition. . . nutritional. . . /60b7d52b. . .
3. www.pulsecanada.com/food. . . nutrition/composition. . . /Canadian-Pea-Co. . .
4. www.wiley.com > . . > Food Types > Functional Foods & Nutraceutical
5. www.iipr.res.in/csciences.html
6. www.aiou.edu.pk/FoodSite/Research%20Papers/48.pdf
7. www.plantphysiol.org/content/124/4/1532.full
8. www.amazon.in/Medicinal-Plants-Chemistry-Properties. . . /1578083958
9. www.tandfonline.com > List of Issues > Table of Contents
10. www.studiauniversitatis.ro/v15/pdf/20-2010/20. . . /SU20-2-10Caunii.pdf
11. www.fao.org/docrep/t0073e/t0073e01.htm
12. www.pulseaus.com.au/. . . /Pulses%20Nutritional%20Value%20and%20Th. .
13. www.researchgate.net/. . . Nutritional. . . oilseeds/. . . /9fcfd50633dab9e5d7.pdf
14. www.uvm.edu/extension/cropsoil/wp-content/uploads/turner_refining.pdf
15. journals.cambridge.org/article_S0021859600040156
16. www.intechopen.com/. . . /pesticides-in-the-modern-world-trends-in-pestic. . .
17. cibrc.nic.in/insecticides_rules.htm
18. www.agcsa.com.au/static/atm_articles/html/3_3c.html
19. www.agf.gov.bc.ca > Agriculture > Pesticide Wise

PBG 351 Plant Genetic Resources : Collection, Conservation and Utilisation (1+1)

Theory

Unit I Concepts in agrobiodiversity : Origin and history of agriculture; conservation and agricultural development; the central role of agrobiodiversity: trends and challenges; centers of crop plant origin and diversity; Crop Wild Relatives and their role in crop domestication; dynamics of domestication; concept of gene pool; agrobiodiversity regions of India- geographical distribution of crops of Indian origin

Unit II: Planning and execution of collection missions : Importance and need for collection missions; planning and execution of collection missions ; Logistics for collection; GIS- Information on collection sites; Passport data and its importance in collection missions; use of flora and herbaria for planning collections; National and international policies and procedures to be adopted in collection missions

Unit III : Concepts in PGR conservation : *In situ* and *ex situ* conservation: concept of biosphere reserves, gene sanctuaries, on-farm conservation, seed genebanks, field genebanks, botanical gardens, herbal gardens, *in vitro* repositories and cryo-genebanks; short-, medium- and long-term conservation, concept of base, active and working collections

Unit IV: International and national policies : International framework and PGR networks; International treaties and policies in relation to agrobiodiversity conservation and sustainable use; CBD , UPOV ; National policies and legal frame work; Organisations; Biodiversity authority; PPV and FR authority; National Biodiversity Authority, IP issues with respect to ITKs and communities safe guarding biodiversity

Practical

Concepts and methods for computing biodiversity; Alpha and beta models; Calculation of species richness and endemism. Field visits to biosphere reserves – *in situ* methods of conservation. Visit to Field gene banks and understanding the modalities of conservation. Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – Seed acquisition, processing, packing, barcoding, viability monitoring, registration and documentation. Visit to clonal gene banks. Biotechnology in conservation - *In vitro* methods of conservation. Exposure to cryoconservation methods. Concepts of PGR documentation and related web resources. Crop genetic diversity - Concepts of core and mini core collections. Molecular methods in PGR documentation and fingerprinting. Planning a Pre breeding programme with adapted and unadapted germplasm. Crop wild relatives - designing a Pre breeding programme with Crop Wild relatives. PGR- Global and National policies. Learning the institutional policies and modalities in exchange and utilization of PGR at TNAU

Lecture schedule:

1. Origin and history of agriculture; conservation and agricultural development-the central role of agrobiodiversity – Methods to estimate biodiversity- trends and challenges
2. Crop diversity - centers of crop plant origin and diversity , Concepts of gene pools
3. Biodiversity hotspots - Global – Indian- Regions of agrobiodiversity
4. Crop wild relatives – domestication of crops
5. Dynamics of crop domestication with special reference to Rice, Wheat, Maize and Tomato
6. Germplasm exploration and collection – Eco-Geographical issues to be considered in planning explorations- – use of GIS and GPS principles during explorations
7. Planning the logistics and execution of collection missions- Global collection missions and achievements

8. Sampling strategies to be adopted in collections – Data recording and handling including passport data, collection of herbaria of samples etc during collection missions
- 9. Mid semester examination**
10. Historical issues related to PGR conservation, scientific basis of PGR conservation – Types : *In situ* and *ex situ* conservation:
11. *In Situ* Conservation methods : concept of biosphere reserves, gene sanctuaries, and on-farm conservation
12. *Ex Situ* conservation methods : Field gene banks and seed gene banks
13. *Ex Situ* conservation methods : Cryo conservation, *in vitro* conservation, DNA banks, conservation of microspores and mega spores
14. Concept of base, active and working collections, core collections and reference sets
15. International framework and PGR networks; International treaties and policies in relation to agrobiodiversity conservation and sustainable use; CBD and UPOV convention
16. National policies: National Biodiversity Authority, PPV & FR authority, IP issues with respect to ITKs and communities safe guarding biodiversity
17. Utilization of Plant Genetic Resources – Pre-breeding concepts for use of adapted and un-adapted germplasm in crop improvement programmes

Practical schedule:

1. Concepts and methods for computing biodiversity; Alpha and beta models;
2. Calculation of species richness and endemism
3. Field visits to biosphere reserves – *in situ* methods of conservation
4. Visit to Field gene banks and understanding the modalities of conservation
5. Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – Seed acquisition, processing, packing, barcoding
6. Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – viability monitoring, registration and documentation
7. Visit to clonal gene banks
8. Biotechnology in conservation - *In vitro* methods of conservation
9. Exposure to cryoconservation methods
10. Concepts of PGR documentation and related web resources
11. Crop genetic diversity - Concepts of core and mini core collections
12. Molecular methods in PGR documentation and fingerprinting
13. Planning a Pre breeding programme with adapted and unadapted germplasm
14. Crop wild relatives - designing a Pre breeding programme with Crop Wild relatives
15. PGR- Global and National policies
16. Learning the institutional policies and modalities in exchange and utilization of PGR at TNAU

17. Practical Examination

Reference

- Engels J. M. and Visser, L. (eds.). 2003. A guide to effective management of germplasm collections. IPGRI Handbook for Genebanks No. 6. IPGRI, Rome, Italy.
- Guarino, L. , Rao, V. R. and Reid, R. (eds.). 1995. Collecting plant genetic diversity. CAB International, Wallingford, UK.

E references:

Food and Agriculture Organization (FAO) Commission on Biodiversity for Food and Agriculture; www.fao.org/biodiversity

http://www.biodiversityinternational.org/uploads/tx_news/Molecular_markers_for_gene_bank_management_1082.pdf

SST 351 Seed entrepreneurship skill development and management (1+1)

Unit I : Current status of National and Global Seed Industry and future perspectives Seed plan - Supply chain management - Significance of Seed Replacement Rate (SRR) Formal and informal seed sector - Seed trade - Role of International agencies in cross border trading (UPOV, OECD, ISTA, IPPC, AOSA, AOSCA) Varietal registration - Seed legislation and regulatory frame works (Seeds Act and Rules, Seed Control Order 1983 and amendments - PPV&FRA 2001).

Unit II : Seed promotional policies and programmes – NPSD 1988 - National seed policy 2002- Seeds Bill 2004 - EXIM policy - Domestic and International organizations involved in seed business - NSC, SSC,NSAI

- International Seed Federation (ISF) - Asia Pacific Seed Association (APSA) - National seed quality regulatory system - Seed Certification Agencies - Notification - Seed Testing Laboratories - Central and Referral laboratories - Uniformity in quality regulation - International Seed quality regulatory system - OECD varietal certification - International Seed analysis certificate - ISTA membership and accreditation system.

Unit III : Seed Export and Import - procedures and guidelines - Germplasm exchange rules and directions - Plant Quarantine system and Sanitary and Phyto Sanitary (SPS) issues and measures for export and import of seeds - NBPGR, FAO - Human resource skill development - Financial requirements and their significance in successful seed company management - Seed company - Corporate affairs – components - Registration and establishment - Grant and issue of license - Seed preference assessment

- Seed Rolling Plan - OPVs and hybrids - Role of Seed multiplication Ratio (SMR) - Varietal Replacement Rate (VRR).

Unit IV : Post harvest handling and machineries - Principles of seed drying, cleaning and upgradation - their significance in seed shelf life preservation - Risk coverage of carry over seeds - factors affecting seed storage - infrastructure facilities - ambient and advanced storage structures - Pre and post seed quality control - management checks and balances - their significance in seed trade - Linkages with various organizations for effective seed trade and business management - Farmers centric and market driven strategies for sustaining seed business and achieving seed security - Seed pricing - pricing policies of public and private agencies - strategies - methods and factors affecting prices.

Lecture schedule

1. Current status of National and Global Seed Industry and future perspectives
2. Seed plan - Supply chain management - Significance of Seed Replacement Rate (SRR) - Formal and informal seed sector
3. Seed trade - Role of International agencies in cross border trading (UPOV,OECD, ISTA IPPC, AOSA, AOSCA)
4. Varietal registration - Seed legislation and regulatory frame works (Seeds Act and Rules, Seed Control Order 1983 and amendments - PPV&FRA 2001)
5. Seed promotional policies and programmes – NPSD 1988 - National seed policy 2002- Seeds Bill 2004 - EXIM policy
6. Domestic and International organizations involved in seed business – NSC, SSC,NSAI- International Seed Federation (ISF) - Asia Pacific Seed Association(APSA)

7. National seed quality regulatory system- Seed Certification Agencies- Notification - Seed Testing Laboratories - Central and Referral laboratories - Uniformity in quality regulation.
8. **Mid semester examination**
9. International Seed quality regulatory system - OECD varietal certification - International Seed analysis certificate - ISTA membership and accreditation system
10. Seed Export and Import - procedures and guidelines - Germplasm exchange rules and directions - Plant Quarantine system and Sanitary and Phyto Sanitary (SPS) issues and measures for export and import of seeds - NBPGR, FAO etc
11. Human Resource skill development - Financial requirements and their significance in successful seed company management
12. Seed company - Corporate affairs - components- Registration and establishment - Grant and issue of license
13. Seed preference assessment - Seed Rolling Plan - OPVs and hybrids - Role of Seed multiplication Ratio (SMR) - Varietal Replacement Rate (VRR)
14. Post harvest handling and machineries - Principles of seed drying, cleaning and upgradation - their significance in seed shelf life preservation
15. Risk coverage of carry over seeds - factors affecting seed storage - infrastructure facilities - ambient and advanced storage structures
16. Pre and post seed quality control - management checks and balances - their significance in seed trade - Linkages with various organizations for effective seed trade and business management
17. Farmers centric and market driven strategies for sustaining seed business and achieving seed security - Seed pricing - pricing policies of public and private agencies - strategies - methods and factors affecting prices

Practical schedule

1. Basic components in establishment of seed company and organizational setup - staffing pattern - Registration procedures - Company - seed producer
2. Visit to public and private sector seed companies
3. Preparation of seed rolling plan
4. Visit to seed production plots - Study on field inspection procedures
5. Lay out and designing of seed processing unit and infrastructure
6. Lay out and designing of large and small scale seed storage facilities
7. Layout and establishment of seed quality control laboratory
8. Financial assistance for seed company establishment - Central Sector Seed Schemes - NABARD - Financing organizations
9. Preparation of projects for financial assistance for establishment of seed company - Micro, small and medium enterprises etc. ,
10. Project preparation on establishment of seed processing units and seed storage infrastructures for financial assistance
11. Visit to seed dealer and retail outlets
12. Value addition techniques for seed preservation, marketing and minimizing post harvest quality losses
13. Management of legal issues related to seeds - stop sale order, punitive action, punishment and appeal - appellate authority
14. Visit to Seed Testing laboratories - Notified and ISTA accredited
15. Preparation of company status report and analysis of critical issues on sustaining seed business
16. Downgrading of seeds - Upgradation and improvement of seed standards of "sales returned seeds" for placing in market
17. **Final practical examination**

References

1. Dadheech, P. K. 1996. Seed Programming. Management system and concepts. Lok Sahitna Kendra. Jodhpur.
2. Gurudev G Singh and Asokan, S. R. , 1997. Management of seed activity. Oxford and IBH Publishing Co. , New Delhi.
3. Ramalingam, C. K. Sivasubramaniam and A. Vijayakumar. 1997. A Guide to seed legislation. Rassi Computers, Madurai
4. Trivedi, R. K. and M. Gunasekaran. 2013. Indian Minimum Seed Certification Standards. Dept. of Agriculture & Cooperation, Ministry of Agriculture, Govt. of India, New Delhi.
5. Compendium of Seed Legislation. 2013. Dept. of Agriculture & Cooperation, Ministry of Agriculture, Govt. of India, New Delhi

Online references

1. www.fao.org
2. www.seednet.gov.in
3. www.agricoop.nic.in
4. www.agri.nic.in

NEM 351 Commercial Production of Nematode Antagonistic bio-agents (1+1)

THEORY

Unit 1 : Concepts and definition of biocontrol agents – Economic yield loss caused by plant parasitic nematodes in agricultural and horticultural crops – Ecofriendly management of plant parasitic nematodes - Types of bioagents – Fungal antagonistic organisms – Applications – Dosage - Commercial formulations available in India – Merits and demerits.

Unit 2: *Paecilomyces lilacinus* – *Pochonia chlamydosporia* - Morphological identification of colony – Conidiaspore and Chlamydospores identification – Isolation from eggs - Mode of action – Host range - virulence – effect of biotic and abiotic factors on growth – compatibility with chemical inputs.

Unit 3: Sterilization – Types and Preparation of sterilizing agents – Equipments used – Principle of autoclave – Principle of Laminar Air Flow Chamber – Fermentor – Laboratory up-keep – Haemocytometer – Assessment of spore load – colony forming units.

Unit 4: Preparation different culture media – Maintenance of pure culture - Mother culture – Subculturing – Mass culturing techniques – solid and liquid formulations - Commercial Formulations – Carrier materials – Packing – Quality control and shelf life.

Unit 5: Market demand analysis - Economics – Establishment of pilot plant – Infrastructure - Budget preparation – Marketing and cost-benefit analysis – risk analysis – Environmental impact test with vertebrate and invertebrate organisms.

Lecture Schedule

1. Introduction - Economic yield loss caused by plant parasitic nematodes in agricultural and horticultural crops. Concepts and definition of biocontrol agents.
2. Types of bioagents – Fungal antagonistic organisms – Applications – Dosage - Commercial formulations available in India – Merits and demerits
3. *Purpureocillium lilacinum* (= *Paecilomyces lilacinus*) – Morphological identification of colony, phialids and conidia spore
4. *Pochonia chlamydosporia* - Morphological identification of colony. Conidia spore and Chlamydospores identification
5. Isolation of *P. lilacinum* and *P. chlamydosporia* from nematode eggs and mode of action.
6. Host range of *P. lilacinum* and *P. chlamydosporia* - virulence and effect of biotic and abiotic factors on growth – compatibility with chemical inputs
7. Sterilization – Types and Preparation of sterilizing agents – Equipments used for production of bioagents.
8. Principle of Autoclave, Laminar Air Flow Chamber and Fermentor
9. Laboratory up-keep – Preparation of cleaning solutions – preparation of stock solutions
10. Haemocytometer – Assessment of spore load – colony forming units.
11. Preparation of different culture media – Maintenance of pure culture - Mother culture – Subculturing
12. Mass culturing techniques – solid and liquid formulations – Shelf life
13. Commercial Formulations – Carrier materials – Packing – Quality control and shelf life.
14. Market demand analysis – Economics
15. Establishment of pilot plant – Infrastructure requirement – cost analysis
16. Budget preparation – Marketing and cost-benefit analysis – risk

analysis

17. Environmental impact test with vertebrate and invertebrate organisms

PRACTICAL:

Plant parasitic nematodes eggs isolation – Eggs parasitization tests with *Paecilomyces lilacinus* and *Pochonia chlamydosporia* – Fungal specific media preparation – Pure culture - Preparation of common culture media – subculturing of *Paecilomyces lilacinus* - subculturing of *Pochonia chlamydosporia* – Preparation of broth – Inoculation – Incubation in mechanical shaker – *In vitro* bioefficacy test on root knot nematode - Fermentation process – Haemocytometer – Assessing spore load in broth – Preparation of commercial formulation – Quality control test - Packing – Analyzing market potential and demand – Conducting environmental impacts test with termites, saprophytes, honeybees, earthworm etc. , - Visit to commercial production unit (HRS, Ooty).

1. Isolation of eggs of plant parasitic nematodes.
2. Eggs parasitization tests with *Purpureocillium lilacinum*
3. Eggs parasitization tests with *Pochonia chlamydosporia*
4. Fungal specific media preparation – Pure culture - Preparation of common culture media
5. Subculturing of *P. lilacinum* and *Pochonia chlamydosporia*
6. Preparation of broth – Inoculation – Incubation in mechanical shaker
7. *In vitro* bioefficacy test on root knot nematode with *Purpureocillium lilacinum*
8. *In vitro* bioefficacy test on root knot nematode with *Pochonia chlamydosporia*
9. Fermentation process
10. Haemocytometer – Assessing spore load in broth
11. Preparation of commercial formulation
12. Quality control test Packing
13. Analyzing market potential and demand
14. Conducting environmental impacts test with termites, saprophytes,
15. Conducting environmental impacts test with honeybees, earthworm
16. Visit to commercial production unit (HRS, Ooty).

17. Practical Examination

V SEMESTER

S. No.	Course code	Course Title	Credit load
1	PBG 301	Fundamentals of Plant Breeding	2+1
2	AEC 301	Agricultural Marketing Trade & Prices	2+1
3	AGM 301	Agricultural Microbiology	1+1
4	PAT 301	Diseases of Field and Horticultural crops and their management - I	1+1
5	ARM 301	Entrepreneurship Development and Business Communication Management	1+1
6	AGR 301	Practical Crop Production - I (<i>Kharif</i> crops)	0+2

7	HOR 311	Post harvest management and value addition of fruits and vegetable crops	1+1
8	SAC 301	Manures, Fertilizers and Soil Fertility Management	2+1
9	APE 311	Protected Cultivation and Secondary Agriculture	1+1
10	AGR 302	Rainfed Agriculture & Watershed Management	1+1
11	AEN 301	Pests of Field crops and stored produces and their management	1+1
11	NCC 101	NCC*	
		Total	13+12=25
		*Non-gradual courses compulsory courses	

PBG 301 Fundamentals of Plant Breeding (2+1)

THEORY

Unit I: Reproductive systems in plant breeding

Objectives and role of plant breeding - historical perspective – activities in Plant Breeding. Centres of origin – contribution of Vavilov, Harlan, Zhukovsky – law of homologous series. Plant genetic resources – importance – germplasm – types – activities – gene erosion - gene bank – collection - conservation – types of conservation. Germplasm: evaluation – use of descriptors, documentation, utilization; Agencies – national and international; germplasm exchange – quarantine. Modes of reproduction – sexual – asexual – mechanisms promoting self and cross pollination – significance of pollination. Self incompatibility – classifications – mechanisms – application – measures to overcome and limitations. Sterility – male sterility – introduction – classification – CMS, GMS, CGMS -inheritance and applications. EGMS - TGMS, PGMS, Gametocides, Transgenic Male sterility and applications. Apomixis – introduction – classification - applications; Parthenocarpy and its types.

Unit II: Breeding methods of self pollinated crops

Polygenic variation-components of variance - phenotypic, genotypic and environmental variance- heritability and genetic advance. Plant introduction as a breeding method – types of introduction – objectives – quarantine - acclimatization – achievements - merits and demerits. Genetic basis of self pollinated crops – Vilmorin's principle of progeny selection - Johannsen's pure line theory. Breeding methods for self pollinated crops without involving artificial hybridization: Pure line selection – procedure – merits and demerits – achievements; Mass selection– procedure - types – merits and demerits-achievements- comparison of mass and pureline selection. Breeding methods of self pollinated crops involving artificial hybridization: Creating variability in self pollinated crops - Hybridization and selection – objectives – steps in hybridization - choice of parents – kinds of emasculation – hybridization-transgressive breeding. Handling segregating generations- Pedigree breeding – procedure – mass pedigree – merits – demerits – achievements; Bulk breeding – procedure – merits – demerits – achievements. Comparison of pedigree and bulk breeding methods. Single Seed Descent (SSD) method – procedure – application – merits and demerits. Backcross breeding – genetic basis — procedures for transferring dominant and recessive genes. Back cross breeding – merits – demerits – multilines- types- procedure- merits and demerits.

Unit III: Breeding methods of cross pollinated crops and clonally propagated crops

Genetic structure of a population in crosses pollinated crops – Hardy Weinberg law – gene frequencies in random mating population. Breeding methods of cross pollinated crops without involving artificial hybridization: Mass selection in cross pollinated crops – modified mass selection – Grid selection – progeny selection. Breeding methods of cross pollinated crops involving artificial hybridization: Recurrent selection principles – types – merits and demerits. Heterosis breeding – theories - genetic basis – hybrid vigour – estimation of heterosis – inbreeding depression. Heterosis breeding – procedure – development of inbreds- evaluation of inbred lines – top cross method and single cross method- prediction of double cross performance- hybrids – single cross- double cross- three way cross hybrids. achievements – merits and demerits. Synthetics and composites - steps in development of synthetics and composites – achievements – merits and demerits. Genetic characters of asexual reproduction – clonal selection – hybridization and clonal selection – merits and demerits – achievements.

Unit IV: Special breeding methods

Polyploidy breeding – classification – induction of polyploidy - achievements – limitations. Wide hybridization-importance-barriers and techniques for overcoming barriers-utilization- Pre-breeding. Mutation breeding: mutation – types – mutagens – breeding procedure – achievements – limitations. Concepts in biotic stress resistance breeding- diseases and pests - gene for gene hypothesis-mechanisms of resistance - sources of resistance- multilines-gene pyramiding-gene deployment-Breeding methods. Concepts in abiotic stress resistance breeding- drought- mechanisms of drought resistance – basis of drought resistance- morphological and physiological characters- sources of drought resistance- Breeding methods.

Unit V: Varietal Release, Seed Production, Markers and IPR

Procedure for release of new varieties-stages in seed multiplication-steps in nucleus and breeder seed production. Introduction to markers – morphological – biochemical- DNA markers – advantages and disadvantages- marker assisted selection in plant breeding. Participatory plant breeding- Intellectual Property Rights- Patenting- Plant Breeders and Farmers Rights.

PRACTICAL

Reproduction in plants - Alternation of generation and life cycle. Mode of pollination - Mechanisms enforcing self and cross pollination in crops- Working out extent of natural out crossing. Breeder's kit and its components. Basic techniques for selfing and crossing in crop plants. Emasculation and pollination techniques in field crops. Emasculation and pollination techniques in horticultural crops. Handling of segregating populations- Layout of different yield trials-maintenance of records. Study of Cytoplasmic genic male sterility system in Rice/horticultural crops. Study of Genic male sterility system in Redgram. Mutagenesis study using physical and chemical mutagens. Germplasm collection and conservation. Experimental designs used in plant breeding-RBD analysis. Calculation of mean, range, PCV, GCV, heritability, genetic advance. Estimation of heterosis and prediction performance of double cross hybrids. Screening techniques for biotic stresses in rice. Screening techniques for abiotic stresses in rice

Theory schedule

52. Objectives and role of plant breeding - historical perspective – activities in Plant Breeding.
53. Centres of origin – contribution of Vavilov, Harlan, Zhukovsky – law of homologous series.
54. Plant genetic resources – importance – germplasm – types – activities – gene erosion - gene bank – collection - conservation – types of conservation.
55. Germplasm: evaluation – use of descriptors, documentation, utilization; Agencies – national and international; germplasm exchange – quarantine.
56. Modes of reproduction – sexual – asexual – mechanisms promoting self and cross pollination – significance of pollination.
57. Self incompatibility – classifications – mechanisms – application – measures to overcome and limitations.
58. Sterility – male sterility – introduction – classification – CMS, GMS, CGMS -inheritance and applications.
59. EGMS - TGMS, PGMS, Gametocides, Transgenic Male sterility and applications.
60. Apomixis – introduction – classification-applications; Parthenocarpy and its types.
35. Polygenic variation-components of variance - phenotypic, genotypic and environmental variance- heritability and genetic advance
36. Plant introduction as a breeding method – types of introduction – objectives – quarantine - acclimatization – achievements - merits and demerits.
37. Genetic basis of self pollinated crops – Vilmorin's principle of progeny selection - Johannsen's pure line theory.

38. Breeding methods for self pollinated crops without involving artificial hybridization: Pure line selection – procedure – merits and demerits – achievements; Mass selection– procedure - types – merits and demerits-achievements- comparison of mass and pureline selection.
39. Breeding methods of self pollinated crops involving artificial hybridization: Creating variability in self pollinated crops - Hybridization and selection – objectives – steps in hybridization - choice of parents – kinds of emasculation – hybridization- transgressive breeding.
40. Handling segregating generations- Pedigree breeding – procedure – mass pedigree – merits – demerits – achievements; Bulk breeding – procedure – merits – demerits – achievements.
41. Comparison of pedigree and bulk breeding methods. Single Seed Descent (SSD) method – procedure – application – merits and demerits.
42. **Mid Semester examination**
43. Backcross breeding – genetic basis — procedures for transferring dominant and recessive genes
44. Back cross breeding – merits – demerits – multilines- types- procedure- merits and demerits.
45. Genetic structure of a population in cross pollinated crops – Hardy Weinberg law – gene frequencies in random mating population.
46. Breeding methods of cross pollinated crops without involving artificial hybridization: Mass selection in cross pollinated crops – modified mass selection – Grid selection – progeny selection
47. Breeding methods of cross pollinated crops involving artificial hybridization: Recurrent selection principles – types – merits and demerits.
48. Heterosis breeding – theories - genetic basis – hybrid vigour – estimation of heterosis – inbreeding depression.
49. Heterosis breeding – procedure – development of inbreds- evaluation of inbred lines – top cross method and single cross method- prediction of double cross performance- hybrids – single cross-double cross- three way cross hybrids. achievements – merits and demerits.
50. Synthetics and composites - steps in development of synthetics and composites – achievements – merits and demerits
51. Genetic characters of asexual reproduction – clonal selection – hybridization and clonal selection – merits and demerits – achievements;
52. Polyploidy breeding – classification – induction of polyploidy - achievements – limitations.
53. Wide hybridization-importance-barriers and techniques for overcoming barriers-utilization- Pre-breeding.
54. Mutation breeding: mutation – types – mutagens – breeding procedure – achievements – limitations.
55. Concepts in biotic stress resistance breeding- diseases and pests - gene for gene hypothesis-mechansims of resistance - sources of resistance- multilines-gene pyramiding-gene deployment-Breeding methods.
56. Concepts in abiotic stress resistance breeding- drought- mechansims of drought resistance – basis of drought resistance- morphological and physiological characters- sources of drought resistance-Breeding methods.
18. Procedure for release of new varieties-stages in seed multiplication-steps in nucleus and breeder seed production.
19. Introduction to markers – morphological – biochemical- DNA markers – advantages and disadvantages- marker assisted selection in plant breeding.
20. Participatory plant breeding- Intellectual Property Rights- Patenting- Plant Breeders and Farmers Rights.

Practical schedule

16. Reproduction in plants - Alternation of generation and life cycle.
17. Mode of pollination - Mechanisms enforcing self and cross pollination in crops- Working out extent of natural out crossing.

18. Breeder's kit and its components
19. Basic techniques for selfing and crossing in crop plants.
20. Emasculation and pollination techniques in field crops.
21. Emasculation and pollination techniques in horticultural crops.
22. Handling of segregating populations- Layout of different yield trials-maintenance of records.
23. Study of Cytoplasmic genic male sterility system in Rice
24. Study of Genic male sterility system in Redgram
25. Mutagenesis study using physical and chemical mutagens
26. Germplasm collection and conservation.
27. Experimental designs used in plant breeding-RBD analysis
28. Calculation of mean, range, PCV, GCV, heritability, genetic advance
29. Estimation of heterosis and prediction performance of double cross hybrids
30. Screening techniques for biotic stresses in rice
31. Screening techniques for abiotic stresses in rice
- 32. Final Practical examination**

References



Singh, B. D. 2005. Plant breeding - Principles and Methods. Kalyani Publishers, New Delhi.



Phundhan Singh. 2001. Essentials of Plant Breeding, Kalyani publishers, New Delhi.



Allard, R. 1989. Principles of Plant Breeding. John Wiley and Sons, New Delhi.



D. N. Bharadwaj. 2012. Breeding Field Crops. Agrobios (India), Jodhpur



Chahal, G. S. and S. S. Gosal. 2002. Principles and Procedures of Plant

6. Breeding: Biotechnological and Conventional Approaches. Narosa Publishing House (India)



Daniel Sundararaj, G. Thulasidas and M. Stephen Dorairaj. 1997. Introduction to Cytogenetics and Plant Breeding. Popular Book Depot. Chennai – 15.



Chopra, V. L. , 1994. Plant breeding theory and practice. Oxford and IBH Publishing Co. Pvt. Ltd. New Delhi



Sharma, J. R. 1994. Principles and practice of plant breeding. Tata McGraw-Hill publishing Co., New Delhi.



Chaudhary, H. K. 1980. Elementary Principles of plant breeding. Oxford and IBH publication Co. , New Delhi



R. K. Singh and B. D. Choudhary. Biometrical methods in quantitative Genetics. Kalyani Publishers, Ludhiana

E- References



<http://www.edugreen.teri.res.in/explore/bio/breed.htm>



<http://cuke.hort.ncsu.edu/gpb/>



<http://www.stumbleupon.com/tag/plant-breeding>



<http://www.iaea.org/>

AEC 301 Agricultural Marketing, Trade and Prices (2+1)

Theory

Unit 1: Agricultural Marketing – Nature and Scope

Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing, market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets; nature and determinants of demand and supply of farm products. Producer's surplus – meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agri-commodities. Approaches to the study of marketing - Market functionaries and Market forces. Marketing of agricultural versus manufactured goods.

Unit 2: Marketing Functions, Pricing and Promotion strategies

Marketing process and functions: Marketing process - concentration, dispersion and equalization; exchange functions – buying and selling; physical functions – storage, transport and processing; facilitating functions – packaging, branding, grading, quality control and labeling (AGMARK); Standardization, Finance, Storage and Warehousing, Processing, Value Addition and Risk Taking - Market Structure, Conduct and Performance paradigm (SCP) – Market Structure: Meaning, Components, Dynamics of Conduct and Performance – Market structure and Price determination under perfect and imperfect competition.

Product Life Cycle (PLC) and competitive strategies: Meaning and stages in PLC; characteristics of PLC; strategies in different stages of PLC; pricing and promotion strategies: pricing considerations and approaches – cost based and competition based pricing; market promotion – advertising, personal selling, sales promotion and publicity – their meaning and merits and demerits.

Unit 3: Marketing Efficiency and Marketing Institutions

Market functionaries and marketing channels: Types and importance of agencies involved in agricultural marketing; meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products; Integration over space, time and form: Meaning, definition and types of market integration; marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing reasons for higher marketing costs of farm commodities; ways of reducing marketing costs; Modern marketing systems versus traditional agricultural marketing systems; Role of Government in agricultural marketing - Public sector institutions - CWC, SWC, FCI and DMI – their objectives and functions; cooperative marketing in India; Market Intelligence -Legal measures for improving agricultural marketing: APMC Act. New EXIM policy of India – Advantages of AEZs, ITPO, Export Promotion Councils, APEDA, MPEDA, and Commodity Boards.

Unit 4: Trade in Agricultural Products

International Trade: Concept of International Trade and its need - Free trade, Autarky and its needs - Theories of Trade: Absolute and comparative advantage; Present status and prospects of Agricultural exports / imports from India and their share - Barriers to Trade: Tariff and non tariff barriers - Trade policy instruments – Terms of Trade - Role of institutions like UNCTAD and GATT - WTO in promoting trade in agricultural products - Free Trade Agreements – AoA and its implications on Indian agriculture: Sanitary and Phyto-sanitary issues, Market Access, Domestic Support and Export Subsidies - IPR.

Unit 5: Agricultural Prices and Risk Analysis

Agricultural Prices and Policy: Meaning and functions of price; administered prices; need for agricultural price policy; Objectives of Price Policy and Price Stabilization – Role of CACP – Concept of MSP, FRP

(SMP) and SAP – Price Parity - Procurement of food grains and buffer stock - Risk in marketing: Meaning and Importance - Types of risk in marketing: Speculation and Hedging and Forward and Futures trading; an overview of futures trading; – Role of Contract Farming in risk mitigation.

Theory Schedule

23. Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing.
24. Market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets.
25. Demand and supply of agri-commodities: meaning, nature and determinants of demand and supply of farm products.
26. Approaches to the study of marketing: Market functionaries and Market forces.
27. Marketing of agricultural versus manufactured goods. Producer surplus – meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agri- commodities.
28. Marketing process and functions: Marketing process - concentration, dispersion and equalization.
29. Exchange functions – buying and selling; physical functions – storage, transport and processing.
30. Facilitating functions – packaging, branding, grading, quality control and labeling (AGMARK).
31. Standardization, Finance, Storage and Warehousing, Processing, Value Addition and Risk Taking.
32. Market Structure, Conduct and Performance paradigm (SCP) – Market Structure: Meaning, Components, Dynamics of Conduct and Performance.
33. Market structure and Price determination under perfect and imperfect competition.
34. Product Life Cycle (PLC) and competitive strategies: Meaning and stages in PLC; characteristics of PLC; strategies in different stages of PLC.
35. Pricing and promotion strategies: pricing considerations and approaches – cost based and competition based pricing.
13. Market promotion – advertising, personal selling, sales promotion and publicity – their meaning and merits and demerits.
14. Market functionaries and marketing channels: Types and importance of agencies involved in agricultural marketing; meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products.
15. Integration over space, time and form: Meaning, definition and types of market integration.
16. **Mid-Semester Examination**
17. Marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing reasons for higher marketing costs of farm commodities; ways of reducing marketing costs.
18. Role of Government in agricultural marketing - Modern marketing systems versus traditional agricultural marketing systems.
19. Public sector institutions- CWC, SWC, FCI, and DMI – their objectives and functions.
20. Co-operative marketing in India.

21. Market Intelligence - Legal measures for improving agricultural marketing: APMC Act. New EXIM policy of India.
22. Advantages of AEZs, ITPO, Export Promotion Councils, APEDA, MPEDA, and Commodity Boards.
23. International Trade: Concept of International Trade and its need - Free trade, Autarky and its needs.
24. Theories of Trade: Absolute and comparative advantage;
25. Present status and prospects of Agricultural exports / imports from India and their share.
26. Barriers to Trade: Tariff and non tariff barriers - Trade policy instruments.
27. Terms of Trade - Role of institutions like UNCTAD and GATT - WTO in promoting trade in agricultural products - Free Trade Agreements.
28. AoA and its implications on Indian agriculture: Sanitary and Phyto-sanitary issues, Market Access, Domestic Support and Export Subsidies - IPR.
29. Agricultural Prices: Meaning and functions of price; administered prices; need for agricultural price policy; Objectives of Price Policy and Price Stabilization – Role of CACP – Concept of MSP, FRP (SMP) and SAP.
30. Price Parity - Procurement of food grains and buffer stock.
31. Risk in marketing: Meaning and Importance - Types of risk in marketing.
32. Speculation and Hedging and Forward and Futures trading: an overview of futures trading.
33. Role of Contract Farming in risk mitigation.

Practical Schedule

21. Preparation of farm survey schedule
22. Visit to a farm to collect information on marketing practices of agricultural commodities and marketing problems.
23. Plotting and study of demand and supply curves and calculation of elasticities.
24. Computation of marketable and marketed surplus of important commodities.
25. Visit to a local market / weekly *shandy* / farmers' market to study various marketing functions performed by different agencies.
26. Study of relationship between market arrivals and prices of some selected commodities.
18. Identification of marketing channels for selected commodity, collection of data regarding marketing costs, margins; price spread estimation for major agricultural and allied agricultural products to assess their marketing efficiency; and presentation of report in the class.
19. Visit to market committee and regulated market to study their organization and functioning.
20. Visit to co-operative marketing society to study its organization and functioning.
21. Visit to market institutions – SWC / CWC to study their organization and functioning.
22. Visit to AGMARK Laboratory / Grading institutions.
23. Farm input marketing: Visit to Farm input dealer to study marketing of farm inputs.
24. Visit to Commodity Boards / AEZ / Export oriented units.
25. Time Series Analysis of prices–TCSI Study of price behaviour over time for some selected commodities.
26. Construction of Index Numbers and their uses.
27. Application of principles of comparative advantage of international trade.
28. **Practical Examination.**

References

37. Acharya S. S. and N. L. Agarwal. 2002. Agricultural Marketing in India. Oxford and IBH Publishing Co. Pvt. Ltd. New Delhi.
38. Acharya S. S. and N. L. Agarwal. 1994. Agricultural Prices - Analysis and Policy. Oxford and IBH Publishing Co. Pvt. Ltd. New Delhi.
39. Kahlon A. S. and S. D. Tyagi, 2000. Agricultural Price Policy in India - Allied Publishers Pvt. Ltd. Bombay.
40. SakOnkvisit. John J. Shaw. 1999. International Marketing Analysis and Strategy. Prentice Hall of India. New Delhi.
41. Sivarama Prasad A. 1999. Agricultural Marketing in India. Mittal Publications, New Delhi.
42. Kohls R. L. and N. Uhl. Joseph. 1980. Marketing of Agricultural Products. Collier Macmillan. New York.

AGM 301 Agricultural Microbiology (1+1)

Theory

Unit I Introduction

Soil microorganisms and their role in soil fertility and crop production. Historical developments in soil microbiology. Diversity of soil microorganisms - culturable (bacteria, actinobacteria, yeasts, molds and algae) and unculturable microorganisms - metagenomic approach; factors influencing soil microbial diversity. Soil enzymes and their role in soil fertility.

Unit II Biogeochemical cycling of nutrients

Carbon cycle; organic matter decomposition in oxygenic and anoxygenic environments; humus formation. Nitrogen cycle – ammonification, nitrification, denitrification and biological nitrogen fixation (BNF) – bacterial and algal nitrogen fixers (diazotrophs) ; free living, associative, symbiotic, endophytic and epiphytic diazotrophs; nodulation in leguminous and non-leguminous plants; biochemistry and molecular biology of BNF. Phosphorus cycle – mineralization, phosphate mobilization and solubilization. Microbial transformation of sulphur, potassium, zinc and silica in soil.

Unit III Microbial degradation and bioremediation

Role of microbes in reclamation of problematic soils. Microbes in solid waste management. Biodegradation of agricultural residues and chemicals – processes involved in remediation.

Unit IV Microbiomes and plant growth

Rhizosphere, spermosphere, phyllosphere, epiphytic and endophytic microbiomes and their significance. Plant growth promoting rhizobacteria. Soil microorganisms and their interactions – positive and negative interactions.

Unit V Microbes in human welfare

An overview of industrially important microorganisms and products. Silage production. Bioinoculants (biofertilizers and biopesticides); types of biofertilisers – nitrogen fixers, phosphate, zinc and silicate solubilizers, potassium releasers, phosphate mobilizers, sulphur oxidizers and Pink Pigmented Facultative Methylophil (PPFM). Biopesticides- types and mechanism of action. Mass production and quality control of bacterial and fungal bioinoculants. Methods of application of bioinoculants. Biofuel production

Practical

Enumeration of soil microbial population - quantitative and qualitative methods. Organic matter decomposition - CO₂ evolution and BOD. Isolation of symbiotic nitrogen fixing bacteria, free living, associative and endophytic nitrogen fixing bacteria. Isolation of phosphobacteria and sulfur oxidizing bacteria. Isolation of zinc and silicate solubilizing/ potassium releasing bacteria. Isolation of plant growth promoting rhizobacteria (*Pseudomonas* sp) and phyllosphere (PPFM) microbes. Examination of AM fungal infection in plants and recovery of AM spores from soil. Examination of Blue Green Algae (BGA) from soil and azolla. Mass production of bacterial bioinoculants, blue green algae, azolla and AM fungi. Visit to biopesticides, silage production and compost yard.

Theory schedule

8. Introduction and historical developments in soil microbiology. Contributions of Beijerinck, Hellriegel, Wilfarth, Frank, Winogradsky, Fleming, Waksman, Doberiner and Mosse
9. Soil microorganisms and their role in soil fertility and crop production
10. Assessment of microbial diversity. Factors influencing the activities of soil microorganisms. Role of soil enzymes in nutrient transformation

11. Carbon cycle. Role of soil microorganisms in the decomposition of organic matter in oxygenic and anoxygenic environments; humus formation.
12. Nitrogen cycle – microbiology and biochemistry of mineralization, ammonification, nitrification and denitrification
6. Biological nitrogen fixation – free living, associative, endophytic, epiphytic and symbiotic diazotrophic microorganisms. Nodulation in *Rhizobium*- legume and *Frankia* – actinorhizal symbioses
33. Biochemistry and molecular biology of nitrogen fixation in different types of diazotrophs
34. Phosphorus cycle and microbial transformation of phosphorus – mineralization, phosphate solubilization and translocation
35. **Mid Semester Examination**
36. Sulphur cycle - sulphur oxidizers; microbial transformation of K, Zn and Si.
37. Role of microbes in reclamation of problem soils. Microbes in solid waste management
38. Biodegradation of agricultural residues and chemicals- processes involved in remediation
39. Importance of soil and plant microbiomes– rhizosphere, spermosphere, phyllosphere, epiphytic and endophytes. Plant growth promoting microbes-types and mechanism of action.
40. Soil microorganisms and their interactions – positive and negative interactions. An overview of industrially important microorganisms and products.
41. Silage production. Bioinoculants – types; biofertilisers - bacterial, fungal (AMF) and algalbiofertilisers. Biopesticides – types and mechanism of action
42. Mass production and quality control of bacterial and fungal bioinoculants. BIS standards– methods of application of bioinoculants.
43. Biofuel production – methane, hydrogen, alcohol and biodiesel production

Practical schedule

- Enumeration of soil microorganisms - quantitative Conn's direct microscopic method – qualitative buried slide technique
32. Enumeration of rhizosphere and bulk soil microorganisms and determination of R:S ratio
33. Studying organic matter decomposition by measurement of CO₂ evolution
34. Isolation of *Rhizobium* from root nodules and *Azotobacter* from soil
35. Isolation of *Azospirillum* by MPN technique
36. Isolation of *Gluconoacetobacter* from sugarcane phyllosphere/rhizosphere and PPFM from phyllosphere
37. Isolation of phosphobacteria, potassium releasing and zinc solubilizing bacteria from soil
38. Isolation of PGPR (*Pseudomonas* sp) and sulphur oxidizing bacteria from soil
39. Examination of AM infection in roots and recovery of spores from soil
40. Mass production of bacterial bioinoculants and AM fungi
41. Examination of blue green algae from soil and azolla
42. Mass multiplication of blue green algae and azolla
37. Methods of application of different bioinoculants
38. Antibiosis in soil – Crowded plate technique
39. Visit to biopesticides production unit
40. Visit to silage production and compost yard
41. **Practical Examination**

Text Books

41. Alexander, M. 1977. Soil Microbiology. John Wiley and Sons. New York
42. Paul, E. A. 2014. Soil Microbiology, Ecology and Biochemistry. 4th Ed. , Academic Press, USA
43. e book: Waksman, S. A 1952. Soil Microbiology John Wiley & Sons, Inc.

44. e book:Paul , E . A. 2007. Soil Microbiology, Ecology and Biochemistry. 3rd Ed. , Academic Press, USA.

Reference :

38. Rangaswamy,G. andBagyaraj, D. J. 1992. Agricultural Microbiology, Asia Publishing House, New Delhi.
39. Subba Rao,N. S. 1999. Soil Microorganisms and plant Growth. Oxford and IBH, New Delhi
40. Osborn, M. , Smith, C. J. 2005. Molecular Microbial Ecology. Taylor and Francis.

E- Reference

1. fire. biol. wwu. edu/hooper/416_05Ncycle1. ppt
2. www. fao. org/docrep/009/a0100e/a0100e05. html

PAT 301 Diseases of Field and Horticultural crops and their management-I (1+1)

Theory

Etiology, symptoms, mode of spread, survival, epidemiology and integrated management of important diseases

Unit-I Diseases of cereals and Millets

Cereals: rice and maize; **Millets:** sorghum, bajra, finger millet and small millets

Unit- II Diseases of Pulses and Oilseeds

Pulses : pigeon pea, urd bean, mung bean, soyabean, cowpea; **Oilseeds:** ground nut, castor and Sesame

Unit- III Diseases of Cash crops: tobacco, jute and mulberry

Unit- IV Diseases of Fruits and vegetables crops

Fruits: banana, guava, papaya, pomegranate; **Vegetables:** tomato, brinjal, okra , cruciferous vegetables, beans, colacasia and sweet potato

Unit- V Diseases of Plantation crops

Plantation: coconut, arecanut, tea, coffee, rubber and cocoa

Practical

Study of symptoms and host parasite relationship of rice, maize, sorghum, bajra, finger millet ,small millets, pigeon pea, urd bean, mung bean, soyabean, cowpea, ground nut, castor Sesame, tobacco, jute , mulberry, banana, guava, papaya, pomegranate, tomato, brinjal, okra , cruciferous vegetables, beans, colacasia , sweet potato, coconut, arecanut, tea, coffee, rubber and cocoa

Theory

Etiology, symptoms, mode of spread, survival, epidemiology and integrated management of

1. Diseases of rice
2. Diseases of maize and sorghum
3. Diseases of pearl millet, finger millet and small millets,
4. Diseases of pigeonpea, urdbean, mungbean, soybean and cowpea
5. Diseases of groundnut , sesame and castor
6. Diseases of tobacco, jute and mulberry
7. Diseases of banana
8. Diseases of guava , papaya and pomegranate
9. Mid semester examination
10. Diseases of tomato
11. Diseases of brinjal and okra
12. Diseases of crucifers
13. Diseases of sweet potato and beans
14. Diseases of coconut and arecanut
15. Diseases of tea
16. Diseases of coffee
17. Diseases of rubber and cocoa

Practical

Study of symptoms and host-parasite relationship of:

1. Diseases of rice
2. Diseases of maize and sorghum
3. Diseases of pearl millet, finger millet and small millets,
4. Diseases of pigeonpea, urdbean, mungbean, soybean and cowpea

5. Diseases of groundnut , sesame and castor
6. Diseases of tobacco, jute and mulberry
7. Diseases of banana
8. Diseases of guava, papaya and pomegranate
9. Diseases of crucifers
10. Field visit/ exposure visit to hilly fruits , vegetables and plantation crops
11. Diseases of tomato, brinjal and okra
12. Diseases of sweet potato and beans
13. Diseases of coconut and arecanut
14. Diseases of tea
15. Diseases of coffee
16. Diseases of rubber and cocoa
17. Final practical examination

Reference Books

1. Arjunan.G. Karthikeyan, G, Dinakaran ,D. Raguchander,T. 1999 Diseases of Horticultural Crops, AE Publications, Coimbatore.
2. Rangasawmi ,G and Mahadevan, A. 1998. Diseases of crop Plants in India, Prentice Hall of India Pvt. Ltd., New Delhi
3. Prakasam, V., Valluvaparidasan, V., Raguchander, T. and K.Prabakar. 1997. Field crop diseases, AE Publication, Coimbatore.

E- Books

2. Agrios, G.N. 2008. Plant Pathology, Academic Press, New York.
3. Rangaswami, G. 2005. Diseases of Crop plants in India. Prentice Hall of India Pvt. Ltd., New Delhi.

Thakur, B.R. 2006. Diseases of field crops and their management

ARM 301 Entrepreneurship Development and Business Management (1+1)

Theory

Unit I – Entrepreneur and Entrepreneurial Process

Concept and Types of Entrepreneurship - Characteristics of Entrepreneurs and Entrepreneurial Skills - Entrepreneurial process – Importance of Entrepreneurship.

Unit II – Entrepreneurship Opportunities

Innovation - principles of innovation - Sources of innovative opportunities - Business environment – Micro and Macro environment - MSME Classification and Opportunities for rural entrepreneurship - KVIC classification, Startup and Business incubators. Agribusiness – Importance, Opportunities and Challenges.

Unit III – Managerial Functions – Planning and Organizing

Management Functions – Planning – Types of Plans and Steps in Planning, Organizing – Principles and Departmentation.

Unit IV – Managerial Functions – Staffing, Directing and Control

Staffing – Job Analysis, Human Resource Planning Process, Recruitment and Selection, Directing- Principles, Techniques and Supervision, Controlling – Process and Types.

Unit V – Functional Areas of Management

Operations Management – Meaning and Scope, Supply Chain Management – Drivers and flows and Total Quality Management – Meaning and Principles, Marketing Management – Market Segmentation and Marketing Mix Financial Management – Meaning, Objectives and Scope.

Practical

Assessment of entrepreneurial traits-Identification of new business opportunities-Exercise on SWOC Analysis of Agribusiness Sector in India -Market survey for understanding customer needs-Starting new business - Visit to firms / discussion with entrepreneurs-Documenting Procedure for Establishing Agribusiness Firms-Government programs and institutions for entrepreneurship development-Financing new agribusiness ventures - Visit to banks / discussion-Exercise on Demand Forecasting for Agricultural Inputs/Products-Preparation of Advertisement and Sales Promotion Measures for Agribusiness-Exercise on Inventory Management – ABC Analysis and EOQ Model-Exercise on discounted measures of Capital Budgeting-Calculation of Break Even Point and its Business Implication-Understanding balance sheet and income statement-Financial Performance Analysis - Ratio Analysis.

Lecture Schedule

13. Concept of Entrepreneurship and Types of Entrepreneurship
14. Characteristics of Entrepreneurs and Entrepreneurial Skills
15. Entrepreneurial process — Importance of Entrepreneurship
16. Innovation - principles of innovation - Sources of innovative opportunities
17. Business environment – Micro and Macro environment
18. MSME Classification and Opportunities for rural entrepreneurship - KVIC classification, Startup and Business incubators
19. Agribusiness – Importance, Opportunities and Challenges
20. Management Functions – Planning – Types of Plans and Steps in Planning
- 21. MID SEMESTER EXAMINATION**

22. Organizing – Principles and Departmentation
23. Staffing – Job Analysis, Human Resource Planning Process, Recruitment and Selection
24. Directing – Principles, Techniques and Supervision
25. Controlling – Process and Types
26. Functional Areas of Management – Operations Management – Meaning and Scope
27. Supply Chain Management – Importance, Drivers and flows and Total Quality Management – Meaning and Principles
28. Marketing Management – Market Segmentation and Marketing Mix
29. Financial Management –Meaning, Objectives and Scope

Practicals schedule

24. Assessment of entrepreneurial traits
25. Identification of new business opportunities
26. Exercise on SWOC Analysis of Agribusiness sector in India
27. Market survey for understanding customer needs
28. Starting new business - Visit to firms / discussion with entrepreneurs
29. Documenting Procedure for Establishing Agribusiness Firms
30. Government programs and institutions for entrepreneurship development
31. Financing new agribusiness ventures - Visit to banks / discussion
32. Exercise on Demand Forecasting for Agricultural Inputs/Products
33. Preparation of Advertisement and Sales Promotion Measures for Agribusiness
34. Exercise on Inventory Management – EOQ Model and ABC Analysis
35. Exercise on discounted measures of capital budgeting
36. Calculation of Break Even Point and its Business Implication
37. Business Plan Preparation
38. Understanding balance sheet and income statement
39. Financial Performance Analysis - Ratio Analysis
40. **PRACTICAL EXAMINATION**

References

52. Chandrasekaran N. and G. Raghuram. Agribusiness Supply Chain Management. 2014. CRC Press, Taylor & Francis Group, Brooklyn.
53. Joseph, L. Massie. 1995. Essentials of Management. Prentice Hall of India Pvt. Ltd. , New Delhi.
54. Mark J Dollinger. 1999. Entrepreneurship Strategies and Resources. Prentice-Hall, Upper Saddle River, New Jersey.
55. Mohanty S K. 2007. Fundamentals of Entrepreneurship. Prentice Hall India Ltd. , New Delhi.
56. Peter F. Drucker, 2006. Innovation and Entrepreneurship. HarperBusiness; Reprint edition, New York.
57. Poornima M. Charantimath 2005. Entrepreneurship Development and Small Business Enterprise, Pearson Education India, New Delhi.
58. Prasanna Chandra, 2007. Financial Management: Theory and Practice, McGraw-Hill Education, New Delhi.
59. Thomas W Zimmer and Norman M Scarborough. 1996. Entrepreneurship. Prentice-Hall, New Jersey.

E- References

21. www.ediindia.org/
22. iie.nic.in/

23. msme.gov.in/
24. niesbudtraining.org
25. www.nimsme.org/
26. www.nsic.co.in/
27. <https://www.nabard.org/>

AGR 301 Practical Crop Production-I (Kharif crop) (0+2)

Practical Schedule:

9. Rice (*Transplanted rice or Direct sown rice*):

Transplanted rice:

9. Rice ecosystems - Climate and weather - Seasons and varieties of Tamil Nadu.
10. Preparation of nursery - Application of manures to nursery - seed treatment - Forming nursery beds and sowing seeds - Weed management and plant protection to nursery.
11. Preparation of main field - Application of organic manures - Green manuring - Bio-fertilizers - Pulling out seedlings and transplanting - Rajarajan 1000 (SRI) - Application of herbicides - Water management - Nutrient management - Plant protection measures - Mechanization in rice cultivation - Recording growth, yield attributes and yield.
12. Harvesting, threshing, drying and cleaning the produce - Working out cost of cultivation and economics.

Practical Schedule:

Transplanted rice:

- 1&2. Study of rice ecosystems, climate, weather, seasons and varieties of Tamil Nadu.
- 3&4. Selection of nursery area, preparation of nursery, application of manures and fertilizer to nursery.
- 5&6. Acquiring skill in seed treatment, seed soaking and incubation, nursery sowing and management.
- 7 & 8. Study and Practice of main field preparation and puddling operations.
- 9&10. Practicing of field preparatory operations - sectioning of field bunds and plastering, leveling and basal application of fertilizers.
- 11 &12. Practicing transplanting techniques in lowland rice.
- 13 &14. Estimation of plant population and acquiring skill in gap filling and thinning.
- 15 &16. Study of weeds and weed management in rice.
- 17 &18. Study and practice of green manuring and bio-fertilizer application in rice.
- 19 & 20. Acquiring skill in nutrient management and practicing top dressing techniques.
- 21 & 22. Study of water management practices for lowland rice.
- 23 & 24. Observation of insect pests and diseases and their management.
- 25 & 26. Recording growth and other related characters of rice.
- 27 & 28. Estimation of yield and yield parameters in rice.
- 29 & 30. Harvesting, threshing and
- 31 & 32. Cleaning, drying and calculating the yield of produce
15. Working out cost of cultivation and economics.
16. **Practical Examination.**

References:

Ahlawat, I. P. S. , Om Prakash and G. S. Saini. 1998. Scientific Crop Production in India. Rama Publishing House, Meerut.

Chidda Singh. 1997. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd. , New Delhi.

Crop Production Guide. 2012. Directorate of Agriculture, Chennai and Tamil Nadu Agricultural University, Coimbatore.

Rajendra Prasad. 2004. Text Book on Field Crop Production, Indian Council of Agrl. Research, New Delhi.

K Annadurai and B Chandrasekaran. 2009. A Text Book Of Rice Science. Scientific Publishers.

Reddy,S. R. 2012. Agronomy of field crops. Kalyani publishers, New Delhi.

ICAR 2015. Hand book of Agriculture. Indian Council of Agriculture, New Delhi.

E-References:

1. WWw. irri. org
2. www. crri. nic. in
3. www. drrindia. org

HOR 311. Postharvest management and value addition of fruits and vegetables (1+1)

Theory

Unit I

Scope and Importance of postharvest technology of fruits and vegetables- factors responsible for postharvest losses – constraints –preharvest factors affecting postharvest quality - postharvest operation – precooling, grading, cleaning, waxing on shelf life of fruits and vegetables.

Unit II

Physiological and biochemical changes occurring during maturity and ripening- Respiration and factors affecting respiration rate - role of ethylene in regulation of ripening.

Unit III

Packaging and storage of fruits and vegetables - heat, chilling and freezing injury - storage (ZECC, cold storage, CA, MA and hypobaric) - cold chain management for fruits and vegetables

Unit IV

Value addition concepts, principles and methods of preservation, intermediate moisture food –Jam, jelly, marmalade, preserve, candy- concepts and standards of fermented and non fermented beverages. Tomato products – Concepts and Standards

Unit V

Drying and dehydration of fruits and vegetables, concepts and methods, osmotic drying. Canning- concepts-processing of canned products-spoilage and prevention. Packaging of products –quality standards- GMP,HACCP, FSSAI, Codex alimentarius and ISO certification.

Practical

Pre harvest operations to improve postharvest shelf life - assessment of maturity indices and harvest criteria of fruits and vegetables-different types of packaging for shelf life extension- of chilling and freezing injury in vegetables and fruits- estimation of ethylene evolution in fruit crops- Identification of postharvest diseases and disorders- Postharvest machineries -extraction and preservation of pulps and juices. Preparation of jam, jelly, RTS, nectar, squash, osmotically dried products, fruit bar and candy and tomato products, canned products. Quality evaluation of products –physio-chemical and sensory evaluation. Visit to cold storage / packaging unit -visit to processing unit/ industry.

Theory schedule

27. Scope and importance of postharvest technology of fruits and vegetables
28. Factors responsible for post harvest losses.
29. Preharvest factors affecting postharvest life of fruits and vegetables
30. Physiological and biochemical changes during maturity and ripening
31. Respiration and ripening and the factors affecting the ripening of fruits and vegetables
32. Role of ethylene in ripening of fruits and vegetables
07. Preharvest operations to extend shelf life of fruits and vegetable crops
08. Postharvest handling of the produce (washing, fungicide treatment, precooling, grading, sorting waxing and nano coating)
20. **Mid -semester examination**
21. Importance of packaging, types of packaging and packaging materials.
22. Methods of storage of fruits and vegetables viz. , Zero energy cool chamber, cold storage, controlled atmosphere, modified atmosphere and hypobaric storage and management of cold chain for export of high value fruits and vegetables

23. Principles and methods of preservation of fruits and vegetables
24. Preservation and value addition of fruits viz. , jam, jelly, marmalade, preserve and candy
25. Concepts and standards in fermented and non fermented beverages from fruits and vegetables
26. Drying, dehydration and osmotic dehydration of fruits and vegetables- concepts and methods
27. Canning of fruits and vegetables- concepts-processing of canned products-spoilage and prevention
28. Packaging of products and standards in value addition of fruits and vegetables viz. , GMP, HACCP, FSSAI, Codex alimentarius and ISO certification.

Practical Schedule

51. Preharvest operations to improve post harvest shelf life of fruits and vegetable crops
52. Assessment of maturity indices and harvest criteria for fruits and vegetable crops
53. Methods of packaging in fruits and vegetables
54. Identification and causes of chilling and freezing injury in vegetables and fruits
55. Estimations of ethylene evolution in fruit crops
56. Identification of postharvest diseases and disorders of fruits and vegetable crops
57. Postharvest machineries for fruits and vegetables crops
58. Postharvest handling of the produce (washing, fungicide treatment, grading, sorting, precooling, waxing and nano coating).
59. Preparation of jam/Jelly and quality evaluation of products
36. Preparation of RTS, nectar, squash and quality evaluation of products
37. Processing of dried and dehydrated fruits and vegetables
38. Preparation of fruit bar and candy and quality evaluation of products
39. Preparation of tomato products
40. Processing of canned fruits and vegetables
- 15 Quality evaluation of products –physio-chemical and sensory evaluation. .
18. Visit to processing unit/ industry and cold storage / packaging unit
19. **Practical examination**

Reference

1. Adel A. Kader. 2002. Post Harvest Technology of Horticultural Crops. University of California Agrl. And Natural Resources Publication.
2. Ashwani. S. and Goel. 2007. Post harvest management and value addition. Daya publishing house, New Delhi.
3. Swati Barche and K. S. Kirad. 2010. Post harvest handling of fruits, vegetables and flowers. Jain Brothers, New Delhi.
4. Sudheer,K. P. and V. Indira. 2007. Post harvest technology of horticultural crops, New India publishing agency, New Delhi.
5. Bhutani, R. C. 2003 Fruit and Vegetable Preservation. Biotech Books, Delhi. 89
6. Pruthi, J. S. 2000. Major Spices and condiments. Productions and post harvest technologies. ICAR publications, New Delhi.
7. Verma, L. R and V. K. Joshi. 2000. Post harvest technology of fruits and vegetables –Handling, Processing, Fermentation and Waste Management. Indus publishing House.

E-References

?? www.fao.org/inpho

?? www.postharvest.ucdavis.edu

?? www.postharvest.ifas.ufl.edu

?? www.postharvest.com.au

?? www.ams.usda.gov

SAC 301 Manures, Fertilizers and Soil Fertility Management (2+1)

Theory

Unit-I : Introduction and importance of organic manures, properties and methods of preparation of bulky and concentrated manures. Green manuring and green leaf manuring. Integrated nutrient management. Carbon sequestration- Carbon trading

Unit-II: Chemical fertilizers: classification, composition and properties of major nitrogenous, phosphatic, potassic fertilizers, secondary & micronutrient fertilizers. Mixed/ Complex/customized/designer fertilizers, water soluble and liquid fertilizers, nano fertilizers & Soil amendments. Fertilizer Storage and Fertilizer Control Order.

Unit-III: History of soil fertility and plant nutrition. Criteria of essentiality. Role, deficiency and toxicity symptoms of essential plant nutrients. Mechanisms of nutrient transport to plants, factors affecting nutrient availability to plants. Chemistry of soil nitrogen, phosphorus, potassium, calcium, magnesium, sulphur and micronutrients.

Unit-IV : Soil fertility evaluation-Soil testing, Critical levels of different nutrients in soil. Forms of nutrients in soil, plant analysis, and rapid plant tissue tests. Indicator plants.

Unit-V L Methods of fertilizer recommendations to crops. Factor influencing nutrient use efficiency (NUE), methods of application under rainfed and irrigated conditions

Practical

Introduction of analytical instruments and their principles, calibration and applications- Colorimetry, Spectrophotometry, Flame Photometry and Atomic Absorption Spectrophotometry. Estimation of available N in soil. Estimation of available P in soil. Estimation of available K in soil. Estimation of available S in soil. Estimation of available Ca and Mg in soil. Estimation of available Micronutrients in soil. Estimation of N in plant. Estimation of P in plant. Estimation of K in plant. Estimation of S in plant. Fertilizer calculations. Visit to STL and FTL/Fertilizer manufacturing or mixing unit.

Lecture Schedule

Organic manures-Importance, Definition, classification, properties and sources- Fortified organics.

Methods of preparation of bulky and concentrated manures - Composting techniques- Aerobic and anaerobic (Bangalore & Coimbatore method) enriched FYM and vermicompost.

Composting of organic waste-Sugarcane trash and coir waste. Green manuring and green leaf

45. Soil carbon sequestration and carbon trading.
46. Fertilizers- Definition, classification of N, P and K fertilizers.
47. N fertilizers- Urea, ammonium sulphate, ammonium nitrate, CAN, properties and their reactions in
48. P fertilizers- Rock phosphate, bone meal, basic slag, single super phosphate, diammonium phosphate, triple super phosphate, properties and their reactions in soil.
49. K fertilizers- MOP and SOP- properties and reactions in soil.
50. Secondary nutrient fertilizers and micronutrient fertilizers.
51. Complex fertilizers- definition, manufacture of ammonium phosphate, nitro phosphate and N,P,K complexes.
73. Mixed fertilizers-definition, preparation and compatibility.
74. Customized/designer fertilizers, water soluble, liquid fertilizers and Nano fertilizers.
75. Micro nutrient mixtures and chelated micronutrients. Soil amendments
76. Fertilizer Storage and Fertilizer Control Order
77. History of soil fertility, productivity plant nutrition and criteria of essentiality. Functions, deficiency and toxicity symptoms of N, P and K.
Functions, deficiency and toxicity symptoms of Secondary, micronutrient and beneficial elements

21. Mid semester Examination

22. Mechanisms of nutrient transport to plants

23. Sources, forms, mobility, transformation, fixation, losses and availability of nitrogen in soil

24. Sources, forms, mobility, transformation, fixation, losses and availability of phosphorous in soil

29. Sources, forms, mobility, transformation, fixation, losses and availability of potassium in soil

Sources, forms, mobility, transformation, fixation, losses and availability of calcium, magnesium and sulphur in soil

Sources, forms, mobility, transformation, fixation, losses and availability of micronutrients in soil

Concepts and approaches of soil fertility evaluation - Liebig's Law, Mitscherlich's law and Bray's nutrient mobility concept. Approaches - Deficiency symptoms, tissue analysis, biological tests and chemical tests.

Techniques/ methods of soil fertility evaluation: Crop logging, critical level, DRIS, Isotopic nutrient availability techniques.

Fertilizer application: Soil and foliar application.

Fertigation - Definition - Types of fertigation and scheduling.

Fertilizer application- specific methods - Seed coating, pelletization, seedling dipping - Nutriseed pack

Nutrient management concepts – 4 R concept-Agronomic approach, Inductive(STCR), SSNM and RTNM

Nutrient management concepts -Deductive, INM, IPNS. Tools - DSSIFER and VDK

Nitrogen use efficiency - Slow release N fertilizers - Significance and enhancement techniques

Nutrient use efficiency of P, K and micronutrients and their enhancement techniques

Nutrient management for dry land and rainfed agriculture.

Long term effect of fertilization on soil health-PME and LTFE.

Practical Schedule

40. Analytical instruments : Principles, calibration and applications – Colorimetry and Spectrophotometry

41. Flame photometry and Atomic absorption spectroscopy

42. Estimation of available N in soil

43. Estimation of available P (Olsen P and Bray P) in soil

44. Estimation of available K in soil

45. Estimation of available sulphur in soil by turbidimetry

46. Estimation of available Ca and Mg in soils.

47. Estimation of DTPA extractable micronutrients in soil

44. Estimation of N content in plant.

15. Estimation of P content in plant.

16. Estimation of K content in plant.

17. Estimation of S content in plant

18. Estimation of N content in manure

19. Estimation of P content in manure

20. Estimation of K content in manure

21. Colloquium on establishment of soil testing laboratories -Fertilizer calculations-Soil test based fertilizer prescription

22. Visit to STL and FTL / Fertilizer manufacturing or mixing unit

23. Practical Examination

References

35. John L. Havlin, James D. Beaton, Samuel L. Tisdale and Werner L. Nelson. 2011. Soil Fertility and Fertilizers- An Introduction to Nutrient Management. PHL Learning Pvt. Ltd. ,New Delhi
36. Gupta, P. K. 2012. A Handbook of Soil, Fertilizer and Manure. Agrobios (India), Jodhpur.
37. Michael, A. M. 2009. Irrigation Theory and Practice. Second Edition. Vikas Publishing House Pvt. Ltd. , New Delhi.
38. Ramesh Chandra and S. K. Singh. 2009. Fundamental and Management of soil quality. Westville Publishing House, New Delhi.

E - references

- 15 [www.fspublishers.org/ijab/past-issues/IJAB Vol_5_No_3/47. pdf](http://www.fspublishers.org/ijab/past-issues/IJAB_Vol_5_No_3/47.pdf)
- 16 [www.springerlink.com/index/IQ11256h8t325054. pdf](http://www.springerlink.com/index/IQ11256h8t325054.pdf)
- 17 [www.ipni.net/ppiweb/bcrops.nsf/\\$webindex/.../Better_Crops_2009-4 J_. pdf](http://www.ipni.net/ppiweb/bcrops.nsf/$webindex/.../Better_Crops_2009-4_J_.pdf)
- 18 [onlinelibrary.wiley.com/doi/10.1002/9780470431771 . index/pdf](http://onlinelibrary.wiley.com/doi/10.1002/9780470431771.index/pdf)
- 19 [agtr.ilri.cgiar.org/agtrweb/Documents/Library/docs/. ./Module4. htm](http://agtr.ilri.cgiar.org/agtrweb/Documents/Library/docs/. ./Module4.htm)
- 20 [www.uoa.edu.er/academics/graduate/.../courses. html -](http://www.uoa.edu.er/academics/graduate/.../courses.html)
- 21 [www.fao.org/wairdocs/ilri/x5546e/x5546e08. htm](http://www.fao.org/wairdocs/ilri/x5546e/x5546e08.htm)
- 22 [www.fao.org/wairdocs/ilri/x5546e/x5546e08. htm](http://www.fao.org/wairdocs/ilri/x5546e/x5546e08.htm)
- 23 [www.uoa.edu.er/academics/graduate/.../courses. html -](http://www.uoa.edu.er/academics/graduate/.../courses.html)
- 24 [www.ncpahindia.com/articles/article17. pdf-Similar](http://www.ncpahindia.com/articles/article17.pdf)
- 25 [www.energy.ca.gov/process/agriculture/ag_pubs/fertigation. pdf -](http://www.energy.ca.gov/process/agriculture/ag_pubs/fertigation.pdf)
- 26 [www.soilandhealth.org/.../010117attrasoilmanual/010117attra. html](http://www.soilandhealth.org/.../010117attrasoilmanual/010117attra.html)
- 27 [goliath.ecnext.com/.../Deficiencies-in-the-soil-quality. html-](http://goliath.ecnext.com/.../Deficiencies-in-the-soil-quality.html)

THEORY**Unit I: Introduction to Protected Cultivation and Green houses**

Protected cultivation – need, advantages and limitations – present status. Green house technology – Introduction – Types of greenhouses- Plant response to greenhouse environment.

Unit II Design of Greenhouses

Planning and design of greenhouses - Design criteria of green house for cooling and heating purposes - Green house equipment - Materials for construction of green houses - Irrigation systems used in greenhouses.

Unit III Applications of Greenhouses

Typical applications - Passive solar greenhouse - Hot air greenhouse heating systems - Greenhouse drying - Cost estimation and economic analysis.

Unit IV: Engineering Properties of Food Materials

Physical properties- size-shape, Aero-hydro dynamic properties, thermal properties- specific heat- thermal conductivity- thermal diffusivity, and their application in PHT equipment design and operation.

Unit V: Drying and Dehydration

Drying and dehydration, Moisture determination- direct method and indirect method of moisture determination, drying rate curves- constant rate period, CMC- Falling rate period, EMC, Drying methods- contact type dryers- convective type dryer- radiation dryer, commercial grain dryer -deep bed dryer-flat bed dryer- tray dryer-fluidized bed dryer. -Recirculatory dryer- solar dryer.

Unit VI: Material Handling

Introduction- selection of material handling machines, Belt conveyor- belt conveyor idlers- idler spacing- belt material- belt tension, Bucket conveyor- head section-Boot section-elevator legs- elevator belts- bucket drive mechanism. Screw conveyor- Details -various shapes screw trough- capacity – horse power, pneumatic conveyor – advantages and limitations.

PRACTICAL

Determination of engineering properties (shape and size, bulk density and porosity of biomaterials). Performance evaluation of screen cleaner. Determination of Moisture content of various grains by direct method and indirect method. Determination of capacity of belt conveyor and bucket conveyor.

LECTURE SCHEDULE

- | | | |
|---|--|------------------------------|
| 1 | Introduction to protected cultivation – need, advantages and limitations and present status – protected cultivation for horticultural crops | TB 1: 1-9 |
| 2 | Green house technology – Definition, History and evolution – Advantages and limitations | TB 1: 77-84 |
| 3 | Types of greenhouses – based on shape, cost, utility and cladding materials | TB 2: 9-24 |
| 4 | Plant response to greenhouse environment – Sunlight, Temperature, Relative Humidity, Carbon dioxide enrichment – Soil / media | TB 5: 61-62
TB 5: 118-124 |
| 5 | Planning and design of greenhouses – Criteria for site selection – orientation – structural design - Design criteria of green house for cooling and heating purposes | TB 2: 25-46 |
| 6 | Equipment and components of a Greenhouse – Summer cooling and winter cooling, natural ventilation and forced ventilation | TB 2: 96-102 |
| 7 | Materials for construction of greenhouses – Wood, iron, glass, polyethylene film | TB 3: 16-28 |
| 8 | Irrigation systems in greenhouses - Rules of watering, hand watering, perimeter watering, overhead sprinklers, boom watering and drip irrigation | TB 4: 61-64
TB 5: 110-116 |
| 9 | Mid semester examination. | |

10	Typical applications of greenhouses – passive solar greenhouse, hot air greenhouse heating system and green house dryng- Cost estimation and economic analysis.	TB 4: 37-38, TB 4: 77-87
11	Physical properties – size- shape- sphericity- density- specific gravity, Aero-hydrodynamic properties – Terminal velocity- drag coefficient	TB 1: 6-11, TB 2: 6-10, TB 3: 2-8
12	Thermal properties- specific heat- thermal conductivity- thermal diffusivity, Application of Engineering properties of cereals, pulses and oilseeds in PHT equipment design and operation	TB 1: 1-6, 15-18, TB 2: 7-9 TB 3: 20-23
13	Moisture measurement, direct method and indirect method of moisture measurments, Drying and dehydration, Drying theory, drying rate curves- constant rate period, CMC- Falling rate period , EMC	TB 1: 107-130 TB 2: 25-49
14	Various drying method, contact type dryers- convective type dryer- radiation dryer	TB 1: 132--161 TB 2: 66-70
15	Commercial grain dryer- deep bed dryer-flat bed dryer- tray dryer-fluidized bed dryer, recirculatory dryer- solar dryer.	TB 1: 143-161 TB 2: 71-90
16	Material handling equipment-introduction, screw conveyor working principle, and selection	TB 1: 289-297 TB 3: 317-332
17	Bucket elevator- head section-Boot section-elevator legs- elevator belts-bucket drive mechanism- Screw conveyor and pneumatic conveyor working principle and selection	TB 1: 297-310 TB 3: 332-347

PRACTICAL SCHEDULE

1. Study of different types of green houses based on shape, etc
2. Measurement of weather data in green houses
3. Computing the rate of air exchange in an active summer and winter cooling systems
4. Experiment on determination of shape and size of the cereal grains
5. Experiment on determination of bulk density and porosity of biomaterials
6. Determination of Moisture content of various grains by direct method
7. Determination of Moisture content of various grains by indirect method
8. Experiment on determination of terminal velocity of different grains
9. Performance evaluation of available screen cleaner
10. Performance evaluation of fluidized bed dryer
11. Performance evaluation of tray dryer
12. Determination of Capacity of a belt conveyor and its performance evaluation
13. Determination of Capacity of a bucket conveyor and its performance evaluation
14. Field visit to greenhouse
15. Visit to Horticulture Research Station, Udthagamandalam
16. Visit to food modern rice mill
17. Final Practical Examination

TEXT BOOK

1. Singh Brahma and Balraj Singh., 2014. Advances in Protected Cultivation, New India Publishing Company.
2. Greenhouse Management for Horticulture crops – S.Prasad&U.Kumar., 2013. AGROBIOS (INDIA).
3. Greenhouse Management for Horticulture crops – Sandhya Sharaf., 2012. Oxford Book Company.
4. Greenhouse for Homeowners and Gardeners - John W. Bartok, Jr., 2000. NRAES
5. Greenhouse Engineering - Robert A. Aldrich and John W. Bartok, Jr., 1994. NRAES
6. Sahay K.M and Singh K .K. Unit operations of agricultural processing. Vikas Publishing house Pvt. Ltd. New Delhi.
7. Chakraverty A. Post-harvest technology of cereals, Pulses and Oil seeds. published by Oxford & IBH publishers. New Delhi.
8. M.N.Dabhi and N.K. Dhamsananiya Agricultural Processing and food engineering (A basic approach). published by Kalyani Publisher. New Delhi.

REFERENCE BOOKS

1. Brennan J.G. Food engineering operations. Second edition. Published by applied science Publisher limited, London.

2. Fellows, P. 2000. Food processing technology Principles and Practice. Second Edition. Published by Woodhead Publishing Limited Abington Hall, Abington Cambridge CB1 6AH, England.
3. Kudra, T. and Mujumdar, A.S. Advanced drying technologies. Marcel Dekker, Inc.

E – BOOKS

[A. Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan, H. S. Ramaswamy.2003. Handbook of Postharvest Technology \(cereals, fruits, vegetables, tea and spices\), Marcel Dekker, Inc.New York, USA.](#)

AGR 302 Rainfed Agriculture and Watershed Management (1+1)

Theory

Unit - I:

Dryland farming - India and Tamil Nadu - Major crops of Dryland in India and Tamil Nadu - rainfed farming - Significance, Characteristics and constraints of dry farming in India - Distribution of Arid and semiarid regions in World, India and Tamil Nadu.

Unit - II:

Rainfall climatology - Length of growing period - Drought - Definition - Types and effects of Drought on crop production - Mechanism of drought tolerance in plants - Drought management - Contingent crop planning - Mid season correction - Mulching - anti transpirants.

Unit - III:

Soil moisture conservation approaches: agronomical, engineering and agrostological measures - In-situ water harvesting, storage and recycling - water harvesting - farm pond, percolation pond.

Unit - IV:

Integrated dry land technologies - Mechanization - Resource management under constraint situation - Cost reduction strategies in crop production - Non-monetary inputs and low cost technologies.

Unit - V:

Watershed management - alternate land use system - Agro forestry systems - Role of institutions - government policies for promotion of dryland farming.

Practical:

Zonation of Dry farming regions of Tamil Nadu, India and World - Characteristics of ACZs of Tamil Nadu and cropping pattern - Study of tools, implements and machineries for tillage, sowing and after cultivation - Rainfall analysis - working out economics - Sustainability Indices - working out LGP - Preparation of contingency crop plan for aberrant rainfall situations - Visit to watershed.

Lecture Schedule:

25. Significance and scope of dry farming in India and history of dryland agriculture.
26. Dry farming and rainfed farming: Definition and Characteristics.
27. Distribution of arid and semi-arid regions in World, India and Tamil Nadu.
28. Major crops of dryland in India and Tamil Nadu.
29. Characteristics of dryland farming and major constraints for crop production.
30. Drought: definition, types and effects of drought on crop production.
31. Drought management strategies and contingent crop planning: mid season correction.
32. Mulching, anti transpirants, in-situ soil moisture conservation techniques and approaches.
- 33. Mid-Semester Examination.**
34. Water harvesting, storage and recycling.
35. Integrated dryland technologies and farm mechanization.
36. Watershed: definition, principles, classification and management.
37. Mechanization in dryland farming.
- 15 Resource management under constraint situations for irrigated and rainfed farming.
- 16 Cost reduction strategies in crop production - cropping system, integrated farming system and dry farming.

- 17 Non-monetary inputs and low cost technologies for crop production.
- 18 Alternate land use systems in dryland - role of institutions – policies.

Practical Schedule:

18. Studies on rainfall pattern in rainfed areas of the country and pattern of onset and withdrawal of monsoons.
19. Agroclimatic, Agro ecological zones and characteristics.
20. Zonation of dry farming regions of Tamil Nadu, India and World.
21. Characteristics of ACZs of Tamil Nadu and cropping pattern.
22. Cropping and farming systems in dryland.
23. Skill development in Seed hardening technique.
24. Input management and efficiency in dryland.
25. Soil erosion and soil conservation practices.
26. Water harvesting structures and their use.
27. Study of methods to reduce evaporation and transpiration.
28. Study of tools, implements, and machineries for tillage, sowing and after cultivation and assessing their efficiencies.
29. Indices in dry farming - working out LGP and planning for cropping system.
30. Drought management technologies in dryland agriculture.
31. Preparation of contingency crop plan for aberrant rainfall situations.
32. Alternate land use system and their merits.
33. Visit to watershed area to study the impact of various soil and moisture conservation methods.
34. **Practical examination.**

References:

- Govindan K. and V. Thirumurugan. 2003. Principles and practice of Dryland Agriculture, Kalyani Publishers, Chennai.
- Rengasamy P. 1990. Dry farming Technology in India. Agri publishing Academy, New Delhi.
- Reddy, G. S. , Reddy, Y. V. R. , Vittal, K. P. R. , Thyagaraj, C. R. , Ramakrishna, Y. S. and L. L Somani. 2008. Dryland Agriculture. Agrotech Publishing Academy, Udaipur
- Jat. , Bharkar. , Sharma and Kothari. 2013. Dryland Technology. Scientific Publishers, Jodhpur
- Pradeep, S. 2014. Dryland Agriculture. Discovery Publishing House Pvt. Ltd, NewDelhi
- Widtsoe, J. A. 2012. Dry Farming for Sustainable Agriculture. Agrobios (India), Jodhpur

- References:

www.tnau.ac.in
www.crida.org
www.icrisat.org

AEN 301 PESTS OF FIELD CROPS AND STORED PRODUCES AND THEIR MANAGEMENT (1+1)

Theory

Unit I: Pests of Cereals and Millets

Distribution, bionomics, symptoms of damage and management strategies for insects and non-insect pests of rice, wheat, maize, sorghum, cumbu, ragi, tenai. Integrated Pest Management - case studies in rice.

Unit II: Pests of Pulses and Oilseeds

Distribution, bionomics, symptoms of damage and management strategies of insects and non-insect pests of pulses (redgram, green gram, black gram, bengal gram, cowpea.), groundnut, castor, gingelly, sunflower, safflower, jatropha, soybean and mustard. Integrated Pest Management - case studies in groundnut.

Unit III. Pests of Cotton and Sugarcane

Distribution, bionomics, symptoms of damage and management strategies of insects and non-insect pests of cotton and sugarcane. Integrated Pest Management - case studies in cotton.

Unit IV: Pests of Green Manures, Forage Crops, Stored Products and Non Insect Pests

Distribution, bionomics, symptoms of damage and management strategies of pests of green manures (Sunnhemp, Sesbania, Daicha. Glyricidia), forage crops (Lucerne and Subabul) and stored products. Rodents and birds of agricultural importance and their management. Locusts and their management.

Theory schedule:

Distribution, bionomics, symptoms of damage and management strategies for insects and non-insect pests of

1. Rice – Sucking pests
2. Rice – Borers and defoliators
3. Maize, sorghum and cumbu
4. Wheat, ragi and tenai
5. Redgram, bengalgram, Blackgram, greengram and cowpea
6. Groundnut, gingelly and sunflower
7. Castor, soybean, safflower, jatropha and mustard
8. Cotton
9. Mid semester examination
10. Sugarcane
11. Green manures- sunnhemp, sesbania, daincha, lucerne, subabul and gliricidia
12. Role of physical, biological, mechanical and chemical factors in deterioration of grain
13. Stored product pests
14. Methods of grain storage and various methods of stored product pest management
15. Mites, slugs and snails, rodents and bird pests
16. Locusts and their management
17. Integrated Pest Management in rice and cotton

Practical schedule:

Identification of symptoms of damage and life stages of pests of

1. Pests of rice (sucking pests)
2. Pests of rice (borers and defoliators)
3. Pests of maize, sorghum and cumbu
4. Pests of wheat, ragi and tenai
5. Pests of redgram and bengalgram
6. Pests blackgram, greengram and cowpea

7. Pests of groundnut, gingelly and sunflower
8. Pests of castor, soybean, safflower, jatropha and mustard
9. Pests of cotton (sucking pests)
10. Pests of cotton (bollworms and defoliators)
11. Pests of sugarcane
12. Pests of green manures-sunn hemp, sesbania, daincha, lucerne, subabul and gliricidia
13. Pests of stored products
14. Gadgets for management of stored product insects
15. Rodents and Birds pests in field and storage
16. Visit to FCI godown and farmer's fields
17. Final practical examination

References:

A. Text Books:

1. Manisegaran, S. and R.P.Soundararajan. 2010. *Pest Management in Field Crops- Principles and Practices*. Agrobios, Jodhpur, India. 316p. {ISBN (10): 81-7754-321-0}
2. David, B.V. and V.V. Ramamurthy. 2011. *Elements of Economic Entomology*, Namrutha Publications, Chennai. 386 p. {ISBN: 978-81-921477-0-3}

E- RESOURCES:

1. <http://www.ncipm.org.in>
2. <http://agritech.tnau.ac.in/>
3. <http://www.nbaii.res.in/>
4. <http://www.nrcg.res.in/>

VII SEMESTER

S No.	Course code	Course Title	Credit Load
1	AEX 401	Student READY programme (Rural Agricultural Work Experience and Agro-industrial Attachment)	0+20
2	AGR 401	Project Report Preparation, presentation and Evaluation	0+1
3.	AEX 402	All India Study Tour*	0+1

Non gradial compulsory

AEX 401 Rural Agricultural Work Experience (RAWE) (0+20)

UNIT I Village Resource Inventory and Planning (using PRA tools, Rich pictures, GIS maps, secondary data, interview, etc.)



Describe the Natural Resources - Village boundaries, topography, historical background, water resources (river, canal, tank, etc.), soil resources, vegetation (trees, crops, etc.), fodder, animal

husbandry (milch cattle, poultry, goatery, fishery, etc.), wild animals, climate, land utilization pattern, etc.



Describe the Agricultural scenario - Cropping pattern, cropping systems, farming systems, area, production and productivity of crops, adoption pattern of recommended varieties / hybrids,

technologies and machinery / implements, organic farming, contract farming, etc.



Explain the Demographic details – population, literacy, land holdings, farmers, farm women, youth, caste, labour, etc.



Analyze the Social factors – social structure, social stratification, social change, social groups, culture, social control, leadership, social processes, migration, social customs, social issues, etc.



Study the Socio-psychological factors – group processes / dynamics, attitude towards innovations, etc.



Assess the Village Infrastructure - Educational institutions, Government institutes / offices, private firms / offices, NGOs, Societies, Banks, Panchayat Union / Grama Panchayat, Clubs, SHGs, FPOs,

Associations, Communication facilities, transport facilities, railway station, police station, hospitals, clinics, veterinary hospital, post office, markets, community centers, religious places of worship, etc.



Analyze the Problems / Constraints – Problem / Constraints related to farming, marketing, processing, transport, communication, access to extension and other services, etc.



Prepare village development plans in consultation with different stakeholders.

Unit II Farm Resource Inventory and Planning (using maps, Rich pictures, farm system modeling, family tree charts, flow diagrams, interview, etc.)



Describe the Farm boundaries, topography, water resources, soil resources, vegetation, animal enterprises, etc.



Describe the cropping pattern, cropping system, farming system, agri-business, etc.



Explore Farmers Practices – Indigenous Technical Knowledge (ITK).



Identify the constraints of the system environment (natural, economic, social, political, legal).



Assess the linkages with Extension agencies, Markets, Input agencies, Media, Development departments, etc.

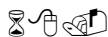


Identify and describe all the people involved in the farm, their work, roles, visions, needs, values, interests and relationships.



Analyze the system in terms of satisfying current needs. What are the critical factors that need to be managed to sustain the system? Are there opportunities for growth and development to satisfy the

future needs of the system? Are there threats that also need to be managed?



Describe the different sub-systems viz., production sub-system, management sub-system, marketing sub-system, human activity sub-system, landscape and natural sub-system, etc., and their

relationships.



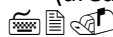
Identify the linkages with the Supra System viz., economic, political, legal and social.



Find out the adoption pattern of recommended varieties / hybrids, technologies, machinery / implements, etc.



Analyze the financial status and performance of the system - Economics of production (area, production, productivity, yield gaps, net returns, cost benefit ratio, etc).



Prepare farm development plans for different types of farmers, by involving them so as to improve their systems.

Unit III Studying activities of State Department of Agriculture

Visit to Office of Assistant Director of Agriculture to study the organizational structure, functions, duties and responsibilities of extension personnel, ATMA, schemes implemented, extension activities

conducted, etc. Involve in different extension activities such as village meetings, demonstrations, campaigns, exhibition, radio / TV programmes and record observations and lessons learnt.

Unit IV Studying activities of an NGO

Visit to an NGO to study the organizational pattern, functions, projects, duties and responsibilities of staff, extension activities, schemes implemented, funding sources, etc.

Unit V Studying activities of an Agri Business Firm

Visit to an Agri-business firm to study the business activities, projects, managerial functions viz., planning, supervision, delegation, communication, budgeting, and related aspects.

AEX 402

All India Study Tour

Syllabus



Visit to important National and International institutes related to agriculture, horticulture, forestry and allied fields in various regions of the country. Exposure to varied agro-climatic zones, crops grown,

cultivation practices, socio-economic and cultural features of the farming community in different parts of the country.

VIII SEMESTER

(Experiential Learning Programme/ HOT)	
Module	Credit Hrs.
1. Module-I	0+10
2. Module-II	0+10
Total	0+20

S.	Course code	Titles of the module	Credits
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No.			
1.	AGM 451	Bio-inoculant production technology	0+10
2.	HOR 451	Hybrid Seed Production in Vegetable Crops	0+10
3.	SAC 451	On Farm Advisory for Soil Health, Water Quality & Plant Nutrition	0+10
4.	AEN 451	Commercial Beekeeping	0+10
5.	SER 451	Commercial Cocoon Production	0+10
6.	ABT 451	Commercial Plant Tissue Culture	0+10
7.	HOR452	Commercial Nursery Technology of Horticultural Crops	0+10
8.	HOR 453	Commercial Landscape Gardening	0+10
9.	PAT 451	Commercial production of Bio-control agents	0+10
10.	PAT 452	Commercial mushroom production	0+10
11.	AMP 451	Commercial broiler and layer production	0+10
12.	SST 451	Commercial seed production	0+10
13.	PBG 451	Hybrid pearl millet seed production	0+10
14.	PBG 452	Hybrid rice parental line seed production	0+10
15.	ARM 451	Managerial skills for Agribusiness	0+10
16.	AGR 451	Development of Integrated Farming system Model	0+10
17.	HOR 454	Protected cultivation of Vegetable crops	0+10
18.	ENS 451	Composting technology	0+10
19.	AGR 452	Organic Agriculture	0+10

AGM 451 Bioinoculants Production Technology (0+10)

Week

Activities

- Biofertilizers - types, production and demand in India; Importance and contribution of biofertilizers in Agriculture and allied sectors. Economics of biofertilizer production. Calculation of commercial production cost – fixed cost- cost of building, equipments and glasswares and variable cost - raw materials, maintenance, labour cost *etc.*,
- Exposure visit to biofertilizer production unit. Facilities and equipments required for laboratory scale, pilot scale and large scale biofertilizer production (liquid and carrier) – principles and specifications. Raw materials required-glass wares, chemicals, printed poly bags and carrier material - specifications of raw materials. Isolation, purification and characterization of nitrogenous biofertilizers – *Azotobacter*, *Azospirillum*, *Rhizobium* and *Gluconoacetobacter*.

- Isolation and purification of nitrogenous biofertilizers – Azolla and Blue Green Algae (BGA). Screening of nitrogen fixers - plant nodulation tests for *Rhizobium*. Use of Gas Chromatography for nitrogenase assay. Nitrogenase activity by ARA (nodule and broth cultures of *Azotobacter*, *Azospirillum* and *Gluconoacetobacter*).
- Isolation, purification and characterization of phosphate solubilizing and potassium releasing (silicate solubilizing) bacteria. Selection of efficient strains by testing their ability under *in vitro* conditions. Isolation of AM spores from soil and morphological characterization of AM spores.
- Selection of efficient AM fungi by plant infection tests. Isolation, purification and characterization of sulphur oxidizing and zinc solubilizing microbes. Screening of efficient sulphur oxidizing and zinc solubilizing microbes.
- Isolation, purification and characterization of plant growth promoting bacteria - Pink Pigmented Facultative Methylophiles (PPFM) and screening of PPFM. Development of markers for easy identification-application of real time PCR in strain identification.
- Preparation of medium and carrier material for large scale production. Mass production of *Azotobacter*.
- Mass production of *Rhizobium*.
- Mass production of *Azospirillum*.
- 41. Mass production of phosphate solubilizer.
- 42. Mass production of *Gluconoacetobacter* and potassium releasing bacterium (silicate solubilizing bacterium).
- 43. Mass production of PPFM.
- 44. Mass production of AM fungi, Azolla and BGA.
- 45. BIS standards / Fertilizer Control Order – Specifications and quality control measures for various biofertilizers. Storage and preservation of various microbial cultures – sub culturing, lyophilization *etc.*, Establishment of Ideal biofertilizer unit; Shelf life and storage of biofertilizers. Constraints in mass production of various biofertilizers. Biofertilizers - Organic certification – processes to be followed – Agencies for Certification. *Rhizobium*, *Azospirillum* and *Azotobacter*. Quality control laboratories in India.
- 46. Application techniques– form, dose, method and time of application of biofertilisers – *Rhizobium*, *Azospirillum*, *Azotobacter*, *Gluconoacetobacter*, phosphate solubilizers, potash releasers and sulphur oxidizers, Azolla, BGA, AM fungi and PPFM. Exposure to advanced techniques in biofertilizer production - Tangential Flow Filtration, lyophilized cells production and automatic packing unit. Visit to biofertilizer inoculated fields in university, farmer's holding and interaction.
Evaluation of plant response to biofertilizer application.
- 47. Financing - credit facilities – assistance – facility available for establishing biofertilizer production & 17 units- licensing required *etc.*, Formulation and presentation of a bankable project for production of
fixed quantity of various biofertilizers.

References

73. Motsara, M.R., Bhattacharyya, P., and Beena Srivatsava. 2004. Biofertiliser Technology, Marketing and Usage – A source book- Cum -Glossary
74. [Somani L.L.](#), 2011. Biofertilisers: Commercial Production Technology and Quality Control Publishers: ATPA. (ISBN-10: 8183211968, ISBN-13: 978-8183211963)
75. NIIR 2012. The Complete Technology Book on Biofertilizer and Organic Farming NIIR Project Consultancy Services, New Delhi. P. 608. (ISBN: 9789381039076)
76. Reeta Khosla 2017. Biofertilizers and Biocontrol Agents for Organic Farming, Publishers: Kojo press. (ISBN-10: 8192756793, ISBN-13: 978-8192756790)

AGR 451 Development of integrated farming system model (0+10)

Conceptual understanding of IFS in the course

Farming system: concepts, scope, objectives and advantages

Cropping systems for different agro climatic zones of India and Tamil Nadu

Crop diversification and intensification in farming system perspective

Integrated Farming Systems

Enterprises selection in Integrated Farming Systems

Integrated Farming System models for wetland ecosystem
Management of different enterprises of wetland IFS
Integrated farming system models for irrigated dry land ecosystem
Management of different enterprises of irrigated dry land IFS
Integrated farming system models for dry land ecosystem
Management of different enterprises of dry land IFS
Interaction between different components of IFS
Resource recycling in integrated farming system
IFS research methodology and evaluation
Carbon foot-printing and green house gas emission studies in IFS models
Farming system characterization for up scaling IFS models to field /
farm. Preparation of bankable projects in IFS under wetland eco-system.
Preparation of bankable projects in IFS under irrigated dry land ecosystem
Preparation of bankable projects in IFS under dryland ecosystem.

ABT 451 Commercial Plant Tissue Culture (0+10)

Practicals (Weekly Schedule)

51. Basics and establishment of Plant Tissue Culture Laboratory

Organization for a plant tissue culture laboratory - Sterilization methods -Equipments and instruments in PTC - Surface sterilization of explants - Handling tissues in aseptic conditions under laminar flow chamber

2. Medium and stock solution preparation-I

Familiarization of different chemicals- inorganic nutrients – carbon sources, vitamins and growth regulators –solidifying agents - Stock solutions preparation for MS medium and B5 medium

3. Medium and stock solution preparation-II

Stock solutions preparation for WPM medium - Medium preparation- MS medium, B5 medium, WPM medium - Sprouting of tubers in potato

4. Meristem and Micropropagation in cassava, rose and chrysanthemum

Meristem tip culture- medium preparation - Meristem tissue culture – cassava - Media preparation for micropropagation in rose and chrysanthemum - Micropropagation in rose and chrysanthemum

5. Micropropagation of banana and neem

Media preparation for micropropagation in banana, neem, eucalyptus, *Aloe vera*, *Phyllanthus* and potato - Micropropagation in banana and neem

6. Micropropagation of eucalyptus, *Aloe vera* and *Phyllanthus*

Micropropagation in eucalyptus, *Aloe vera*, *Phyllanthus* - Media preparation for micropropagation in sugarcane and bamboo - Inoculation of potato sprouts

7. Micropropagation of Sugarcane, bamboo and sub culturing

Micropropagation- sugarcane, Bamboo - Medium preparation for subculturing in rose, chrysanthemum, banana, neem, eucalyptus and *Phyllanthus*

8. Sub culturing -I

Medium preparation for subculturing- meristem tip culture, sugarcane and *Aloe vera*, bamboo, microtuber induction in potato - Subculturing in rose and chrysanthemum

9. Sub culturing -II

Subculturing in banana, neem and eucalyptus -**Mid semester Examination-** Subculturing in *Phyllanthus*, *Aloe vera* and cassava

10. Callus induction in *Phyllanthus* and *Coleus* and rooting

Subculturing in sugarcane and bamboo - Medium preparation for callus induction in *Phyllanthus* and *Coleus* - Inoculation of explants for callus induction in *Phyllanthus* and *Coleus* - Medium preparation for rooting in rose and chrysanthemum

11. Media preparation and inoculation for rooting of microshoots

Medium preparation for rooting in banana, neem, eucalyptus, *Aloe vera*, *Phyllanthus*, bamboo, cassava and sugarcane - Inoculation of micro shoots for rooting in rose, chrysanthemum, banana and neem

12. Inoculation for rooting of microshoots and hardening

Inoculation of microshoots for rooting in eucalyptus, aloe vera, phyllanthus, bamboo, cassava and sugarcane - Observations on microtuber induction in potato - Hardening chambers- mist-chamber, glasshouse, polyhouse and tunnel house - Hardening procedures, visit to any hardening facility

13. Synthetic seed preparation and Establishment of cell suspensions

Subculturing for proliferation of callus-medium preparation - Synthetic seed preparation-stocks preparation - Subculturing of callus and synthetic seed preparation. Cost-effective methods in PTC - Establishment of suspensions-medium preparation

14. Secondary metabolite production and analysis

Suspension culture in *Phyllanthus* and *Coleus*- Hairy root cultures with *Agrobacterium rhizogenes* - Preparation of stocks, medium for hairy root infection - Growth parameters for suspension- Fresh and dry weight, PCV and viability assay - Extraction of secondary metabolites and analysis through HPLC-GC-MS. Bioassay of secondary metabolites- anti-bacterial and anti-fungal activity.

15. Field transfer of TC plants

Field transfer of tissue culture plants – Hardening procedures and maintenance of regenerated plants. National certification system for tissue culture plants-application procedures. Visit to a field planted with TC plants. Visit to an Accredited Test Lab/National Certification Centre- NRCB, Trichy.

16. Entrepreneurship development I

Visit to a commercial tissue culture laboratory - Meeting the entrepreneur - Guest lecture from experts from financial institutions-funding opportunities.

17. Project preparation

Project preparation for Plant tissue culture - **Practical Examination**

References



Razdan, M.K. 2003. **Introduction to Plant Tissue Culture**. Enfield: Science Publishers Inc. USA



Dixon, R. A. 2003. **Plant Cell Culture – A Practical Approach**, IRL Press. Oxford. London



Gamborg OL, Phillips GC (2004) **Plant cell tissue and organ culture. Fundamental methods**. Narosa Publishing House, New Delhi



George E.F., Hall, M.A. and De Klerk, G.J. 2008. **Plant Propagation by Tissue Culture. Volume 1. The Background**. 3rd edition. Springer. Netherlands



Robert N. Trigano and Dennis J. Gray, 2000. **Plant Tissue Culture. Concepts and laboratory exercises**. Second edition. CRC press. London.

E-References

1. www-pub.iaea.org/mtcd/publications/pdf/te_1384_web.pdf
2. dbtindia.nic.in/NCS/Guidelines.pdf
3. dbtmicropropagation.nic.in/surveytcp.pdf
4. www.agritechpublications.com/article.htm

AGR 452 Organic Agriculture (0 +10)

Organic nutrient and weed management

Quantification of cow dung and cow urine recovery per animal - analysis their nutrients constituents- Methods of storing of cow dung - analysis of temperature, nutrients and microbial load in different layers of manure pit -Quantification of yield and nutrient content of fodder crops grown organically- Quantification of biomass for different green manures and green leaf manures grown organically- Experiencing mulching and other techniques in weed management.

Biofertilizers preparation

Introduction to biofertilizers and equipments – Isolation of bacterial biofertilizers – Method of application of biofertilizers – Arbuscular Mycorrhizal Fungi – Production and Assessment of infective propagules – Cyanobacterial biofertilizer – PPFM and liquid bioinoculants

Organic manure preparation

Biological wastes, farm wastes: collection, segregation, pre digestion of wastes, Biocompost preparation in pit method and vermicompost bed formation - Preparation of Panchagavya, Jeevamruth and EM and characterization – Harvest of biocompost and vermicompost – Compost maturity indices – FAO standard – Enriched compost preparation – Preparation of bankable project on establishment of organic input production unit.

Eco-friendly Pest Management

Establishment of model pest repellants cafeteria - Preparation and application of herbal leaf extracts in pest management - Monitoring of insect pests through traps and lures - Fruit fly trapping survey in horticultural crops - Case study on Agro-Eco System Analysis (AESAs) - Push and Pull Strategies in organic crop protection.

Non chemical diseases management

Diagnosis of disease symptoms and pathogens ,Preparation of enriched farm yard manure and methods of application of bio control agents - Cultural methods of disease management-Disease assessment and scoring - Removal of pathogens like ergot by mechanical methods- Preparation and foliar spraying of Arappu butter milk extract - Preparation and foliar spraying of pseudomonas butter milk extract- Preparation and foliar spraying of garlic vasambu extracts - Preparation and foliar spraying of cowdung 20% extract for BLB management- Preparation and foliar spraying of anti viral principles - Role of milk, curd and buttermilk in disease management.

Organic Certification and Preparation of Bankable Projects

Organic certification – Importance and scope – Procedure for obtaining certification — Post harvest management and value addition, supply chain management -Preparation of bankable projects – Visit to Tamil Nadu Organic Certification Department and organic outlets – Visit to Nationalized Banks to learn about funding for projects.

Experiential Learning –Organic Agriculture (0 +10)

Class Schedule

1 week	Quantification of cow dung and cow urine recovery per animal and analysis their nutrients constituent. Methods of storing of cow dung and analysis of temperature, nutrients and microbial load in different layers
2 week	Quantification of yield and nutrient content of fodder crops grown organically. Quantification of biomass for different green manures and green leaf manures grown organically.
3 week	Experiencing mulching techniques in weed management
4 week	Introduction to biofertilizers, equipments and Good Laboratory practices Preparation of culture media for biofertilizers Isolation of <i>Rhizobium</i> from root nodules of leguminous plants Isolation of <i>Azospirillum</i> from roots of cereal crops/ grasses Isolation of phosphobacteria from soil Microscopic observation of biofertilizer cultures
5 week	Population assessment of bacterial biofertilizers Method of application of bacterial biofertilizers Mass production of Arbuscular Mycorrhizal Fungi Identification of AM propagules in roots and soil Mass production of <i>Azolla</i> and method of application PPFM and Liquid bioinoculants
6 week	Collection, segregation, shredding and quantification of biological wastes/ farm wastes for biocompost and vermicompost preparation and initiating the pre digestion process (15 days)
7 week	Procuring inputs for preparing the formulations of <i>Panchagavya</i> , Jeevamruth and Effective Microorganisms (EM) -EM to be prepared from mother culture obtained from progressive organic farmers for multiplication Formation of beds and digging of compost pit of required size based on the availability of the farm wastes. Filling the pit and bed for biocompost
8 week	Vermicompost process respectively. Release of earthworms onto the compost bed Monitoring the composting process for moisture and temperature for efficient composting. Sampling of partially decomposed material for determining the nutritive value especially carbon build up
9 week	Harvest of matured composts, quantification and assessment of compost maturity indices and comparing with FAO standards for marketability. Characterization of <i>Panchagavya</i> , Jeevamruth and EM formulations Preparation of enriched biocompost, vermicompost and FYM using <i>Azospirillum</i> and <i>Azotobacter</i> or <i>Azophos</i>
10 week	Establishment of model pest repellants cafeteria. Preparation and application of herbal leaf extracts in pest management.
11 week	Monitoring of insect pests through traps and lures. Fruit fly trapping survey in horticultural crops.

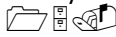
12 week	Case study on Agro-Eco System Analysis (AESA). Push and Pull Strategies in organic crop protection.
13 week	Preparation of a bankable project on Establishment of a pilot scale organic manure production unit for obtaining bank loans Diagnosis of disease symptoms and pathogens and cultural methods of disease management Disease assessment and scoring.
14 week	Removal of pathogens like ergot by mechanical methods. Preparation and foliar spraying of Arappu butter milk extract and fliar spraying of pseudomonas butter milk extract Preparation and foliar spraying of garlic vasambu extracts Preparation and foliar spraying of cowdung 20% extract for BLB management
15 week	Organic certification – Importance and scope Procedure for obtaining certification Post harvest management and value addition
16 week	Supply chain management in Organic Farming Exposure visit to Tamil Nadu Organic Certification Directorate and organic outlets
17 week	Preparation of bankable projects Evaluation of individual and group assignments and report submission Visit to Nationalized Banks to learn about funding for projects. Final Practical Examination

PBG 452 Hybrid rice: Parental line seed production technique (0+10)

Activities:



1st week : Studying botany of Rice, Hybrids and their development , breeding methods used in hybrid rice parental line development, Impact of Hybrid Rice in Tamil Nadu.



2nd week : Selection of field based upon the land with adequate fertility, drainage, irrigation, sun light and free air with adequate isolation distance (100 m distance isolation or 25 days time isolation).



3rd week and 4th week: Seed treatment with Carbendazim, *Pseudomonas fluorescens* and *Azospirillum*

Staggered sowing of A x B line seed production in CORH 3 rice Hybrid

CORH 3

A - Female (Male
sterile) TNAU CMS 2A

B - Male (Maintainer
line) TNAU CMS 2B

Staggered sowing of parents

First sowing of Male line (B1) 3 days before A line sowing – 3kg

Single sowing of the entire female (A) line seeds - 20kg and second sowing of male line (B2) 3kg
on the same day

Third sowing of Male line (B3) 3 days after A line sowing – 4kg

A line seed production in CORH 3



5th week

Seedlings pest and disease management in nursery

Main field preparation



6th week

Transplanting the seedlings in the main field

A x B

Ratio - Female: Male	=	6 : 2
Seedlings/hill for Female (A) line	=	1 (with two to three tillers)
Seedlings / hill for Male (B) line	=	2-3
Spacing in Female (A) line	=	10 x 15 cm
Spacing in Male (B) line	=	30 x 15 cm
Spacing between 'A' and 'B' lines	=	20 cm

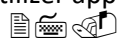


7th week and 8th week

Weedicide application : Three days after planting, application of Butacholor @ 2.5 litres / hectare with 50 kg sand retaining 1 cm water in the main field. After weedicide application,the water should not be drained from the field for two days

Gap filling : Within 7 - 10 days after planting.

Fertilizer application



9th week and 10th week

Crop protection measures to be followed

Fertilizer application (Top dressing)

Panicle initiation and flowering

Flowering: 'A' line should be earlier by one or two days.

Adjustment of flowering date

If the flowering is to be delayed, spray 2% urea solution with Knapsack sprayer to induce vegetative growth.

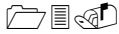
If the flowering is to be hastened, apply 2% DAP solution to arrest vegetative growth.

By this method 3-4 days difference in flowering can be adjusted.

Copious irrigation hastens the flowering in Male (B) line

Draining the water will delay the flowering in Male (B) line

If there is early flowering in Male (B) line than the Female (A) lines those ear heads may be jerked (removed)



11th week and 12th week

Fertilizer application (Top dressing)

Rogueing

Genetically different plants in both 'A' and 'B' lines should be removed periodically from tillering stage onwards
Pollen shedders in 'A' line should be removed from flowering to grain filling stage daily

GA3 application

Panicle exertion is incomplete in 'A' line

Two times of GA3 spray with Knapsack sprayer is needed to make the complete panicle exertion

First spray of 60g GA3 in 500 litres of water/ha at the time of 20 percent flowering

Second spray 40g in 500 litres of water/ha on the next day within 24 hours after the first spray

Dissolve GA3 first in Methyl alcohol 1g in 10 cc and then in water

Spray the solution at 8.00 to 10.00 a.m. or 4.00 to 6.00 p.m.



13th week

Supplementary pollination

It is done to promote higher cross pollination.

Rope pulling or stick shaking for 10 days from 20 per cent flowering Best time is 10.00 to 1.00 a.m.



14th week

Harvest and storage

Male (B) line should be harvested first and removed

Final rogueing should be done before the harvest of female (A) line

Harvested produce should be threshed, cleaned, dried and stored properly at 12% moisture.

❖ 15th week

Monitoring of farmers field for hybrid performance



16th week

Calculation of Economics of seed production

Deliverables

Students can learn the method of genetically pure parental line seed production techniques in rice hybrid CORH 3

The Hybrid rice seed production and parental line seed production techniques learned from this course will be much useful for the students to become entrepreneur in the future.

PBG 451 Hybrid pearl millet seed production (0+10)

Activities :



1st week

Studying the botany of Pearl millet, Flowering behavior and mode of pollination. Hybridization techniques, characteristics of A, B & R lines. Significance of hybrid Breeding in pearl millet. Maintenance breeding of A, B and R lines.



2nd week

TNAU Cumbu hybrid CO 9 seed production; (ICMA 99111 x PT 0029-30 R)

Duration 75-80 days

Selection of field based on the isolation distance (500 m)

Seed treatment

Staggered sowing : Male parent (PT 6029-30) R line sowing has to be taken up seven days earlier than female parent).

Ist sowing : PT 6029 – 30 R (male parent – R line)

Seed rate 1.5 kg/ha R line

Row Ratio : 4:2 (4 A : 2 R)

Spacing : 45 x 15 cm

Fertilizer application

Herbicide application



3rd week

IInd sowing : ICMA 93111 (Female parent A line) After 7 days of male line sowing

Sowing of border Rows : R line - 4 Rows.



4th week

Thinning of seedlings in both A and R lines



5th week & 6th week

Weeding and irrigation

Top dressing of fertilizer



7th week

IInd weeding and pest and disease control measures

Monitoring of flowering and rogueing



8th and 9th week



10th week



11th week

Thorough roguing of the A line and harvesting

Threshing in the a separate threshing floor

Seed cleaning



12th

week Germination test

Seed treatment of hybrid seed



13th week

Seed packing



14th week



15th week



16th week and 17th

week Report preparation and submission

Deliverables

Students can learn the method of genetically pure hybrid seed production techniques in pearl millet.

SAC 451 On Farm Advisory for Soil Health, Water Quality and Plant Nutrition (0+10)

Practical Schedule /week

29. Identification and Selection of farm holdings growing different crops
30. Studying the land features and collection of soil samples
31. Assessment of soil physical and chemical quality indices of collected soil samples
32. Assessment of soil biological quality indices and interpretation (Dept. of Agrl. Microbiology)
33. Interpretation of analytical results of collected soil samples for their quality
34. Identifying soil constraints - Interpretation of results (Soil physics)
35. Problem solving management techniques, Calculation of ameliorants.
36. Assessing the Land suitability for agricultural, horticultural and tree crops (Dept.of RS&GIS)
37. Water sample collection, quality assessment
38. Assessing the land suitability for irrigation
39. Fertilizer prescription calculations for important crops - Nutrient equivalent basis -Soil Test Crop Response based recommendation for targeted yields
40. Deriving the nutrient requirement using DSSIFER soft ware for different crops (STCR)
41. Issue of Soil Health Card and Fertilizer prescription using DSSIFER software
42. Diagnosis of nutrient deficiencies using VDK software and corrective measures
43. Formulating the plan for the selected farm holding for the existing crops
44. Formulating the most viable farm plan for the selected farm holding and Development of Soil Constraint Management Package (SCMP)
45. Recap and Practical examination

References

60. Baruah J.C and D.K.Patgiri. 1996. Physics and Chemistry of Soils, New Age International Publications
61. Garison Sposito, 2008. The Chemistry of Soils. Oxford University Press, USA
62. Black, C.A. 1965. Agronomy Monograph. Methods of Soil Analysis. Part 1. Physical and Mineralogical properties including Statistics of Measurement and Sampling. Wiley, New York.
63. Hesse, P.R. 1971. A Text book of Soil Chemical Analysis. John Murray (Publishers) Ltd. London
64. Indian Society of Soil Science 1991. Soil related constraints in crop production Bulletin No.15.ISSS, New Delhi.
41. Indian Society of Soil Science 1996. Soil management relation to land degradation and environment. Bulletin No.17.ISSS, New Delhi.
42. Jackson, M.L. 1973. Soil Chemical Analysis. Prentice Hall Pvt.Ltd
43. Mani.A.K., R.Santhi, K.M.Sellamuthu. 2007. A handbook of Laboratory Analysis.A.E.Publications, Coimbatore
44. Motsara.M.R and R.N.Roy.2008.Guide to laboratory establishment for plant nutrient analysis. Food and Agriculture organization of the United Nations.Rome.pp.220 (also available online <ftp: /ftp.fao.org/agl/ agll/docs /fpnb 19. pdf>)
45. Piper, C.S 1942. Soil and plant analysis: Inter science *Publishers*, New York.
46. Sehgal,J.2005. A text book of Pedology- Concepts and applications. Kalyani Publishers, Ludhiana, New Delhi.
47. Subramanian,S.S, G.V.Kothandaraman, S.Natarajan and P.P.Ramaswami. ed. 1987. Soil survey and land use planning for watershed management. Directorate of Soil and Crop Management Studies, TNAU, Coimbatore
48. Tandon, H.L.S.1994. Fertilizers, organic manures, recyclable wastes and
49. biofertilizers. Fertilizer Development and Consultant Organization, New Delhi

50. USDA 1954. Diagnosis and Improvements of Saline and Alkali Soils. (Ed) L.A.Richards. Handbook No.60. USDA Washington DC.
51. Werner Bergmann.Ed.1992.Nutritional Disorders of Plants - Development, Visual and analytical Diagnosis. Gustav Fischer Verlag Jena.Stuttgart. NewYork

E-References

20. <ftp://ftp.fao.org/agl/agll/docs/fpnb19.pdf>
21. <http://www.agric.gov.mt/soil-and-irrigation-water-lab>
22. www.soiltesting.okstate.edu/
23. www.texasplantandsoillab.com/
24. www.ulm.edu/spal/
25. soilhealth.cals.cornell.edu/extension/.../managing_constraints.pdf
26. cnal.cals.cornell.edu/
27. http://agritech.tnau.ac.in/agriculture/agri_reosurcemgt_soil_soilconstraints.html
28. <http://edis.ifas.ufl.edu/topicjertilization>

HOR 451 Hybrid seed production in vegetable crops (0+10)

CONTENT

Unit I - Introduction to quality seed production, principles and practices

Scope and importance of vegetable seed industry and vegetable seed production - principles and practices of seed production - generation system of seed multiplication - pollination behaviour - tools employed in hybrid seed production - study of morphological characters of varieties, parental line and hybrids - designing of planting ratio and border rows - physical and genetic contaminants - isolation distance.

Unit II - Seed production planning and pre sowing seed treatments

Planning of seed production - season and land selection - assessment of seed source and seed selection - pre sowing seed invigouration treatments - dormancy breaking treatments - seed priming - pelleting - polymer coating. Practicing nursery and main field preparation - practicing the sowing of seeds in the nursery - types of nursery - media preparation for protray nursery - sowing -nursery management.

Unit III - Seed crop management and hybrid seed production techniques

Main field preparation - layout - formation of beds - transplanting - fertilizer and nutrient management - weed management - irrigation management - special cultural practices - pest and disease management - identification and removal of off-types and volunteer plants - practicing hybridization techniques (emasculatation and pollination) - identification of physiological disorders and management - exposure visit to seed certification department - seed certification procedures - registration and sowing report - field inspection - field counting - visit to seed production plots.

Unit IV - Pre and post harvest operations

Pre-harvest sanitation spray - identification of physiological and harvestable maturity indices - harvesting methods - post harvest verification – fruit grading - extraction of seeds - processing sequence

26. seed drying - seed cleaning - grading - pre-storage seed treatment - seed packing - seed storage - visit to seed processing unit and seed storage godown and learning sanitation measures. Economics of hybrid seed production (cost benefit ratio) - visit to private seed industry.

Unit V - Seed testing and marketing

Seed sampling procedure and submission of samples - seed testing procedure – estimation of seed moisture - physical purity analysis - germination test - visit to grow out test field and DNA finger printing laboratory for genetic purity assessment - visit to seed retail shop - seed marketing - project preparation.

Crops: Tomato, brinjal, chillies, bhendi, and gourds.

Deliverables:

Students who complete this course will gain enough confidence to establish seed industry as a successful business venture.

References

Arya, P.S., 1995. Vegetable seed production principles. Kalyani Publishers. New Delhi.
Geetharani, P., V.Swaminathan and V.Ponnuswami. 2012. Seed Technology of Horticultural Crops. Narendra Publishing House, Delhi - 6.

Kulkarni, G.N. 2011. Principles of seed technology, Kalyani publishers, Ludhiana, New Delhi
Singh, N, and Vishal Nath, 2011. Varieties and hybrids of vegetables, Satish serial publishing house, Delhi.
Sumesh Chandra Gaur, 2013. A hand book of seed processing and marketing Agrobios (India), Jodhpur Udai R.
Bishnoi and R.P.S. Kharb, 2012. Fundamentals of seed production and testing, Oxford book
company, Jaipur.
Vanangamudi, K. *et al.*, 2010. Vegetable hybrid seed production and management, Agro bios (India), Jodhpur.

E-References

7. Seednet.gov.in
8. www.dare.gov.in
9. [http:sfci.nic.in](http://sfci.nic.in)
10. www.iar.org.in/Directorate1.htm
11. www.apsa.org
12. www.seedassociationofindia.com
13. www.apaseed.com
14. www.apaseed.org

Hybrid seed production in vegetable crops (0+10)

Practical schedule

Crops: Tomato, brinjal, chillies, bhendi, and gourds.

Week	Classes
1.	Scope and importance of vegetable seed industry and vegetable seed production - principles and practices of seed production - generation system of seed multiplication.
2.	Pollination behaviour - tools employed in hybrid seed production - study of morphological characters of varieties, parental line and hybrids.
3.	Designing of planting ratio and border rows - physical and genetic contaminants - isolation distance.
4.	Planning of seed production - season and land selection - assessment of seed source and seed selection.
5.	Pre sowing seed invigouration treatments - dormancy breaking treatments - seed priming - pelleting - polymer coating.
6.	Practicing nursery and main field preparation - practicing the sowing of seeds in the nursery - types of nursery - media preparation for protray nursery - sowing -nursery management.
7.	Main field preparation - layout - formation of beds - transplanting - fertilizer and nutrient management.
8.	Weed management - irrigation management - special cultural practices - pest and disease management.
9.	Identification and removal of off-types and volunteer plants - practicing hybridization techniques (emasculation and pollination) - Identification of physiological disorders and management.
10.	Exposure visit to seed certification department - seed certification procedures - registration and sowing report - field inspection - field counting - visit to seed production plots – project preparation.
11.	Pre-harvest sanitation spray - identification of physiological and harvestable maturity indices - harvesting methods - project preparation - project preparation.
12.	Post harvest verification – fruit grading - extraction of seeds - processing sequence - seed drying - seed cleaning - grading - pre-storage seed treatment - seed packing - seed storage.
13.	Visit to seed processing unit and seed storage godown and learning sanitation measures - project preparation.
14.	Economics of hybrid seed production (cost benefit ratio) - visit to private seed industry.
15.	Seed sampling procedure and submission of samples - project preparation.
16.	Seed testing procedure - estimation of seed moisture - physical purity analysis - germination test - visit to grow out test field and DNA finger printing laboratory for genetic purity

	assessment.
22.	Visit to seed retail shop - seed marketing - project preparation and submission.

SST 451 Commercial seed production (0+10)

CONTENT

Unit I - Introduction to quality seed production, principles and practices

Scope and importance of seed industry and seed production - principles and practices of seed production - generation system of seed multiplication - pollination behaviour - tools employed in hybrid seed production - study of morphological characters of varieties, parental line and hybrids - designing of planting ratio and border rows - physical and genetic contaminants - isolation distance.

Unit II - Seed production planning and pre sowing seed treatments

Planning of seed production - season and land selection - assessment of seed source and seed selection - pre sowing seed invigouration treatments - dormancy breaking treatments - seed priming - pelleting - polymer coating. Practicing nursery and main field preparation - practicing the sowing of seeds in the nursery - protrait nursery - sowing - nursery management.

Unit III - Seed crop management and hybrid seed production techniques

Main field preparation - layout - formation of beds - transplanting - fertilizer and nutrient management - weed management - irrigation management - special cultural practices - pest and disease management - identification and removal of off-types and volunteer plants - practicing hybridization techniques (emasculation and pollination and detasseling) - identification of physiological disorders and management - exposure visit to seed certification department - seed certification procedures - registration and sowing report - field inspection – field counting - visit to seed production plots.

Unit IV - Pre and post harvest operations

Pre-harvest sanitation spray - identification of physiological and harvestable maturity indices - harvesting methods - post harvest verification - kapas sorting, cob sorting and pod verification - threshing / extraction of seeds - processing sequence - seed drying - seed cleaning - grading - pre-storage seed treatment - seed packing - seed storage -visit to seed processing unit and seed storage godown and learning sanitation measures. Economics of variety and hybrid seed production (cost benefit ratio) - visit to private seed industry.

Unit V - Seed testing and marketing

Seed sampling procedure and submission of samples - seed testing procedure - estimation of seed moisture - physical purity analysis - germination test - visit to grow out test field and DNA finger printing laboratory for genetic purity assessment - visit to seed retail shop - seed marketing - project preparation.

Crops

Cereals, pulses, oilseeds, cotton and commercially important vegetable crops.

Deliverables:

Students who complete this course will gain enough confidence to establish seed industry as a successful business venture

Reference

36. Agrawal, R.L. 1996. Seed Technology, Oxford & IBH Publishing Co., New Delhi.
37. Bhaskaran, M. A.Bharathi, K.Vanangamudi, N.Natarajan, P.Natesan, R.Jerlin and K.Prabakar. 2003. Principles of seed production. Kaisher Graphics, Coimbatore.
38. Copeland LO & McDonald MB. 2001. Principles of Seed Science and Technology. 4th Ed. Chapman & Hall.
39. Singhal NC. 2003. Hybrid Seed Production in Field Crops. Kalyani Publishers, New Delhi.
40. Vanangamudi, K. 2014. Seed Science and Technology. An Illustrated Text Book. New India Publishing Agency, New Delhi.

E References

1. Seednet.gov.in
2. www.iar.org.in/Directorate1.htm
3. www.apsa.org
4. www.seedassociationofindia.com
5. www.apaseed.com
6. www.apaseed.org

Commercial seed production (0+10)

Schedule of Activities

Week	Classes
1.	Scope and importance of seed industry and seed production - principles and practices of seed production - generation system of seed multiplication.
2.	Pollination behaviour - tools employed in hybrid seed production - study of morphological characters of varieties, parental line and hybrids.
3.	Designing of planting ratio and border rows - physical and genetic contaminants - isolation distance.
4.	Planning of seed production - season and land selection - assessment of seed source and seed selection.
5.	Pre sowing seed invigouration treatments - dormancy breaking treatments - seed priming - pelleting - polymer coating.
6.	Practicing nursery and main field preparation - practicing the sowing of seeds in the nursery - protray nursery - sowing - nursery management.
7.	Main field preparation - layout - formation of beds - transplanting - fertilizer and nutrient management.
8.	Weed management - irrigation management - special cultural practices - pest and disease management.
9.	Identification and removal of off-types and volunteer plants - practicing hybridization techniques (emasculatation and pollination and detasseling) - identification of physiological disorders and management.
10.	Exposure visit to seed certification department - seed certification procedures - registration and sowing report - field inspection - field counting - visit to seed production plots - project preparation.
11.	Pre-harvest sanitation spray - identification of physiological and harvestable maturity indices - harvesting methods - project preparation.
12.	Post harvest verification - kapas sorting, cob sorting and pod verification - threshing / extraction of seeds - processing sequence - seed drying - seed cleaning - grading - pre-storage seed treatment - seed packing - seed storage.
13.	Visit to seed processing unit and seed storage godown and learning sanitation measures - project preparation.
14.	Economics of variety and hybrid seed production (cost benefit ratio) - visit to private seed industry.
15.	Seed sampling procedure and submission of samples - project preparation.
16.	Seed testing procedure - estimation of seed moisture - physical purity analysis - germination test - visit to grow out test field and DNA finger printing laboratory for genetic purity assessment.
17.	Visit to seed retail shop - seed marketing - project preparation and submission.

ENS 451 COMPOSTING TECHNOLOGY

Activities

Collection and characterization of solid-wastes – analyzing physical and chemical properties – site selection for composting – infrastructure required for compost making – processing of solid waste for composting – carbon : nitrogen ratio maintenance – selection of microbial inoculum for composting - compost bed formation – windrow method – heap method – application of microbial inoculum – recording compost heap temperature- thermophilic phase and mesophilic phase – turning of compost pile for uniform composting – moisture maintenance in compost pile – assessing reduction in carbon and nitrogen ratio – compost maturity assessment – curing of compost material – value addition through beneficial microbes - Assessing nutritive value of compost – national and international standards for compost quality parameters – project preparation for compost making facility – Marketing of compost products – working out cost benefit ratio for compost production – Record maintenance in compost making.

Deliverables/Out come

The students who are undergoing this experiential learning will have independent skill to manage large quantity of solid waste through composting technology. They know how to prepare a project on solid waste management and it will create a self enterprising activity for them.

References:

1. Kelly Smith. 2012. How to build, maintain and use a compost system. Atlantic publishers, Florida.
25. John Berry. 2014. How to make compost? A quick and easy guide. Hewel Trading, USA
26. Roland Ulrich. 2014. Creating humus on farm – The controller heat method of composting. Outskirts press
27. Augustine Afulio, 2014. Integrated solid waste management. Hand book for Beginners, Planners, Environmentalists, Students and policy makers. Warmra Twechoprise, Nairobi, Kenya.

E -References:

1. <http://www.eartheasy.com>
2. <http://www.composting council.org>
3. <http://www.Epa.gov/compost>
4. <http://www.Compost.css.cornnell.edu>

HOR 452 Commercial Nursery Technology of Horticultural Crops (0+10)

Deliverables

Students who undergo this course will gain practical knowledge and hands on experience in different aspects of a commercial fruit nursery.

Students' attitude in leadership quality, managerial skill and professionalism will be enriched.

References

1. Sadhu, M.K. 1989. Plant Propagation. Wiley Eastern Ltd., New Delhi
2. Bose, T.K., S.K. Mitra, M. K. Sadhu and B. Mitra. 1991. Propagation of Tropical and Subtropical Horticultural crops. Naya Prakash Publishers, Culcutta, India.
3. Hartmann, H.T., D. E. Kester, F.T. davies and R. L. Grene. 2010. Plant Propagation – Principles and Practices. Prentice Hall of India Private Ltd., New Delhi.
4. Nanda, K. K and V. K. Kochhar. 1995. Vegetative Propagation of Plants. Kalyani Publishers, Ludhiana.

References

1. <http://www.oer.nios.ac.in>
2. <http://www.sas.upenn.edu>
3. <http://www.tmnehs.gov.in>
4. <http://www.instructables.com>

HOR 453 COMMERCIAL LANDSCAPE GARDENING (0 + 10)

Activities

Understanding the concept of experiential learning and identifying the broad area for experiential learning project activity in commercial Landscape Gardening - Fixing the area of interest for individual or group experiential learning project activity in commercial Landscape Gardening (Green consultancy, Green wall fixtures, Green showcases, Green wall hangings, Green furniture, Cacti buckets and flower bouquets, Green glasses, Trees indoor, Smart garden *etc.*,) - Rationale for selecting the activity in commercial Landscape Gardening and formulating the anticipated methodology for execution - Preparation of the project with budget for execution and marketing - Nurturing the students potential and innovativeness in their area of interest and facilitating the project activity (planning, development and execution) - Concept of advertising the product and developing market strategies for efficient selling - Working out the cost economics / balance sheet involved in the project - Generating a reflective report about the project and the student's potential in academic and personal development.

Deliverables

24. Entrepreneurship skill and buoyancy in handling commercial ventures in the domain of landscape gardening is assured
25. Student's attitude in leadership quality, managerial skill and professionalism will be enriched

HOR 454 PROTECTED CULTIVATION OF VEGETABLE CROPS (0 + 10)

Practical content

Understanding the concept of experiential learning and identifying the broad area for experiential learning project activity in protected cultivation in vegetable crops - Fixing the area of interest for individual or group in project activity in protected cultivation (Establishment and operation of protected structures - types of growing structures - construction of poly house and shade net house - manipulation of environmental factors - practical learning in nursery raising - growing systems - growing media - sterilization - preparation of beds- planting- and cultivation practices - harvesting practices - post harvest handling – storage - project preparation and analysis of cost economics etc.,) - Rationale for selecting the activity in protected cultivation and formulating the anticipated methodology for execution - Preparation of the project with budget for execution and marketing - Nurturing the students potential and innovativeness in their area of interest and facilitating the project activity (planning, development and execution) - Concept of advertising the product and developing market strategies for efficient selling - Working out the cost economics / balance sheet involved in the project - Generating a reflective report about the project and the student's potential in academic and personal development.

Crops: Tomato / Capsicum / Cucumber

68. References

- ¹ Prasad, S. and U. Kumar. 2005. Green house management for horticultural crops. 2nd ed. Agrobios.
- ² Tiwari, G.N. 2003. Green house technology for controlled environment . Narosa Publ. House.

References

1. Harry Tomilson, 2010. The complete book on bonsai. Abbevellie press, Glasgow. Phaidon, 2014. The garden book, Phaidon book, London
2. Judith blacklock, 2012. The complete guide for beginners. Flower fresh publisher, U.K.
3. Nancy Norris, 2011. Miniature garden guide book. Kalmbach Publishing company, Netherlands
4. Bose.T.K, R.G. Maiti, R.S. Dhua and P.Das. 1999. Floriculture and Landscaping. Naya prakash, Calcutta. Booth. N.K. 1983. Basic elements of landscape architectural design.
5. Randhawa, G.S. and A. Mukhopadhyay, 2001. Floriculture in India. Allied Publishers Limited, New Delhi. Foja Singh, 1997. Advances in Floriculture. Media Today Pvt Ltd., New Delhi-17
6. Chattopadhyay, S.K. 2007. Commercial Floriculture. Gene-Tech Books, New Delhi

References

<http://floracultureinternational.com>
<http://www.wvu.edu/~agexten/hortcult/greenhou/>
http://www.umass.edu/umext/floriculture/fact_sheets/greenhouse_management.html
<https://sharepoint.agriculture.purdue.edu/agriculture/flowers/GHguides.aspx>
<http://www.ag.auburn.edu/hort/landscape/structures.html>
www.bonsaiempire.com
www.gardenweb.com
www.my-garden-school.com
www.florista.in
www.realsimple.com

PAT 451 Commercial production of Bio-control agents 0+10
(Team teaching by Entomologists, Pathologists and Economists)

Reference

1. Kennedy, J.S and Zadda Kavitha. 2006. Manual on commercial Production of biocontrol agents. Department of Agricultural Entomology, TNAU, Coimbatore. 156p
2. Gautam, R.D. 1994. Biological Pest Suppression. Westville Publishing House, New Delhi. 221 p
3. Course plan: TB: Text Book

Units / Practicals	Topics to be covered	Chapter [#]
1 st week	Establishment of a biocontrol unit, Mass production of <i>Corcyra cephalonica</i> and <i>Maconellicoccus hirsutus</i>	Chap#1,2,3,6 (TB1)
2 nd week	Mass production of parasitoids viz., <i>Trichogramma</i> sp., <i>Chelonus blackburnii</i> , <i>Bracon</i> sp	Chap#7,8,10 (TB1)
3 rd week	Mass production of parasitoids viz., <i>Goniozus nephantidis</i> and <i>Nesolynx thymus</i>	Chap#11 (TB1)
4 th week	Mass production of predators viz., <i>Cryptolaemus montrouzieri</i> and <i>Chrysoperla carnea</i> .	Chap#13,14 (TB1)
5 th week	Rearing of host insects viz., <i>Helicoverpa armigera</i> and <i>Spodoptera litura</i>	Chap#4,5 (TB1)
6 th week	Mass production of nuclear polyhedrosis virus of <i>Helicoverpa armigera</i> and <i>Spodoptera litura</i>	Chap#15,16 (TB1)
7 th week	Mass production of entomopathogenic fungi viz. <i>Metarhizium anisopliae</i> , <i>Beauveria bassiana</i> and <i>Verticillium lecanii</i>	Chap#18,19,20 (TB1)
8 th week	Processing and standardization of microbial pathogens	Chap#21 (TB1)

29. References

<http://www.mycologia.Org>
<http://www.nysaes.cornell.edu>
http://www.Eduwebs.org/bugs/mealybug_destroyers.htm
[http:// plant.disease.ippc.orst.adw/articles](http://plant.disease.ippc.orst.adw/articles)
[http:// www.nbaii.res.in](http://www.nbaii.res.in)

PRACTICAL SYLLABUS

Unit 3:

Importance of biological control in plant disease management – Handling of equipments – sterilization techniques –Preparation of media

Collection of soil sample and Isolation of antagonists - *Trichoderma*, *Chaetomium*, *Beauveria*, *Pseudomonas fluorescens*, and *Bacillus subtilis* - Maintenance of pure cultures - Morphological and molecular characterization of antagonists

Unit 4

Keys for the identification of lab contaminants - Assessing the efficacy *in vitro* - mode of action of antagonists - Fermentation systems and different kinds of formulations - Mass multiplication

Methods of delivery of biocontrol agents - Bio efficacy against plant diseases – Container content compatibility - packaging methods and shelf life studies of bio control agents -Guidelines and requirements to establish a commercial bio control lab - energy requirements to establish a commercial

bio control lab

Unit 5

Legal issues involved in the establishment of commercial bio control lab and registration (Small scale and large scale) - Cost Analysis and project preparation - Principles of enterprise management. Exposure visit to commercial bio control units

PRACTICAL SCHEDULE

9 week

14. Bio-control agents and their significance in plant disease management
15. Safety procedures for handling of equipments (Autoclave, Laminar Air Flow Chamber, Hot air oven, pH meter)
16. Safety procedures for handling of equipments (Electronic balance, Fermentor, Distillation unit, Spectrophotometer, Microscopes and Spiral Kneader)
17. Good laboratory practices of a bio control lab
18. Sterilization techniques

10 week

16. Preparation of PDA and Rose Bengal agar medium
17. Preparation of *Trichoderma* selective medium,
18. Preparation of Kings B medium and Nutrient Agar medium
19. Preparation of Actinomycetes and *Chaetomium* selective medium.

20. week

- Collection of soil samples and isolation of *Trichoderma*, *Beauveria* and *Chaetomium* **11**
29. Collection of soil samples and isolation of *Pseudomonas fluorescens* and *Bacillus subtilis* and maintenance of pure cultures of biocontrol agents
 30. Morphological and molecular characterization of *Trichoderma*
 31. Morphological and molecular characterization of *Pseudomonas fluorescens*
 32. Morphological and molecular characterization of *Bacillus subtilis*

12 week

16. Keys for the identification of lab contaminants (*Salmonella*, *Shigella*, *Vibrio*, *Aspergillus*, *Penicillium*, *Rhizopus* etc.,)
17. Assessing the efficacy of *Trichoderma* under *in vitro* condition.
18. Assessing the efficacy of *Pseudomonas* and *Bacillus* under *in vitro* condition.
19. Studies on the mode of action of *Trichoderma* against soil-borne, foliar and Post harvest pathogens
20. Studies on mode of action of *Pseudomonas* and *Bacillus* against soil-borne, foliar and Post harvest pathogens

13 week

38. Fermentation systems
39. Different kinds of formulations- solid , liquid oil invert formulation etc.
40. Mass multiplication of *Trichoderma*
41. Mass multiplication of *Trichoderma*
42. Quality analysis of *Trichoderma*

14 week

- 19 Mass multiplication of *Pseudomonas*
- 20 Mass multiplication of *Bacillus*
- 21 Quality analysis of *Pseudomonas* and *Bacillus*
- 22 Methods of delivery of bio control agents - *Trichoderma*
- 23 Methods of delivery of bio control agents - *Pseudomonas* and *Bacillus*

15 week

35. Bioefficacy of *Trichoderma* against plant diseases
36. Bioefficacy of *Pseudomonas* and *Bacillus* against plant diseases
37. Biocontrol agents in pipeline –*Chaetomium*
38. Biocontrol agents- *Beauveria*
39. Container content compatibility, packaging methods and shelf life studies of bio control agents.
40. Guidelines and requirements to establish a commercial bio control lab

16 week

- Studies on energy requirements to establish a commercial bio control lab
- Legal issues involved in the establishment of commercial bio control lab and registration (Small scale and large scale).
- Legal issues involved in the establishment of commercial bio control lab and registration (Small scale and large scale).
- Exposure visit to commercial bio control units
- Exposure visit to commercial bio control units

17 week

18. Cost Analysis and project preparation: Principles of enterprise management.
19. Financial management – Agricultural Finance – Source of finance– Acquisition – Ratio analysis.

Economics of Mass Production of Biocontrol agents**Mass Production of *Trichoderma viride* talc formulation (500kg for 30 students)**

S. No.	Particulars	Quantity	Amount (Rs.)
I.Non-Recurring or Capital Investment *			
1.	Fermentor – 100 lit	1	6,00,000
2.	Autoclave	1	40,000
3.	Hot air Oven	1	25,000
4.	Laminar Air flow Chamber-2’/3’/4’	1	60,000
5.	Electronic Balance	1	10,000
6.	Racks and Cabinet	1	15,000
7.	Plastic tray and glasswares	1	20,000
8.	Sealing machine	1	3000
9.	Refrigerator	1	20,000
10.	Gas connection, Cooker and burner	1	5,000
Total			7,98,000

S. No.	Particulars	Quantity	Amount (Rs.)
II. Recurring or Working Expenditure			
1.	Talc Powder	500kgs	7500
2.	Chemicals		2500
3.	Polybags		750
4.	Electricity and gas refilling		1000
5.	Labour charges		5000
6.	Miscellaneous expenditures		2000
Total			18,750.00

III. Income		
1.	500 kgs of product @Rs. 75/-	37,500.00
2.	Total Expenditure	18,750.00
	Net Profit Rs.	18,750.00

11. **Non-Recurring:** One time investment

Mass Production of *Pseudomonas fluorescens* talc formulation (1000kg for 30 students)

S. No.	Particulars	Quantity	Amount (Rs.)
I.Non-Recurring or Capital Investment			
1.	Fermentor	1	3,00,000
2.	Autoclave	1	40,000
3.	Hot air Oven	1	25,000
4.	Laminar Air flow Chamber	1	60,000
5.	Electronic Balance	1	10,000
6.	Racks and Cabinet	1	15,000
7.	Plastic tray and glasswares	1	20,000
8.	Sealing machine	1	3000
9.	Refrigerator	1	20,000
10.	Gas connection, Cooker and burner	1	5,000
		Total	4,98,000

S. No.	Particulars	Quantity	Amount (Rs.)
II. Recurring or Working Expenditure			
1.	Talc Powder	1000kgs	15000
2.	Chemicals		5000
3.	Polybags		1500
4.	Electricity and gas refilling		2000
5.	Labour charges		10000
6.	Miscellaneous expenditures		5000
		Total	38,500.00

III. Income		
1.	1000 kgs of product @Rs. 75/-	75,000.00
2.	Total Expenditure	38,500.00
	Net Profit Rs.	36,500.00

K Non-Recurring: One time investment

REFERENCE

- KK Baker, K.F. and Cook, R.J. 1974. Biological control of plant pathogens. W.H. Freeman and Co. San Francisco, U.S.A.
- LL Chet, I. 1987. Innovative approaches to plant disease control, John wiley and Sons, New York.
- MM Dinakaran, D, G.Arjunan & G.Karthikeyan 2003. Biological control of crop diseases.
- NN Papavizas, G.C. 1985. *Trichoderma* and *Gliocladium* : biology, ecology and potential for biocontrol. Annu. Rev. Phytopathol. 23 : 23-54.
- OO Maheswari ,D.K and R.C Dubey 2008 .Potential micro organisms for sustainable agriculture. I.K International Publishing House Pvt. Lts , New Delhi
- PP Prakasam, V., Raguchander, T. and Prabakar, K. 1998. Plant Disease Management. AE Publications, Coimbatore, India.

- QQ Ahamed S and Narain U 2007 . Eco friendly management of plant diseases. Daya Publishing house , New Delhi
- RR Utkhede, R.S. and Gupta, V.K. 1996. Management of soil borne diseases. Kalyani Publishers, New Delhi.

PAT 452 COMMERCIAL MUSHROOM PRODUCTION (0+10)

(Team Teaching)

Unit 1 : Different types of mushroom , Morphology - **Edible and poisonous type - edible mushrooms- *Pleurotus*, *Agaricus*, *Volvariella* and *Calocybe*** – nutritional values - and pharmacological values-**preparation of culture media- pure culture techniques- sterilizing techniques-media - glassware - maintenance of culture**

Unit 2: Mother spawn **production-type of spawn-Multiplication of bed spawn** – Substrates for mushroom cultivation **and their preparation -mushroom cultivation techniques for *Agaricus*, *Pleurotus*, *Calocybe* and *Volvariella*- maintenance of spawn running and cropping room-harvest-packing and storage of *Pleurotus*, *Agaricus* and *Calocybe*.**

Unit 3 : Problems in cultivation of *Agaricus*, *Pleurotus*, *Calocybe* and *Volvariella* – pests, diseases and weed moulds, abiotic disorders – management strategies -- Biodegradation of coir pith - cost estimation

Unit 4 : Post harvest technology of *Agaricus*, *Pleurotus*, *Calocybe* and *Volvariella* – methods of preservation –**Drying: solar, cabinet, fluidized bed and freeze drying** – Packing methods and storage - Controlled atmospheric storage- modified atmospheric storage and canning – Cost analysis.

Unit 5 : Mushroom recipes of *Agaricus*, *Pleurotus*, *Calocybe* and *Volvariella* - Cooking methods- value added products – instant food mixes –Cost analysis. Project preparation- principles of mushroom farm enterprise management – cost estimation

Practical schedule

1 week

Studying the general characters of mushrooms
Different types of mushrooms and their morphology
Identification of edible and poisonous mushrooms
Morphological characters of *Pleurotus*, *Agaricus*, *Volvariella* and *Calocybe*
Equipments required for culture media preparation and tissue culture - their operation

2 week

Equipments required for spawn preparation - their operation
Equipments required for substrate sterilization - their operation
Preparation of different types of culture media- Potato Dextrose Agar (PDA) , Oats meal agar (OMA), Malt extract Agar medium (MEA)
Pure culture technique –Tissue isolation methodology
Pure culture technique –**Tissue isolation in PDA, OMA and MEA medium 3**

week

Sub culturing of fungal cultures and maintenance.
Spawn preparation- laboratory requirements, essentials required for mother spawn and bed spawn preparation and their usage.
Oyster mushroom: mother spawn preparation – Cooking of cholam grains , packing in polybags and autoclaving
Oyster mushroom: mother spawn preparation - inoculation
Oyster mushroom: first generation bed spawn preparation - Cooking of cholam, packing in polybags and autoclaving

4 week

Oyster mushroom: first generation bed spawn preparation - inoculation
Observe the spawn contaminants.
Oyster mushroom: second generation bed spawn preparation – Cooking of cholam, packing in polybags and autoclaving
Oyster mushroom: second generation bed spawn preparation - inoculation
Management of contaminants in mother spawn and bed spawn

5 week

Oyster mushroom cultivation – essentials required, cropping room requirement
Oyster mushroom: preparation of substrates for bed preparation

Oyster mushroom – Bed preparation

Oyster mushroom – Maintenance of beds, harvest and storing

Oyster mushroom – pest and their management

6 week

Oyster mushroom – moulds and disease management

Visit to oyster mushroom farm (spawn lab and mushroom farm)

Visit to ulavar sandai markets (Farmers' Market) and observing the marketing pattern of oyster mushroom

Milky mushroom: mother spawn preparation – Cooking of cholam grains, packing in polybags and autoclaving

Milky mushroom: mother spawn preparation - **inoculation 7**

week

Milky mushroom: first generation bed spawn preparation – Cooking of cholam, packing in polybags and autoclaving

Observing the spawn contaminants, their management

Milky mushroom: first generation bed spawn preparation - inoculation

Milky mushroom: second generation bed spawn preparation – Cooking of cholam, packing in polybags and autoclaving

Milky mushroom: second generation bed spawn preparation - inoculation **8**

week

Milky mushroom cultivation – essentials required, cropping room requirement

Milky mushroom:– substrates for bed preparation

Milky mushroom – Bed preparation

Milky mushroom- casing

Milky mushroom – Maintenance of beds, harvest and storing **9**

week

Milky mushroom – pest and their management
Milky mushroom – moulds and disease management

Visit to Milky mushroom farm (spawn lab and mushroom farm)

Visit to ulavar sandai, markets and observing the marketing pattern of milky mushroom
Button mushroom: visiting units and learning – tissue isolation, spawn preparation

10 week

Button mushroom: visiting units and learning compost preparation

Button mushroom: visiting units and learning: cropping, harvest and storage

Visit to ulavar sandai, markets and observing the marketing pattern of button mushroom

Paddy straw mushroom: tissue isolation

Paddy straw mushroom: **spawn preparation**

11 week

Paddy straw mushroom: Substrate preparation for beds

Paddy straw mushroom: bed preparation

Paddy straw mushroom cultivation – cropping room requirement, Maintenance of beds, harvest and storing

Paddy straw mushroom – pest and disease management

Abiotic disorders and their management

12 week

Integrated pest and disease management in Mushrooms
Biodegradation of agrowastes using mushroom spawn
Biodegradation of agrowastes using mushroom spawn- continuation
Mushroom as a component in Integrated Farming System
Interaction with successful spawn producers – TNAU community radio

13 week

Interaction with successful mushroom producers- TNAU community radio
 Short term post harvest processing of oyster mushroom
 Long term post harvest processing of oyster mushroom
 Packing methods of oyster mushrooms

Short term post harvest processing of milky mushroom **14**

week

Long term post harvest processing of milky mushroom
 Packing methods of milky mushrooms
 Short term post harvest processing of button mushroom
 Long term post harvest processing of button mushroom
 Packing methods of button mushrooms

15 week

Canning of button mushroom
 Recipe and value added products from oyster mushroom
 Recipe and value added products from oyster mushroom continued
 Recipe and value added products from milky mushroom
 Recipe and value added products from milky mushroom continued

16 week

Recipe and value added products from button mushroom
 Recipe and value added products from button mushroom continued
 Recipe and value added products from paddy straw mushroom
 Project preparation on oyster -spawn production and economics
 Project preparation on oyster mushroom production and cost estimation

17 week

Project preparation on milky spawn production and cost estimation
 Project preparation on milky mushroom production and cost estimation
 Project preparation on button -spawn production and cost estimation
 Project preparation on button - mushroom production and cost estimation
 Practical examination

BUDGET

Economics of Spawn Production (100 spawn bags per day) for oyster and milky mushroom mother spawn and bed spawn

Total working days for spawn: 25

Sl.No.	Item	Quantity	Rate (Rs.)	Total (Rs.)
	Recurring cost (100 spawn x 25 days)			
1.	Polypropylene bags	18 kg	120/kg	14,400
2.	Cholam grain	700 Kg	30/kg	21,000
3.	Calcium carbonate (commercial grade)	50	25/kg	1,250
4.	Non-absorbent cotton (400 g rolls)	100	80/roll	8,000
5.	Fungicides & Fumigants	--	--	3,000
6.	Electricity & Fuel	--	--	20,000
7.	Labour @ 2 women per day for 25 days	2 nos	190/person	9,500
8.	Glass wares and chemicals for preparing mother spawn	--	5,000	5,000

9.	Miscellaneous	--	--	2,000
	Total			84,150
	Overall total			84,150

Economics of Oyster mushroom production and milky mushroom (each 5 Kg/day)

Total working days: 10

Sl.No.	Item	Quantity	Rate (Rs.)	Total (Rs.)
	Recurring Cost			
1.	Paddy straw	100 kg	5/kg	5,000
2.	Spawn	-	-	-
3.	Polythene bags for bed & packing	2 kg	120/kg	240
4.	Fungicides, Fumigants & Chemicals	--	--	5000
5.	Labour @ 2 Per day	2	190 /day	3,800
6.	Miscellaneous (gunny bags, rope,racks)	--		6000
II.	Non recurring-Oyster Mushroom Shed	800 sq.ft.	75,000	75,000
Total (Rs.)				95,040

REFERENCES

Agarwal, R.K. and C.L.Jandaik.1986. Mushroom cultivation in India. Indian Mushroom Growers Association, Solan, Himachal Pradesh.p-83.

Bahl, N.1988. Hand book of Mushroom II Edn. Oxford & IBM Publishing Co. New Delhi.

Reference books- further reading

Marimuthu, T., A.S Krishnamoorthy, K.Sivaprakasam and R.Jeyarajan, 1989. Oyster Mushroom Production. The Vijay Books. Sivakasi, India.P.57.

A.S Krishnamoorthy, Marimuthu, T., and S. Nakkeran . 2005 . Mushroom Biotechnology ,The Vijay Books. Sivakasi, India., Pub.ODL, TNAU, Cbe-3

Pathak,V.N. Nagendra Yadav and Maneekas Gaur.2000.Mushroom production and processing Technology. Agribios (India) Ltd., New Delhi

E-REFERENCES

1. www.mushroomcouncil.com/grow/grow.html
2. www.krishiworl.com/html/mushroom.html
3. www.gmushrooms.com/pots.html.
4. www.mushworld.com/home/
5. www.mushroomcouncil.org

AMP 451 Commercial broiler and layer production (0+10)

	Unit – I - Introduction
	Poultry Industry in India - Current status of broiler and layer industry – Scope of broiler and layer production in India - Introduction to Broilers and Layers – Commercial strains of broilers and layers.
	Unit – II – Housing Management
	Housing management – Location and layout of commercial broiler and layer farm – Preparation of poultry house - Equipments used in broiler and layer farm - Different system of Management - Deep litter system, Cage system of management, Raised housing - Litter management - Preparation of brooder house – Brooder Management –Chick management – Grower management – Layer management -Summer management of broiler -Winter management of broiler – Lighting management – Common procedures followed in broiler and layer farm.
	Unit – III – Feeding Management
	Water – Requirement, quality analysis and its maintenance - Feeding Management of broilers and layers – Types of feed – Feed ingredients – Quality assessment of feed ingredients and feed – Additive and supplementation of additives - Storage of feed-Feeding Methods – Nutrient requirement of different stages of broilers and layers – Various standards for broilers and layers - Feed formulation – Least cost formulation –Preparation of compound feed –Components in feed mill –Feed mill operations - Hatchery Management – Hatchery layout and design – Hatcher and Setter – Collection and handling of egg – Setting and hatching of eggs and chicks quality assessment.
	Unit – IV – Flock Health Management
	Common disease of broilers and layers – Control and Prevention - Medication and Vaccination in broilers and layers – Different vaccination methods – Cold chain for vaccine – Vaccination schedule for broilers and layers -Postmortem inspection –Waste management -Disposal of dead birds and Manure management - Biosecurity measures.
	Unit – V – Processing and Marketing
	Processing of broilers - Slaughtering of broilers and cut up parts of broilers – Evaluation of egg for its quality - Record maintenance- Marketing Channels in broilers and layers - Export of egg and poultry meat - Integration method of broilers marketing–Team teaching along with Department of Economics on Economics of broiler and layers farming and Project preparation for broiler and layer farm unit for bank loan–Role of NECC and BCC in marketing of poultry and its products - Visit to commercial broiler farm, layer farm, feed plant, hatchery unit and processing plant.

Practical schedule

Week	Topic to be covered
D	Current status of broiler and layer industry Scope of broiler and layer production in India Commercial strains of broilers and layers Location and layout of commercial broiler and layer farm Preparation of poultry house
2	Equipments used in broiler and layer farm Different system of Management Deep litter system Cage system of management, Raised housing Litter management

3	Preparation of brooder house Brooder Management, Grower management and Layer management Summer management of broiler and layer Winter management of broiler and layer Lighting management
4	Common procedures followed in broiler and layer farm. Water – Requirement Quality analysis and its maintenance Feeding Management of broilers and layers Types of feed
5	Feed ingredients Quality assessment of feed ingredients and feed Additive and supplementation of additives Storage of feed-Feeding Methods Nutrient requirement of different stages of broilers and layers
6	Various standards for broilers and layers Feed formulation Least cost formulation Preparation of compound feed Components in feed mill
7	Feed mill operations Hatchery Management Hatchery layout and design Hatcher and Setter Collection and handling of egg
8	Setting and hatching of eggs and chicks quality assessment Common disease of broilers and layers Control and Prevention Medication and Vaccination in broilers and layers Different vaccination methods
9	Cold chain for vaccine Vaccination schedule for broilers and layers Postmortem inspection Disposal of waste -dead birds and manure
10	Biosecurity measures.
11	Processing of broilers Slaughtering of broilers and cut up parts of broilers

D	Evaluation of egg for its quality Record maintenance Marketing Channels in broilers and layers
E	Export of egg and poultry meat Integration method of broilers marketing
F	Team teaching along with Department of Economics on Economics of broiler and layers farming and Project preparation for broiler and layer farm unit for bank loan
G	Role of NECC and BCC in marketing of poultry and its products
H	Visit to commercial broiler farm, layer farm, feed plant, hatchery unit and processing plant
I	Examination

AEN 451 Commercial Beekeeping (0+10)

Activities

Honey bee species, castes, social biology and communication in honey bees - Bee pasturage and preparation of bee floral calendar - Honey bees for crop pollination and seed production. - Stingless bees, little bees, rock bees conservation and honey harvest - Beehives, beekeeping equipments specification and uses, visit to manufacturing unit - Hiving feral Indian bee colony, site selection for apiary, visit to migratory bee keeping sites, visit to commercial cerana bee farm, - Honey extraction, processing, purity testing and value addition, visit to honey processing unit - Hive inspection, maintenance of hive records, management in nectar flow season, dearth period, management of swarming, absconding and laying workers - Dividing, uniting bee colonies, artificial feeding, protecting bees from pesticides - Insect, mite and bird enemies of honeybees, brood and adult diseases - Mass queen rearing and production of mating nucleus, visit to beekeeping society - Methods of collection of bees wax, bee pollen, propolis, bee venom, royal jelly - Visit to commercial mellifera bee farm - Marketing and economics of honey and bee products, preparation of bee keeping projects for bank funding

Reference

- (l) Atwal, A.S. 2013. Mellifera Bee Keeping and Pollination. *Kalyani Publishers, Ludhiana*. 394 p.
- (m) Ted Hooper, 1991. Guide to Bees and Honey (Thrid Edition), *BAS printers ltd. Over Wallop, Hampshire* 271 p.
- (n) Roger A. Morse, 1994. The new complete guide to beekeeping. *The Countryman Press, Woodstock, Vermont*. 207p.
- (o) Thomas D. Seeley. 1995. The Wisdom of the Hive, Harvard University Press, Cambridge, 295p.

Sl. No.	Practical classes	Activity	Ref Book
1.	1-10	Honey bee species, castes, social biology and communication in honey bees	Chapters 3,5,6 and 10 of TB Chapters 2 to 5 of RB 1 Chapter 2,10 of RB 4
2.	11-20	Bee pasturage and preparation of bee floral calendar	Chapter 11 of TB Chapter 12 of RB 1 Chapter 10 of RB 2
3.	21-30	Honey bees for crop pollination and seed production.	Chapter 17 and 18 of TB Chapter 15 of RB 1
4.	31-40	Stingless bees, little bees, rock bees conservation and honey harvest	Chapter 5 of TB Chapter 2 of RB 1
5.	41-50	Beehives, beekeeping equipments specification and uses, visit to manufacturing unit	Chapter 9 of TB Chapter 1 of RB 1 Chapter 4 of RB 2 Chapter 1 of RB 3 Chapter 4 of RB 4
6.	51-70	Hiving feral Indian bee colony, site selection for apiary, visit to migratory bee keeping sites, visit to commercial cerana bee farm	Chapter 8 of TB Chapter 1 and 8 of RB 1 Chapter 2 of RB 3
7.	71-80	Honey extraction, processing, purity testing and value addition, visit to honey processing unit	Chapter 12, 20, 29 of TB Chapter 10 of RB 1 Chapter 11 of RB 2 Chapter 5 of RB 3
8.	81-100	Hive inspection, maintenance of hive records, management in nectar flow season, dearth period, management of swarming, absconding and laying workers	Chapter 12 of TB Chapter 6, 9 of RB 1 Chapter 7 of RB 2 Chapter 3 to 7 of RB 3
9.	101-120	Dividing, uniting bee colonies, artificial feeding, protecting bees from pesticides	Chapter 12, 14, 25 of TB Chapter 9 of RB 1
10.	121-130	Insect, mite and bird enemies of honeybees, brood and adult diseases	Chapter 21 of TB Chapter 13 of RB 1 Chapter 8 of RB 3 Chapter 9 of RB 2
11.	131-140	Mass queen rearing and production of mating nucleus, visit to beekeeping society	Chapter 16, 22 of TB Chapter 10 of RB 1 Chapter 8 of RB 2
12.	141-150	Methods of collection of bees wax, bee pollen, propolis, bee venom, royal jelly - Visit to commercial mellifera bee farm	Chapter 19 of TB Chapter 14 of RB 1
13.	151-170	Marketing and economics of honey and bee products, preparation of bee keeping projects for bank funding	Chapter 28, 29 of TB Chapter 14 of RB 1

E- Reference

- i. http://agritech.tnau.ac.in/farm_enterprises/fe_apiculture_home.html
<http://agdev.anr.udel.edu/maarec/>
<http://www.aragriculture.org/insects/beekeeping.htm>
<http://tiwanabeefarm.com/>
- ii. <http://beekeeping.com/>
- iii. <http://www.apimondia.com/en>

SER 451 Commercial Cocoon Production (0+10)

UNIT I : MULBERRY PRODUCTION AND MANAGEMENT

- Area and distribution of mulberry –Popular Varieties – climatic requirements and soils
- Propagation of nursery - Selection of planting material - Nursery bed Preparation - planting - management – Economics.
- Main field preparation - manuring – planting - Irrigation –Weeding- fertilizers – Intercropping – Training and pruning. Shoot harvest - Transporting - preservation – Economics - project preparation.
- Pruning methods –farm machinery implements.
- Insect pests and diseases of mulberry –management.

UNIT II: SILKWORM REARING AND MANAGEMENT

-  Authorized Silkworm Races – crossbreed and bivoltine. Rearing houses – plan and maintenance. Rearing appliances - disinfection.
-  Agencies involved in egg production - procurement - transportation - preservation– incubation - black boxing – hatching. Brushing of eggs – rearing of chawki worms – leaf selection – feeding – moulting - bed cleaning - bed disinfectants.
-  Chawki Garden – maintenance and management
-  Estimation of population of chawki - establishing Chawki Rearing Centres - Record maintenance – Transport - Fixation of rate. Visit to Chawki Rearing Centre.
-  Late age rearing – tray and shoot rearing methods - leaf selection – feeding - spacing - bed cleaning
-  Moulting care - bed disinfectants. Mounting and mountages. Spinning care and Harvesting.
-  Calculation of Effective rate of rearing - Transporting and marketing of cocoons- Economics of rearing silkworms.
-  Project preparation for establishing Late age rearing centres. Large scale sericulture farming and contract farming.

Unit III: Silk Reeling

-  Physical and commercial properties of cocoons and silk. Cocoon sorting - defective cocoons - cocoon drying - stifling – cooking - brushing - reeling machines - parts and their functions.
-  Study of silk reeling - re-reeling - Skein preparation – packing.
-  Eri silk spinning – spinning - methods.
-  Sampling and testing procedure for winding, size, strength test, condition cohesion and seriplane test. Standards for grading raw silk. Economics of establishing reeling units. Visit to silk reeling units automatic silk reeling units.

PRACTICAL SCHEDULE (WEEKLY)

Week	Syllabus to be covered and Expected Learning experience
1	Area and distribution of mulberry – Popular Varieties – climatic requirements and soil requirement. Preparation of nursery - planting material - manure application.
2	Nursery management – irrigation, weeding, fertilizer application and plant protection. Economics of nursery management.
3	Main field preparation, manuring - Planting methods - Irrigation – Weeding – Intercropping
4	Training and pruning the mulberry crop. Chawki rearing garden - Pruning methods – schedule of operations. Visit to Chawki rearing garden.
5	Late age silkworm rearing - Harvest of leaf and shoot- methods of harvest. Transporting – preservation of leaves and shoots – methods. Visit to Late age silkworm rearing garden - Calculation of brushing capacity.
6.	Farm machinery implements – mulberry pruner, stem cutter and power weeder. Insect pests and diseases of mulberry –natural enemies- IPM. Economics of Mulberry leaf production. Preparation of project proposals.
7.	Authorized Silkworm Races –crossbreed and bivoltine silkworm rearing. Requisites, inspection and selection of site for rearing house.
8.	Rearing houses – plan and maintenance. Rearing appliances – Disinfection. Agencies involved in egg production - procurement of eggs. Transportation – preservation of eggs – incubation.
9.	Black boxing of eggs. - hatching - estimation of hatching percentage. Brushing of eggs – practicing brushing. Rearing of chawki worms – methods. Leaf selection and feeding for young age silkworms.
10.	Moulting, Bed cleaning and bed disinfectants for chawki worms. Estimation of population of chawki worms. Establishing Chawki Rearing Centres. Record maintenance and logistics at Chawki Rearing Centres. Transport of Chawki worms. Visit to Chawki Rearing Centre.
11.	Late age rearing – tray and shoot rearing methods. Leaf selection and feeding for late age silkworms. Spacing of late age worms and bed cleaning.
12.	Moulting care, application of bed disinfectants and its importance. Mounting of worms and mountages. Spinning care and Harvesting.
13.	Calculation of Effective rate of rearing. Maintenance of rearing records, rearing environment for successful rearing. Transporting and marketing of cocoons. Economics of rearing silkworms and maintenance of rearing records.
14	Project preparation for establishing Late age rearing centres. Large scale sericulture farming and contract farming. Visit to Chawki Rearing Centre and late age rearing centres.

18.	Physical and commercial properties of cocoons and silk. Study of cocoon sorting – defective cocoons - drying - stifling - cooking – brushing. Study of reeling machines parts and their functions. Study of silk reeling - re-reeling - Skein preparation - packing.
19.	Study of tasar and muga cocoons – characteristics - cooking and reeling. Study of eri silk spinning - methods of spinning. Sampling and testing procedure for winding, size, strength test. Sampling and testing procedures for condition cohesion and seriplane test.
20.	Standards for grading raw silk. Economics of establishing reeling units. Visit to cocoon market and silk reeling units.

References

- Dandin S.B. Jayant Jayswal and K. Giridhar. 2003. Hand book of Sericulture Technologies. Central Silk Board, Bangalore.
- Krishnaswami, S., M.N. Narasimhanna, S.K Suryanarayan and S.Kumararaj. 1978. Sericulture Manual 2 – Silkworm Rearing . FAO Agricultural Services Bulletin 15/2. Food and Agriculture Organisation of the United Nations, Rome, 131 p.
- Somashekar, T.H. and K. Kawakami. 2003. Manual on Bivoltine Silk Reeling Technology. Central Silk Board, Bangalore. 122 p.

E- References

1. www.silkbase.org
2. www.papilo.ab.a.u.tokyo.ac.jp

ARM 451 MANAGERIAL SKILLS FOR AGRIBUSINESS (0+10)

Week 1

11. Sectors of Agribusiness – Seed, Fertilizer, PP Chemicals, Poultry, Bio inputs, Food Processing, Nursery, Logistics, Warehousing, Retail, Consultancy etc
12. Discussion on Agribusiness – Input sector
13. Discussion on Agribusiness – Processing sector
14. Discussion on Agribusiness – Service sector
15. Presentation on identified agribusiness sector – growth and future directions

18. Institutions promoting agribusiness in India
19. Government schemes promoting Agribusiness – Start up India, Make in India
20. Visit to District Industries Centre
21. Visit to MSME
22. Presentation on promotional schemes for identified agribusiness sector

Week 3

4. Business incubation – Types, Process
5. Business incubation models
6. Visit to Directorate of Agribusiness Development
7. Discussion with the TNAU incubatees
8. Visit to an Agribusiness firm

17. Functions of management
18. Functional areas of management - Operations
19. Functional areas of management – Human resources
20. Functional areas of management - Marketing
21. Functional areas of management - Finance

1. Location decision for a business
2. Layout – Goods and Services
3. Demand forecasting
4. Planning the operations
5. Scheduling the operations

Week 6

1. Inventory management decisions
2. Warehousing management
3. Transportation management
4. Packaging management
5. Presentation of plan of operations

Week 7

3. Market segmentation
4. Targeting and positioning
5. Marketing mix – 4Ps
6. Product - Features, brand name, uses
7. Place – Distribution strategies

Week 8

11. Price – Pricing strategies
12. Promotion – Advertising and Sales promotion

13. Planning display, preparation of floor layout plan
14. Preparation of a newspaper advertisement, selection of advertising media
15. Visit to retail outlets to understand the retail formats

Week 9

18. Visit to local shandy
19. Visit to Farmers market
20. Visit to Regulated market
21. Market survey –
22. Presentation on the survey conducted

Week 10

18. Forms of business organization
19. Farmer Producer Organizations
20. Visit to FPO
21. Financial Assistance for promoting FPOs
22. Presentation on the activities carried out by FPO

Week 11

1. Human Relations skills required for business
2. Leadership – Good and Bad cases
3. Communication– Verbal and written communication strategies
4. Emotions – Emotional Intelligence
5. Business Etiquettes

2. Human Resource Management Policy of Firms
3. Human Resource Planning
4. Recruitment and Selection
5. Training
6. Negotiation

- 2 Company Vision and Mission statement
- 3 SWOT / TOWS Analysis
- 4 BCG / Portfolio Matrix
- 5 Levels of Management
- 6 Company – Strategy formulation

Week 14

9. Source of funds
10. Capital Budgeting Techniques
11. Analyzing Financial Statements
12. Analyzing Financial Statements
13. Discussion on a Case Study

Week 15

3. Business Plan – components, types
4. Preparation of model business plan
5. Preparation of model business plan
6. Presentation of business plan
7. Presentation of business plan

Week 16

18. Entrepreneur – Qualities, Types of Entrepreneurship
19. Institutes promoting Entrepreneurship
20. Writing Biography of an agribusiness entrepreneur
21. Writing Biography of an agribusiness entrepreneur
22. Finding Entrepreneurial competency level

Week 17

- Term paper presentation and Evaluation
Conducting Final Practical Examination

Optional courses for 2 credits			
S.N.	Code No.	Courses	Credit Hours
1.	CRP 351	Physiological Techniques in crop production	2 (1+1)
2.	SAC351	Designer fertilizer Production	2 (1+1)
3.	SAC352	Rejuvenation of Deteriorated lands	2 (1+1)
4.	SAC353	Soilless crop production	2 (1+1)
5.	SAC354	Instrumental methods of analysis	2 (1+1)
6.	SST 351	Seed entrepreneurship skill development and management	2 (1+1)
7.	AGR 351	Weed and water management	2 (1+1)
8.	PGP 351	Plant Genetic Resources Collection, Conservation and Utilization	2 (1+1)
9.	NEM 351	Commercial Production of Nematode Antagonistic bio-agents	2 (1+1)
10.	AGM351	Downstream Processing for Industrially Important Microbial Products	2 (1+1)
11.	AGM 352	Microbial Enzymes	2 (1+1)
12.	AGM 353	Microbial Quality and Safety of Foods	2 (1+1)
13.	AGM 354	Plant –Microbe Interaction	2 (1+1)
14.	AGM 355	Quality Control of Bio-inoculants	2 (1+1)
15.	SAC 355	Crop and Pesticide Chemistry	2 (1+1)

CRP 351 Physiological Techniques in crop production (1+1)

Aim

To impart basic knowledge on various functions and processes related to stress physiology, nutritive physiology, hormonal physiology, production physiology, post harvest physiology etc., and their application in crop production.

Unit I Stress physiology

Abiotic Stress: Water deficit – impact of drought on crop productivity -characteristic features of drought tolerant plants, Drought resistance and tolerance mechanisms, Osmotic adjustment and osmoregulation, stress proteins, Water use efficiency – carbon isotope discrimination- Physiological traits associated with drought and Mitigation techniques. Flooding: Physiological mechanism of adaptation, Mitigation techniques, Temperature stress: High and Low Temperature, Tolerance mechanisms - Functions of HSPs and CSPs, Oxidative stress: Reactive Oxygen Species (ROS). Role of scavenging systems – Enzymatic and Non-Enzymatic, Physiological traits associated with high and low temperature, Mitigation techniques. Salt stress: Physiological basis of tolerance mechanisms, Physiological traits associated with salt stress, Mitigation techniques. UV stress and climate change: Physiological adaptation of crops to UV radiation and tolerance mechanisms.

Unit II Nutriophysiology

Diagnosis and correction measures for nutritional disorders including Macro, Micro and secondary nutrients in Cereals, Pulses, Oilseeds, Fibre and Sugar crops; Fruits, Vegetables, Flowers, Spices, Plantation and Aromatic crops. Impact of heavy metals on physiology and productivity of crops, Phytoremediation, Importance of beneficial elements – Na, Si, Se, Co.

Unit III Hormonal physiology

Role of hormones in plant growth and yield enhancement, stress management and quality improvement - Auxins, Gibberellins, Cytokinins, Abscissic acid, Ethylene and Brassinosteroids. Role of other phytohormones in crop production-triacontanol, polyamines, jasmonates and salicylic acid, New generation PGRs - 1- MCP, Triazoles, strigalactone, pro-hexadione Ca.

Unit IV Production Physiology

Physiological limitations of crop productivity, Physiological and genetic basis of crop environment interaction, Plant architecture – Ideotype concept, Crop photosynthetic efficiency –

C₃ , C₄ and CAM Strategies to improve the crop photosynthesis, Source- sink balance and Harvest Index.

Unit V Postharvest physiology

Environmental factors influencing senescence, ripening and postharvest life of fruits, flowers, vegetables and seeds. Physiological and biochemical aspects of senescence and fruit ripening. Regulatory role of ethylene in senescence and ripening, Pre and post harvest measures to influence shelf life.

Theory Lecture schedule

1. Classification of abiotic stresses - Drought – types-Drought resistance and tolerance mechanisms- adaptations-Physiological traits associated with drought -osmotic adjustment.
2. Reactive Oxygen Species- scavenging enzymes- stress proteins-water use efficiency – carbon isotope discrimination- concept -mitigation techniques.
3. Flooding - Physiological mechanism of adaptation- physiological traits associated with flooding- Role of ethylene.
4. Temperature stress-High and Low Temperature -Tolerance mechanisms-Functions of HSPs and CSPs - Physiological traits associated with high and low temperature.
5. Salt stress - Physiological basis of tolerance mechanisms –adaptations- Physiological traits associated with salt stress.
6. Physiological adaptation of crops to high and low light and UV radiation.
7. Impact of heavy metals on physiology and productivity of crops –Phytoremediation.

8. Mid Semester examination

9. Diagnosis and correction measures for nutritional disorders in Cereals, Millets, Pulses, Oilseeds.
10. Diagnosis and correction measures for nutritional disorders in Fibre, Sugar crops, Fruits and Vegetable, Flowers, Spices, Plantation, Medicinal and Aromatic crops.
11. Importance of beneficial elements – Na, Si, Se, Co- Effect of crop specific application of beneficial elements.
12. Role of hormones in plant growth and yield enhancement, stress management and quality improvement – Auxins, Gibberellins and Cytokinins.
13. Role of hormones in plant growth and yield enhancement, stress management and quality improvement –Absciscic acid, Ethylene and Brassinosteroids.

14. Role of other phytohormones -triacontanol, polyamines, jasmonates and salicylic acid. New generation PGRs - 1- MCP, Triazoles, strigalactone, pro-hexadione Ca.
15. Physiological limitations of crop productivity, Physiological and genetic basis of crop environment interaction, Plant architecture – Ideotype concept.
16. Crop photosynthetic efficiency – C₃, C₄ and CAM Strategies to improve the crop photosynthesis, Source- sink balance and harvest index.
17. Environmental factors influencing senescence and ripening-Physiological and biochemical aspects of senescence and fruit ripening-Factors affecting post harvest life of fruits- measures for enhancing the shelf life of fruits, vegetables and flowers.

Practical schedule

1. Determination of osmotic potential
2. Gas Exchange measurements using Infra Red Gas Analyzer
3. Stress induction response techniques
4. Water Use Efficiency or Transpiration Efficiency of crops
5. Estimation of anti oxidant enzymes activity – Catalase and peroxidase
6. Estimation of Membrane thermal stability
7. Bioassay for Cytokinins
8. Effect of PGRs on flowering
9. Effect of PGRs on fruit ripening
10. Influence of ABA on stomatal regulation
11. Diagnosis of nutritional disorders and their amelioration measures
12. Nutrient application techniques
13. Estimation of Macro nutrients in plant samples
14. Estimation of micro and secondary nutrients using AAS
15. Field Visit I- Diagnosis of nutrient disorders
16. Field Visit II-Postharvest storage facilities
17. Final practical

Outcome

Students will acquire basic knowledge on various functions and processes related to stress physiology, nutritio physiology, hormonal physiology, production physiology and post harvest physiology. In addition, hands on exposure to determine osmotic potential, influence of PGRs in

fruit ripening, stomatal regulation, Stress induction response techniques and estimation of macro and micro nutrient will help the student gain confidence in skill oriented education. Student will also get exposure in diagnosis of nutrient disorders which will augment in recommending remedial measures to nutrient disorders and extending the storage life of harvested produce.

Text books

5. Barker AB & Pilbeam DJ. 2007. Handbook of Plant Nutrition. 2nd Edition, CRC Press, Taylor & Francis Group. Pp 1-773.
6. Bewley, J.D. and Black, M. 1985. Seed Physiology of Development and Germination. Plenum Publishing Corp. New York, NY. pp 70-73.
7. Bleasdale, J.K.A. 1984. Plant Physiology in Relation to Horticulture. 2nd Ed. MacMillan & Avi Publishing Company, USA.
8. Leopold, A.C. and Kriedemann, P.E. 1985. Plant Growth and Development. 3rd Ed. McGraw-Hill, New York, pp 545.
9. [Madhava Rao](#) KV, [Raghavendra](#), AS and [Janardhan Reddy](#) K. 2006. Physiology and Molecular Biology of Stress Tolerance in Plants. Springer publishers, Netherland.
10. Taiz. L. and Zeiger. E., 2015 (Sixth edition). Plant Physiology and Development. Publishers: Sinauer Associates, Inc., Massachusetts, USA.
11. Wilkins MB. 1969. Physiology of Plant Growth and Development. Tata McGraw-Hill, New York, pp 127-162.

e- books and e-references

1. <http://www.plantphys.net>
2. <http://www.plantstress.com>
3. <http://www.ipni.net>
4. <http://www.edis.ifas.ufl.edu>
5. <http://www.greenair.com/plantlnk.htm>
6. <http://www.tvdsb.on.ca>

AGR 351 WEED AND WATER MANAGEMENT (1+1)

Unit - I

Weeds: Introduction, Definitions; harmful and beneficial effects, classification;; crop weed competition and allelopathy. Methods of weed control: physical, cultural, chemical and biological methods. Integrated weed management.

Unit - II

Herbicides - Classification, characteristics, formulations, methods of application; advantages-.Weed management in major field crops - aquatic and problematic weeds and their control.

Unit – III

Role of water in plant growth - Importance of irrigation- Soil - water - plant relationship - Soil Plant Atmospheric Continuum (SPAC) - Hydrological cycle - Soil water movement - soil moisture constants - Moisture extraction pattern - Absorption of water.

Unit IV

Crop water requirement- Factors affecting water requirement- Factors affecting water requirement. Scheduling of irrigation – Water use efficiency

Unit V

Methods of irrigation: surface, sub-surface sprinkler and drip irrigation – Micro irrigation: layout, suitability, merits and scope.

Practical

Identification of weeds in wetlands, gardenland and drylands – Practicing different methods of weed control -Calculations on weed control efficiency and weed index; Classification and characteristics of herbicides - Computation of herbicide doses- Study of herbicide application equipment and calibration; Demonstration of methods of herbicide application;

Estimation of soil moisture - Measurement of irrigation water through water measuring devices (flumes, weirs and water meter) - Calculation of irrigation water requirement (problems)- Acquiring skill in land shaping for different surface irrigation methods - Operation and economics of drip and sprinkler irrigation systems - Estimation of crop water requirement - Irrigation efficiency (problems) –

Theory

Lecture Schedule :

1. Weeds - Definition, classification
2. Characteristics of weeds, harmful and beneficial effect of weeds.
3. Crop weed interactions - Critical crop weed competition, competitive and allelopathic effects of weeds and crops.
4. Methods of weed control: physical, cultural
5. Methods of weed control: chemical and biological methods. Integrated weed management.
6. Herbicides - Classification, characteristics, formulations,
7. Methods of herbicide application
8. Weed management in major field crops - aquatic and problematic weeds and their control.

9. Mid semester examination

10. Role of water in plants - Importance of irrigation
11. Soil - Plant -water relationship - Soil-plant-atmospheric continuum - Hydrologic cycle - absorption of water and evapotranspiration.
12. Soil water movement - saturated and unsaturated flow and vapour movement - soil moisture constants and their importance in irrigation.
13. Available soil moisture - definition and importance - moisture extraction pattern
14. Crop water requirement - factors affecting crop water requirement -Critical stages for irrigation - water requirement for different field crops.
15. Scheduling of irrigation - Different approaches- Water use efficiency
16. Methods of irrigation - surface (flooding, beds and channels, border strip, ridges and furrows, broad bed and furrows, surge irrigation) and sub-surface methods.
17. Micro irrigation system (drip and sprinkler irrigation) - suitability, components, layout, operation, advantage and disadvantage.

Practical schedule:

1. Identification, classification and characterization of wetland weeds.
2. Identification, classification and characterization of gardenland and dryland weeds.
3. Practicing skill development on cultural and non chemical weed management.
4. Identification, classification and characterization of herbicides.
5. Practicing skill development on herbicide application techniques.
6. Practicing Skill development on spray equipment's and spray fluid calibration.
7. Calculation of herbicide quantity and recommendation for different eco systems

8. Working out weed index, weed control efficiency and economics in weed management studies.
9. Estimation of soil moisture by gravimetric method and tensiometer.
10. Estimation of soil moisture by resistance blocks and neutron probe and other improved devices.
11. Measurement of irrigation water with flumes and weirs.
12. Calculation of irrigation water based on source, water flow, soil moisture status and depth of irrigation.
13. Land leveling and land shaping - Beds and channels - ridges and furrows.
14. Layout, operation and maintenance of drip and sprinkler irrigation systems.
15. Estimation of crop water requirement by direct and indirect methods.
16. Calculations on irrigation efficiency parameters.
- 17. Practical examination**

References:

- Gupta, O. P. 1998. Modern Weed Management. Agro Botanica Bikaner, India.
- Subramanian, S. A. Mohammed Ali and R. Jayakumar. 1991. All about Weed Control. Kalyani Publishers, New Delhi.
- Jaganathan R., and R. Jayakumar. 2003. Weed Science Principles, Kalyani Publishers, New Delhi.
- Das, P.C. 2015. Weed Science. New India Publishing Agency, Jaipur
- Michael, A.M. 1997. Irrigation: Theory and Practice Vikas Publishers.
- Sankara Reddy, G.H. and T. Yellamanda Reddy. 1997. Efficient use of irrigation water. Kalyani Publishers
- Reddy, S.R. 2012. Irrigation Agronomy. Kalyani publishers, New Delhi
- Nirmal Tripathy. 2014. Irrigation, Watershed and Drainage. Agrotech Press, Jaipur
- Gupta, A.K. 2014. Irrigation and Drainage. Oxford Book Company, Jaipur
- Israelsen, O.W. 2012. Irrigation Principles and Practices. Axis Books (India), Jodhpur.
- Michael, A.M. 2009. Irrigation- Theory and Practice, Vikas Publishing House, Noida

E-References:

- www.tnau.ac.in
- www.fao.org
- www.tnau.ac.in/agriportal
- www.irri.org
- www.wcc.nrcs.usda.gov/nrcsirrig
- www.wcc.nrcs.usda.gov/irrig.info.html
- www.croinfo.net/irrigschedule.html

AGM 351 DOWNSTREAM PROCESSING FOR INDUSTRIALLY IMPORTANT MICROBIAL PRODUCTS (1+1)

Aim : This course aims to develop the skills of the students in the area of Downstream processing of important microbial products.

Objective: At the end of the course, student would have learnt about various purification techniques available for industrially important microbial products like, antibiotics, amino acids, proteins, enzymes, organic acids etc

Unit I - Introduction to Bio separation Processes

Role and importance of bioseparations in biotechnological processes. Problems and requirements of bioproduct purification. Cost-cutting strategies. Characteristics of biological mixtures. Classification of bioproducts - Biological activity, Analysis of purity-Process economics- Capital and operating cost analysis

Unit II - Physical Methods of Separation

Centrifugation and filtration. Cell disruption methods; enrichment operations: precipitation methods- with salts, organic solvents, and polymers; extractive separations - aqueous two-phase extraction, supercritical extraction; adsorption methods; membrane based separation theory - types of membranes, types of membrane processes - dialysis; ultrafiltration; microfiltration and reverse osmosis.

Unit III Physical and chemical methods - Isolation of products

Physico-chemical basis of bio-separation processes. Removal of particulate matter, biomass and insolubles: flocculation; sedimentation; adsorption: isotherm, batch, continuous and scale-up of adsorption; extraction: solvent separation, equipment and modes of extraction. Aqueous-two-phase extraction process, supercritical fluid extraction. Precipitation of proteins-methods and scale-up.

Unit IV Isolation of Products - Chromatographic methods of purification

Chromatography - principles, instruments and performance parameters. Paper, TLC, adsorption, gel filtration, reverse phase, ion-exchange, hydrophobic interaction, bioaffinity, pseudo affinity chromatographic techniques; GC, HPLC, HPTLC, FPLC, perfusion chromatography and membrane based chromatographic techniques and sample preparation. Electrophoretic separations.

Unit – V Finishing Operations

Products polishing: Crystallization and drying; Purification of antibiotics, amino acids, enzymes and organic acids, proteins, pigments, plant growth hormones, monoclonal antibodies, human growth hormones – Insulin etc.

Practical

Kinetics of a bacterium / yeast in batch culture- doubling time, specific growth rate and growth curve. Batch cultivation of microbes for product formulation; calibration of pH probe, medium preparation, sterilization, and calibration of dO₂ probe. Inoculation of a bacterial / yeast strain in fermentor for batch production of an enzyme / biomass/ ; estimation of kinetic parameters of product formation, biomass production and substrate utilization. Laboratory centrifugation for separation of biomass from fermentation broth. Filtration and microfiltration for solid-liquid separation. Extraction of crude enzyme from microbial biomass and enzyme assay. Production of microbial secondary metabolites and product recovery. Mechanical cell disruption by homogenization: determination of product release kinetics. Sonication of microbial cell suspensions and determination of product release kinetics. Salting out of proteins by precipitation with ammonium sulphate. Precipitation of proteins with acids or bases: determination of isoelectric point. Batch adsorption of an acid on activated charcoal. Solvent extraction and aqueous two-phase extraction of proteins. Gel-filtration chromatography of a protein - salt mixture. Ion-exchange chromatography of a positively charged protein. Gas chromatography for quantitative estimation of volatiles- ethanol. SDS-PAGE of proteins and determination of molecular weight.

Lecture Schedule

1. Role and importance of bio separations in biotechnological processes. Problems and requirements of bioproduct purification
2. Cost- cutting strategies. Characteristics of biological mixtures. Classification of bio products
3. Biological activity, Analysis of purity. Process economics. Capital and operating cost analysis
4. Centrifugation and filtration-Cell disruption methods
5. Enrichment Operations: precipitation methods(with salts; organic solvents and polymers)
6. Extractive separations; aqueous two-phase extraction; supercritical extraction and adsorption methods

7. Membrane based separation theory - types of membranes; types of membrane processes – dialysis, ultrafiltration, microfiltration and reverse osmosis
8. Physico-chemical basis of bio-separation processes. Removal of particulate matter, biomass and insolubles: flocculation; sedimentation

9. Mid semester evaluation

10. Adsorption – Isotherm, batch, continuous and scale-up of adsorption
11. Extraction: solvent separation, equipment and modes of extraction. Aqueous two-phase extraction process, supercritical fluid extraction.
12. Precipitation of proteins: methods and scale-up. Chromatography - principles, instruments and practice
13. Paper; TLC, adsorption, gel filtration; reverse phase, ion-exchange, hydrophobic interaction, bioaffinity ; pseudo affinity chromatographic techniques
14. GC, HPLC, FPLC HPTLC, parafusion chromatography and membrane based chromatographic techniques and sample preparation. and electrophoretic separation
15. Products polishing. Crystallization and drying. Purification of antibiotics
16. Purification of amino acids, organic acids, enzymes and proteins
17. Monoclonal antibodies; human growth hormones – Insulin etc

Practical Schedule

1. Kinetics of a bacterium / yeast in batch culture: calculation of doubling time, specific growth rate, and plotting of growth curve.
2. Preparation of fermentor and accessories for batch cultivation of microbes: calibration of pH probe, medium preparation, sterilization, and calibration of dO_2 probe.
3. Inoculation of a bacterial / yeast strain in fermentor for batch production of an enzyme / biomass: estimation of kinetic parameters of product formation, biomass production and substrate utilization.
4. Laboratory centrifugation for separation of biomass from fermentation broths. Filtration and microfiltration for solid-liquid separation.
5. Extraction of crude enzyme and assay of enzyme
6. Production of secondary metabolites from microbes and recovery
7. Mechanical cell disruption by homogenization: determination of product release kinetics.
8. Sonication of microbial cell suspensions and determination of product release kinetics.

9. Salting out of proteins by precipitation with ammonium sulphate.
10. Precipitation of proteins with acids or bases: determination of isoelectric point.
11. Batch adsorption of an acid on activated charcoal.
12. Solvent extraction and aqueous two-phase extraction of proteins.
13. Gel-filtration chromatography of a protein-salt mixture.
14. Ion-exchange chromatography of a positively charged protein.
15. Gas chromatography for quantitative estimation of volatiles: ethanol.
16. SDS-PAGE of proteins and determination of molecular weight

17. Final Practical Examination

Text Books

1. Sivasankar B., 2010. Bioseparations: Principles and Techniques, PHI, New Delhi .
2. Stanbury, P.F., Whitaker, A. and Hall, S.J. 2016. Principles of Fermentation Technology, BH Elsevier Publications, Third Edition.

Reference Books

1. Harrison, R.G. Todd, P., Rudge, S.R. and Petrides, D.P., 2003. Bioseparation Science and Engineering, Oxford University Press.
2. Costa, C.A. and Cabral, J.S. 1991. Chromatographic and Membrane Processes in Biotechnology, Publisher: Kluwer Academic Publishers, The Netherlands.
3. Harrison et al. 2006. Bioseparation Science and Engineering. Oxford Univ. Press.
4. Nooralabettu Krishna Prasad, 2010. Downstream Process Technology: A New Horizon in Biotechnology, PHI, New Delhi.
5. Jenkins, R.O. 1992. Product Recovery in Bioprocess Technology, Biotechnology by Open Learning Series, Butterworth-Heinemann, London, Second Edition.

AGM 352 MICROBIAL ENZYMES (1+1)

Aim: Main aim of the course is to impart knowledge on microbial sources of enzymes and their utility in different industries.

Objectives: Student will learn about the microbial enzymes, their characteristics, techniques of microbial enzyme production and use in different industries.

Unit I: Introduction

Introduction and Scope, General distinctive features and industrial applications. Enzymes: Historical perspectives, Nomenclature and classification of enzymes. Isozymes, biological roles, activation energy, chemical nature of enzymes, characteristics of enzymes, 3'D' structure of enzymes, active site, factors affecting enzyme activity, modifiers of enzyme activity, enzyme activators, enzyme inhibitors and allosteric enzymes.

Unit II: Enzyme kinetics

Kinetics of single substrate reactions, Estimation of *Michaelis - Menten* parameters, multi substrate reactions, mechanisms and kinetics, turnover number, types of inhibition, kinetic models: substrate and product inhibition, allosteric regulation of enzymes, *Monod ChangeuxWyman* model, pH and temperature effect on enzymes and deactivation kinetics. Vitamins and their co-enzymes: structure and functions with suitable examples. Metallo enzymes and metal ions as co-factors and enzyme activators.

Unit III: Extraction and purification of microbial enzymes

Importance of enzyme purification, different sources of enzymes. Extracellular and intracellular enzymes. Physical and chemical methods used for cell disintegration. Enzyme fractionation by precipitation - using temperature, salt, solvent, pH, *etc.*, Liquid-liquid extraction -ultra filtration, ionic exchange, gel chromatography, affinity chromatography and other special purification methods. Enzyme crystallization techniques. Criteria of purity of enzymes.

Unit IV: Industrial applications of enzyme and enzyme engineering

Industrial applications: Microbial enzymes in textile, leather, wood industries and detergents. Enzymes in clinical diagnostics. Enzyme sensors for clinical processes and environmental analyses. Enzymes as therapeutic agents. Enzyme engineering: Chemical modification and site-directed mutagenesis to study the structure-function relationship of industrially important enzymes. Cloning- strain improvement.

Unit V: Enzyme formulation

Physical and Chemical techniques for enzyme immobilization – adsorption - matrix entrapment encapsulation - cross-linking - covalent binding - examples; advantages and disadvantages of different immobilization techniques. Freeze drying and spray drying of immobilized enzymes.

Practical

Amylase production using *Bacillus amyloliquefaciens* and its assay. Protease production by using *Bacillus* isolate and its quantification. Production of cellulase by solid state fermentation (SSF) of rice straw through lignocellulolytic fungi: (a) Estimation of filter paper lyase activity (b) Estimation of carboxy methyl cellulase activity. Production and estimation of xylanase from rice straw through submerged fermentation. Immobilization of microbial cells for enzyme production. Protease production from *Bacillus subtilis* using soybean meal. Purification of fungal α - amylase or bacterial protease by fractionation, chromatographic techniques and electrophoretic separation. Studies on enzyme kinetics of alpha amylase / protease- optimization of parameters viz., substrate, enzyme concentration, reaction temperature, reaction pH, K_m , V_{max} and metal ions as activators and inhibitors. Enzyme extraction- concentration- ultrafiltration- chromatography- microencapsulation.

Theory schedule

1. Introduction and scope, general distinctive features and industrial applications.
2. Enzymes: historical perspectives, nomenclature and classification .
3. Isozymes, biological roles, activation energy, chemical nature of enzymes.
4. Characteristics of enzymes, 3'D' structure of enzymes, active site, factors affecting enzyme activity.
5. Modifiers of enzyme activity, enzyme activators, enzyme inhibitors; allosteric enzymes.
6. Kinetics of single substrate reactions, estimation of *Michaelis - Menten* parameters, multi substrate reactions, mechanisms and kinetics; turnover number.
7. Types of inhibition; Kinetic models: substrate and product inhibition; Allosteric regulation of enzymes.
8. *Monod Changeux Wyman* model, pH and temperature effect on enzymes and deactivation kinetics.

9. Mid Semester Examination

10. Vitamins and their co-enzymes: structure and functions with suitable examples. Metallo enzymes and metal ions as co-factors and enzyme activators.

11. Importance of enzyme purification, different sources of enzymes. Extracellular and intracellular enzymes.
12. Physical and chemical methods used for cell disintegration. Enzyme fractionation by precipitation - using temperature, salt, solvent, pH, *etc.*, liquid-liquid extraction.
13. Ionic exchange, gel chromatography, affinity chromatography and other special purification methods. Enzyme crystallization techniques. Criteria of purity of enzymes.
14. Industrial applications: Microbial enzymes in textile, leather, wood industries and detergents. Enzymes in clinical diagnostics. Enzyme sensors for clinical processes and environmental analyses. Enzymes as therapeutic agents.
15. Enzyme engineering: Chemical modification and site-directed mutagenesis to study the structure-function relationship of industrially important enzymes. Cloning- strain improvement.
16. Physical and Chemical techniques for enzyme immobilization – adsorption - matrix entrapment encapsulation - cross-linking - covalent binding - examples;
17. Advantages and disadvantages of different Immobilization techniques - overview of applications of immobilized enzyme systems

Practical schedule

1. & 2. Amylase production using *Bacillus amyloliquefaciens* in submerged and solid state fermentation and its assay.
3. & 4. Protease production using *Bacillus* isolate and its quantification.
5. Production of cellulase by solid state fermentation (SSF) of rice straw through lignocellulolytic fungi - Estimation of filter paper lyase activity.
6. Production of cellulase by SSF of rice straw through lignocellulolytic fungi- Estimation of carboxy methyl cellulase activity.
7. Production and estimation of xylanase from rice straw through submerged fermentation.
8. Immobilization of microbial cells for enzyme production.
9. Protease production from *Bacillus subtilis* using soybean meal.
10. -12. Purification of fungal alpha-amylase or bacterial protease by fractionation, chromatographic techniques and electrophoretic separation.
- 13-14. Studies on enzyme kinetics of alpha amylase / protease - optimization of parameters viz., substrate, enzyme concentration, reaction temperature, reaction pH, K_m , V_{max} and metal ions as activators and inhibitors.

15-16. Enzyme extraction- concentration-ultrafiltration-chromatography-
microencapsulation

17. Final Practical Examination

Text Books

1. Alexander N. Glazer, Hiroshi Nikaido. 2007. Microbial Biotechnology, Second Edition, ISBN 9780521842105, Cambridge University Press.
2. Trevor Palmer and Philip L. Bonner, 2004. Enzymes: Biochemistry, Biotechnology, Clinical Chemistry, East- West Press.

Reference Books

1. Shuler, M.L. and F. Kargi, 2002. Bioprocess Engineering : Basic Concepts, Second Edition, Pearson.
2. Blanch, H.W and D.S. Clark, 1997. Biochemical Engineering, Marcel & Dekker, Inc.,.
3. Bailey, J.E and D.F. Ollis, 1986. Biochemical Engineering Fundamentals, Second Edition, McGraw-Hill.
4. Nicholas C. Price and Lewis Stevens, 1982. Fundamentals of Enzymology, Oxford University Press.
5. Alan Wiseman, 1999. Handbook of Enzyme Biotechnology, Third Edition, Ellis Harwood Publications.

E-Book

1. James Lee, M., Biochemical Engineering, PHI, USA, e-Book Version 2.1, 2002.

AGM 353 MICROBIAL QUALITY AND SAFETY OF FOODS (1+1)

Aim: This course aims in imparting knowledge on microbial quality and safety of foods.

Objectives : At the end of the course the student will be able to learn the techniques of microbial quality assessment and food safety standards.

Unit I Introduction

Food safety risks and assessment– characteristic features of spoilage - significance of spoilage of different groups of foods - cereal and cereal products, vegetables and fruits, meat, poultry, sea foods, milk and milk products, packed and canned foods.

Unit II Food borne infections and intoxications

Food borne pathogens- food infections and intoxications of food borne diseases – bacteria, fungal, protozoa and viral. Investigation and management of food borne diseases.

Unit III Techniques for detection of pathogens and toxins

Advanced laboratory techniques for food-borne pathogens: principle, working and application of GC-MS, HPLC, LC/MS, inductively coupled Plasma Mass Spectroscopy, TOF and PCR-real time PCR. DGGE. Metagenomics, proteomics and immunological methods

Unit IV Food standards

Food standards – HACCP concepts, principles – EU, FDA and WHO standards - CODEX – Food Law & regulations - HACCP principles & applications – GM foods – SSOP – GMP & GAP – Food traceability- Food Audit - functions, duties and responsibilities of food safety regulators.

Practical

Food sampling procedures – Preparation & plan - Examination of microorganisms of by aerobic plate count -Assessing *Bacillus cereus*, coliforms, *Campylobacter*, *Salmonella* and *Staphylococcus* in various food samples. Microbiological examination of canned foods. Rapid detection of pathogens and toxins in foods– HACCP of fruits and vegetables - visit to food processing industry.

Lecture Schedule

1. Food Safety Risks - biological, chemical, physical risks; risk assessment.
2. Physical and chemical changes during food spoilage.
3. Significance of microbial spoilage of different groups of foods - cereals, vegetables and fruits, packed and canned foods.
4. Significance of microbial spoilage of different groups of foods-meat, poultry, sea foods, milk and milk products.
5. Food borne pathogens – food infection – intoxication - bacterial food borne diseases – viral and protozoa.
6. Mycotoxicoses in foods – occurrence and economic significance and food control measures.
7. Analytical techniques and their working principles for the detection of toxin in foods – GC-MS, HPLC and HPTLC.

8. Analytical techniques and their working principles for the detection of toxin in foods - LC/MS, ICP – MS and TOF.

9. Mid Semester Examination

10. Molecular Detection of Food borne Pathogens - PCR, real time PCR, DGGE, metagenomics, proteomics and immunological methods
11. Food standards – India, EU, FDA and WHO Standards of Food Safety
12. Food laws & Regulations - CODEX
13. HACCP concepts, principles and applications
14. GM foods and current guidelines for production and labeling
15. Food Safety – Standard Sanitation Operating Procedures; GMP, GAP for food safety
16. Food traceability– significance - Food safety Audit
17. Functions, duties and responsibilities of food safety regulators

Practical schedule

1. Different sampling plan in food and preparation for various foods
2. Determination of aerobic plate counts in food
3. Enumeration of yeasts and molds in food
4. Detection and confirmation of *Bacillus cereus* in food
5. Microbiological hazard analysis in processed fruit product
6. Microbiological hazard analysis in water - water quality control
7. Microbiological hazard analysis in fresh vegetables – *Campylobacter* and *Salmonella*
8. Microbiological hazard analysis in beverages - Detection of coliforms
9. Microbiological techniques for cereal based food analysis
10. Microbiological hazard analysis in meat products - *Staphylococcus aureus*
11. Rapid detection of toxin producing *Escherichia coli* (STEC) in food products –PCR method
12. Detection of aflatoxin in groundnut and maize kernels
13. Hazard Analysis and Critical Control Point (HACCP) of Fruits / vegetables
14. Hazard Analysis and Critical Control Point (HACCP) of processed foods
15. Visit to Food Processing industry
16. Microbiological examination of canned foods
17. **Final Practical Examination**

Text Books

1. Adams, M.R. and Moss, M.O.1995. Food Microbiology. The Royal Society of Chemistry, Cambridge.
2. Frazier, W.C. and Westhoff, D.C.1988. Food Microbiology. TATA McGraw Hill Publishing Company Ltd., New Delhi.

Reference Books

1. Jay, J.M.1987. Modern Food Microbiology. CBS Publishers and distributors, New Delhi.
2. Banwart, G.J.1989. Basic Food Microbiology. Chapman & Hall New York.
3. Board, R.C.1983. A Modern Introduction to Food Microbiology. Blackwell Scientific Publications, Oxford.
4. Robinson, R.K.1990. Dairy Microbiology Elsevier Applied Science, London.
5. Hobbs, B.C. and Roberts, D.1993. Food Poisoning and Food Hygiene. Edward Arnold.
6. Lund B.M., Baird Parker A.C., and Gould G.W. 2000. The Microbiological Safety and Quality of Foods. Vol. 1-2, ASPEN Publication, Gaithersberg, MD.
7. Gould G.W. 1995. New Methods of Food Preservation. Blackie Academic and Professional, London.

AGM 354 PLANT-MICROBE INTERACTIONS (1+1)

Aim: To allow students to explore various ways in which microbes interact with plants, and the outcomes of that interplay. The focus is on examination of the physiological, biochemical and genetic basis of these interactions, using comparisons to other prokaryotic and eukaryotic model systems.

Objectives: At the end of this course students will be able to:

- recognize and distinguish between the different types of plant-microbe interactions
- explain the physiological and biochemical processes underlying major symbiotic and pathogenic relationships
- analyse the design and content of current research studies published in this area of study
- draw connections between the biology of plant-microbe relationships, the impacts of those relationships on plant growth

Unit I Introduction

Introduction to plant - microbe interactions; types of interaction -positive and negative.Plants as microbial habitat.Spermosphere, phyllosphere and endophyticmicroorganisms.Bacterial secretion systems, gene regulation and quorum sensing in bacterial - plant interactions.Role of plant-microbial interactions in soil health and plant growth promotion. Signalling - effect of microbial signalling on plant productivity and plant signalling on microbial diversity and activity in the soil.

Unit II Plant- microbe interaction -I

Beneficial bacteria – *Arthrobacter*, *Azospirillum*, *Azoarcus* *Bacillus*, *Burkholderia*, *Frankia*, *Gluconacetobacter*, *Herbaspirillum*,*Paenibacillus*, *Pseudomonas*, *Rhizobium*, *Streptomyces* and *Xanthomonas* - host interactions and plant growth promotion. *Agrobacterium* induced tumorigenesis and rhizogenesis. *Azolla* - *Anabaena* symbiosis.

Unit III Plant– microbe interaction -II

Mutualistic fungal symbionts.Ectomycorrhizal and endomycorrhizal fungi. Infection processes. Interactions with host plants and other soil microbes. Infection processes- fungi and toxins. Induced resistance.Secondary product responses and fungal virulence.Lichens, algal and bacterial interaction.

Unit IV Applications – Agriculture and industries

Plant growth promoting bacteria-types and mechanism of plant growth promotion.Biocontrol agents- types and mechanism of action.Induced resistance: Protein defense responses and systemic responses.Phytoremediation.Industrial application and medicinal applications – antiviral, anti-cancerous, immunosuppressive and antioxidants.

Unit V Techniques to study plant–microbe interactions

Techniques to study plant- microbe interactions - Phytotron, Rhizotron and Confocal Laser Scanning Microscope.

Practical

Collection and assay of root exudates. Characterization of root exudates. Studying the effect of root exudates on selected bacterial population. Isolation of ecto and endorhizosphere microorganisms.Isolation of spermospheremicorganisms from germinating seeds. Isolation and purification of endophytic and phyllosphere microorganisms from rice. Visit to plantations and collection of ectomycorrhizal fruiting bodies, orchidaceous and ericoid mycorrhizal root samples

and lichens. Examination of endomycorrhizal infection in orchids and ericaceous plants. Examination of lichens associated with trees. Collection and examination of *Anabaena azollae* associated with Azolla. Testing rhizogenesis by *A. rhizogenes* in laboratory conditions.

Lecture Schedule

1. Introduction to plant - microbe interactions; types of interaction - positive and negative. Plants as microbial habitat.
2. Role of microbial diversity in soil health and plant growth promotion.
3. Signalling – effect of microbial signalling on plant productivity.
4. Effect of plant signalling on microbial diversity and activity in the soil.
5. Spherosphere and phyllosphere microorganisms; endophytic microorganisms.
6. Bacterial secretion systems, gene regulation and quorum sensing in bacterial - plant interaction
- 7.&8. Beneficial bacteria – *Arthrobacter*, *Azospirillum*, *Azoarcus*, *Bacillus*, *Burkholderia*, *Frankia*, *Gluconacetobacter*, *Herbaspirillum*, *Paenibacillus*, *Pseudomonas*, *Rhizobium*, *Streptomyces* and *Xanthomonas* - host interactions and plant growth promotion
9. **Mid Semester Examination**
10. *Agrobacterium* induced tumour formation and root proliferation; exploitation of tumorigenesis and rhizogenesis. Mutualistic fungal symbionts
11. Ectomycorrhizal and endomycorrhizal fungi. Infection processes. Interactions with host plants and other soil microbes
12. Infection processes - fungi and toxins. Secondary product and fungal virulence
13. Azolla - *Anabaena* symbiosis. Association of lichens with trees
14. Plant growth promoting bacteria- types and mechanism of plant growth promotion
15. Biocontrol agents- types and mechanism of action
16. Induce Resistance-Induced Systemic Resistance (ISR) and Systemic Acquired Resistance (SAR)
17. Protein defense responses and systemic responses

Practical Schedule

1. Collection of root exudates
2. Assay of root exudates – growth regulator/sugars
3. Studying the effect of root exudates on selected bacterial population – well diffusion assay

4. Isolation and purification of ectorhizosphere – rhizosphere and rhizoplane microorganisms
5. Isolation and purification of endophytic microorganisms from rice
6. Isolation and purification of phyllosphere microorganisms from rice
7. Isolation and purification of *Rhizobium*
8. Isolation and purification of *Frankia* from *Casuarina equisetifolia*
9. Visit to plantations and collection of ectomycorrhizal fruiting bodies, lichens, orchid and ericaceous mycorrhizae infected root samples
10. Examination of ectomycorrhizal fruiting bodies and isolation of ectomycorrhizal fungi
11. Examination of lichens associated with trees; orchidaceous and ericoid mycorrhizae fungal infection in plants
12. Examination of AM fungal infection in plants
13. Collection and examination of AM spores from soil
14. Testing *Agrobacterium tumefaciens* induced tumour formation in dicotyledonous plants in laboratory conditions
15. Testing *A. rhizogenes* induced root proliferation in laboratory conditions
16. Collection and examination of endosymbiont associated with *Azolla*
17. **Final Practical Examination**

Text Books

1. Kamal B., Normand B. and Fouad D. 2009. Plant-microbe interactions.
2. Chrispeels M.J. and D.E. Sadava. 2003. Plants, Genes, and Crop Biotechnology. Jones & Bartlett Publishers, Boston.

Reference Books

1. Susan I. 1992. Fungal-Plant Interactions, Chapman Hall,
2. George A. 2005. Plant Pathology, Academic Press, Fifth Edition.

E books

1. Nautiyal C.S. and Patrice, D. 2008. Molecular mechanism of Plant and Microbe Coexistence, Springer-Verlag Berlin Heidelberg
2. Lugtenberg, B. 2015. Principles of Plant-Microbe Interactions. Microbes for Sustainable Agriculture, Springer International Publishing Switzerland

AGM 355 QUALITY CONTROL OF BIOINOCULANTS (1+1)

Aim: This course aims in imparting knowledge on techniques of quality control of bioinoculants especially nitrogen fixers, nutrient solubilizers and mobilizers and plant growth promoters

Objectives

At the end of the course, the student will be able to learn the techniques of quality control of microbial inoculants.

UNIT I

Overview of bioinoculant production and quality control. Nitrogen fixers-types. P solubilizers and mobilizers. Potassium releasing bacteria. Microbial transformation of micronutrients zinc, sulphur, iron *etc.* PPFM – PGPR. Mass production methods. Introduction to quality control standards. FCO standards.

UNIT II

Quality control of *Rhizobium* inoculants-Purity checking of mother culture-morphological, biochemical and cultural characteristics of *Rhizobium*-Cross inoculation groups-recent classification-methods for testing nodulation-Estimation of nitrogen fixation – direct & indirect methods - Carrier based and liquid inoculants - FCO standards-scope for new formulations-quality control at different stages of production.

UNIT III

Quality control of *Azospirillum* and *Azotobacter* inoculants-purity checking of mother culture-morphological, biochemical and cultural characteristics-different species-Nitrogen fixation-ARA and microkjeldhal method-Quality control at different stages of production-Quality control of PSB- Purity checking of mother culture-Morphological, biochemical and cultural characteristics-Quantitative and qualitative assay for P solubilization-FCO standards.

UNIT IV

Endophytic nitrogen fixation in sugarcane-*Gluconacetobacter diazotrophicus*- Potash releasing bacteria-Characteristics-Mechanism of K release – PPFM - PGPR and bioinoculants for micronutrients-Quality control of Mycorrhizae-Ecto and Endo mycorrhiza-AMF-Morphological and cultural characteristics-root infection test and spore count-IP by MPN technique-mechanism of P mobilization-Hairy root organ culture.

UNIT V

Cyanobacterial biofertilizers-types-nitrogen fixing cyanobacteria-Heterocysts-role of akinetes in survival-soil based composite culture-new formulations-quality testing-population by MPN technique-*Azolla-Anabaena* symbiosis-spore inoculum production-rapid methods for quality

control of biofertilizers-molecular and immunological methods. Quality of biocontrol agents- *Pseudomonas* and *Trichoderma*.

Practical

Quality control of *Rhizobium*- *Azospirillum*- *Azotobacter*- morphological and biochemical characterization-Nodulation by Roll paper towel technique-N₂ fixation by ARA- sampling methods-population estimation in broth and inoculants by SPC/MPN methods- P solubilizing bacteria- Available P estimation in Pikosviskya's broth - Organic acid production by titrable acidity-Acid and alkaline phosphatase activity - Quantitative estimation- enumeration of population in hydroxy appetite medium- AM fungi- spore count by wet sieving method- root infection studies by staining with tryphan blue-IP estimation by MPN- hairy root culture Potassium releasing bacteria-quantification of K release-population estimation- PPFM, PGPR, SOB, Zn solubilizing bacteria-qualitative and quantitative assays-Cyanobacteria- Composite culture-enumeration of population by haemocytometer count and MPN technique-*Azolla*-determination of heterocyst frequency-sporocarps-spore inoculum production- rapid methods for quality control- Molecular methods –immunological methods.

Lecture schedule

1. Overview of bioinoculant production and quality control
2. Nitrogen fixing microbes-symbiotic , associative symbiotic, nonsymbiotic and endophytic nitrogen fixation-Mechanism of nitrogen fixation
3. Phosphate solubilizing and mobilizing microbes-Mechanism of action
4. Potash releasing bacteria -PPFM-PGPR- Zinc solubilizers- Sulphur oxidizers-mechanism
5. Mass production methods of bacterial, fungal and algal biofertilizers
6. Introduction to quality control standards-BIS-FCO standards - Sampling methods
7. Purity checking of *Rhizobium* mother culture-Morphological, biochemical and cultural characteristics-cross inoculation groups-recent classification
8. *In vitro* methods for testing nodulation- Roll paper towel technique-MPN counts, growth pouches/tubular pots -N₂ fixation-direct & indirect methods –N estimation by Microkjeldahl method,¹⁵N technique & ARA-principles & methods-enumeration of population at different stages of production
9. **Mid Semester Examination**
10. Criteria for selection of carrier material-FCO standards (2011) for carrier based and liquid Inoculants-scope for new formulations

11. Purity checking of *Azospirillum* and *Azotobacter* mother culture- morphological, biochemical and cultural characteristics-different species
12. Phosphobacteria-organisms involved-purity checking of mother culture- morphological, biochemical and cultural characteristics-quantification-available P estimation in broth-organic acid production-acid and alkaline phosphatase
13. FCO standards for *Azospirillum*, *Azotobacter* and phosphobacteria-quality control of inoculants at different stages of production- Rapid methods for quality control-molecular and immunological methods. Quality of biocontrol agents-*Pseudomonas* and *Trichoderma*.
14. Potassium releasing bacteria- organisms involved-morphological, biochemical and cultural Characteristics-mechanism and quantification of K release
15. Endophytic nitrogen fixation-*Glucanacetobacter diazotrophicus*-PPFM, PGPR-bioinoculants for micronutrients viz., sulphur, zinc and iron
16. Ecto and Endo mycorrhizae-AM fungi-morphological and cultural characteristics-mechanism of P mobilization-quality control-root infection-spore count and IP by MPN
17. Nitrogen fixing cyanobacteria-heterocysts-role of akinetes in survival-soil based composite culture-quality checking by MPN technique-*Azolla*-*Anabaena* symbiosis-heterocyst frequency-sporocarps-spore inoculum production-storage and shelf life

Practical schedule

1. Sampling methods and study of different inoculants-carrier based and liquid inoculants
2. Morphological, cultural and biochemical characterization of *Rhizobium*
3. Purity checking of *Rhizobium* strains-Leonard Jar, germination paper roll and tubular pot methods-MPN counts
4. Morphological, cultural and biochemical characterization of *Azospirillum*
5. Morphological, cultural and biochemical characterization of *Azotobacter*
6. Estimation of N₂ fixation- Acetylene Reduction Assay of nodules/cultures
7. Estimation of N₂ fixation in broth under *in vitro* conditions-Microkjeldahl method
8. Quantitative and qualitative assay for phosphobacteria-Available P estimation in Pikosviskya's broth by Olsen's method and organic acid production by titrable acidity
9. Assay of acid and alkaline phosphatase activity of phosphobacterial cultures
10. Enumeration of population of *Rhizobium* at different stages of production with Congo red

YEMA by SPC method

11. Enumeration of population of *Azospirillum* different stages of production with N free bromothymol blue (Nfb) semi solid medium by MPN method
12. Enumeration of population of *Azotobacter* at different stages of production with Waksman No.77 medium by SPC method
13. Enumeration of population of phosphobacteria at different stages of production with Sperber's hydroxy appetite medium by SPC method
14. Quality control of AM fungi-Spore count by wet sieving, root infection by Tryphan blue staining and Infective propagules by MPN
15. Potassium releasing bacteria- quantification of K release
16. Cyanobacterial inoculants-microscopic examination, determination of heterocyst frequency & population estimation by haemocytometer count and MPN method

17. Final Practical Examination

Text books

1. Maheshwari, D. K. and R.C.Dubey, 2008. Potential Microorganisms for Sustainable Agriculture - A techno-Commercial Perspective. I.K.International Publishing House Pvt.Ltd., New Delhi and Bangalore
2. Rai, M.K. 2006. Hand book of microbial biofertilizers. CRC press.

Reference Books

1. Bagyaraj, D.J. and A. Manjunath. 1990. Mycorrhizal symbiosis and plant growth, Univ. of Agricultural Sciences, Bangalore, India.
2. Kannaiyan S. 2002. Biotechnology of Biofertilisers. Kluwer Academic publishers & Narosa Publishing House.
3. Motsara M.R., Bhattacharya P., and Srivastava B. 1995. In: Biofertilizer Technology, Marketing and Uses -A Source Book cum Glossary. Fertilizer Development and Consultancy Organization, New Delhi.
4. Somasegaran, P. and H.J.hoben.1985. Methods in Legume-*Rhizobium* Technology, NifTAL Project and MIRCEN, University of Hawaii, Paia, USA
5. SubbaRao, N. S. 1993. Biofertilizers in Agriculture and Forestry. Oxford and IBH Publishing Co. Ltd., New Delhi.

e-books/ materials

1. Biofertilizers and Organic Fertilizers in Fertilizer (Control) Order, 1985
2. FNCA biofertilizer project group, 2006. Biofertilizer Manual, Japan Atomic Industrial Forum, MEXT, Tokyo, Japan.

SAC 351 DESIGNER FERTILIZER PRODUCTION (1+1)

Aim

This course is aimed to impart knowledge on the production, characterization, evaluation and economics of various designer fertilizers. Further, it also provides guidelines for establishing a designer fertilizer production unit.

Theory- Syllabus

UNIT I

Designer Fertilizers - Definitions - Concepts - Historical development - Scope and Need - Scenario of Multi nutrient deficiencies in soils and plants.

UNIT II

Designer Fertilizers – Classification- Production and Characterisation - Speciality / Customized fertilizer mixtures - Fortified fertilizers - Pelleted fertilizers - Multi nutrient liquid formulations.

UNIT III

Foliar Formulations - Leaf nutrient analysis -Value added fertilizers- enriched with organics/chelates- methods and guidelines for preparing designer Fertilizers- Filler materials- Industries and approved formulations.

UNIT IV

Quality of Designer fertilizers- Compatibility of fertilizer materials - issues in storability, hygroscopicity, clogging, etc. - Toxicity - Advantages and Disadvantages - Key Challenges - Crop response to designer fertilizers - Agricultural, Horticultural, high value crops- yield and quality - Soil health - Nutrient use efficiencies

UNIT V

Feasibility of using designer fertilizers for drip fertigation- Poly houses - roof gardening- Quality Standards- Specifications - Guidelines for Patenting, Licensing and Registration of newer products

Lecture Schedule

1. Designer Fertilizers - Definitions - Concepts -Historical development
2. Scope and need for Designer Fertilizers –Multi nutrient deficiencies in soils and plants - Critical limits - current scenario of multi nutrient disorders
3. Classification-Types- Speciality / Customised, Fortified and Pelleted fertilizers, Multi-nutrient liquid formulations
4. Speciality / Customised Fertilisers- Definitions- Production-characteristics- sources - suitability for crops -Merits and Demerits
5. Fortified fertilizers-Definitions- Production-characteristics-sources-suitability for crops-Merits and Demerits
6. Pelleted fertilizers - Definitions- Production- characteristics- sources- suitability for crops -Merits and Demerits
7. Multi nutrient liquid formulations - Definitions- Production- characteristics- sources - suitability for crops -Merits and Demerits
8. Foliar Formulations - Leaf nutrient analysis - organic and synthetic chelates
9. Mid semester examination
10. Value added fertilizers- enriched with organics/chelates
11. Methods and guidelines for preparing designer Fertilizers - Filler materials
12. Industries and Approved formulations and mixtures - Advantages and Disadvantages - Key Challenges
13. Quality of designer fertilizers - Compatibility of fertilizer materials - issues in storability, hygroscopicity, clogging, etc - Toxicity
14. Crop response to designer fertilizers - Agricultural and Horticultural crops- Yield and Quality- Soil health and Nutrient use efficiencies
15. Feasibility of using designer fertilizers for drip fertigation - Poly houses - roof gardening
16. Quality Standards-Specifications for designer fertilisers
17. Guidelines for Patenting, Licensing and Registration of newer products

Practical Syllabus

Preparation of Designer Fertilizer Mixtures for major agricultural, Horticultural and High value crops- Preparation of multi nutrient liquid formulations for drip fertigation, poly houses, roof gardening - Preparation of pelletised fertilizer mixtures for high value crops and roof gardening- Preparation of fortified fertilizer mixtures for major agricultural and horticultural crops- Preparation of value added fertilizers - Assessing the storability of the formulations and mixtures -Machineries in designer fertilizer production - Computation of cost effectiveness of the designer fertilizers - Visit to Designer Fertilizer

manufacturing Unit- Protocols for establishing a Designer Fertilizer Production Unit -
Procedures for Licensing, registration and Patenting.

Practical schedule

1. Preparation of Designer Fertilizer Mixtures for major agricultural crops: Rice/
Pulse
2. Preparation of Designer Fertilizer Mixtures for major horticultural crops : Banana/
Tapioca
3. Preparation of Designer Fertilizer Mixtures for high value crops : Turmeric/Cotton
4. Preparation of multi nutrient liquid formulations for drip fertigation :
Sugarcane/Turmeric
5. Preparation of multi nutrient liquid formulations for poly houses: Tomato/
Cucumber/Capsicum
6. Preparation of pelletised fertilizer mixtures for high value crops : Maize/
Turmeric/Cotton
7. Preparation of pelletised fertilizer mixtures for roof gardening : Chillies/ Brinjal/
Greens
8. Preparation of fortified fertilizer mixtures for major agricultural crops: Maize,
Groundnut
9. Preparation of fortified fertilizer mixtures for major horticultural crops: Onion,
Bhendi
10. Preparation of value added fertilizers
11. Assessing the storability of the mixtures and formulations
12. Machineries in designer fertilizer production
13. Computation of cost effectiveness of the designer fertilizers
14. Visit to Designer Fertilizer manufacturing Unit
15. Protocols for establishing a Designer Fertilizer Production Unit - Guidelines and
budget
16. Procedures for Licensing, registration and Patenting
17. Final practical examination

Outcome

- Acquiring technical knowhow and skills on preparing various designer fertilizer mixtures and formulations to establish a designer fertilizer production unit as an entrepreneur

References

1. Hagin, J. and B. Tucker (2012). Fertilization of Dry land and Irrigated Soils, Springer Science & Business Media, 06-Dec-2012 - Technology & Engineering, 190 pages
2. John Havlin, Samuel L. Tisdale, James D. Beaton, Werner L. Nelson (2013) Soil Fertility and Fertilizers: An Introduction to Nutrient Management, Pearson, 2013 - Technology & Engineering - 516 pages
3. Hari Lal Singh Tandon (2012) Fertilizer Management: Balance-efficiency-profitability Fertilizer Development and Consultation Organization, Arid regions agriculture - 187 pages
4. Anac,D. and P.Martin Pravel (1999).Improved crop quality by nutrient management , Kluwer Academic Publications, London Springer Science & Business Media, 30-Sep,1999 - Science - 310 pages
5. Gustafson,A.F. (2013) Handbook of Fertilizers - Their Sources, Make-Up, Effects, And Use,Read Books Ltd, 16-Apr-2013 - Technology & Engineering - 170 pages
6. Kolay,A.K.(2007) Manures and Fertilizers Atlantic Publishers & Dist, Fertilizers - 173 pages
7. UN Industrial Development Organization(1998), Fertilizer Manual, Int'l Fertilizer Development Center, Springer Science & Business Media, 31-Mar-1998 - Nature - 616 pages
8. Casper, M. S. (1973).Liquid fertilizers, Noyes Data Corp., 1973 - Technology & Engineering - 268 pages

SAC 353 Soilless Crop Production (1+1)

Aim : To introduce the techniques of growing plants without soil or minimal soil and outline the field of application of soilless cultivation systems in commercial greenhouse production. The course will provide a sufficient background on the physical and chemical properties of substrates and necessary knowledge on the chemistry of nutrient solutions to the students and enable them to calculate nutrient solutions of any desirable composition. Further, provide advanced knowledge on automated methods of nutrient and water cycling in various soilless cultivation systems.

Unit I

Protected agriculture overview and scope - Introduction to soilless cultivation of plants - History of solution culture - Present status of hydroponics - Nutrient requirements - Absorption of nutrients by the roots and interaction between the roots and the soil solution - Nutrient ratios - Effects of pH, EC and nutrient ratios on plant growth, yield and quality

Unit II

Containers - Grow bag / container media – formulations - their properties - Systems with aggregates as substrate - bag culture, container culture, trough culture, thin layer systems, other alternative systems - effect of volume and shape of container. Hydroponic systems - Systems involving solely water as a substrate - deep water culture, floating hydroponics, Nutrient Film Technique, plant plane hydroponics, aeroponics.

Unit III

Container media - Description of substrates -sand, gravel, rockwool, expanded minerals, pumice, zeolite, pyroclastic materials, peat, coir, tree bark, sawdust, wood fibres, etc. - Physical properties – impact of physical properties on irrigation management. Chemical properties - Container media analyses - Total and available nutrients - Microbiology and phytosanitation in container media

Unit IV

Composition of nutrient solution: Calculation of nutrient solutions for open systems/ closed systems: concept of drainage solution plus fresh water - Management of nutrient solution - Nutrient solution recycling - Irrigation control - characteristics of irrigation systems : capacity, uniformity - Delivery Systems : overhead systems, drip irrigation,

sub-irrigation. Irrigation scheduling : preset schedule, sensor-based schedule, transpiration-based schedule

Unit V

Equipments in Soilless culture - Automated delivery of nutrient solution - sensor based monitoring - moisture, nutrient, temperature and humidity sensors - Integrated system development for electronic control of equipments for irrigation and nutrient solution recycling. Nutrient solution disinfection - heating, UV-irradiation, chemical treatments by means of ozone, hydrogen peroxide, chlorine, iodine, etc. - membrane filtration - slow sand filtration

Lectures

1. Importance and scope of protected agriculture; Factors affecting crop growth under protected cultivation – temperature, light intensity, CO₂ and humidity
2. Introduction to soilless cultivation of plants; Various systems of soil-less crop production - bag culture, container culture, trough culture, thin layer systems, other alternative systems..
3. History of solution culture and Present methods of hydroponics - deep water culture, floating hydroponics, Nutrient Film Technique, plant plane hydroponics, aeroponics. - Basic needs and suitable crops for hydroponics.
4. Calculation of nutrient solution – mixing of nutrients, concentration and method, dose and time of application; Management of nutrient solution.
5. Fertilization - Nutrient requirements by crops, absorption of nutrients, nutrient ratios and its effect on crop growth and yield; Sensor based nutrient management.
6. Containers – kinds of substrates - sand, gravel, rockwool, expanded minerals, pumice, zeolite, pyroclastic materials, peat, coir, tree bark, sawdust, wood fibres, etc. – resources and methods of preparation.
7. Physical properties of substrates - air to water ratios, bulk density, particle size distribution, porosity, water release curves, hydraulic conductivity
8. Chemical properties of substrates - pH, electrical conductivity, ion sorption, ion exchange, concentration and composition of ions, cation exchange capacity.
9. Mid semester examination
10. Standardization of soil-less media and solution culture

11. Irrigation systems – Drip irrigation, sub-irrigation, matric suction irrigation and irrigation management.
12. Automated irrigation system – sensor based, transpiration based schedule
13. Equipments used in soil-less cultivation – various sensors used
14. Sterilization of substrate in grow bag media and disinfection of nutrient solution in hydroponics.
15. Day to day maintenance of soil-less system of crop production.
16. Suitability of crops for growing under green house cultivation; specific technology for raising vegetable crops under protected cultivation.
17. Economics and Business opportunities in soil-less system of crop production.

Practical

Propagation of plants for culture by hydroponics - testing seeds or cutting using media for adaptation to soilless culture - Growing crops in solution culture observation on growth and maturity phases - Growing crops in water culture and identification of plant nutrient deficiencies / nutritional stresses - Control of acidity of solutions by regulation of pH of nutrient solutions or by control of the sources of N - Formulation of nutrient solutions - Factors governing stability of Nutrient solutions - solubility of salts, purity of constituents, buffering of acidity, quality of water - Regulation of salinity in nutrient solutions – record of change in concentration of soluble salts in nutrient solutions - Culture of plants by nutrient film technique – growing plants using thin layers of water which flow by roots in a designed chamber -Evaluation of solid media in solution culture – growth of plants in solid media - sand, gravel, sawdust, perlite, etc.- Visit to commercial hydroponics greenhouse

Schedule

1. Preparation of soilless media using different substrates for grow bag method and sowing crop.
2. Preparation of solution culture for hydroponics and sowing crop.
3. Estimation of physical characteristics of grow bag media (cocopeat, vermiculite)
4. Determination of chemical properties like pH, EC, CN ratio of grow bag media.
5. Determination of water soluble and exchangeable nutrients in grow bag media.

6. Evaluation of porosity of medium based on moisture characteristic/ water retention curve.
7. Computation of container capacity, air filled porosity of grow bag media
8. Estimation of moisture constants by pressure plate apparatus and computation of available water and water holding capacity.
9. Preparation of common nutrient solution for hydroponics and drip system.
10. Preparation of fertilizer pellet packs for crops under matric suction irrigation.
11. Nutrient monitoring study using sensors in grow bag media.
12. Nutrient monitoring study using sensors in continuous recycling solution culture.
13. Assembling water lines and measurement of water consumption under drip / matrix suction irrigation.
14. Recording of operations involved in devices – peristaltic pump, filter pump, injectors, solenoid valves
15. Study of circuitry for sensor based nutrient monitoring system with wireless controls.
16. Visit to successful greenhouse cultivation system
17. Practical examination

Outcome

As population increases and arable land declines, people will turn to new technologies like soil-less crop production to feed the growing population. Soil-less culture will help to improve the yield and quality of the produce so that we can ensure food security of our country.

References

- Burt, C., K. O'Connor, and T. Ruehr. 1998. Fertigation. ITRC, California Polytechnic State University, San Luis Obispo, CA.
- FAO. 1990. Soilless Culture for Horticultural Crop Production.
- Jones, J. Benton. 1983. A Guide for the Hydroponic and Soilless Culture Grower. Timber Press.
- Jones, J. Benton. 2005. Hydroponics: A Practical Guide for the Soilless Grower. 2nd ed. CRC Press. Boca Raton.
- Kipp, J.A., G. Wever and C. de Kreij. 2000. International Substrate Manual. Elsevier, the Netherlands.
- Marschner, H. 1995. Mineral Nutrition of Higher Plants (2nd ed.). Academic Press, London.

Mengel, K. and Kirkby, E.A. 2001. Principles of Plant Nutrition (5th ed). Kluwer Academic Publishers, Dordrecht.

Raviv, M and Lieth, J.H. (eds.) 2008. Soilless Culture, Theory and Practice. Elsevier, London.

Schwartz, M. 1995. Soilless Culture Management. Springer-Verlag, Berlin.
Proceedings of the World Congress on Soilless Culture: Agriculture in the coming millennium. Editors, A. Bar-Tal, Z. Plau. 2001.

SAC 354 Instrumental Methods of Analysis (1+1)

Objective

To familiarize students with the design, operational principles and practical applications of modern instruments used for the quantitative analysis of soil, plant water, manure and fertilizer samples

Unit I

Principles of instrumentation- classification of instrumental methods – selection of instruments - Principles involved in digestion methods – dry ashing and wet digestion – open vs closed digestion - Block digester , microwave digester and IR digestion systems - components -operation - special consideration

Unit II

Automated methods – Principle and applications - Total N analyser, CN analyser

Unit III

Optical methods – spectrophotometry – visible, ultraviolet and infrared spectrometry - Principle - Instrumentation – sample handling and measurement - method development and validation - accuracy

Unit IV

Emission and absorption Spectroscopy - principles and applications - flame photometry, atomic absorption spectrophotometry, inductively coupled plasma emission spectrometry - instrumentation - features and operation of components - sample handling - errors - fault finding - trouble shooting

Unit V

Chromatography techniques – classification - paper chromatography, TLC - Gas chromatography- HPLC , GC – MS - principles - Instrumentation - sample preparation and handling - errors – trouble shooting

THEORY SCHEDULE

1. Basic principles in instrumental method of analysis
2. Principle and practice of digestion methods
3. Principle and practice of N analyser and CN analyser
4. Spectrophotometry : Types, Principle and instrumentation

5. Spectrophotometry: Sample handling and measurement, method development and validation and checking for accuracy
6. Emission spectroscopy (Flame Photometer) : Instrumentation, interferences, trouble shooting and maintenance
7. Absorption spectroscopy (Atomic Absorption Spectrophotometer) : Principle, instrumentation, features and operation of components
8. Absorption spectroscopy (Atomic Absorption Spectrophotometer) : Sample handling and measurement, errors due to molecular and ionic species, matrix effect and other interferences, trouble shooting and maintenance
9. Mid semester examination
10. Absorption spectroscopy (Inductively Coupled Plasma Emission Spectrometer) : Concepts and instrumentation
11. Absorption spectroscopy (Inductively Coupled Plasma Emission Spectrometer) : Preparation of samples and standards, interferences, trouble shooting and maintenance
12. Principle and practice of paper chromatography, Thin layer chromatography
13. Gas chromatography : Principle, Types and instrumentation
14. Gas chromatography : Operation ,sample handling, maintenance and trouble shooting and applications
15. High Performance Liquid Chromatography : Principle, Instrumentation and operation
16. High Performance Liquid Chromatography : Sample preparation , method development, maintenance and troubleshooting
17. GC – MS : Principle , instrumentation, Sample preparation , method development, maintenance and troubleshooting

PRACTICAL

Collection and processing of samples - Digestion of samples – block digester and microwave digester methods - N analyser – Spectrophotometry - UV –Vis Spectrophotometer -- Emission spectroscopy - Flame photometer – Absorption spectroscopy – Atomic Absorption Spectrophotometer (AAS) and Inductively Coupled Plasma Emission Spectrometer (ICP) – Chromatography - Gas Chromatography and High Performance Liquid Chromatography- Procedures for establishing a analytical laboratory

PRACTICAL SCHEDULE

1. Collection and processing of samples (Soil, plant, water, manure and fertilizer)
2. Digestion of samples by block digester/microwave digester and sample preparation for different analysis
3. N analyser : Calibration , sample estimation and results interpretation
4. UV -Vis spectrophotometer : Getting acquainted with parts of UV -Vis spectrophotometer and preparation of standards
5. UV -Vis spectrophotometer : Calibration, sample estimation (P/S/B) and results interpretation
6. Flame photometer : Getting acquainted with components of flame photometer and preparation of standards, calibration, sample estimation (P/S/B) and results interpretation

7. Getting acquainted with components of AAS and standards (micronutrients and heavy metal) preparation
8. Calibration and sample estimation for micronutrients and heavy metals by AAS
9. Getting acquainted with components of ICP
10. Calibration, sample estimation for elements by ICP and results interpretation
11. Analyte extraction from sample and separation for GC
12. Calibration, sample introduction and interpretation of results in GC
13. Analyte extraction from sample and separation for HPLC
14. Calibration, sample introduction and interpretation of results in HPLC
15. Visit to a analytical laboratory (Government/Private)
16. Procedures for establishing a analytical laboratory – Guidelines and budget
17. Practical examination

Outcome : At the end of the course the student will be able to handle sophisticated instruments. The knowledge earned will help in establishing a analytical laboratory for analyzing soil/plant/water/fertilizer/manure samples.

References

Chatwal, G. and Anand. 2005. *Instrumental Methods of Chemical Analysis*. Academic Press.

Muralidhar Rao, D., A.V.N Swamy and D. Dharaneeswara Reddy. 2013. *Instrumental Methods of Analysis*. CBS Publishers & Distributors Pvt.Ltd., New Delhi.

Skoog, A., Holler, F and Nieman, A. 2007. *Principles of Instrumental Analysis* (6th Ed.). Thomson Brooks/Cole publishing. ISBN: 9780030020780.

Willard M, Merritt LL & Dean JA. 1989. *Instrumental Methods of Analysis*. (7th Ed). Wadsworth Publishing Company, Boston, USA. ISBN :9780534001423.

e resources

<http://blogs.rediff.com/ticconcvibi1985/2013/05/12/download-ebook-modern-hplc-for-practicing-scientists-free/>

<http://cdn.intechopen.com/pdfs-wm/26275.pdf>

<http://faculty.ksu.edu.sa/Dr.almajed/Books/practical%20HPLC.pdf>

<http://web.uni-plovdiv.bg/plamenpenchev/mag/books/spectroscopy/PRIMER.PDF>

SAC 352 Rejuvenation of Deteriorated Lands (1+1)

Objective To impart knowledge related to various factors and processes of land deterioration and their rejuvenization techniques.

Theory syllabus

Unit I

Type, factors and processes of soil / land deterioration and its impact on soil productivity, including soil fauna, bio deterioration and environment. Land rejuvenization and conservation / management techniques; afforestation and silviculture and soil carbon restoration.

Unit II

Causes, reclamation and management of soil physical deterioration - surface and sub surface hard pans, shallow, slowly permeable and highly permeable and fluffy paddy soils; soil erosion, ravine and sand dune, coastal and seasonally inundated soils and mined land.

Unit III

Causes, reclamation and management of salt-affected soils – saline, sodic and saline sodic soils; acid and acid sulphate soils; laterite soils.

Unit IV

Extent, diagnosis and mapping of land deterioration by conventional and modern RS-GIS tools. Monitoring land deterioration by fast assessment and modern tools.

Unit V

Land use policy, incentives and participatory approach for reversing land deterioration; global issues for twenty first century.

Practical syllabus

Determination of gypsum requirement in sodic soils, lime requirement in acid soils and soil erodibility indices – field reclamation of saline, saline-sodic, sodic and acid soils. . Characterization of soil physical constraints - surface and sub surface hard pans, shallow, slowly permeable soils, and fluffy paddy soils - characterization of mined out and ravine lands, laterite, sand dune, coastal and seasonally inundated soils.

Theory-Lecture schedule

1. Type, factors and processes of soil / land deterioration.
2. Soil / land deterioration impact on soil productivity, including soil fauna, bio deterioration and environment.
3. Land rejuvenization and conservation techniques; Land configuration techniques; Surface / vertical mulching.
4. Afforestation and silviculture methods; Soil carbon restoration – use of industrial C- rich by products.
5. Causes, reclamation and management of soil physical deterioration - surface and sub surface hard pans, shallow, slowly permeable and highly permeable and fluffy paddy soils.
6. Causes, management of soil erosion.
7. Causes, reclamation and management of mined and ravine lands.
8. Causes, reclamation and management of sand dunes, coastal and seasonally inundated soils.
9. Mid semester examination
10. Causes, reclamation and management of saline and saline sodic soils
11. Causes, reclamation and management of sodic soils
12. Causes, reclamation and management of acid and acid sulphate soils.
13. Causes, reclamation and management of laterite soils.
14. Extent, diagnosis and mapping of land deterioration by conventional and modern RS-GIS tools.
15. Monitoring land deterioration by fast assessment and modern tools.
16. Land use policy, incentives and participatory approach for reversing land deterioration.
17. Global issues for twenty first century.

Practical schedule

1. Practicing field reclamation of saline soils and assessing its impact and cost benefit.
2. Practicing field reclamation of saline-sodic soils and assessing its impact and cost benefit.
3. Practicing field reclamation of sodic soils by gypsum application and assessing its impact and cost benefit.
4. Practicing field reclamation of sodic soils by press mud application and assessing its impact and cost benefit.
5. Practicing field reclamation of sodic soils by spent wash application and assessing its impact and cost benefit.
6. Practicing field reclamation of acid soils and assessing its impact and cost benefit.
7. Practicing management of surface and sub surface hard pans and assessing its impact and cost benefit.
8. Practicing management of slowly permeable and highly permeable and fluffy paddy soils and assessing its impact and cost benefit.
- 9 to 13. Field visit and characterization of eroded, ravine and mined out lands, sand dune, coastal, seasonally inundated and laterite soils and assessing its management impact and cost benefit.
14. Field visit to agro forestry and silviculture farms and assessing its impact on soil conservation.

15. Assessing the suitability of industrial byproducts for eco friendly recycling through soil conservation and rejuvenization.
16. Diagnosis and mapping of land degradation using RS and GIS tools.
17. Final Practical examination.

Outcome:

Students will gain knowledge to manage deteriorated lands by suitable ameliorative measures

References

- Biswas TD and Narayanasamy G. (Eds.). 1996. Soil Management in Relation to Land Degradation and Environment. Bull. Indian Soc. Soil Sci.17, New Delhi.
- Doran JW and Jones AJ. 1996. Methods of Assessing Soil Quality. Soil Science Society of America, Madison.
- Greenland DJ and Szabolcs I. 1994. Soil Resilience and Sustainable Land Use. CABI.
- Lal R, Blum WEH, Vailentine C and Stewart BA. 1997. Methods for Assessment of Soil Degradation.
- Sehgal J & Abrol IP. 1994. *Soil Degradation in India - Status and Impact*. Oxford & IBH.

SAC 355 Crop and Pesticide Chemistry (1+1)

Aim

To impart knowledge on the chemistry and nutritional significance of various field and horticultural crops so as to include them in the breeding and biofortification programmes towards nutritional security. This course will also impart knowledge on different pesticides, their nature and mode of action and their fate in soil so as to monitor their effect on the environment

Unit-I

Chemistry of Agricultural Crops: Chemical constituents of plants - Proximate and ultimate constituents - Chemical composition and nutritional quality of cereals, pulses and forage crops. Chemical composition and nutritional quality of oilseeds and sugarcane. Post harvest changes in Sugarcane.

Unit-II

Chemistry of Horticultural Crops, alkaloids and Essential oils: Chemical composition and nutritional quality of fruits, vegetables, spices, condiments, tuber crops and beverages. Post harvest changes in fruits. Chemistry of medicinal and aromatic plants.

Unit-III

Pesticides and its Formulations: Pesticides - Definition - Classification-Trends in pesticide use. Pesticide formulations -dusts, wettable powders, emulsifiable concentrate, granules. Insecticides - classification-. Characteristics, Mode of action and use of Organophosphates - Carbamates - Pyrethroids , Botanicals, Insect Growth Regulators and Newer insecticides.

Unit-IV

Fungicides and Herbicides : Fungicides - classification of fungicides -properties, mode of action of inorganic, organic and systemic fungicides - Rhodenticides- Zinc phosphide - Aluminium phosphide - Bromodiolone Herbicides - classification - properties - mode of action of inorganic and organic herbicides like phenoxy compounds, substituted ureas, amides, thiocarbamates, triazines, pyridines, imidazolines and sulphonyl ureas.

Unit-V

Pesticides and Environment : Insecticide Act and Insecticide Rules - Fate of pesticides in soil- Impact of pesticides on environment

Practical

Estimation of moisture, ash, crude protein, P, K and crude fibre in plant samples - Determination of reducing and non-reducing sugars in jaggery — Oil content in Groundnut- Estimation of total solids, ascorbic acid, titratable acidity in fruits- Phenols/ Mucilages in Vegetables - HCN content in Tapioca/ Sorghum - Analysis of pesticides - Physical tests - Bulk density, wettability, suspensibility, Emulsion stability -. Estimation of pesticide residues in soil, water, vegetables, fruits and pesticidal calculations. Visit to Pesticide Testing Laboratory.

Lecture Schedule

1. Chemical composition and nutritional quality of cereals and pulses - Rice, wheat, maize, minor millets, Red gram, blackgram, and soybean. Starch synthesis and protein synthesis
2. Chemical composition and nutritional quality of oil seed crops - Groundnut, sesame, sunflower, castor, coconut and palm.
3. Chemical composition and nutritional quality of sugarcane -Sucrose synthesis - Post harvest changes in sugarcane. Nutritional quality of forage crops.
4. Chemical composition and nutritional quality of fruits - Mango, banana, papaya, grapes, guava, apple and pomegranate. Chemistry of post harvest changes in fruits.
5. Chemical composition and nutritional quality of vegetables- Tomato, bhendi, brinjal, moringa, greens, cauliflower, radish and peas.
6. Chemical composition of spices and condiments (Turmeric, chillies, pepper, ginger, onion, garlic and Beverages (tea and coffee).Tuber crops- Potato & Tapioca
7. Alkaloids in medicinal plants (Cinchona, Gloriosa, Coleus and Aloe vera)
8. Pesticides - Definition - Classification-Trends in pesticide use
9. Mid semester Examination
10. Pesticide formulations - dusts - wettable powders, flowables, sprays –Emulsion concentrates – water soluble liquids - granules, fumigants and aerosols - characteristics and uses.
11. Insecticides classification -Characteristics, Mode of action and use of Organophosphates(Chlorpyrifos, Phorate, Dimethoate, Quinalphos and Profenophos)

12. Characteristics, Mode of action and use of Carbamates (Carbaryl, carbofuran, carbosulfan, aldicarb) and synthetic pyrethroids (Deltamethrin, Fenvalerate, Cypermethrin and Lambda-cyhalothrin)
13. Characteristics, Mode of action and use of Botanicals (nicotine and neem), Insect Growth Regulators (Novaluron, Buprobasin and GABA inhibitors) and *newer insecticides* (Neonicotinoids - Imidachloprid, Thiacloprid, Acetamiprid, Flupendiamide, Fipronil, Emamectin, Thiomethoxam, Indoxacarb, Chlorantraniliprole)
14. Fungicides - Classification – Inorganics (sulfur) and Organic fungicides (Chlorobenzene and Chlorothalnil) - Characteristics, mode of action and use
15. Characteristics, mode of action and use of Systemic fungicides (Benomyl, Carbendazim, Metalaxyl, Quinones, Diclonex, Dicarboximides –vincozolin).
16. Herbicides - Classification of herbicides - Characteristics, Mode of action and use of 2, 4-D, Sulfonyl ureas - Metsulfuron, Pyrosulfuron, Imidazoline, Alachlor, Butachlor, Oxyfluorfen, Fulchloralin, Pendimethalin, Atrazine, Paraquat and Glyphosate. Bisperipac sodium.
17. Fate of pesticides in soil-Impact of pesticides on the environment , Highlights of Insecticide Act -1968 and Insecticide Rules -1971

Practical schedule

1. Sampling, processing and storage of plant materials for chemical analysis - Estimation of moisture and ash content
2. Preparation of tri acid extracts of plant samples -Estimation of P and K in triple acid extract
3. Estimation of crude protein
4. Estimation of crude fibre
5. Estimation of reducing and non-reducing sugars in jaggery
6. Estimation of oil content in groundnut
7. Estimation of total solids, ascorbic acid and titrable acidity in fruit samples
8. Estimation of phenols in vegetables / Mucilages in Bhendi
9. Determination of HCN content in Tapioca/ forage sorghum

10. Determination of bulk density in dust formulation, wettability and suspensibility test in wettable powder formulations
11. Estimation of emulsion stability in EC formulation
12. Estimation of pesticide residues in soil using GC/HPLC
13. Estimation of pesticide residues in Water / Soft drinks using GC/HPLC
14. Estimation of pesticide residues in Vegetables using GC/HPLC
15. Estimation of pesticide residues in Fruits using GC/HPLC & Pesticide requirement calculations
16. Visit to Pesticide Testing Laboratory
17. Practical Examination

Text books

1. Konard Mengel *et al.*, 2006. Principles of Plant Nutrition. 5th Edition, Springer International.
2. Vasanthi ,D, T.Chitdeshwari, M.R.Latha, C.Sudhalakshmi and A.Vadivel,2014. Text book on Crop and Pesticide Chemistry Pp.310
3. Dhakshinamoorthy, M. 2000. An Introduction to Plant Biochemistry and Chemistry of Crops, Suri Associates, Coimbatore Pp.192
4. Handa, S, K,2004. Principles of Pesticides Chemistry, Agrobios (INDIA), Jodhpur.
5. Roy,N.K, 2002. Chemistry of Pesticides.CBSPublishers &Distributors, New Delhi.

References

1. Brijesh Tiwari and Narpinder Singh. 2012. Pulse Chemistry and Technology. Scientific Publishers, Jodhpur, India.
2. Hand book of Agriculture, 2009. Published by Indian Council of Agricultural Research, New Delhi 110 012.Pp.1583
3. Paul H.Moore and Frederik C.Botha. 2014. Sugarcane : Physiology, Biochemistry and Functional Biology (World Agriculture Series). Amazon Publishers, India.
4. Petra Marschner. 2012. Marschner's mineral nutrition of higher plants.^{3rd} Edition. ISBN: 978-0-12-384905-2.Elsevier publications.
5. Yash P. Kalra, 1998, Handbook of Reference Methods for Plant Analysis, Taylor & Francis Group, LLC, New york, London

6. Dodia, D.A., I.S.Patel and G.M.Patel. 2010. Botanical Pesticides for Pest Management. Amazon Publishers, India.
7. Gupta,A.2006. Pesticide Residue in Food commodities. Agrobios Publishers, Jodhpur.
8. Hassall, K.A. 2013. The Chemistry of pesticides, their metabolism, mode of action and uses in crop production. Scientific Publishers, Jodhpur, India.
9. Koul, O., G.S.Dhaliwal, S.Khohar and R.Singh. 2014. Biopesticides in Sustainable Agriculture. Progress and Potential. Amazon Publishers, India.
10. Mac Bean, C. 2013. The Pesticide Manual: A World Compendium. Amazon Publishers, India.
11. Sree Ramulu, U.S. 1979. Chemistry of Insecticides and Fungicides - Oxford and IBM Publishing Co, New Delhi.

e-references

1. www.apo-tokyo.org/OOe-books/AG-12_Leg
2. www.researchgate.net/...Chemical_composition...nutritional.../60b7d52b...
3. www.pulsecanada.com/food...nutrition/composition.../Canadian-Pea-Co...
4. www.wiley.com >... > Food Types > Functional Foods & Nutraceutical
5. www.iipr.res.in/csciences.html
6. www.aiou.edu.pk/FoodSite/Research%20Papers/48.pdf
7. www.plantphysiol.org/content/124/4/1532.full
8. www.amazon.in/Medicinal-Plants-Chemistry-Properties.../1578083958
9. www.tandfonline.com > List of Issues > Table of Contents
10. www.studiauniversitatis.ro/v15/pdf/20-2010/20.../SU20-2-10Caunii.pdf
11. www.fao.org/docrep/t0073e/t0073e01 .htm
12. www.pulseaus.com.au/.../Pulses%20Nutritional%20Value%20and%20Th..
13. www.researchgate.net/...Nutritional...oilseeds/.../9fcfd50633dab9e5d7.pd
14. www.uvm.edu/extension/cropsoil/wp-content/uploads/turner_refining.pdf
15. journals.cambridge.org/article_S0021859600040156
16. www.intechopen.com/.../pesticides-in-the-modern-world-trends-in-pestic...
17. cibrc.nic.in/insecticides_rules.htm
18. www.agcsa.com.au/static/atm_articles/html/3_3c.html

19. www.agf.gov.bc.ca > Agriculture > Pesticide Wise

Out come:

The students of undergraduate will gain knowledge on chemical composition and nutritional quality of various agricultural and horticultural crops. Proper understanding of chemistry of pesticides will be inculcated among the students. The students will acquire the skills on quality monitoring of crops and pesticides through practices.

PBG 351 Plant Genetic Resources : Collection, Conservation and Utilisation (1+1)

Aim:

To inculcate the knowledge and to expose the students for experiencing on the current status and trends of the science, practices and policies of plant genetic resources with respect to collection, conservation and beneficial use for food and agriculture

Syllabus for theory

Unit I Concepts in agrobiodiversity

Origin and history of agriculture; conservation and agricultural development; the central role of agrobiodiversity: trends and challenges; centers of crop plant origin and diversity; Crop Wild Relatives and their role in crop domestication; dynamics of domestication; concept of gene pool; agrobiodiversity regions of India- geographical distribution of crops of Indian origin

Unit II: Planning and execution of collection missions

Importance and need for collection missions; planning and execution of collection missions ; Logistics for collection; GIS- Information on collection sites; Passport data and its importance in collection missions; use of flora and herbaria for planning collections; National and international policies and procedures to be adopted in collection missions

Unit III : Concepts in PGR conservation

In situ and *ex situ* conservation: concept of biosphere reserves, gene sanctuaries, on-farm conservation, seed genebanks, field genebanks, botanical gardens, herbal gardens, *in vitro* repositories and cryo-genebanks; short-, medium- and long-term conservation, concept of base, active and working collections

Unit IV: International and national policies

International framework and PGR networks; International treaties and policies in relation to agrobiodiversity conservation and sustainable use; CBD , UPOV ; National policies and legal frame work; Organisations; Biodiversity authority; PPV and FR authority; National Biodiversity Authority, IP issues with respect to ITKs and communities safe guarding biodiversity

Syllabus for practical

Concepts and methods for computing biodiversity; Alpha and beta models; Calculation of species richness and endemism. Field visits to biosphere reserves – *in situ* methods of conservation. Visit to Field gene banks and understanding the modalities of conservation.

Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – Seed acquisition, processing, packing, barcoding, viability monitoring, registration and documentation. Visit to clonal gene banks. Biotechnology in conservation - *In vitro* methods of conservation. Exposure to cryoconservation methods. Concepts of PGR documentation and related web resources. Crop genetic diversity - Concepts of core and mini core collections. Molecular methods in PGR documentation and fingerprinting. Planning a Pre breeding programme with adapted and unadapted germplasm. Crop wild relatives - designing a Pre breeding programme with Crop Wild relatives. PGR-

Global and National policies. Learning the institutional policies and modalities in exchange and utilization of PGR at TNAU

Lecture schedule:

1. Origin and history of agriculture; conservation and agricultural development-the central role of agrobiodiversity – Methods to estimate biodiversity- trends and challenges
2. Crop diversity - centers of crop plant origin and diversity , Concepts of gene pools
3. Biodiversity hotspots - Global – Indian- Regions of agobiodiversity
4. Crop wild relatives – domestication of crops
5. Dynamics of crop domestication with special reference to Rice, Wheat, Maize and Tomato
6. Germplasm exploration and collection – Eco-Geographical issues to be considered in planning explorations- – use of GIS and GPS principles during explorations
7. Planning the logistics and execution of collection missions- Global collection missions and achievements
8. Sampling strategies to be adopted in collections – Data recording and handling including passport data, collection of herbaria of samples etc during collection missions
9. Mid semester examination
10. Historical issues related to PGR conservation, scientific basis of PGR conservation – Types :*In situ* and *ex situ* conservation:
11. *In Situ* Conservation methods : concept of biosphere reserves, gene sanctuaries, and on-farm conservation
12. *Ex Situ* conservation methods : Field gene banks and seed gene banks
13. *Ex Situ* conservation methods : Cryo conservation, *in vitro* conservation, DNA banks, conservation of microspores and mega spores
14. Concept of base, active and working collections, core collections and reference sets
15. International framework and PGR networks; International treaties and policies in relation to agrobiodiversity conservation and sustainable use; CBD and UPOV convention
16. National policies: National Biodiversity Authority, PPV & FR authority, IP issues with respect to ITKs and communities safe guarding biodiversity
17. Utilization of Plant Genetic Resources – Pre-breeding concepts for use of adapted and un-adapted germplasm in crop improvement programmes

Final theory Examination

Practical schedule:

1. Concepts and methods for computing biodiversity; Alpha and beta models;
2. Calculation of species richness and endemism
3. Field visits to biosphere reserves – *in situ* methods of conservation
4. Visit to Field gene banks and understanding the modalities of conservation

5. Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – Seed acquisition, processing, packing, barcoding
6. Visit to Ramiah Gene Bank to understand the concepts of medium and long term storage in seed gene banks – viability monitoring, registration and documentation
7. Visit to clonal gene banks
8. Biotechnology in conservation - *In vitro* methods of conservation
9. Exposure to cryoconservation methods
10. Concepts of PGR documentation and related web resources
11. Crop genetic diversity - Concepts of core and mini core collections
12. Molecular methods in PGR documentation and fingerprinting
13. Planning a Pre breeding programme with adapted and unadapted germplasm
14. Crop wild relatives - designing a Pre breeding programme with Crop Wild relatives
15. PGR- Global and National policies
16. Learning the institutional policies and modalities in exchange and utilization of PGR at TNAU
17. Practical Examination

1. Reference Text Books (one or two only):

1. Engels J.M. and Visser, L. (eds.). 2003. A guide to effective management of germplasm collections. IPGRI Handbook for Genebanks No.6. IPGRI, Rome, Italy.
2. Guarino, L., Rao, V.R. and Reid, R. (eds.). 1995. Collecting plant genetic diversity. CAB International, Wallingford, UK.

2. E references:

1. Food and Agriculture Organization (FAO) Commission on Biodiversity for Food and Agriculture; www.fao.org/biodiversity
2. http://www.bioversityinternational.org/uploads/tx_news/Molecular_markers_for_gene_bank_management_1082.pdf

SST 351 Seed entrepreneurship skill development and management 2 (1+1)

Objective

To update the knowledge and skills of students about the seed entrepreneurship, establishment and management of seed enterprise.

Unit I

Current status of National and Global Seed Industry and future perspectives Seed plan - Supply chain management - Significance of Seed Replacement Rate (SRR) Formal and informal seed sector - Seed trade - Role of International agencies in cross border trading (UPOV, OECD, ISTA, IPPC, AOSA, AOSCA) Varietal registration - Seed legislation and regulatory frame works (Seeds Act and Rules, Seed Control Order 1983 and amendments - PPV&FRA 2001).

Unit II

Seed promotional policies and programmes – NPSD 1988 - National seed policy 2002- Seeds Bill 2004 - EXIM policy - Domestic and International organizations involved in seed business - NSC, SSC,NSAI - International Seed Federation (ISF) - Asia Pacific Seed Association (APSA) - National seed quality regulatory system - Seed Certification Agencies - Notification - Seed Testing Laboratories - Central and Referral laboratories - Uniformity in quality regulation - International Seed quality regulatory system - OECD varietal certification - International Seed analysis certificate - ISTA membership and accreditation system.

Unit III

Seed Export and Import - procedures and guidelines - Germplasm exchange rules and directions - Plant Quarantine system and Sanitary and Phyto Sanitary (SPS) issues and measures for export and import of seeds - NBPGR, FAO - Human resource skill development - Financial requirements and their significance in successful seed company management - Seed company - Corporate affairs – components - Registration and establishment - Grant and issue of license - Seed preference assessment - Seed Rolling Plan - OPVs and hybrids - Role of Seed multiplication Ratio (SMR) - Varietal Replacement Rate (VRR).

Unit IV

Post harvest handling and machineries - Principles of seed drying, cleaning and upgradation - their significance in seed shelf life preservation - Risk coverage of carry over seeds - factors affecting seed storage - infrastructure facilities - ambient and advanced storage structures - Pre and post seed quality control - management checks and balances - their significance in seed trade - Linkages with various organizations for effective seed trade and business management - Farmers centric and market driven strategies for sustaining seed business and achieving seed security - Seed pricing - pricing policies of public and private agencies - strategies - methods and factors affecting prices.

Lecture schedule

1. Current status of National and Global Seed Industry and future perspectives
2. Seed plan - Supply chain management - Significance of Seed Replacement Rate (SRR) - Formal and informal seed sector
3. Seed trade - Role of International agencies in cross border trading (UPOV,OECD, ISTA IPPC, AOSA, AOSCA)
4. Varietal registration - Seed legislation and regulatory frame works (Seeds Act and Rules, Seed Control Order 1983 and amendments - PPV&FRA 2001)
5. Seed promotional policies and programmes – NPSD 1988 - National seed policy 2002- Seeds Bill 2004 - EXIM policy
6. Domestic and International organizations involved in seed business – NSC, SSC,NSAI- International Seed Federation (ISF) - Asia Pacific Seed Association(APSA)
7. National seed quality regulatory system- Seed Certification Agencies- Notification - Seed Testing Laboratories - Central and Referral laboratories - Uniformity in quality regulation.
8. **Mid semester examination**
9. International Seed quality regulatory system - OECD varietal certification - International Seed analysis certificate - ISTA membership and accreditation system
10. Seed Export and Import - procedures and guidelines - Germplasm exchange rules and directions - Plant Quarantine system and Sanitary and Phyto Sanitary (SPS) issues and measures for export and import of seeds - NBPGR, FAO etc
11. Human Resource skill development - Financial requirements and their significance in successful seed company management
12. Seed company - Corporate affairs - components- Registration and establishment - Grant and issue of license

13. Seed preference assessment - Seed Rolling Plan - OPVs and hybrids - Role of Seed multiplication Ratio (SMR) - Varietal Replacement Rate (VRR)
14. Post harvest handling and machineries - Principles of seed drying, cleaning and upgradation - their significance in seed shelf life preservation
15. Risk coverage of carry over seeds - factors affecting seed storage - infrastructure facilities - ambient and advanced storage structures
16. Pre and post seed quality control - management checks and balances - their significance in seed trade - Linkages with various organizations for effective seed trade and business management
17. Farmers centric and market driven strategies for sustaining seed business and achieving seed security - Seed pricing - pricing policies of public and private agencies - strategies - methods and factors affecting prices

Practical schedule

1. Basic components in establishment of seed company and organizational setup - staffing pattern - Registration procedures - Company - seed producer
2. Visit to public and private sector seed companies
3. Preparation of seed rolling plan
4. Visit to seed production plots - Study on field inspection procedures
5. Lay out and designing of seed processing unit and infrastructure
6. Lay out and designing of large and small scale seed storage facilities
7. Layout and establishment of seed quality control laboratory
8. Financial assistance for seed company establishment - Central Sector Seed Schemes - NABARD - Financing organizations
9. Preparation of projects for financial assistance for establishment of seed company - Micro, small and medium enterprises etc.,
10. Project preparation on establishment of seed processing units and seed storage infrastructures for financial assistance
11. Visit to seed dealer and retail outlets
12. Value addition techniques for seed preservation, marketing and minimizing post harvest quality losses
13. Management of legal issues related to seeds - stop sale order, punitive action, punishment and appeal - appellate authority
14. Visit to Seed Testing laboratories - Notified and ISTA accredited
15. Preparation of company status report and analysis of critical issues on sustaining seed business

16. Downgrading of seeds - Upgradation and improvement of seed standards of "sales returned seeds" for placing in market

17. Final practical examination

Out come

The students will gain knowledge about the National and International seed trade and seed quality regulation system.

References

1. Dadheech, P.K. 1996. Seed Programming. Management system and concepts. Lok Sahitna Kendra. Jodhpur.
2. Gurudev G Singh and Asokan, S.R., 1997. Management of seed activity. Oxford and IBH Publishing Co., New Delhi.
3. Ramalingam, C. K.Sivasubramaniam and A. Vijayakumar. 1997. A Guide to seed legislation. Rassi Computers, Madurai
4. Trivedi, R.K. and M. Gunasekaran. 2013. Indian Minimum Seed Certification Standards. Dept. of Agriculture & Cooperation, Ministry of Agriculture, Govt. of India, New Delhi.
5. Compendium of Seed Legislation. 2013. Dept. of Agriculture & Cooperation, Ministry of Agriculture, Govt. of India, New Delhi

Online references

1. www.fao.org
2. www.seednet.gov.in
3. www.agricoop.nic.in
4. www.agri.nic.in

NEM 351 Commercial Production of Nematode Antagonistic bio-agents 2 (1+1)

Aim: The course is aimed to inculcate entrepreneurial skill to students and expose them to different aspects in commercial production of biocontrol agents, marketing and economics. The course will also provide hands on training to students in mass culturing and marketing of nematode antagonistic fungal organisms.

SYLLABUS

THEORY:

Unit 1

Concepts and definition of biocontrol agents – Economic yield loss caused by plant parasitic nematodes in agricultural and horticultural crops – Ecofriendly management of plant parasitic nematodes - Types of bioagents – Fungal antagonistic organisms – Applications – Dosage - Commercial formulations available in India – Merits and demerits.

Unit 2:

Paecilomyces lilacinus – *Pochonia chlamydosporia* - Morphological identification of colony – Conidiaspore and Chlamydospores identification – Isolation from eggs - Mode of action – Host range - virulence – effect of biotic and abiotic factors on growth – compatibility with chemical inputs.

Unit 3:

Sterilization – Types and Preparation of sterilizing agents – Equipments used – Principle of autoclave – Principle of Laminar Air Flow Chamber – Fermentor – Laboratory up-keep – Haemocytometer – Assessment of spore load – colony forming units.

Unit 4:

Preparation different culture media – Maintenance of pure culture - Mother culture – Subculturing – Mass culturing techniques – solid and liquid formulations - Commercial Formulations – Carrier materials – Packing – Quality control and shelf life.

Unit 5:

Market demand analysis - Economics – Establishment of pilot plant – Infrastructure - Budget preparation – Marketing and cost-benefit analysis – risk analysis – Environmental impact test with vertebrate and invertebrate organisms.

LECTURE SCHEDULE

1. Introduction - Economic yield loss caused by plant parasitic nematodes in agricultural and horticultural crops. Concepts and definition of biocontrol agents.
2. Types of bioagents – Fungal antagonistic organisms – Applications – Dosage - Commercial formulations available in India – Merits and demerits
3. *Purpureocillium lilacinum* (= *Paecilomyces lilacinus*) – Morphological identification of colony, phialids and conidia spore
4. *Pochonia chlamydosporia* - Morphological identification of colony. Conidia spore and Chlamydospores identification
5. Isolation of *P. lilacinum* and *P. chlamydosporia* from nematode eggs and mode of action.
6. Host range of *P. lilacinum* and *P. chlamydosporia* - virulence and effect of biotic and abiotic factors on growth – compatibility with chemical inputs
7. Sterilization – Types and Preparation of sterilizing agents – Equipments used for production of bioagents.
8. Principle of Autoclave, Laminar Air Flow Chamber and Fermentor
9. Laboratory up-keep – Preparation of cleaning solutions – preparation of stock solutions
10. Haemocytometer – Assessment of spore load – colony forming units.
11. Preparation of different culture media – Maintenance of pure culture - Mother culture – Subculturing
12. Mass culturing techniques – solid and liquid formulations – Shelf life
13. Commercial Formulations – Carrier materials – Packing – Quality control and shelf life.

14. Market demand analysis – Economics
15. Establishment of pilot plant – Infrastructure requirement – cost analysis
16. Budget preparation – Marketing and cost-benefit analysis – risk analysis
17. Environmental impact test with vertebrate and invertebrate organisms

PRACTICAL:

Plant parasitic nematodes eggs isolation – Eggs parasitization tests with *Paecilomyces lilacinus* and *Pochonia chlamydosporia* – Fungal specific media preparation – Pure culture - Preparation of common culture media – subculturing of *Paecilomyces lilacinus* - subculturing of *Pochonia chlamydosporia* – Preparation of broth – Inoculation – Incubation in mechanical shaker – *In vitro* bioefficacy test on root knot nematode - Fermentation process – Haemocytometer – Assessing spore load in broth – Preparation of commercial formulation – Quality control test - Packing – Analyzing market potential and demand – Conducting environmental impacts test with termites, saprophytes, honeybees, earthworm etc., - Visit to commercial production unit (HRS, Ooty).

1. Isolation of eggs of plant parasitic nematodes.
2. Eggs parasitization tests with *Purpureocillium lilacinum*
3. Eggs parasitization tests with *Pochonia chlamydosporia*
4. Fungal specific media preparation – Pure culture - Preparation of common culture media
5. Subculturing of *P. lilacinum* and *Pochonia chlamydosporia*
6. Preparation of broth – Inoculation – Incubation in mechanical shaker
7. *In vitro* bioefficacy test on root knot nematode with *Purpureocillium lilacinum*
8. *In vitro* bioefficacy test on root knot nematode with *Pochonia chlamydosporia*
9. Fermentation process
10. Haemocytometer – Assessing spore load in broth
11. Preparation of commercial formulation
12. Quality control test Packing
13. Analyzing market potential and demand
14. Conducting environmental impacts test with termites, saprophytes,
15. Conducting environmental impacts test with honeybees, earthworm
16. Visit to commercial production unit (HRS, Ooty).
17. Practical Examination

